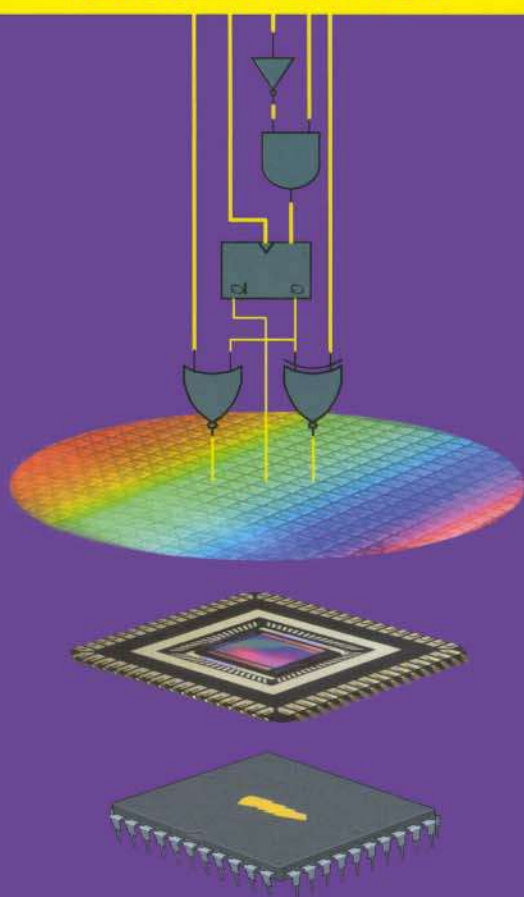


ADVANCED ASIC CHIP SYNTHESIS

Using Synopsys® Design Compiler™
Physical Compiler™ and PrimeTime®

SECOND EDITION



Himanshu Bhatnagar

Kluwer Academic Publishers

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**Using Synopsys® Design Compiler™
Physical Compiler™ and PrimeTime®**

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KLUWER ACADEMIC PUBLISHERS
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*To my wife Nivedita
and my daughter Nayan*

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Foreword

The semiconductor industry has a proven track record of quickly reducing reference to IC design scale to the ridiculously irrelevant. We, as a group, quickly saturated our terminology to refer to levels of integration as we applied the term "Large Scale Integration" (LSI) in the mid 80's to chips containing more than 1,000 transistors and moved to the more progressive "Very Large Scale" (VLSI) as we passed into the 10,000 to 100,000 transistor territory only a year or two later. A few more attempts at renaming our design output with terms such as ULSI (Ultra-Large Scale Integration) were fortunately left in the annals of history as the more insightful realized that the consequences of Moore's Law would quickly require us to move beyond the confines of the English language to create appropriate superlatives. We, however, could not resist changing our conception of the design task by coining the phrase "System on a Chip" in the early to mid 90's to convey the understanding that these levels of integration allowed for the development of more than complex electronic components but self contained information processing systems. Once again, however, we find ourselves struggling with the reality that the "systems" referred to only 3 to 4 years ago are today barely enough to fill the pad ring of a modest pin count device.

We should not be surprised, therefore, that some in the design community are recognizing the need to again rethink and redefine the metrics that scope the modern IC design task. Now, however, instead of focusing on the collection

of transistors or functions as a metric of design production, this group has moved to focus on our most precious commodity, time. For them, today's design task is being defined by the window of opportunity in which the design output is relevant, usually a period that cannot extend beyond 12 - 18 months. This group, therefore, is focused on the tools and techniques that can raise the design productivity to the point that the transistor counts, functions and subsystems that can fill the silicon can be reliably designed and characterized in this amount of time. We should not be caught completely by surprise therefore if our lexicon begins to define levels of integration with terms such as SMSI ("Six Month Scale Integration") or TMSI ("Twelve Month Scale Integration") perhaps.

This book sets itself squarely in the middle of this effort as it explores and conveys a collection of tools and techniques focused on dramatically reducing the time required to complete the IC design task and get an IC product to market. The author, Mr. Bhatnager, takes a set of today's most productive IC design tools and exposes ways in which these tools can be applied to further streamline the full design process. These techniques challenge the designer to move beyond linear high-level design flows that utilize HDL languages for design description, synthesis to create gate and transistor implementation and timing analysis. This book exposes practical techniques by which more information can either be introduced sooner in the design flow or fed back quicker in order to both reduce the number of iterations and the complexity associated with each one. The result is techniques that lead to better quality designs sooner.

Today's semiconductor business operates in the world of compressed time and hyper-competition. To compete effectively in this world every designer and every design team is well advised to focus on continually improving their time-to-market metric. This book will serve the advanced student in VLSI design as well as the seasoned practitioner in this quest.

Mr. F. Matthew Rhodes
Sr. Vice President and General Manager
Personal Computing Division
Conexant Systems, Inc.

Preface

This second edition of this book describes the advanced concepts and techniques used towards ASIC chip synthesis, physical synthesis, formal verification and static timing analysis, using the Synopsys suite of tools. In addition, the entire ASIC design flow methodology targeted for VDSM (Very-Deep-Sub-Micron) technologies is covered in detail.

The emphasis of this book is on real-time application of Synopsys tools, used to combat various problems seen at VDSM geometries. Readers will be exposed to an effective design methodology for handling complex, sub-micron ASIC designs. Significance is placed on HDL coding styles, synthesis and optimization, dynamic simulation, formal verification, DFT scan insertion, links to layout, physical synthesis, and static timing analysis. At each step, problems related to each phase of the design flow are identified, with solutions and work-around described in detail. In addition, crucial issues related to layout, which includes clock tree synthesis and back-end integration (links to layout) are also discussed at length. Furthermore, the book contains in-depth discussions on the basics of Synopsys technology libraries and HDL coding styles, targeted towards optimal synthesis solution.

Target audiences for this book are practicing ASIC design engineers and masters level students undertaking advanced VLSI courses on ASIC chip design and DFT techniques.

This book is not intended as a substitute or a replacement for the Synopsys reference manual, but is meant for anyone who is involved in the ASIC design flow. Also, it is useful for those designers (and companies) who do not have layout capability, or their own technology libraries, but rely on outside vendors for back-end integration and final fabrication of the device. The book provides alternatives to traditional method of netlist hand-off to outside vendors because of various problems related to VDSM technologies. It also addresses solutions to common problems faced by designers when interfacing various tools from different EDA tool vendors.

All commands have been updated to Tcl version of Design Compiler in this edition of the book. The commands have been changed to reflect the most up-to-date version (2000.11—SP1) of Synopsys suite of tools.

Overview of the Chapters

Chapter 1 presents an overview to various stages involved in the ASIC design flow using Synopsys tools. The entire design flow is briefly described, starting from concept to chip tape-out. This chapter is useful for designers who have not delved in the full process of chip design and integration, but would like to learn the full process of ASIC design flow.

Chapter 2, outlines the practical aspects of the ASIC design flow as described in Chapter 1. Beginners may use this chapter as a tutorial. Advanced users of Synopsys tools may benefit by using this chapter as a reference. Users with no prior experience in synthesis using Synopsys tools should skip this chapter and return to it later after reading the rest of the chapters.

The basic concepts related to synthesis are described in detail in Chapter 3. These concepts introduce the reader to synthesis terminology used throughout the later chapters. Readers will find the information provided here useful by gaining a basic understanding of these tools and their environment. In addition to describing the purpose of each tool and their setup, this chapter also focuses on defining objects, variables, attributes and compiler directives used by the Design Compiler.

Chapter 4 describes the basics of the Synopsys technology library. Designers usually do not concern themselves with the full details of the technology

library, as long as the library contains a variety of cells with different drive strengths. However, a rich library usually determines the quality of synthesis. Therefore, the intent of this chapter is to describe the Synopsys technology library from the designer's perspective. Focus is provided on delay calculation method and other techniques that designers may use in order to alter the behavior of the technology library, hence the quality of the synthesized design.

Proper partitioning and good coding style is essential in obtaining quality results. Chapter 5 provides guidelines to various techniques that may be used to correctly partition the design in order to achieve the optimal solution. In addition, the HDL coding styles is covered in this chapter that illustrates numerous examples and provides recommendations to designers on how to code the design in order to produce faster logic and minimum area.

The Design Compiler commands used for synthesis and optimization are described in Chapter 6. This chapter contains information that is useful for the novice and the advanced users of Synopsys tools. The chapter focuses on real-world applications by taking into account deviations from the ideal situation i.e., "Not all designs or designers, follow Synopsys recommendations". The chapter illustrates numerous examples that help guide the user in real-time application of the commands.

Chapter 7 discusses optimization techniques in order to meet timing and area requirements. Comparison between older version of Design Compiler and the new version is highlighted. Emphasis is provided on the new optimization technique employed by Design Compiler called "TNS". Also, detailed analysis on various methods used for optimizing logic is presented. In addition, different compilation strategies, each with advantages and disadvantages are discussed in detail.

DFT techniques are increasingly gaining momentum among ASIC design engineers. Chapter 8 provides a brief overview of the different types of DFT techniques that are in use today, followed by detailed description on how devices can be made scannable using Synopsys's Test Compiler. It describes commands used for inserting scan through Design Compiler. A multitude of guidelines is presented in order to alleviate the problems related to DFT scan insertion on a design.

Chapter 9 discusses the links to layout feature of Design Compiler. It describes the interface between the front-end and back-end tools. Also, this chapter provides different strategies used for post-layout optimization of designs. This includes in-place and location based optimization techniques. Furthermore, a section is devoted to clock tree insertion and problems related to clock tree transfer to Design Compiler. Various solutions to this common problem are described. This chapter is extremely valuable for designers (and companies) who do not possess their own layout tool, but would like to learn the place and route process along with full chip integration techniques.

The introduction of Physical Compiler dramatically changed the traditional approach to synthesis. Chapter 10 describes this flow in detail. The chapter describes various methods of achieving optimal results using Physical Compiler. In order to understand the Physical Compiler flow, readers are advised to read all chapters related to the traditional flow (especially Chapter 9) before reading this chapter. This will help correlate the topics described in this chapter to the traditional flow. Various example scripts are provided in this chapter illustrating the usage of this novel tool.

Chapter 11, titled “SDF Generation: for Dynamic Timing Simulation” describes the process of generating the SDF file from Design Compiler or PrimeTime. A section is devoted to the syntax of SDF format, followed by detailed discussion on the process of SDF generation, both for pre and post-layout phases of the design. In addition, few innovative ideas and suggestions are provided to facilitate designers in performing successful simulation. This chapter is useful for those designers who prefer dynamic simulation method to formal verification techniques, in order to verify the functionality of the design.

Chapter 12 introduces to the reader, the basics of static timing analysis, using PrimeTime. This includes a brief section devoted to Tcl language that is utilized by PrimeTime. Also described in this chapter are selected PrimeTime commands that are used to perform static timing analysis, and also facilitate the designer in debugging the design for possible timing violations.

The key to working silicon usually lies in successful completion of static timing analysis performed on a particular design. This capability makes static

timing analysis one of the most important steps in the entire design flow and is used by many designers as a sign-off criterion to the ASIC vendor. Chapter 13 is devoted to several basic and advanced topics on static timing analysis, using PrimeTime. It effectively illustrates the usage of PrimeTime, both for the pre and the post-layout phases of the ASIC design flow process. In addition, numerous examples on analyzing reports and suggestions on various scenarios are provided. This chapter is useful to those who would like to migrate from traditional methods of dynamic simulation to the method of analyzing designs statically. It is also helpful for those readers who would like to perform in-depth analysis of the design through PrimeTime.

Conventions Used in the Book

All Synopsys commands are typed in “Ariel” font. This includes all examples that contain synthesis and timing analysis scripts.

The command line prompt is typed in “Courier New” font. For example:

```
dc_shell>          and,          pt_shell>
```

Option values for some of the commands are enclosed in < and >. In general, these values need to be replaced before the command can be used. For example:

```
set_false_path -from <from list> -to <to list>
```

The “\” character is used to denote line continuation, whereas the “|” character represents the “OR” function. For example:

```
compile -map_effort low | medium | high \
      -incremental-mapping
```

Wherever possible, keywords are *italicized*. Topics or points, that need emphasis are underlined or highlighted through **bold** font.

Acknowledgements

I would like to express my heartfelt gratitude to a number of people who contributed their time and effort towards this book. Without their help, it would have been impossible to take this enormous undertaking.

First and foremost, a special thanks to my family, who gave me continuous support and encouragement that kept me constantly motivated towards the completion of this project. My wife Nivedita, who patiently withstood my nocturnal and weekend writing activities, spent enormous amount of time towards proofreading the manuscript and correcting my "Engineers English". I could not have accomplished this task without her help and understanding.

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During SNUG 2000, I met Cliff Cummings (President & Consultant, Sunburst Designs). Cliff is very well known in this industry as an expert in Verilog RTL coding and synthesis. I asked him to help me review certain chapters of my book. I would like to thank him for providing valuable suggestions, which I incorporated in Chapter 5.

Writing the second edition of this book took longer than previously anticipated. The main reason was the introduction of Physical Compiler. I wanted to enhance the book but did not want to write about something that was not mature. Carl Harris of Kluwer Academic Publishers understood this and supported me throughout the project. His understanding even when I kept on delaying the book is appreciated.

A final word, “Thank you Mom and Dad for your endless faith in me”.

Himanshu Bhatnagar
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About The Author

Himanshu Bhatnagar is an ASIC Design Group Leader at Conexant Systems, Inc. based in Newport Beach, California U.S.A. Conexant Systems Inc. is the world's largest independent company focused exclusively on providing semiconductor products for communication electronics. Himanshu has been instrumental in defining the next generation ASIC design flow methodologies using latest high performance tools from Synopsys and other EDA tool vendors.

Before Joining Conexant, Himanshu worked for ST Microelectronics in Singapore and the corporate headquarters based in Grenoble, France. He completed his undergraduate degree in Electronics and Computer Science from Swansea University (Wales, U.K), and his masters degree in VLSI design from Clemson University, (South Carolina, USA).

ASIC DESIGN METHODOLOGY

As deep sub-micron semiconductor geometries shrink, traditional methods of chip design have become increasingly difficult. In addition, an increasing numbers of transistors are being packed into the same die-size, making validation of the design extremely hard, if not impossible. Furthermore, under critical “time-to-market” pressure the chip design cycle has remained the same, or is constantly being reduced. To counteract these problems, new methods and tools have evolved to facilitate the ASIC design methodology.

The main function of this chapter is to bring to the forefront different stages involved in chip design as we move deeper into the sub-micron realm. Various techniques that improve the design flow are also discussed.

Since the last edition of this book, Synopsys introduced another tool called Physical Compiler. In the tool, synthesis and placement now are more tightly coupled. Consequently, there is a dramatic change in the traditional design flow. This chapter stresses the importance of the new techniques to the reader, and explains the necessity of these techniques in the design flow to achieve the maximum benefit, by reducing the overall cycle time. Since the tool is fairly new to the IC design world, and as yet, not embraced 100% by

the ASIC design community, both the traditional and the new flows are discussed.

This chapter focuses on the entire synthesis based ASIC design flow methodology, from RTL coding to the final tape-out. Both the traditional and the Physical Compiler based flow are discussed.

1.1 Traditional Design Flow

The traditional ASIC design flow contains the steps outlined below. Figure 1-1 illustrates the flow chart relating to the design flow described below. Subsequent chapters describe in detail synthesis related topics.

1. Architectural and electrical specification.
2. RTL coding in HDL.
3. DFT memory BIST insertion, for designs containing memory elements.
4. Exhaustive dynamic simulation of the design, in order to verify the functionality of the design.
5. Design environment setting. This includes the technology library to be used, along with other environmental attributes.
6. Constraining and synthesizing the design with scan insertion (and optional JTAG) using Design Compiler.
7. Block level static timing analysis, using Design Compiler's built-in static timing analysis engine.
8. Formal verification of the design. RTL compared against the synthesized netlist, using Formality.
9. Pre-layout static timing analysis on the full design through PrimeTime.
10. Forward annotation of timing constraints to the layout tool.
11. Initial floorplanning with timing driven placement of cells, clock tree insertion and global routing
12. Transfer of clock tree to the original design (netlist) residing in Design Compiler.
13. In-place optimization of the design in Design Compiler.
14. Formal verification between the synthesized netlist and clock tree inserted netlist, using Formality.

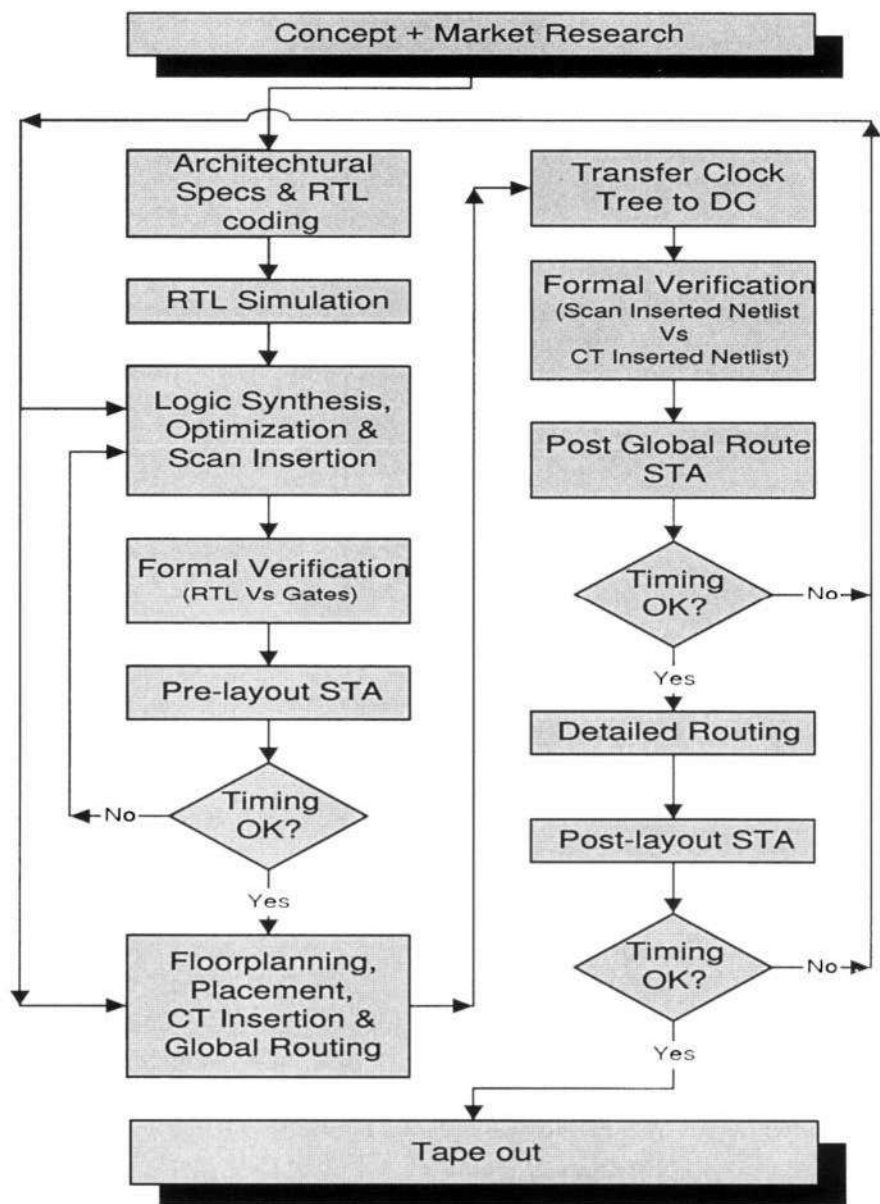


Figure 1-1. Traditional ASIC Design Flow

15. Extraction of estimated timing delays from the layout after the global routing step (step 11).
16. Back annotation of estimated timing data from the global routed design, to PrimeTime.
17. Static timing analysis in PrimeTime, using the estimated delays extracted after performing global route.
18. Detailed routing of the design.
19. Extraction of real timing delays from the detailed routed design.
20. Back annotation of the real extracted timing data to PrimeTime.
21. Post-layout static timing analysis using PrimeTime.
22. Functional gate-level simulation of the design with post-layout timing (if desired).
23. Tape out after LVS and DRC verification.

Figure 1-1, graphically illustrates the typical ASIC design flow discussed above. The acronyms STA and CT represent static timing analysis and clock tree respectively. DC represents Design Compiler.

1.1.1 Specification and RTL Coding

Chip design commences with the conception of an idea dictated by the market. These ideas are then translated into architectural and electrical specifications. The architectural specifications define the functionality and partitioning of the chip into several manageable blocks, while the electrical specifications define the relationship between the blocks in terms of timing information.

The next phase involves the implementation of these specifications. In the past this was achieved by manually drawing the schematics, utilizing the components found in a cell library. This process was time consuming and was impractical for design reuse. To overcome this problem, hardware description languages (HDL) were developed. As the name suggests, the functionality of the design is coded using the HDL. There are two main HDLs in use today, Verilog and VHDL. Both languages perform the same function, each having their own advantages and disadvantages.

There are three levels of abstraction that may be used to represent the design; Behavioral, RTL (Register Transfer Level) and Structural. The Behavioral

level code is at a higher level of abstraction. It is used primarily for translating the architectural specification, to a code that can be simulated. Behavioral coding is initially performed to explore the authenticity and feasibility of the chosen implementation for the design. Conversely, the RTL coding actually describes and infers the structural components and their connections. This type of coding is used to describe the functionality of the design and is synthesizable to form a structural netlist. This netlist comprises of the components from a target library and their respective connections; very similar to the schematic based approach.

The design is coded using the RTL style, in either Verilog or VHDL, or both. It can also be partitioned if necessary, into a number of smaller blocks to form a hierarchy, with a top-level block connecting all lower level blocks.

☛ Synopsys recently introduced Behavior Compiler, capable of synthesizing Behavior level style of coding. Since this is a major topic of discussion and is not relevant to this book, only RTL related synthesis is covered in this book.

1.1.2 Dynamic Simulation

The next step is to check the functionality of the design by simulating the RTL code. All currently available simulators are capable of simulating the behavior level as well as RTL level coding styles. In addition, they are also used to simulate the mapped gate-level design.

Figure 1-2, illustrates a partitioned design surrounded by a test bench ready for simulation. This test bench is normally written in behavior HDL while the actual design is coded in RTL.

Usually the simulators are language dependent (either Verilog or VHDL), although there are a few simulators in the market, capable of simulating a mixed HDL design.

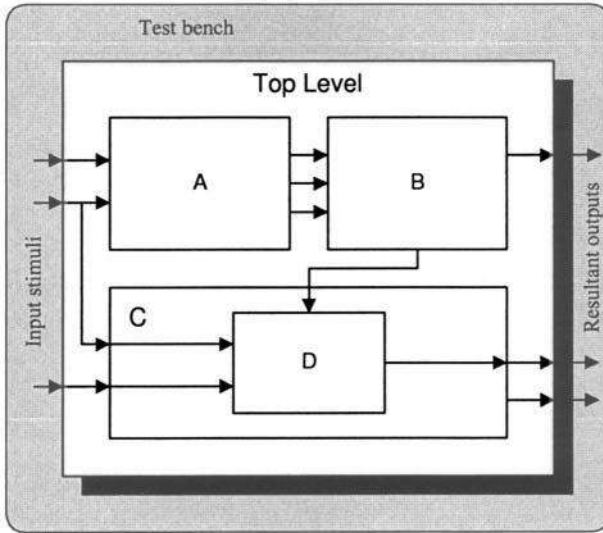


Figure 1-2. Design Hierarchy Example

The purpose of the test bench is to provide necessary stimuli to the design. It is important to note that the coverage of the design is totally dependent on the number of tests performed and the quality of the test bench. This is the reason why a sound test bench is extremely critical to the design. During the simulation of the RTL, the component (or gate) timing is not considered. Therefore, to minimize the difference between the RTL simulation and the synthesized gate-level simulation at a later stage, the delays are usually coded within the RTL source, usually for sequential elements.

1.1.3 Constraints, Synthesis and Scan Insertion

For a long time, the HDLs were used for logic verification. Designers would manually translate the HDL into schematics and draw the interconnections between the components to produce a gate-level netlist. With the advent of synthesis tools, this manual task has been rendered obsolete. The tool has taken over and performs the task of reducing the RTL to the gate-level netlist. This process is termed as synthesis.

Synopsys's Design Compiler (from now on termed as, DC) is the de-facto standard and by far the most popular synthesis tool in the ASIC industry today.

Synthesizing a design is an iterative process and begins with defining timing constraints for each block of the design. These timing constraints define the relationship of each signal with respect to the clock input for a particular block. In addition to the constraints, a file defining the synthesis environment is also needed. The environment file specifies the technology cell libraries and other relevant information that DC uses during synthesis.

DC reads the RTL code of the design and using the timing constraints, synthesizes the code to structural level, thereby producing a mapped gate-level netlist. This concept is shown in Figure 1-3.

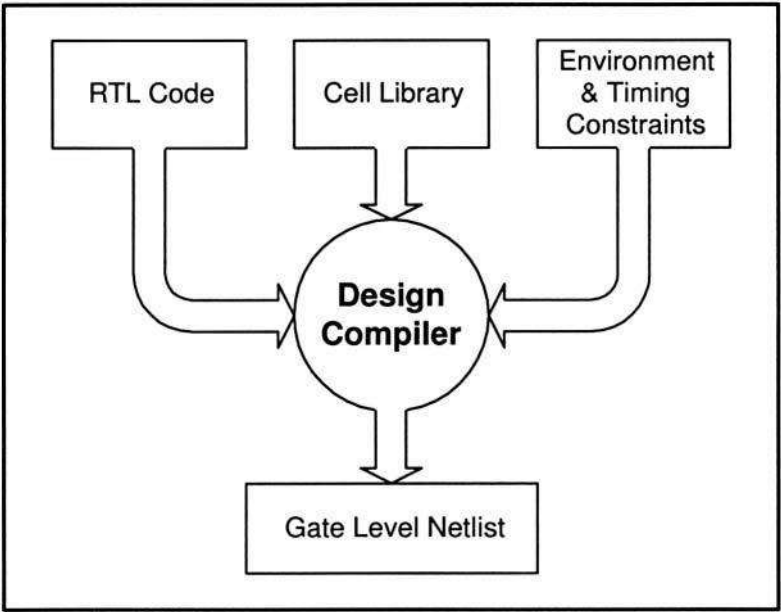


Figure 1-3. Design Compiler Inputs and Outputs

Usually, for small blocks of a design, DC's internal static timing analysis is used for reporting the timing information of the synthesized design. DC tries to optimize the design to meet the specified timing constraints. Further steps may be necessary if timing requirements are not met.

Most designs today, incorporate design-for-test (DFT) logic to test their functionality, after the chip is fabricated. The DFT consists of logic and memory BIST (built-in-self-test), scan logic and Boundary Scan logic (JTAG) etc.

The logic and memory BIST comprises of synthesizable RTL that is based upon controller logic and is incorporated in the design before synthesis. There are tools available in the market that may be used to generate the BIST controller and surrounding logic. Unfortunately, Synopsys does not provide this capability.

The scan insertion may be performed using the test ready compile feature of DC. This procedure maps the RTL directly to scan-flops, before linking them in a scan-chain. An advantage of using this feature is its ability to enable DC to take the scan-flop timing into account while synthesizing. This technique is important since the scan-flops generally have different delays associated with them as compared to their non-scan equivalent flops (or normal flops).

JTAG or boundary scan is primarily used for testing the board connections, without unplugging the chip from the board. The JTAG controller and surrounding logic may also be generated directly by DC.

1.1.4 Formal Verification

The concept of formal verification is fairly new to the ASIC design community. Formal verification techniques perform validation of a design using mathematical methods without the need for technological considerations, such as timing and physical effects. They check for logical functions of a design by comparing it against the reference design.

A number of EDA tool vendors have developed the formal verification tools. However, only recently, Synopsys also introduced to the market its own formal verification tool called Formality.

The main difference between formal methods and dynamic simulation is that former technique verifies the design by proving that the structure and functionality of two designs are logically equivalent. Dynamic simulation methods can only probe certain paths of the design that are sensitized, thus may not catch a problem present elsewhere. In addition, formal methods consume negligible amount of time as compared to dynamic simulation.

The purpose of the formal verification in the design flow is to validate the RTL against RTL, gate-level netlist against the RTL code, or the comparison between gate-level to gate-level netlists.

The RTL to RTL verification is used to validate the new RTL against the old functionally correct RTL. This is usually performed for designs that are subject to frequent changes in order to accommodate additional features. When these features are added to the source RTL, there is always a risk of breaking the old functionally correct feature. To prevent this, formal verification may be performed between the old RTL and the new RTL to check the validity of the old functionality.

The RTL to gate-level verification is used to ascertain that the logic has been synthesized accurately by DC. Since the RTL is dynamically simulated to be functionally correct, the formal verification of the design between the RTL and the scan inserted gate-level netlist assures us that the gate-level also has the same functionality. In this instance if we were to use the dynamic simulation method to verify the gate-level, it would have taken a long time (days and weeks, depending on the size of the design) to verify the design. In comparison, the formal method would take a few hours to perform a similar verification.

The last part involves verifying the gate-level netlist against the gate-level netlist. This too is a significant step for the verification process, since it is mainly used to verify – what has gone into the layout versus what has come out of the layout. What comes out of the layout is obviously the clock tree inserted netlist (flat or hierarchical). This means that the original netlist that

goes into the layout tool is modified. The formal technique is used to verify the logic equivalency of the modified netlist against the original netlist.

1.1.5 Static Timing Analysis using PrimeTime

As previously mentioned, the block level static timing analysis is done using DC. Although, the chip-level static timing can be performed using the above approach, it is recommended that PrimeTime, be used instead. PrimeTime is the Synopsys stand-alone sign-off quality static timing analysis tool that is capable of performing extremely fast static timing analysis on full chip-level designs. It provides a Tcl interface that provides a powerful environment for analysis and debugging of designs.

The static timing analysis, to some extent, is the most important step in the whole ASIC design process. This analysis allows the user to exhaustively analyze all critical paths of the design and express it in an orderly report. Furthermore, the report can also contain other debugging information like the fanout or capacitive loading of each net.

The static timing is performed both for the pre and post-layout gate-level netlist. In the pre-layout mode, PrimeTime uses the wire load models specified in the library to estimate the net delays. During this, the same timing constraints that were fed to DC previously are also fed to PrimeTime, specifying the relationship between the primary I/O signals and the clock. If the timing for all critical paths is acceptable, then a constraints file may be written out from PrimeTime or DC for the purpose of forward annotation to the layout tool. This constraint file in SDF format specifies the timing between each group of logic that the layout tool uses, in order to perform the timing driven placement of cells.

In the post-layout mode, the actual extracted delays are back annotated to PrimeTime to provide realistic delay calculation. These delays consist of the net capacitances and interconnect RC delays.

Similar to synthesis, static timing analysis is also an iterative process. It is closely linked with the placement and routing of the chip. This operation is

usually performed a number of times until the timing requirements are satisfied.

1.1.6 Placement, Routing and Verification

As the name suggests, the layout tool performs the placement and routing. There are a number of methods in which this step could be performed. However, only issues related to synthesis are discussed in this section.

The quality of floorplan and placement is more critical than the actual routing. Optimal cell placement location, not only speeds up the final routing, but also produces superior results in terms of timing and reduced congestion. As explained previously, the constraint file is used to perform timing driven placement. The timing driven placement method forces the layout tool to place the cells according to the criticality of the timing between the cells.

After the placement of cells, the clock tree is inserted in the design by the layout tool. The clock tree insertion is optional and depends solely on the design and user's preference. Users may opt to use more traditional methods of routing the clock network, for example, using fishbone/spine structure for the clocks in order to reduce the total delay and skew of the clock. As technologies shrink, the spine approach is getting more difficult to implement due to the increase in resistance (thus, RC delays) of the interconnect wires. It is therefore the intent of this section (and the entire book) to stress solely on the clock tree synthesis approach.

At this stage an additional step is necessary to complete the clock tree insertion. As mentioned above, the layout tool inserted the clock tree in the design after the placement of cells. Therefore, the original netlist that was generated from DC (and fed to the layout tool), lacks the clock tree information (essentially the whole clock tree network, including buffers and nets). Therefore, the clock tree must be re-inserted in the original netlist and formally verified. Some layout tools provide direct interface to DC to perform this step. Chapter 9 introduces some of these steps, both traditional and not-so-traditional approaches. For the sake of simplicity, let's assume that the clock tree insertion to the original netlist has been performed.

The layout tool generally performs routing in two phases – global routing and detailed routing. After placement, the design is globally routed to determine the quality of placement, and to provide estimated delays approximating the real delay values of the post-routed (after detailed routing) design. If the cell placement is not optimal, the global routing will take a longer time to complete, as compared to placing the cells. Bad placement also affects the overall timing of the design. Therefore, to minimize the number of synthesis-layout iterations and improve placement quality, the timing information is extracted from the layout, after the global routing phase. Although, these delay numbers are not as accurate as the numbers extracted after detailed routing, they do provide a fair idea of the post-routed timing. The estimated delays are back annotated to PrimeTime for analysis, and only when the timing is considered satisfactory, the remaining process is allowed to proceed.

Detailed routing is the final step that is performed by the layout tool. After detailed route is complete, the real timing delays of the chip are extracted, and plugged into PrimeTime for analysis.

These steps are iterative and depend on the timing margins of the design. If the design fails timing requirements, post-layout optimization is performed on the design before undergoing another iteration of layout. If the design passes static timing analysis, it is ready to undergo LVS (layout versus schematic) and DRC (design rule checking) before tape-out.

It must be noted that all steps discussed above can also be applied for hierarchical place and route. In other words, one can repeat these steps for each sub-block of the design before placing the sub-blocks together in the final layout and routing between the sub-blocks.

1.1.7 Engineering Change Order

This step is an exception to the normal design flow and should not be confused with the regular design cycle. Therefore, this step will not be explained in subsequent chapters.

Many designers regard engineering change order (ECO) as the change required in the netlist at the very last stage of the ASIC design flow. For instance, ECO is performed when there is a hardware bug encountered in the design at the very last stage (say, after tape-out), and it is necessary to perform a metal mask change by re-routing a small portion of the design.

As a result ECO is performed on a small portion of the chip to prevent disturbing the placement and routing of the rest of the chip, thereby preserving the rest of the chip's timing. Only the part that is affected is modified. This can be achieved, either by targeting the spare gates incorporated in the chip, or by routing only some of the metal layers. This process is termed as metal mask change.

Normally, this procedure is executed for changes that require less than 10% modification of the whole chip (or a block, if doing hierarchical place and route). If the bug fix requires more than 10% change then it is best to repeat the whole procedure and re-route the chip (or the block).

The latest version of DC incorporates the ECO compiler. It makes use of the mathematical algorithms (also used by the formal verification techniques), to automatically implement the required changes. Making use of the ECO compiler provides designers an alternative to the tedium of manually inserting the required changes in the netlist, thus minimizing the turn-around time of the chip.

Some layout tools have incorporated the ECO algorithm within their tool. The layout tool has a built-in advantage that it does not suffer from the limitation of crossing the hierarchical boundaries associated with a design. Also, the layout tool benefits from knowing the placement location of the spare cells (normally included by the designers in the design), thus can target the nearest location of spare cells in order to implement the required ECO changes and achieve minimized routing.

1.2 Physical Compiler Flow

With shrinking semiconductor geometries, synthesis results based on wire-load models are getting too inaccurate and unpredictable. Physical Compiler

a new tool from Synopsys bypasses this issue by integrating synthesis and placement within one common engine, thus avoiding the delay computation based on wire-load models.

The basic Physical Compiler design flow contains the steps outlined below. Figure 1-4 illustrates the flow chart relating to the design flow described below. Some commonality exists between the traditional flow and the Physical Compiler based flow, therefore only steps relevant to the Physical Compiler flow are outlined.

1. Design environment setting. This includes both the technology library and the physical library to be used, along with other environmental attributes.
2. Floorplan the design.
3. Constrain, synthesize (with scan insertion) and generate placement of the design using Physical Compiler.
4. Pre-layout static timing analysis using PrimeTime (delay numbers based on placement rather than wire-load models).
5. Formal verification of the design. RTL against the synthesized netlist, using Formality.
6. Port the netlist and the placement information over to the layout tool.
7. Insert clock tree in the design using the layout tool.
8. Formal verification between clock tree inserted netlist and the original scan inserted netlist.
9. Perform detailed routing using the layout tool.
10. Extract real timing delays from the detailed routed design.
11. Back-annotate the real extracted data to PrimeTime.
12. Post-layout static timing analysis using PrimeTime.
13. Functional gate-level simulation of the design with post-layout timing (if desired).
14. Tape out after LVS and DRC verification.

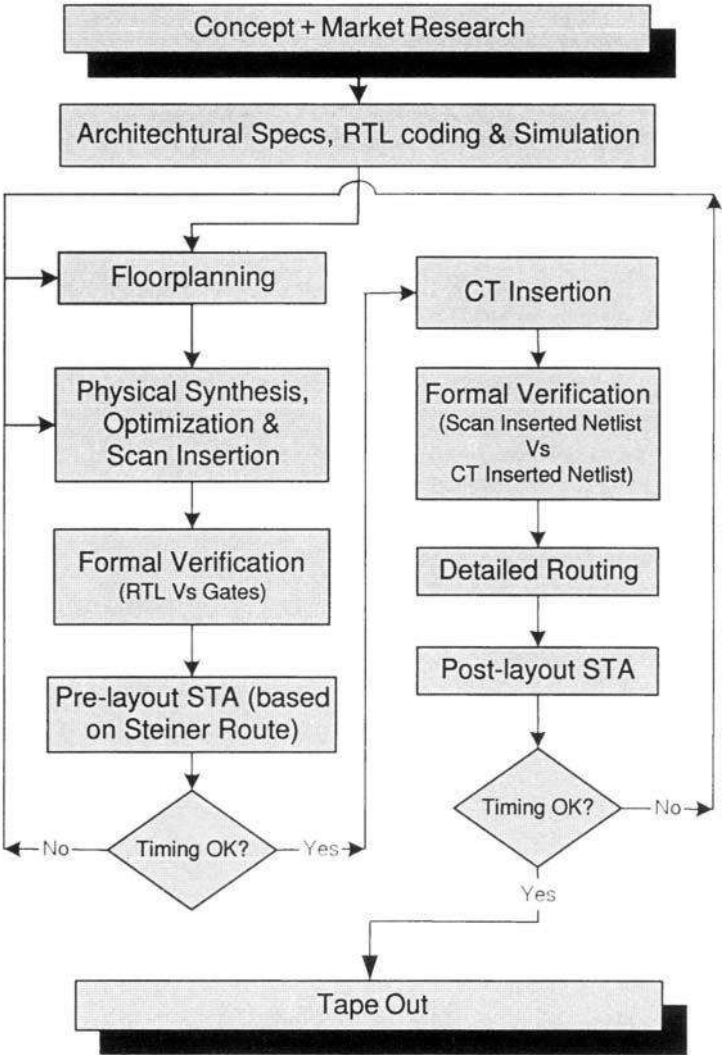


Figure 1-4. Physical Compiler Flow

1.2.1 Physical Synthesis

Traditionally synthesis methods are based on using the wire-load models. The basic nature of the wire-load models is such that they are fanout based. In other words, the delay computation of cells is performed based on the number of fanouts a cell drives. While this method was ideal for larger geometries ($>0.35\mu\text{m}$), it is not suitable for smaller geometries. The resistance of wires is dominating the cell delays causing the fanout based delay computation to be unreliable and totally unpredictable.

The concept of physical synthesis was recently introduced by Synopsys in the form of Physical Compiler (henceforth, called PhyC) as a solution to the above problem. The previous capability of DC is retained and the PhyC enhancements have been added on top of DC thus making PhyC a superset of DC.

PhyC does not use the wire-load models; instead the delay computation is based on the placement rather than fanout. In other words, the synthesis and optimization is based on the placement of cells. By incorporating scan chain re-ordering capability in the current release of PhyC (2000.11), it indeed makes it an extremely powerful and useful tool.

Figure 1-4 illustrates this approach in a very generic form. PhyC can be used in two modes: RTL-to-placed-gates (rtl2pg) or Gates-to-placed-gates (g2pg). For the former mode, the input to PhyC is the RTL, the floorplan information, along with the necessary setup to include logical and physical libraries. The output produced by PhyC is a structural netlist and the placed gates information in PDEF3.0 format. The second mode of g2pg is provided that can be used for optimizing an existing gate level netlist based on the floorplan information. In this case, instead of the RTL, the input to PhyC is the gate level netlist. The rest of the setup and I/O files remain the same.

One important point to note is that in the current release of PhyC (version 2000.11), does not have the capability of synthesizing clock trees in the design. Synopsys has recently announced the availability of the Clock Tree Compiler tool that is an add-on to PhyC. Users who do not have access to this tool have no option but to use their layout tool to insert clock tree in the design database.

A number of steps must be performed in order to perform successful synthesis. These will be discussed later in subsequent chapters. For the moment, however, the process illustrated above is sufficient for the purpose of explaining the design flow.

1.3 Chapter Summary

In this chapter the ASIC design flows incorporating the latest tools and technology for very deep sub-micron (VDSM) technologies were reviewed. The flow started with the definition of specification, and ended with physical layout. The significance was placed on logic and physical synthesis related topics.

Also introduced was a new concept of physical synthesis as applicable to the design flow to shorten the design cycle of the chip. The need to perform physical synthesis was emphasized to get a better estimation of delays and shorten the time-to-market.

TUTORIAL

Synthesis and Static Timing Analysis

This chapter is intended both for beginners and advanced users of Synopsys tools. Novices with no prior experience in synthesis using Synopsys tools are advised to skip this chapter and return to it after reading rest of the book. Beginners with minimal experience in synthesis may use this chapter as a jump-start to learn the ASIC design process, using Synopsys tools. Advanced users will benefit by using this chapter as a reference.

The chapter offers minimal or no explanation for Synopsys commands (they are explained in subsequent chapters). The emphasis is on outlining the practical aspects of the ASIC design flow described in Chapter 1, with Synopsys synthesis in the center. This helps the reader correlate the theoretical concepts with its practical application.

In order to describe both the traditional and the physical compiler based flows, all scripts related to the former are maintained (the commands have been changed to the Tcl format). A separate section based on the Physical Compiler (or PhyC) flow has been added. This provides the users the ability to choose whichever flow best suits their design needs.

Although, the previous chapter stressed skipping the gate-level simulation in favor of formal verification techniques, many designers are reluctant to forego the former step. Due to this reason, this chapter also covers the SDF generation from DC, to be used for simulation purposes. Also, the chapter includes static timing analysis using PrimeTime (PT), in addition to application of formal verification methods, using Formality.

Synthesis and optimization may be performed using any number of approaches. This solely depends upon the methodology you prefer, or are most comfortable using. This chapter uses one such approach that is most commonly used by the Synopsys user's community. You may cater this approach to suit your individual requirements with relative ease.

For the sake of clarity and ease of explanation, the bottom-up compile methodology (described later) is used in all examples and scripts, relating to the synthesis process presented in this chapter. Also, it must be noted that the entire ASIC flow is extremely iterative and one should not assume that the process described in this chapter is suitable for all designs. Later chapters discuss each topic in detail that can be tailored to your designs and methodology.

2.1 Example Design

The best way to start this topic is to go through the whole process on an example design. A tap controller design, coded in Verilog HDL and consisting of one level of hierarchy as shown below is chosen for this purpose:

```
tap_controller.v
  tap_bypass.v
  tap_instruction.v
  tap_state.v
```

The top level of the design is called *tap_controller* which instantiates three modules called *tap_bypass*, *tap_instruction* and *tap_state*. This design contains a single 30 MHz clock called “tk” and a reset called “trst”. Timing specifications for this design dictate that the setup-time needed for all input

signals with respect to “tck” is 10ns, while the hold-time is 0ns. Furthermore, all output signals must be delayed by 10ns with respect to the clock.

The process technology targeted for this design is 0.25 micron. In order to achieve greater accuracy due of variance in process, two Synopsys standard cell technology libraries, characterized for worst-case and the best-case process parameters are used. The libraries are called *ex25_worst.db* and *ex25_best.db*, with a corresponding symbol library containing schematic representations, called *ex25.sdb*. The name of the operating conditions defined in the *ex25_worst.db* library is WORST, while the name of the operating conditions in the *ex25_best.db* library is BEST.

It is assumed that the functionality of the design has been verified by dynamically simulating it at the RTL level.

2.2 Initial Setup

The next step is to synthesize the design, i.e., map the design to the gates belonging to the specified technology library. Before we begin synthesis, several setup files must be created as follows:

- a) *.synopsys_dc.setup* file for DC & PhyC.
- b) *.synopsys_pt.setup* file for PT.

The first file is the setup file for DC & PhyC and is used for logic synthesis as well as physical synthesis, while the second file is associated with PT and defines the required setup to be used for static timing analysis.

Create both of these files with the following contents, assuming that the libraries are kept in the directory – */usr/golden/library/std_cells/*

DC & PhyC .synopsys_dc.setup file

```

set search_path      [list . /usr/golden/library/std_cells]
set target_library   [list ex25_worst.db]
set link_library      [list {*} ex25_worst.db ex25_best.db]
set symbol_library    [list ex25.sdb]
set physical_library  [list ex25_worst.pdb]

define_name_rules BORG -allowed {A-Za-z0-9_} \
                    -first_restricted "_" -last_restricted "_" \
                    -max_length 30 \
                    -map {"*cell*", "mycell"}, {"*-return", "myreturn"}}

set bus_naming_style          %s[%d]
set verilogout_no_tri         true
set verilogout_show_unconnected_pins true
set test_default_scan_style    multiplexed_flip_flop

```

PT .synopsys_pt.setup file

```

set search_path [list . /usr/golden/library/std_cells]
set link_library [list {*} ex25_worst.db ex25_best.db]

```

2.3 Traditional Flow

The following steps outline the traditional flow. Here DC is used for logic synthesis while the layout tool handles the rest of the back-end that includes placement and routing.

2.3.1 Pre-Layout Steps

The following sub-sections illustrate the steps involved during the pre-layout phase. This includes one-pass logic synthesis with scan insertion, static timing analysis, SDF generation to perform functional gate-level simulation, and finally formal verification between the source RTL and synthesized netlist.

2.3.1.1 Synthesis

The pre-layout logic synthesis involves optimizing the design for maximum setup-time, utilizing the statistical wire-load models and the worst-case operating conditions from the *ex25_worst.db* technology library. In order to maximize the setup-time, you may constrain the design by defining clock uncertainty for the setup-time. In general, a 10% over-constrain is usually sufficient, in order to minimize the synthesis-layout iterations.

After initial synthesis if gross hold-time violations are detected, they should be fixed at the pre-layout level. This also helps in reducing the synthesis-layout iterations. However, it is preferable to fix minor hold-time violations after the layout, with real delays back annotated.

In this tutorial, we assume that minor hold-time violations exist, therefore these violations will be fixed during the post-layout optimization. Fixing hold-time violations involves back annotation of the extracted delays from the layout to DC. In addition, hold-time fixes require usage of the best-case operating conditions from the *ex25_best.db* library.

Generic synthesis script for sub-modules

```
set active_design tap_bypass

analyze -format verilog $active_design.v
elaborate $active_design

current_design $active_design
link

uniquify

set_wire_load_model -name SMALL
set_wire_load_mode top
set_operating_conditions WORST

create_clock -period 33 -waveform [list 0 16.5] tck
set_clock_latency 2.0 [get_clocks tck]
```

```

set_clock_uncertainty -setup 3.0 [get_clocks tck]
set_clock_transition 0.1 [get_clocks tck]
set_dont_touch_network [list tck trst]

set_driving_cell -cell BUFF1X -pin Z [all_inputs]
set_drive 0 [list tck trst]

set_input_delay 20.0 -clock tck -max [all_inputs]
set_output_delay 10.0 -clock tck -max [all_outputs]

set_max_area 0

set_fix_multiple_port_nets -buffer_constants -all
compile -scan

check_test

remove_unconnected_ports [find -hierarchy cell {"**"}]

change_names -h -rules BORG

set_dont_touch current_design

write-hierarchy -output $active_design.db
write-format verilog -hierarchy \
                    -output $active_design.sv

```

The above script contains a user-defined variable called *active_design* that defines the name of the module to be synthesized. This variable is used throughout the script, thus making the rest of the script generic. By re-defining the value of *active_design* to other sub-modules (*tap_instruction* and *tap_state*), the same script may be used to synthesize the sub-modules. Users can apply the same concept to clock names, clock periods, etc. in order to parameterize the scripts.

Lets assume that you have successfully synthesized three sub-blocks, namely *tap_bypass*, *tap_instruction* and *tap_state*. The one-pass synthesis was done with no scan chain stitching. Only the flops were directly mapped to scan

flops. We can apply the same synthesis script to synthesize the top level, with the exception that we have to include the mapped "db" files for the sub-blocks, before reading the *tap_controller.v* file. In addition, this time we have to perform scan insertion also in order to stitch the scan chains. Also, the wire-load mode may need to be changed to **enclosed** for proper modeling of the interconnect wires. Since the sub-modules contain the **dont_touch** attribute, the top-level synthesis will not optimize across boundaries, and may violate the design rule constraints. To remove these violations, you must re-synthesize/optimize the design with the **dont_touch** attribute removed from the sub-blocks.

DFT scan insertion at the top-level is another reason for removing the **dont_touch** attribute from the sub-blocks. This is due to the fact that the DFT scan insertion cannot be implemented at the top-level, if the sub-blocks contain the **dont_touch** attribute. The following script exemplifies this process by performing initial synthesis with scan enabled, before re-compiling (`compile -only_design_rule`) the design with **dont_touch** attribute removed from all the sub-blocks.

Synthesis Script for the top-level

```
set active_design tap_controller

set sub_modules {tap_bypass tap_instruction tap_state}

foreach module $sub_modules {
    set syn_db $module.db
    read_db syn_db
}

analyze -format verilog $active_design.v
elaborate $active_design

current-design $active_design
link

uniquify
```

```
set_wire_load_model -name LARGE
set_wire_load_mode enclosed
set_operating_conditions WORST
```

```
create_clock-period 33 -waveform [list 0 16.5] tck
set_clock_latency 2.0 [get_clocks tck]
set_clock_uncertainty -setup 3.0 [get_clocks tck]
set_clock_transition 0.1 [get_clocks tck]
set_dont_touch_network [list tck trst]
```

```
set_driving_cell -cell BUFF1X -pinZ [all_inputs]
set_drive 0 [list tck trst]
```

```
set_input_delay 20.0-clock tck-max [all_inputs]
set_output_delay 10.0-clock tck-max [all_outputs]
```

```
set_max_area 0
```

```
set_fix_multiple_port_nets -all -buffer_constants
```

```
compile -scan
```

```
remove_attribute [find -hierarchy design {"*"}] dont_touch
```

```
current_design $active_design
uniquify
```

```
check_test
create_test_patterns -sample 10
preview_scan
insert_scan
check_test
```

```
compile -only_design_rule
```

```
remove_unconnected_ports [find -hierarchy cell {"*"}]
change_names-hierarchy-rules BORG
```



```
set_dont_touch current_design

write -hierarchy -output $active_design.db
write-format verilog -hierarchy \
    -output $active_design.sv
```

2.3.1.2 Static Timing Analysis using PrimeTime

After successful synthesis, the netlist obtained must be analyzed to check for timing violations. The timing violations may consist of either setup and/or hold-time violations.

The design was synthesized with emphasis on maximizing the setup-time, therefore you may encounter very few setup-time violations, if any. However, the hold-time violations will generally occur at this stage. This is due to the data arriving too fast at the input of sequential cells (data changing its value before being latched by the sequential cells), thereby violating the hold-time requirements.

If the design is failing setup-time requirements, then you have no other option but to re-synthesize the design, targeting the violating path for further optimization. This may involve grouping the violating paths or over-constraining the entire sub-block, which had violations. However, if the design is failing hold-time requirements, you may either fix these violations at the pre-layout level, or may postpone this step until after layout. Many designers prefer the latter approach for minor hold-time violations (also used here), since the pre-layout synthesis and timing analysis uses the statistical wire-load models and fixing the hold-time violations at the pre-layout level may result in setup-time violations for the same path, after layout. It must be noted that gross hold-time violations should be fixed at the pre-layout level, in order to minimize the number of hold-time fixes, which may result after the layout.

PT script for pre-layout setup-time analysis

```
set active_design tap_controller

read_db -netlist_only $active_design.db

current_design $active_design

set_wire_load_model -name large
set_wire_load_mode top

set_operating_conditions WORST

set_load 50.0 [all_outputs]
set_driving_cell -cell BUFF1X -pinZ [all_inputs]

create_clock -period 33 -waveform [0 16.5] tck
set_clock_latency 2.0 [get_clocks tck]
set_clock_transition 0.2 [get_clocks tck]
set_clock_uncertainty 3.0 -setup [get_clocks tck]

set_input_delay 20.0 -clock tck [all_inputs]
set_output_delay 10.0 -clock tck [all_outputs]

report_constraint -all_violators

report_timing -to [all_registers -data_pins]
report_timing -to [all_outputs]

write_sdf -context verilog -output $active_design.sdf
```

The above PT script performs the static timing analysis for the *tap_controller* design. Notice that the clock latency and transition are fixed in the above example, because at the pre-layout level the clock tree has not been inserted. Therefore, it is necessary to define a certain amount of delay that approximates the final delay associated with the clock tree. Also, the clock transition is specified because of the high fanout associated with the clock

network. The high fanout suggests that the clock network is driving many flip-flops, each having a certain amount of pin capacitance. This gives rise to slow input ramp time for the clock. The fixed transition value (again approximating the final clock tree number) of clock prevents PT from calculating incorrect delay values, that are based upon the slow input ramp to the flops.

The script to perform the hold-time analysis at the pre-layout level is shown below. To check for hold-time violations, the analysis must be performed utilizing the best-case operating conditions, specified in the *ex25_best.db* library. In addition, an extra argument (`–delay_type min`) is specified in the `report_timing` command, as follows:

PT script for pre-layout hold-time analysis

```
set active_design tap_controller

read_db –netlist_only $active_design.db

current_design $active_design

set_wire_load large
set_wire_load_mode top

set_operating_conditions BEST

set_load 50.0 [all_outputs]
set_driving_cell –cell BUFF1X –pin Z [all_inputs]

create_clock –period 33 –waveform [0 16.5] tck
set_clock_latency 2.0 [get_clocks tck]
set_clock_transition 0.2 [get_clocks tck]
set_clock_uncertainty 0.2 –hold [get_clocks tck]

set_input_delay 0.0–clock tck [all_inputs]
set_output_delay 0.0–clock tck [all_outputs]
```

```
report_constraint -all_violators

report_timing -to [all_registers -data_pins] \
               -delay_type min
report_timing -to [all_outputs] -delay_type min

write_sdf -context verilog -output $active_design.sdf
```

2.3.1.3 SDF Generation

To perform timing simulation, you will need the SDF file for back annotation. The static timing was performed using PT; therefore it is prudent that the SDF file be generated from PT itself as shown in the previous scripts. However, some designers feel comfortable in using DC to generate the SDF file. We will therefore use DC to generate the SDF in this section.

In addition, depending on the design, the resultant SDF file may require a certain amount of “massaging” before it can be used to perform timing simulation of the design. The reason for massaging is explained in detail in Chapter 11.

The following script may be used to generate the pre-layout SDF for the *tap_controller* design. This SDF file is targeted for simulating the design dynamically with timing. In addition, the script also generates the timing constraints file. Though this file is also in SDF format, it is solely used for forward annotating the timing information to the layout tool in order to perform timing driven layout using the traditional approach.

DC script for pre-layout SDF generation

```
set active_design tap_controller

read_db $active_design.db

current_design $active_design
link
```

```
set_wire_load_model LARGE
set_wire_load_mode top
set_operating_conditions WORST

create_clock -period 33--waveform [list 0 16.5] tck
set_clock_latency 2.0 [get_clocks tck]
set_clock_transition 0.2 [get_clocks tck]
set_clock_uncertainty 3.0 -setup [get_clocks tck]

set_driving_cell -cell BUFF1X -pin Z [all_inputs]
set_drive 0 [list tck trst]

set_load 50 [all_outputs]

set_input_delay 20.0--clock tck--max [all_inputs]
set_output_delay 10.0--clock tck--max [all_outputs]

write_sdf -output $active_design.sdf

write_constraints -format sdf -cover_design \
                -output constraints.sdf
```

2.3.1.4 Floorplanning and Routing

The floorplanning step involves physical placement of cells and clock tree synthesis. Both these steps are performed within the layout tool. The placement step may include timing driven placement of the cells, which is performed by annotating the *constraints.sdf* file (generated by DC) to the layout tool. This file consists of path delays that include the cell-to-cell timing information. This information is used by the layout tool to place cells with timing as the main criterion i.e., the layout tool will place timing critical cells closer to each other in order to minimize the path delay.

Let's assume that the design has been floorplanned. Also, the clock tree has been inserted in the design by the layout tool. The clock tree insertion modifies the existing structure of the design. In other words, the netlist in the layout tool is different from the original netlist present in DC. This is because of the fact that the design present in the layout tool contains the clock tree,

whereas the original design in DC does not contain this information. Therefore, the clock tree information should somehow be transferred to the design residing in DC or PT. The new netlist (containing the clock tree information) should be formally verified against the original netlist to ensure that the transfer of clock tree did not break the functionality of the original logic. Various methods of transferring the clock tree information to the design are explored in detail in Chapter 9. For the sake of simplicity, let us assume that the clock tree information is present in the *tap_controller* design.

The design is now ready for routing. In a broad sense, routing is performed in two phases – global route and detailed route. During global route, the router divides the layout surface into separate regions and performs a point-to-point “loose” routing without actually placing the geometric wires. The final routing is performed by the detailed router, which physically places the geometric wires and routes within the regions. Full explanations of these types of routing are explained in Chapter 9. Lets assume that the design has been global routed.

The next step involves extracting the estimated parasitic capacitances, and RC delays from the global routed design. This step reduces the synthesis-layout iteration time, especially since cell placement and global routing may take much less time than detailed routing the entire chip. However, if the cells are placed optimally with minimal congestion, detailed routing is also very fast. In any case, extraction of delays after the global route phase (albeit estimates) provides a faster method of getting closer to the real delay values that are extracted from the layout database after the detailed routing phase.

Back annotate the estimates to the design in PT for setup and hold-time static timing analysis, using the following scripts.

PT script for setup-time analysis, using estimated delays

```
set active_design tap_controller

read_db -netlist_only $active_design.db

current_design $active_design
```

```
set_operating_conditions WORST

set_load 50.0 [all_outputs]
set_driving_cell -cell BUFF1X -pin Z [all_inputs]

source capacitance.pt # estimated parasitic capacitances
read_sdf rc_delays.sdf # estimated RC delays

create_clock -period 33 -waveform [0 16.5] tck
set_propagated_clock [get_clocks tck]
set_clock_uncertainty 0.5 -setup [get_clocks tck]

set_input_delay 20.0-clock tck [all_inputs]
set_output_delay 10.0-clock tck [all_outputs]

report_Constraint -all_violators

report_timing -to [all_registers -data_pins]
report_timing -to [all_outputs]
```

PT script for hold-time analysis, using estimated delays

```
set_active_design tap_controller

read_db -netlist_only $active_design.db

current_design $active_design

set_operating_conditions BEST

set_load 20.0 [all_outputs]
set_driving_cell -cell BUFF1X -pin Z [all_inputs]

source capacitance.pt # estimated parasitic capacitances
read_sdf rc_delays.sdf # estimated RC delays

create_clock -period 33 -waveform [0 16.5] tck
```

```

set_propagated_clock [get_clocks tck]
set_clock_uncertainty 0.05 -hold [get_clocks tck]

set_input_delay 0.0-clock tck [all_inputs]
set_output_delay 0.0-clock tck [all_outputs]

report_constraint -all_violators

report_timing -to [all_registers -data_pins] \
               -delay_type min
report_timing -to [all_outputs] -delay_type min

```

The above script back annotates *capacitance.pt* and *rc_delays.sdf* file. The *capacitance.pt* file contains the capacitive loading per net of the design in **set_load** format, while the *rc_delays.sdf* file contains point-to-point interconnect RC delays of individual nets. DC (and PT) performs the calculation of cell delay, based upon the output net loading and input slope of each cell in the design. The reason for using this approach is explained in detail in Chapter 9.

If the design fails setup-time requirements, you may re-synthesize the design with adjusted constraints or re-floorplan the design. If the design is failing hold-time requirements, then depending on the degree of violation you may decide to proceed to the final step of detailed routing the design, or re-optimize the design with adjusted constraints.

If re-synthesis is desired then the floorplan (placement) information consisting of the physical clusters and cell locations, should be back annotated to DC. This step is desired because up till now, DC did not know the physical placement information of cells. By annotating the placement information to DC, the post-layout optimization of the design within DC is vastly improved. The layout tool generates the physical information in PDEF format that can be read by DC, using the following command:

```
read_clusters <file name in PDEF format>
```


The script to perform this is similar to the initial synthesis script, with the exception of the annotated data and incremental compilation of the design, as illustrated below:

Script for incremental synthesis of the design

```
set active_design tap_controller

read_db $active_design.db

current_design $active_design
link

source capacitance.dc /* estimated parasitic capacitances */
read_timing -f sdf rc_delays.sdf /* estimated RC delays */
read_clusters clusters.pdef /* physical information */

create_wire_load -hierarchy      \
                  -percentile 80  \
                  -output cwl.m.txt

create_clock -period 33 -waveform [0 16.5] tck
set_propagated_clock [get_clocks tck]
set_clock_transition 0.2 [get_clocks tck]
set_clock_uncertainty 3.0 -setup [get_clocks tck]

set_dont_touch_network [list tck trst]

set_driving_cell -cell BUFF1X -pin Z [list all_inputs]
set_drive 0 [list tck trst]

set_input_delay 20.0-clock tck -max [all_inputs]
set_output_delay 10.0-clock tck -max [all_outputs]

set_max_area 0

set_fix_multiple_port_nets -all -buffer_constants
```

```
reoptimize_design -in_place

write -hierarchy -output $active_design.db
write -format verilog -hierarchy \
    -output $active_design.sv
```

The `create_wire_load` command used in the above script creates a custom wire-load model for the *tap_controller* design. The initial synthesis run used the wire-load models present in the technology library that are not design specific. Therefore, in order to achieve better accuracy for the next synthesis iteration, the custom wire-load models specific to the design should be used.

The following command may be used to update the technology library present in DC's memory to reflect the new custom wire-load models. For example:

```
dc_shell-t> update_lib ex25_worst.db cwl.m.txt
```

Let's assume that the design has been re-analyzed and is now passing both setup and hold-time requirements. The next step is to detail route the design. This is a layout dependent feature, therefore will not be discussed here.

2.3.2 Post-Layout Steps

The post-layout steps involve, verifying the design for timing with actual delays back annotated; functional simulation of the design; and lastly, performing LVS and DRC.

Let us just presume that the design has been fully routed with minimal congestion and area. The finished layout surface must then be extracted to get the actual parasitic capacitances and interconnect RC delays. Depending upon the layout tool and the type of extraction, the extracted values are generally written out in the SDF format for the interconnect RC delays, while the parasitic information is generated as a string of `set_load` commands for each net in the design. In addition, if a hierarchical place and route has been performed, the physical placement location of cells in the PDEF format should also be generated.

2.3.2.1 Post-Layout Static Timing Analysis using PrimeTime

The first step after layout is to perform static timing on the design, using the actual delays. Similar to post-placement, the post-route timing analysis uses the same commands, except that this time the actual delays are back annotated to the design.

Predominantly, the timing of the design is dependent upon clock latency and skew. It is therefore prudent to perform the clock skew analysis before attempting to analyze the whole design. A useful Tcl script is provided by Synopsys through their on-line support on the web, called SolvNET. You may download this script and run the analysis before proceeding. Let us assume that the clock latency and skew is within limits. The next step is to perform the static timing on the design, to check the setup and hold-time violations (if any) using the following scripts:

PT script for setup-time analysis, using actual delays

```
set active_design tap_controller

read_db -netlist_only $active_design.db

current_design $active_design

set_operating_conditions WORST

set_load 50.0 [all_outputs]
set_driving_cell -cell BUFF1X -pin Z [all_inputs]

source capacitance.pt # actual parasitic capacitances
read_sdf rc_delays.sdf # actual RC delays
read_parasitics clock_info_wrst.spf # for clocks etc.

create_clock -period 33 -waveform [0 16.5] tck
set_propagated_clock [get_clocks tck]
set_clock_uncertainty 0.5 -setup [get_clocks tck]
```

```

set_input_delay 20.0-clock tck [all_inputs]
set_output_delay 10.0-clock tck [all_outputs]

report_constraint -all_violators

report_timing -to [all_registers -data_pins]
report_timing -to [all_outputs]

```

PT script for hold-time analysis, using actual delays

```

set active_design tap_controller

read_db -netlist_only $active_design.db

current_design $active_design

set_operating_conditions BEST

source capacitance.pt # actual parasitic capacitances
read_sdf rc_delays.sdf # actual RC delays
read_parasitics clock_info_best.spf # for clocks etc.

set_load 50.0 [all_outputs]
set_driving_cell -cell BUFF1X -pin Z [all_inputs]

create_clock -period 33 -waveform [0 16.5] tck
set_propagated_clock [get_clocks tck]
set_clock_uncertainty 0.05 -hold [get_clocks tck]

set_input_delay 0.0-clock tck [all_inputs]
set_output_delay 0.0-clock tck [all_outputs]

report_constraint -all_violators
report_timing -to [all_registers -data_pins] \
               -delay_type min
report_timing -to [all_outputs] -delay_type min

```

2.3.2.2 Post-Layout Optimization

The post-layout optimization or PLO may be performed on the design to improve or fix the timing requirements. DC provides several methods of fixing timing violations, through the in place optimization (or IPO) feature. As before, DC also makes use of the physical placement information to perform location based optimization (LBO). In this example, we will use the cell resizing and buffer insertion feature of the IPO to fix the hold-time violations.

2.3.2.2.1 Hold-Time Fixes

The design was synthesized for maximum setup-time requirements. Timing was verified at each step (after synthesis and then, after the global route phase), therefore in all probability the routed design will pass the setup-time requirements. However, some parts of the design may fail hold-time requirements at various endpoints.

If the design fails the hold-time requirements then you should fix the violations by adding buffers to delay the arrival time of the failing signals, with respect to the clock. Let's assume that the design is failing hold-time requirements at multiple endpoints.

There are various approaches to fix the hold-time violations. Such methods are discussed in detail in Chapter 9. In this example, we will utilize the `dc_shell-t` commands to fix the hold time violations, as illustrated below:

DC Script to fix the hold-time violations

```
set active_design tap_controller

read_db $active_design.db

current_design $active_design
link

source capacitance.dc /*actual parasitic capacitances */
```

```

read_timing -f sdf rc_delays.sdf    /*actual RC delays */
read_clusters clusters.pdef        /*physical hierarchy info */

create_clock -period 33 -waveform {0 16.5} tck
set_propagated_clock [get_clocks tck]
set_clock_uncertainty -hold 0.05 tck

set_dont_touch_network [list tck trst

set_driving_cell -cell BUFF1X -pin Z [all_inputs]
set_drive 0 [list tck trst]

set_input_delay _min 0.0-clock tck-max [all_inputs]
set_output_delay -min 0.0-clock tck-max [all_outputs]

set_fix_hold tck /* fix hold-time violations w.r.t. tck */

reoptimize_design -in_place

write -hierarchy -output $active_design.db
write-format verilog -hierarchy \
                        -output $active_design.sv

```

In the above script, the **set_fix_hold** command instructs DC to fix hold-time violations with respect to the clock *tck*. The **-in_place** argument of the **reoptimize_design** command is the IPO command, which is regulated by various variables that are described in Chapter 9. Making use of these variables, DC inserts or resizes the gates to fix the hold time violations. The LBO variables are helpful in inserting the buffers at the correct location, so as to minimize its impact on some other logic path, leading off from the violating path.

After IPO, the design should again be analyzed through PT to ensure that the violations have been fixed using the post-layout PT script illustrated before.

Once the design passes all timing requirements, the post-layout SDF may be generated (from PT or DC) for simulation purposes, if needed. We will use DC to generate the worst-case post-layout SDF using the script provided

below. A similar script may be used to generate the best-case SDF. Obviously, you need to back annotate the best-case extracted numbers from the layout tool to generate the best-case SDF from DC. This solely depends on the layout tool and the methodology being used.

DC script for worst-case post-layout SDF generation

```
set active_design tap_controller

read_db $active_design.db

current_design $active_design
link

set_operating_conditions WORST

source capacitance.dc /* actual parasitic capacitances */
read_timing rc_delays.sdf /* actual RC delays */

create_clock-period 33 -waveform {0 16.5} tck
set_propagated_clock [get_clocks tck]
set_clock_uncertainty -setup 0.5 [get_clocks tck]

set_driving_cell -cell BUFF1X -pin Z [all_inputs]
set_drive 0 [list tck trst]

set_load 50 [all_outputs]

set_input_delay 20.0-clock tck-max [all_inputs]
set_output_delay 10.0-clock tck-max [all_outputs]

write_sdf -output $active_design.sdf
```

It is recommended that formal verification be performed again between the source RTL and the final netlist, to check for any errors that may have been unintentionally introduced during the whole process. This is the final step; the design is now ready for LVS and DRC checks, before tape-out.

2.4 Physical Compiler Flow

The Physical Compiler (or PhyC) flow provides an integrated approach to synthesis and placement combined. It does not utilize the traditional approach of using the wire-load models, thus minimizing the discrepancy between the pre-layout synthesis and post-layout delays.

Due to the importance of this technique and its novel nature, a complete chapter has been devoted to this flow. Instead of duplicating the scripts here, all the scripts related to this flow have been illustrated in that chapter. Readers are advised to read Chapter10 to learn about PhyC flow in order to understand how PhyC is used and where it fits within the synthesis and layout approach we have been so used to.

2.5 Chapter Summary

This chapter highlighted the practical side of the ASIC design methodology in the form of a tutorial. An example design was used to guide the reader from start to finish. At each stage, brief explanation and relevant scripts were provided.

The chapter started with basics of setting up the Synopsys environment and technical specification of the example design. Further sections were divided into pre-layout, floorplanning and routing, and finally the post-layout steps.

The pre-layout steps included initial synthesis and scan insertion of the design, along with static timing analysis, and SDF generation for dynamic simulation. In order to minimize the synthesis-layout iterations, the floorplanning and routing section stressed upon the placement of cells, with emphasis on back annotating to DC, the estimated delays extracted after global routing the design. The final section used post-layout optimization techniques to fix the hold-time violations, and to generate the final SDF for simulation.

The application of formal verification method using Synopsys Formality was also included. This section did not contain any scripts, but the reader was

made aware of the usefulness of formal techniques and where they are applied.

Finally the Physical Synthesis approach was discussed that eliminates the need for wire-load models and integrates the synthesis with placement. All scripts related to this approach are provided in Chapter 10.

BASIC CONCEPTS

This chapter covers the basic concepts related to synthesis, using Synopsys suite of tools. These concepts introduce the reader to synthesis terminology used throughout the later chapters. These terms provide the necessary framework for Synopsys synthesis and static timing analysis.

Although this chapter is a good reference, advanced users of Synopsys tools, already familiar with Synopsys terminology, may skip this chapter.

3.1 Synopsys Products

This section briefly describes all relevant Synopsys products related to this book.

- a) Library Compiler
- b) Design Compiler and Design Vision
- c) Physical Compiler
- d) PrimeTime
- e) DFT Compiler
- f) Formality

Library Compiler

The core of any ASIC design is the technology library containing a set of logic cells. The library may contain functional description, timing, area and other pertinent information of each cell. Library Compiler (LC) parses this textual information for completeness and correctness, before converting it to a format, used globally by all Synopsys applications.

Library Compiler is invoked by typing `lc_shell` in a UNIX shell. All the capabilities of the LC can also be utilized within `dc_shell`.

Design Compiler and Design Vision

The Synopsys Design Compiler (DC) and Design Vision (DV) comprise a powerful suite of logic synthesis products, designed to provide an optimal gate-level synthesized netlist based on the design specifications, and timing constraints. In addition to high-level synthesis capabilities, it also incorporates a static timing analysis engine, along with solutions for FPGA synthesis and links-to-layout (LTL).

Design Compiler is the command line interface of Synopsys synthesis tool and is invoked by either typing `dc_shell` or `dc_shell-t` in a UNIX shell. The `dc_shell` is the original format that is based on Synopsys's own language while `dc_shell-t` uses the standard Tcl language. This book focuses only on the Tcl version of DC because of the commonality with other Synopsys tools, like PrimeTime.

The Design Vision is the graphical front-end version of DC and is launched by typing `design_vision`. Design Vision also supports schematic generation, with critical path analysis through point-to-point highlighting.

Although, beginners may initially prefer using DV, they quickly migrate to using DC, as they become more familiar with Synopsys commands.

Physical Compiler

Physical Compiler (or PhyC) is a new tool by Synopsys that is a superset of DC. In addition to incorporating all the synthesis and optimization capabilities of DC, it also provides the ability to concurrently place cells optimally, based on the timing and/or area constraints of the design.

PhyC is invoked by typing `psyn_shell`. A separate GUI version is also available, which is launched by typing `psyn_gui`. Although slow by comparison, `psyn_gui` provides the users the ability to traverse between the logical and the schematic view of the design.

It must be noted that PhyC being the superset of DC, all `dc_shell` commands are available within `psyn_shell`. The reverse is not true. You cannot use `psyn_shell` commands within `dc_shell`.

PrimeTime

PrimeTime (PT) is the Synopsys sign-off quality, full chip, gate-level static timing analysis tool. In addition, it also allows for comprehensive modeling capabilities, often required by large designs.

PT is faster compared to DC's internal static timing analysis engine. It also provides enhanced analysis capabilities, both textually and graphically. In contrast to the rest of Synopsys tools, this tool is Tcl language based, therefore providing powerful features of that language to promote the analysis and debugging of the design.

PT is a stand-alone tool and can be invoked as a command line interface or graphically. To use the command line interface, type `pt_shell` in the UNIX window, or type `primetime` for the graphical version.

DFT Compiler

The DFT Compiler (DFTC) is the Synopsys test insertion tool that is incorporated within the DC suite of tools. The DFTC is used to insert DFT features like scan insertion and boundary scan, to the design. All DFTC commands are directly invoked from `dc_shell` or `psyn_shell`.

Formality

Formality is the Synopsys formal verification or more precisely a logic equivalence checking tool. The tool features enhanced graphical debugging capabilities that include schematic representation of logic under verification, and visual suggestions annotated to the schematic as pointers of possible incorrect logic. It also provides suggestions for possible fixes to the design.

3.2 Synthesis Environment

As with most EDA products, Synopsys tools require a setup file that specifies the technology library location and other parameters used for synthesis. Synopsys also defines its own format for storing and processing the information. This section highlights such details.

3.2.1 Startup Files

There is a common startup file called “.synopsys_dc.setup” for all tools of the DC and PhyC family. A separate startup file is required for PT which is named “.synopsys_pt.setup”. These files are in Tcl format, and contain path information to the technology libraries and other environment variables.

The default startup files for PhyC, DC and PT reside in the Synopsys installation directory, and are automatically loaded upon invocation of these tools. These default files do not contain the design dependent data. Their function is to load the Synopsys technology independent libraries and other parameters. The user in the startup files specifies the design dependent data. During startup, these tools read the files in the following order:

1. Synopsys installation directory.
2. Users home directory.
3. Project working directory.

The settings specified in the startup files, residing in the project working directory, override the ones specified in the home directory and so forth, i.e., the configuration specified in the project working directory takes precedence over all other settings.

It is up to the discretion of the user to keep these files wherever it is convenient. However, it is recommended that the design dependent startup files be kept in the working directory.

The minimum information required by DC is the `search_path`, `target_library`, `link_library` and the `symbol_library`. PhyC requires the `physical_library` information in addition to the DC related setup. PT requires

the `search_path` and `link_library` information only. Typical startup files are shown in Example 3.1

Example 3.1: Setup Files

PhyC & DC .synopsys_dc.setup file

```
set search_path      [list. /usr/golden/library/std_cells \
                      /usr/golden/library/pads]

set target_library   [list std_cells_lib.db]
set physical_library [list std_cells_lib.pdb pad_lib.pdb]
set link_library     [list {*} std_cells_lib.db pad_lib.db]
set symbol_library   [list std_cells_lib.sdb pad_lib.sdb]
```

PT .synopsys_pt.setup file

```
set search_path      [list. /usr/golden/library/std_cells \
                      /usr/golden/library/pads]

set link_path        [list {*} std_cells_lib.db pad_lib.db]
```

3.2.2 System Library Variables

At this time, it is worth explaining the difference between the `target_library` and the `link_library` system variables. The `target_library` specifies the name of the technology library that corresponds to the library whose cells the designers want DC to infer and finally map to. The `link_library` defines the name of the library that refers to the library of cells used solely for reference, i.e., cells in the `link_library` are not inferred by DC. For example, you may specify a standard cell technology library as the `target_library`, while specifying the pad technology library name and all other macros (RAMs, ROMs etc.) in the `link_library` list. This means that the user would synthesize the design that targets the cells present in the standard cell library, while linking to the pads and macros that are instantiated in the design. If the pad

library is included in the `target_library` list, then DC may use the pads to synthesize the core logic.

The target library name should also be included in the `link_library` list, as shown in Example 3.1. This is important while reading the gate-level netlist in DC. DC will not be able to link to the mapped cells in the netlist, if the target library name is not included in the link library list. For this case, DC generates a warning stating that it was unable to resolve reference for the cells present in the netlist.

The `target_library` and `link_library` system variables allow the designer to better control the mapping of cells. These variables also provide a useful means to re-map a gate-level netlist from one technology to the other. In this case, the `link_library` may contain the old technology library name, while the `target_library` may contain the new technology library. Re-mapping can be performed by using the `translate` command in `dc_shell`.

The `symbol_library` system variable holds the name of the library, containing graphical representation of the cells in the technology library. It is used, to represent the gates schematically, while using the graphical front-end tool, DA. The symbol libraries are identified with a “`sdb`” extension. If this variable is omitted from the setup file, DA will use a generic symbol library called “`generic.sdb`” to create schematics. Generally, all technology libraries provided by the library vendor include a corresponding symbol library. It is imperative that there be an exact match of the cell names and the pin names, between the technology and the symbol library. Any mismatch in a cell will cause DA to reject the cell from the symbol library, and use the cell from the generic library.

In addition to specifying logical libraries, physical libraries also need to be identified if using `psyn_shell`. These libraries contain the physical information (like physical dimensions of cells, orientation, layer information etc.) needed by PhyC. The physical libraries are identified with a “`pdb`” extension and are referenced by the `physical_library` system variable as shown in the above example.

It must be noted that DC uses the `link_library` variable, whereas PT calls it the `link_path`. Apart from the difference in name and the format, the

application of both these variables is identical. Since PT is a gate-level static timing analyzer, it only works on the structural gate-level netlists. Thus, PT does not utilize the `target_library` variable.

3.3 Objects, Variables and Attributes

Synopsys supports a number of objects, variables and attributes in order to streamline the synthesis process. Using these, designers can write powerful scripts to automate the synthesis process. It is therefore essential for designers to familiarize themselves with these terms.

3.3.1 Design Objects

There are eight different types of design objects categorized by DC. These are:

- *Design*: It corresponds to the circuit description that performs some logical function. The design may be stand-alone or may include other sub-designs. Although, sub-designs may be part of the design, it is treated as another design by Synopsys.
- *Cell*: It is the instantiated name of the sub-design in the design. In Synopsys terminology, there is no differentiation between the cell and instance; both are treated as cell.
- *Reference*: This is the definition of the original design to which the cell or instance refers. For example, a leaf cell in the netlist must be referenced from the link library, which contains the functional description of the cell. Similarly, a sub-design instantiated (called cell by Synopsys) must be referenced in the design, which contains functional description of the instantiated sub-design.
- *Port*: These are the primary inputs, outputs or IO's of the design.

- *Pin*: It corresponds to the inputs, outputs or IO's of the cells in the design (Note the difference between port and pin)
- *Net*: These are the signal names, i.e., the wires that hook up the design together by connecting ports to pins and/or pins to each other.
- *Clock*: The port or pin that is identified as a clock source. The identification may be internal to the library or it may be done using `dc_shell-t` commands.
- *Library*: Corresponds to the collection of technology specific cells that the design is targeting for synthesis; or linking for reference.

3.3.2 Variables

Variables are placeholders used by DC for the purpose of storing information. The information may relate to instructions for tailoring the final netlist, or it may contain user-defined value to be used for automating the synthesis process. Some variables are pre-defined by DC and may be used by the designer to obtain the current value stored in the variable. For example, the variable called “`bus_naming_style`” has a special meaning to DC, while “`captain_picard`” has no meaning to DC. The latter may be used to hold any user-defined value for scripting purposes.

All variables are global and last only during the session. They are not saved along with the design database. Upon completion of the `dc_shell-t` session, the value of the variables is lost. Most `dc_shell-t` variables have a default value associated with them, which they inherit at the start of the session. For instance, the following variable uses “`SYNOPSIS_UNCONNECTED_`” as its default value. You may change it to “`MY_DANGLE_`” and DC will write out the verilog netlist with the prefix of all unconnected nets as “`MY_DANGLE_`”

```
dc_shell-t> set verilogout_unconnected_prefix “MY_DANGLE_”
```

A list of all DC variables may be obtained by using the following DC command:

```
dc_shell-t> printvar *
```

For a particular type of variable (say test related) use the following and DC will return all variables which have the “test” string in them.

```
dc_shell-t> printvar *test*
```

3.3.3 Attributes

Attributes are similar in nature to variables. Both store information. However, attributes store information on a particular design object such as nets, cells or clocks.

Generally, attributes are pre-defined and have special meaning to DC, though designers may set their own attributes if desired. For example, the `set_dont_touch` is a pre-defined attribute, used to set a `dont_touch` on a design, thereby disabling DC optimization on that design.

Attributes are set on and retrieved from the design object by using the following commands:

```
set_attribute    <object list>  
                 <attribute name>  
                 <attribute value>  
  
get_attribute    <object list>  
                 <attribute name>
```

For example, one may use the following command to find the `max_transition` value set in the library (called “STD_LIB”) and the `area` attribute set for an inverter (INV2X) in that library:

```
dc_shell-t> get_attribute STD_LIB default_max_transition
```

```
dc_shell-t> get_attribute STD_LIB/INV2X area
```

Attributes may be removed from DC by using the following `dc_shell-t` command:

```
remove_attribute <attribute name>
```



3.4 Finding Design Objects

One of the most useful commands provided by DC & PT is the `get_*` commands. Sometimes, it becomes necessary to locate objects in `dc_shell-t` for the purpose of scripting or automating the synthesis process. The `get_*` commands are used to locate a list of designs or library objects in DC. Several types of `get_*` commands are provided. Examples are:

`get_ports`, `get_nets`, `get_designs`, `get_lib_cells`, `get_cells`, `get_clocks` etc.

A full list of get commands may be found by typing “`help get_*`” in the `dc_shell-t` command line.

Some examples of the `get_*` command are:

```
dc_shell-t> set_dont_touch [get_designs blockA]
```

☺ Applies the `dont_touch` attribute on the design called `blockA`.

```
dc_shell-t> remove_attribute [get_designs *] dont_touch
```

☺ Removes the `dont_touch` attribute from the whole design.

```
dc_shell-t> get_lib_cells stdcells_lib/*
```

☺ Lists all library cells of the library called `stdcells_lib`.

```
dc_shell-t> get_pins stdcells_lib/DFF1/*
```

☺ Lists all the pins of `DFF1` cell present in the library called `stdcells_lib`.

```
dc_shell-t> set_dont_touch_network [get_ports [list clk scan_en] ]
```

☺ Applies the `dont_touch_network` attribute on the specified ports.

3.5 Synopsys Formats

Most Synopsys products support and share, a common internal structure, called the “db” format. The db files are the binary compiled forms representing the text data, be it the RTL code, the mapped gate-level designs, or the Synopsys library itself. The db files may also contain any constraints that have been applied to the design.

In addition, all Synopsys tools understand the following formats of HDL. DC is capable of reading or writing any of these formats.

1. Verilog
2. VHDL
3. EDIF

Today, Verilog and VHDL are the two main HDLs in use, for coding a design. EDIF (Electronic Design Interchange Format) is primarily utilized for porting the gate level netlist, from one tool to another. EDIF was a popular choice a few years back. However, recently Verilog has gained popularity and dominance prompted by its simple to read format and description. Most of the EDA tools today, support both Verilog and EDIF.

VHDL in general is not used for porting the netlist from one vendor tool to another, since it requires the use of IEEE packages, which may vary between different tools. This language is essentially used for the purpose of coding the design and system level verification.

3.6 Data Organization

It is a good practice to organize files according to their formats. This facilitates automating the synthesis process. A common practice is to organize them using the following file extensions.

Script files:	<filename>.scr
RTL Verilog file:	<filename>.v
Synthesized Verilog netlist:	<filename>.sv
RTL VHDL file:	<filename>.vhd

Synthesized VHDL netlist:	<filename>.svhd
EDIF file:	<filename>.edf
Synopsys database file:	<filename>.db
Reports:	<filename>.rpt
Log files:	<filename>.log

3.7 Design Entry

Before synthesis, the design must be entered in DC in the RTL format (although other formats also exist). DC provides the following two methods of design entry:

- a) “read” command
- b) “analyze/elaborate” command

Synopsys initially introduced the **read** command, which was then followed by the **analyze/elaborate** commands. The latter commands for design entry provide a fast and powerful method over the **read** command and are recommended for RTL design entry.

The **analyze** and **elaborate** commands are two different commands, allowing designers to initially analyze the design for syntax errors and RTL translation before building the generic logic for the design. The generic logic or GTECH components are part of the Synopsys generic technology independent library. They are unmapped representations of boolean functions and serve as place holders for the technology dependent library.

The **analyze** command also stores the result of the translation in the specified design library (UNIX directory) that may be used later. For example, a design analyzed previously may not need re-analysis and can merely be elaborated, thus saving time. Conversely, the **read** command performs the function of both **analyze** and **elaborate** commands but does not store the analyzed results, therefore making the process slow in comparison.

Parameterized designs (such as usage of *generic* statement in VHDL) must use **analyze** and **elaborate** commands in order to pass required parameters,

while elaborating the design. The read command should be used for entering pre-compiled designs or netlists in DC.

The following table lists major differences between the read and analyze/elaborate commands for various categories:

Table 3-1. Difference between analyze/elaborate and read commands

Category	analyze/elaborate	read
Input format	RTL in Verilog or VHDL.	All formats: Verilog, VHDL, EDIF, db etc.
Recommended usage	Synthesizing the RTL in Verilog or VHDL format.	Reading netlists, pre-compiled designs etc.
Design Libraries	Use <code>-library</code> option to specify design library other than the directory from which <code>dc_shell</code> was invoked.	Storing of analyzed results is not possible.
Generics (used in VHDL)	Parameters of the generic statements may be set during elaboration of the design.	Cannot be used to pass parameters.
Architecture (in VHDL)	Can specify architecture in VHDL to be elaborated.	Cannot specify architecture in VHDL to be elaborated.

In contrast to DC, PT uses different commands for design entry. PT, being a static timing analyzer, only works on the mapped structural netlists. The design entry commands used by PT are described in Chapter 12.

3.8 Compiler Directives

Sometimes it is necessary to control the synthesis process from the HDL source itself. This control is primarily needed because of differences that may exist between the synthesis and the simulation environments. Other times, the control is needed simply to direct DC to map to certain types of components; or for embedding the constraints and attributes directly in the HDL source code.

DC provides a number of compiler directives targeted specifically for Verilog and VHDL design entry formats. These directives provide the means to control the outcome of synthesis, directly from the HDL source code. The

directives are specified as “comments” in the HDL code, but have specific meaning for DC. These special comments alter the synthesis process, but have no effect on the simulation.

The following sub-sections describe some of the most commonly used directives, both for Verilog and VHDL formats. For a complete list of directives, users are advised to refer to the Design Compiler Reference Manual.

3.8.1 HDL Compiler Directives

The HDL compiler directives refer to the translation process of RTL in Verilog format to the internal format used by Design Compiler. As stated above, specific aspects of the translation are controlled by “comments” within the Verilog source code. At the beginning of each directive is the regular Verilog comment `//` or `/*` followed by the keyword “**synopsys**” (all in lower case). Generally, users prefer the former style to specify HDL compiler directives. Therefore, to keep it simple, only `//` style of comments for HDL directives are discussed in this section.

All comments beginning with `// synopsys` are assumed to only contain HDL compiler directives. DC displays an error if anything apart from HDL compiler directives (say other comments or parts of Verilog code) are present after the `// synopsys` statement.

3.8.1.1 `translate_off` and `translate_on` Directives

These are some of the most useful and frequently used directives. They provide the means to instruct DC to stop translation of the Verilog source code from the start of “`// synopsys translate_off`”, and start the translation again after it reaches the next directive, “`// synopsys translate_on`”. These directives must be used in pairs, with the `translate_off` directive taking the lead.

Consider a scenario where parts of the code present in the source RTL is meant solely for the purpose of dynamic simulation (or maybe the test bench

is structured to make use of these statements). Example 3.2 illustrates such a scenario, which contains the Verilog *`ifdef* statement to facilitate setting parameters at the command line during simulation. Such a code is clearly unsynthesizable, since the `VENDOR_ID` depends on the mode specified during simulation. Furthermore, since the HDL compiler cannot handle this statement, it issues an error stating that the design could not be read due to “Undefined macro ‘ifdef’”

Example 3.2

```
`ifdef MY_COMPANY
  `define VENDOR_ID 16'h0083
`else
  `define VENDOR_ID 16'h0036
`endif
```

The `translate_off` and `translate_on` HDL directives may be used in this case to bypass the “simulation only” parts of the verilog code as illustrated in Example 3.3. The resulting logic will contain the `VENDOR_ID` values pertaining to `MY_COMPANY` only. To change it to the other value, the user has to edit the code and move the HDL directives to make the other `VENDOR_ID` value visible.

Example 3.3

```
// synopsys translate_off
  `ifdef MY_COMPANY
// synopsys translate_on

  `define VENDOR_ID 16'h0083

// synopsys translate_off
  `else
    `define VENDOR_ID 16'h0036
  `endif
// synopsys translate_on
```


3.8.2 VHDL Compiler Directives

Similar to the HDL compiler, the VHDL compiler directives are special VHDL comments that affect the actions of the VHDL compiler. All VHDL compiler directives start with the VHDL comment (--), followed either by synopsys or pragma statements. This provides a special meaning to the compiler and compels it to perform specified task.

3.8.2.1 `translate_off` and `translate_on` Directives

These directives work in the similar fashion as the ones described previously for the HDL compiler, with the exception that these require the VHDL comments as follows:

```
-- synopsys translate_off
-- synopsys translate_on
-- pragma translate_off
-- pragma translate_on
```

The VHDL compiler ignores any RTL code between the `translate_off/on` directives, however it does perform a syntax check on the embedded code. In order to refrain the compiler from conducting syntax checks; the code must be made completely transparent. This can be achieved by setting the following variable to true:

```
hdlin_translate_off_skip_text = true
```

These directives are primarily used to block simulation specific constructs in the VHDL code. For example, the user may have a library statement present in the netlist, which specifies the name of the library that contains the VITAL models of the gates present in the netlist. This means that for the purpose of simulation, the gates present in the netlist are being referenced from this library. Upon reading the VHDL code, DC produces an error, since the library statement is specific to simulation only. To circumvent this problem,

one may envelop the library statement with the above directives to force DC to completely ignore the library statement.

3.8.2.2 **synthesis_off and synthesis_on Directives**

The `synthesis_off/on` directives work in a manner similar to the `translate_on/off` directives. The behavior of `synthesis_off/on` directives itself, is not affected by the value of the `hdlin_translate_off_skip_text` variable. However, the `translate_off/on` directives perform exactly the same function if the value of the variable specified above is set to false.

The above directives are the preferred approach to hide the simulation only constructs. Though, the VHDL compiler performs the syntax checks of the code present within these directives, it ignores the code for the purpose of synthesis.

Syntactically, these variables may be used as follows:

```
-- pragma synthesis_off
    <VHDL code goes here, used only for simulation>
-- pragma synthesis_on
```

3.9 Chapter Summary

This chapter introduced the reader to various terminology and concepts used by Synopsys.

Starting from a brief description and purpose of some of the tools provided by Synopsys, the chapter covered the Synopsys environment that included examples of startup files needed for PhyC, DC and PT, followed by the concepts of Objects, Variables and Attributes.

A brief introduction was also provided for the `find` command and its usefulness. Different Synopsys formats were discussed along with Design entry methods. The advantages and disadvantages of using `read` versus `analyze/elaborate` command were also covered.

Finally, the chapter concluded by describing some of the most useful directives used by DC for the purpose of hiding simulation only constructs.

Throughout the chapter, various examples were provided to facilitate the user in understanding these concepts.

SYNOPSYS TECHNOLOGY LIBRARY

Synopsys technology library format has almost become the de-facto library standard. Its compact yet informative format allows adequate representation of the deep sub-micron technologies. The popularity of the Synopsys library format is evident from the fact that most place and route tools provide a direct translation of the Synopsys libraries, with almost a one-to-one mapping between the timing models in Synopsys libraries, and the place and route timing models. A basic understanding of the library format and delay calculation methods is the key for successful synthesis.

Designers usually do not concern themselves with full details of the technology library as long as the library contains a variety of cells, each with different drive strengths. However, in order to optimize the design successfully, it is essential for designers to have a clear understanding of the delay calculation method used by DC along with the wire-load modeling and cell descriptions. It is therefore, the intent of this chapter to describe the Synopsys technology library from the designer's perspective, rather than discussing details about the structural and functional syntax of the library.

4.1 Technology Libraries

The Synopsys technology libraries can be separated in two broad classes:

1. Logic library
2. Physical library

4.1.1 Logic Library

The logic library contains information relevant only to the synthesis process and is used by DC for synthesis and optimization of the design. This information may include pin-to-pin timing, area, pin types and power along with other necessary data needed by DC. No physical information is present in the logic library.

The logic library is a text file (usually with extension “.lib”), which is compiled using the Library Compiler (LC) to generate a binary format with “.db” extension.

4.1.2 Physical Library

The physical library contains the physical characteristics of the cell along with other necessary information relevant to Physical Compiler. Such information may contain data relating to the physical dimensions of cells, layer information, orientation of cells etc. For each logical cell, a corresponding physical cell should also be present.

The physical library is also a text file (usually with extension “.plib”) and is compiled by LC to generate a binary format with a “.pdb” extension. Synopsys have provided a useful utility called “lef2pdb” that takes the standard LEF (Library Exchange Format) file and the process technology file (also in LEF format) as input and converts it to the “.pdb” format. The former file contains physical information about each cell in the design, whereas the process technology file contains information specific to a process such as number of layers, pitch, resistance, capacitance etc.

Following is the usage of this command.

```
lc_shell> lef2pdb -t tech.lef -l standard_cells.lef
```

Full explanation of the physical library and its syntax is beyond the scope of this book. Readers are advised to refer to the Physical Library Reference Manual for details.

4.2 Logic Library Basics

The logic library contains the following information:

- a) Library group
- b) Library level attributes
- c) Environment description
- d) Cell description

4.2.1 Library Group

The library group statement specifies the name of the library, followed by an open brace. The closing brace is the last entry in the library file. Anything between the open and closing brace, is part of the entire library group description.

```
library (ex25) { /* start of library */  
...  
  < library description >  
...  
}/* end of library*/
```

It is recommended that the file name and the technology library name be the same.

4.2.2 Library Level Attributes

The library level attributes are statements that apply to library as a whole. These generally contain library features such as technology type, date, revision, and default values that apply to the entire library.

```
library (ex25) {
    technology (cmos);
    delay_model          : table_lookup;
    date                 : "Feb 29,2000";
    revision             : "1.0";
    current_unit         : "1 A";
    time_unit            : "1 ns";
    voltage_unit         : "1 V";
    pulling_resistance_unit : "1 kohm";
    capacitive_load_unit (1.0, pf);
    default_inout_pin_cap : 1.5 ;
    default_input_pin_cap : 1.0 ;
    default_output_pin_cap : 0.0 ;
    default_max_fanout    : 10.0 ;
    default_max_transition : 3.0;
    default_operating_conditions: NOMINAL
    in_place_swap_mode    : match_footprint;
    .....
    .....
}
```

4.2.3 Environment Description

Environment attributes are defined in the library to model the variations of temperature, voltage and manufacturing processes. These consist of scaling factors (derating), timing range models and operation conditions. In addition, the environment description also contains wire-load models that are used by DC to estimate interconnect wiring delays.

4.2.3.1 Scaling Factors

The scaling factors or K-factors are multipliers that provide means for derating the delay values based on the variations in process, voltage and temperature, or simply PVT. Only some of the K-factor statements are shown below as an example. Please refer to the library compiler reference manual for full details.

```
k_process_fall_transition      : 1.0 ;
k_process_rise_transition      : 1.2 ;
k_process_fall_propagation    : 0.4 ;
k_process_rise_propagation     : 0.4 ;
k_temp_fall_transition         : 0.03 ;
k_temp_rise_transition         : 0.04 ;
k_temp_fall_propagation        : 1.2 ;
k_temp_rise_propagation        : 1.24;
k_volt_fall_transition         : 0.02 ;
k_volt_rise_transition         : 0.5 ;
k_volt_fall_propagation        : 0.9 ;
k_volt_rise_propagation        : 0.85 ;
```

4.2.3.2 Operating Conditions

Sets of operating conditions defined in the library specify the process, temperature, voltage and the RC tree model. These are used during synthesis and timing analysis of the design. A library is characterized using one set of operating conditions. During synthesis or timing analysis, if another set of operating conditions is specified, then DC uses the K-factors to derate the delay values based upon the specified operating conditions. Library developers may define any number of operating conditions in the library. Typically the following operating conditions are defined in the technology library:

```
operating_conditions (WORST) {
    process : 1.3 ;
    temperature: 100.0 ;
    voltage: 2.75 ;
```



```
    tree_type : worst_case_tree ;  
}  
operating_conditions (NOMINAL) {  
    process : 1.0 ;  
    temperature: 25.0 ;  
    voltage : 3.00 ;  
    tree_type : balanced_tree ;  
}  
operating_conditions (BEST) {  
    process : 0.7 ;  
    temperature: 0.0 ;  
    voltage: 3.25 ;  
    tree_type : best_case_tree ;  
}
```

The process, temperature and voltage attributes have already been explained previously. The **tree_type** attribute defines the environmental interconnect model to be used. DC uses the value of this attribute to select the appropriate formula while calculating interconnect delays. The **worst_case_tree** attribute models the extreme case when the load pin is at the most distant end of a net, from the driver. In this case the load pin incurs the full net capacitance and resistance. The **balanced_tree** model uses the case where all load pins are on separate but equal interconnect wires from the driver. The load pin in this case, incurs an equal portion of net capacitance and resistance. The **best_case_tree** models the case where the load pin is sitting right next to the driver. The load pin incurs only the net capacitance, without any net resistance.

4.2.3.3 Timing Range Models

The **timing_range** models provide additional capability of computing arrival times of signals, based upon the specified operating conditions. This capability is provided by Synopsys to accommodate the fluctuations in operating conditions for which the design has been optimized. DC uses the timing ranges to evaluate the arrival times of the signals during timing analysis.

```

    timing_range (BEST) {
        faster_factor : 0.5 ;
        slower_factor : 0.6 ;
    }
    timing_range (WORST) {
        faster_factor : 1.2 ;
        slower_factor : 1.3 ;
    }

```

4.2.3.4 Wire-Load Models

The `wire_load` group contains information that DC utilizes to estimate interconnect wiring delays during the pre-layout phase of the design. Usually, several models appropriate to different sizes of the logic are included in the technology library. These models define the **capacitance**, **resistance** and **area** factors. In addition, the `wire_load` group also specifies **slope** and **fanout_length** for the logic under consideration.

The **capacitance**, **resistance** and **area** factors represent the wire resistance, capacitance and area respectively, per unit length of interconnect wire. The **fanout_length** attribute specifies values for the length of the wire associated with the number of fanouts. Along with fanout and length, this attribute may also contain values for other parameters, such as **average_capacitance**, **standard_deviation** and **number_of_nets**. These attributes and their values are written out automatically, when generating wire-load models through DC. For manual creation, only the values for fanout and length are needed, using the **fanout_length** attribute. For nets exceeding the longest length specified in the **fanout_length** attribute, the slope value is used to linearly interpolate the existing **fanout_length** value, in order to determine its value.

```

wire_load (SMALL) {
    resistance : 0.2 ;
    capacitance : 1.0 ;
    area : 0 ;
    slope : 0.5 ;
    fanout_length( 1, 0.020) ;
    fanout_length( 2, 0.042) ;
    fanout_length( 3, 0.064) ;

```

```

        fanout_length( 4, 0.087 ) ;
        * * * *
        fanout_length(1000,20.0) ;
    }
    wire_load(MEDIUM) {
        resistance      : 0.2 ;
        capacitance     : 1.0 ;
        area            : 0 ;
        slope           : 1.0 ;
        fanout_length( 1, 0.022 ) ;
        fanout_length( 2, 0.046 ) ;
        fanout_length( 3, 0.070 ) ;
        fanout_length( 4, 0.095 ) ;
        * * * *
        fanout_length(1000,30.0) ;
    }
    wire_load(LARGE) {
        resistance      : 0.2 ;
        capacitance     : 1.0 ;
        area            : 0 ;
        slope           : 1.5 ;
        fanout_length( 1, 0.025 ) ;
        fanout_length( 2, 0.053 ) ;
        fanout_length( 3, 0.080 ) ;
        fanout_length( 4, 0.110 ) ;
        * * * *
        fanout_length( 1000, 40.0 ) ;
    }
}

```

In addition to the `wire_load` groups, other attributes are defined in the library to automatically select the appropriate `wire_load` group, based on the total cell area of the logic under consideration.

```

wire_load_selection(AUTO_WL) {
    wire_load_from_area ( 0, 5000, "SMALL" ) ;
    wire_load_from_area ( 5000, 10000, "MEDIUM" ) ;
    wire_load_from_area ( 10000, 15000, "LARGE" ) ;
}

```

```

default_wire_load_selection  : AUTO_WL ;
default_wire_load_mode      : enclosed ;

```

It is recommended that the value of the `default_wire_load_mode` be set to “enclosed” or “segmented” instead of “top”. The wire load modes and their application are described in detail in Chapter 6.

4.2.4 Cell Description

Each cell in the library contains a variety of attributes describing the function, timing and other information related to each cell. Rather than going into detail and describing all the attributes possible, only the relevant attributes and related information useful to designers are shown in the example below:

```

cell (BUFFD0) {
    area : 5.0 ;
    pin (Z) {
        max_capacitance : 2.2 ;
        max_fanout : 4.0 ;
        function : “I” ;
        direction : output ;
        timing () {
            . . . .
        }
        timing () {
            . . . .
        }
        related_pin : “I” ;
    }
    pin (I) {
        direction : input ;
        capacitance : 0.04 ;
        fanout_load : 2.0 ;
        max_transition : 1.5 ;
    }
}

```

The **area** attribute defines the cell area as a floating-point number without any units followed by pin description and their related timing.

In addition, several design rule checking (DRC) attributes may be associated with each pin of the cell. These are:

- **fanout_load** attribute for input pins.
- **max_fanout** attribute for output pins.
- **max_transition** attribute for input or output pins.
- **max_capacitance** attribute for output or inout pins.

The DRC conditions are based on the vendor's process technology and should not be violated. The DRC attributes define the conditions in which the cells of the library operate safely. In other words, cells are characterized under certain conditions (output loading, input slope etc.). Designs violating these conditions may have a severe impact on the normal operation of the cells, thereby causing the fabricated chip to fail.

Even though, the previous example contains all four attributes, generally only two are used. In most cases, either the **fanout_load** along with **max_fanout**, or **max_transition** with **max_capacitance** are used.

The **fanout_load** and **max_fanout** DRC attributes are related to each other, in such that the **max_fanout** value at the output of the driver pin cannot exceed the sum of all **fanout_load** values at each input pin of the driven cells. Consider the cell (BUFFD0) shown in the previous example. This cell contains a **max_fanout** value of 4.0 associated to the output pin Z, while the **fanout_load** value at its input is 2.0. This cell therefore, cannot drive more than 2 of its own kind (BUFFD0) cells, since

$$\text{max_fanout (4)} = \text{fanout_load (2) of 1}^{\text{st}} \text{ cell} + \text{fanout_load (2) of 2}^{\text{nd}} \text{ cell}$$

If the DRC violations occur, then DC replaces the driving cell with another that has a higher **max_fanout** value.

The **max_transition** attribute is generally applied to the input pin, whereas the **max_capacitance** is applied to the output pin. Both attributes perform

the similar function as the `max_fanout` and `fanout_load` attributes. The difference being, that the `max_transition` attribute defines that any net that has a transition time greater than the specified `max_transition` value of the load pin, cannot be connected to that pin. The `max_capacitance` at the output pin specifies that the output pin of the driver cell cannot connect to any net that has the total capacitance (interconnect and load pin capacitance) greater than, or equal to the maximum value defined at the output pin.

If DRC violations occur, then DC replaces the driving cell with another that has a higher `max_capacitance` value.

In addition, the output pin contains attributes defining the function of the pin, and the delay values related to the input pin. The input pin defines its' pin capacitance and the direction. The `capacitance` attribute should not be confused with the `max_capacitance` attribute. DC uses the `capacitance` attribute to perform delay calculations only, while the `max_capacitance`, as explained above, is used for design rule checking.

It is also worthwhile to mention here that for sequential cells, the clock input pin uses another attribute (`clock : true`) that specifies that the input pin is of type "clock". More details can be found in the Library Compiler Reference Manual.

The cell's DRC attributes are often the most criticized part of the cell library. Library developers often find it impossible to satisfy everyone and are often blamed for not implementing the "right" numbers for these attributes. The problem is caused because the library, to a certain extent is dependent upon the coding style and chosen methodology. What works perfectly for one design may produce inadequate results for another design. It is therefore the intent of this section to briefly explain the solutions that designers may use to tailor the library to suit their needs.

In order to accommodate the design requirements, it is possible to change the values of the above DRC attributes on a per cell basis. However, it must be noted that the DRC attributes set in the library can only be tightened, they cannot be loosened. This can only be done, if the attributes are pre-specified in the cell description. Users should realize that if these attributes are not

already specified on the pin of the cell in the technology library, it is not possible to add these attributes on the pin, from `dc_shell`.

For instance, to change the `max_fanout` value specified for pin Z of cell `BUFFD0` (of library `ex25` described previously), from 4.0 to 2.0; the following `dc_shell-t` command may be used:

```
dc_shell-t> set_attribute [get_pins ex25/BUFFD0/Z] max_fanout 2
```

One may also use wildcards in the above command to cover a variety of cells. This is useful for cases where a global change is required. For example, users may use the following command to change the `max_fanout` value on all cells with 0 drive strengths in the technology library:

```
dc_shell-t> set_attribute [get_pins ex25/*D0/Z] max_fanout 2
```

Similarly, the `set_attribute` command may be used to alter the value of other DRC attributes. The above command may be specified in the `.synopsys_dc.setup` file for global implementation.

4.3 Delay Calculation

Synopsys supports several delay models. These include the CMOS generic delay model, CMOS piecewise linear delay model and the CMOS non-linear table lookup model. Presently, the first two models are not in common use, due to their inefficiencies in representing the true delays caused by VDSM geometries. The non-linear delay model is the most prevalent delay model used in the ASIC world.

4.3.1 Delay Model

The non-linear delay model (NLDM) method uses a circuit simulator to characterize a cell's transistors with a variety of input slew rates, and output load capacitances. The results form a table, with input transition and output load capacitance as the deciding factor for calculating the resultant cell delay.

Figure 4-1, shown below depicts the resulting delays and slew rates, interpolated to produce a non-linear delay model. The model's accuracy depends on the precision and range, of the chosen input slew rates and load capacitances.

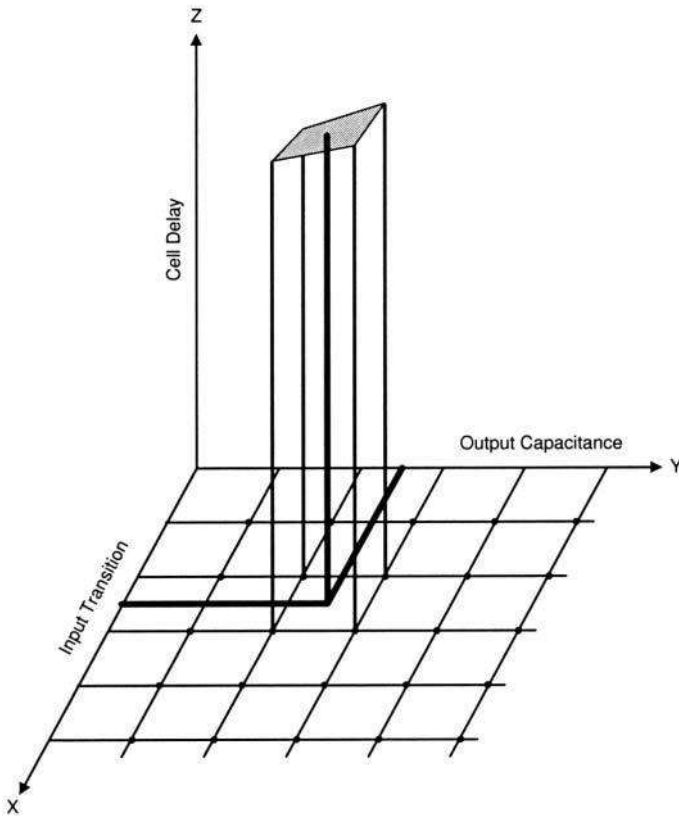


Figure 4-1. NLDM Table

If the delay number falls within the square (table in the library), then the delay is computed using interpolation techniques. The values of the surrounding four points are used to determine the delay value, using numerical methods. The problem arises, when any of the parameters fall outside the table. DC is best designed to extrapolate the resulting delay, but

often ends up with an extremely high value. This may be a blessing in disguise, since a high value is easily noticeable during static timing analysis, providing designers an opportunity to correct the situation.

4.3.2 Delay Calculation Problems

The delay calculation of a cell is performed using the input transition time and the capacitive loading seen at the output. The input transition time of a cell is evaluated based upon the transition delay of the driving cell (previous cell). If the driving cell contains more than one timing arc, then the worst transition time is used, as input to the driven cell. This directly impacts the static timing analysis and the generated SDF file for a design.

Consider the logic shown in Figure 4-2. The signals, *reset* and *signal_a* are inputs to the instance U1. Let us presume that the *reset* signal is non critical as compared to *signal_a*. The *reset* signal is a slow signal, therefore, the transition time of this signal is high as compared to *signal_a*. This causes two transition delays to be computed for cell U1 (2 ns from A to Z, and 0.3 ns from B to Z). When generating SDF, the two values will be written out separately as part of the cell delay, for the cell U1. However, the question now arises, which of the two values does DC use to compute the input transition time for cell U2? DC uses the worst (maximum) transition value of the preceding gate (U1) as the input transition time for the driven gate (U2). Since the transition time of *reset* signal is more compared to *signal_a*, the 2ns value will be used as input transition time for U2. This causes a large delay value to be computed for cell U2 (shaded cell).

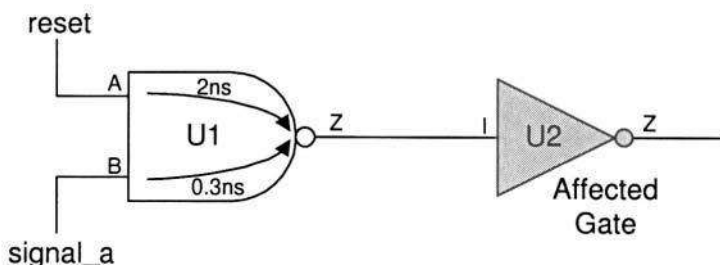


Figure 4-2. Delay Computation

To avoid this problem one needs to inform DC, not to perform the delay calculation for the timing arc – pin A to pin Z of cell U1. This step should be performed before writing out the SDF. The following `dc_shell` command may be used for this purpose:

```
dc_shell-t> set_disable_timing U1 -from A -to Z
```

Unfortunately, this problem also arises during static timing analysis. Failure to disable the timing computation of the false path leads to large delay values computed for the driven cell.

4.4 What is a Good Library?

Cell libraries determine the overall performance of the synthesized logic. A good cell library will result in fast design with smallest area, whereas a poor library will degrade the final result.

Historically, the cell libraries were schematic based. Designers would choose the appropriate cell and connect them manually to produce a netlist for the design. When the automatic synthesis engines became prevalent, the same schematic based libraries were converted and used for synthesis. However, since the synthesis engine relies on a number of factors for optimization, this approach almost always resulted in poor performance of the synthesized

logic. It is therefore imperative that the cell library be designed catered solely towards the synthesis approach.

The following guidelines outline, the specific kind of cells in the technology library desired by the synthesis engine.

- a) A variety of drive strengths for all cells.
- b) Larger varieties of drive strengths for inverters and buffers.
- c) Cells with balanced rise and fall delays (used for clock tree buffers and gated clocks).
- d) Same logical function and its inversion as separate outputs, within the same physical cell (e.g., OR gate and NOR gate, as a single cell), again with a variety of drive strengths.
- e) Same logical function and its inversion as separate cells (e.g., AND gate and NAND gate as two separate cells), with a variety of drive strengths.
- f) Complex cells (e.g., AOI, OAI or NAND gate with one input inverted etc) with a variety of high drive strengths.
- g) High fanin cells (e.g., AOI with 6 inputs and one output) with a range of different drive strengths.
- h) Variety of flip-flops with different drive strengths, both positive and negative-edge triggered.
- i) Single or Multiple outputs available for each flip-flop (e.g., Q only, or QN only, or both), each with a variety of drive strengths.
- j) Flops to contain different inputs for Set and Reset (e.g., Set only, Reset only, no Set or Reset, both Set and Reset).
- k) Variety of latches, both positive and negative-edge enabled each with different drive strengths.

1) Several delay cells. These are useful when fixing the hold-time violations.

Using the above guideline will result in a library optimized to handle the synthesis algorithm. This provides DC with the means to choose from a variety of cells to implement the best possible logic for the design.

It is worthwhile to note that the usage of high fanin cells, although useful in reducing the overall cell area, may cause routing congestion, which may inadvertently cause timing degradation, and/or increase in the area of the routed design. It is therefore recommended that these cells be used with caution.

Some designers prefer to exclude the low drive strengths for high fanin cells from the technology library. This is again based on the algorithm used by the routing engine and the type of placement (timing driven etc.) used by designers. If the router is not constrained, then it uses a method by which it associates a weight to each net of the design while placing cells. Depending upon the weight of the net, the cells are pulled towards the source having the highest weight. High fanin cells have a larger weight associated to its inputs (because of the number of inputs) compared to the weight associated with their outputs (single output). Therefore, the router will place these cells near the gates that are driving it. This will result in the high fanin cell being pulled away from the cell it is supposed to be driving, causing a long net to be driven by the high fanin cell. If the high fanin cell is not strong enough to drive this long net (large capacitance) then the result will be the computation of large cell delay for the high fanin cell, as well as the driven gate (because of slow input transition time). By eliminating the low drive strengths of the high-fanin cells from the technology library, this problem can be prevented after layout.

4.5 Chapter Summary

To summarize, this chapter described the contents of the Synopsys logic library from the designer's perspective. The emphasis was placed upon the correct usage and understanding of the logic library, rather than focusing on details that are relevant only to library developers.

A brief discussion was also provided for the physical library that is used by the Physical Compiler. Emphasis was not placed on describing the syntax and functionality of this library due to fact that the topic of discussion is beyond the scope of this book.

The chapter started with basics of the logic library, with separate groups within the library. The relevant portions of each group were explained in detail. This included explanation of all attributes that the library uses to perform its task.

Special emphasis was given to describing the delay calculation method, along with operating conditions, wire-load modeling and cell description. At each step, problems associated and workarounds were explained in detail.

Finally, suggestions were provided to the user as to what constitutes a good library optimized for synthesis engine. This includes helpful hints by taking into account the router behavior of the layout tool.

PARTITIONING AND CODING STYLES

Successful synthesis depends strongly on proper partitioning of the design, together with a good HDL coding style.

Logical partitioning is the key to successful synthesis (and place and route, if layout is hierarchical). Traditionally, designers partitioned the design in accordance with the functionality of each block, giving no thought to the synthesis process. As a result of incorrect partitioning, the inflexible boundaries degrade the synthesis results, which makes optimization difficult. Partitioning the design correctly can significantly enhance the synthesized result. In addition, reduced compile time and simplified script management is also achieved.

A good coding style is imperative, not only for the synthesis process, but also for easy readability of the HDL code. Today, many designers only stress verifying the functionality of the design. Driven by time restriction and/or lack of communication between the team members, designers do not have the luxury of carefully scrutinizing the HDL coding style. The fact remains, however, that a good coding style not only results in reduction of chip area and aids in top-level timing, but also produces faster logic.

5.1 Partitioning for Synthesis

Partitioning can be viewed as, utilizing the “Divide and Conquer” concept to reduce complex designs into simpler and manageable blocks. Promoting design reuse is one of the most significant advantages to partitioning the design.

Apart from the ease in meeting timing constraints for a properly partitioned design, it is also convenient to distribute and manage different blocks of the design between team members.

The following recommendations achieve best synthesis results and reduction in compile time.

- a) Keep related combinational logic in the same module.
- b) Partition for design reuse.
- c) Separate modules according to their functionality.
- d) Separate structural logic from random logic.
- e) Limit a reasonable block size (depends on the memory capacity of the machine)
- f) Partition the top level (separate I/O Pads, Boundary Scan and core logic).
- g) Do not add glue-logic at the top level.
- h) Isolate state-machine from other logic.
- i) Avoid multiple clocks within a block.
- j) Isolate the block that is used for synchronizing multiple clocks.
- k) *WHILE PARTITIONING, THINK OF YOUR LAYOUT STYLE.*

The **group** and **ungroup** commands provide the designer with the capability of altering the partitions in DC, after the design hierarchy has already been defined by the previously written HDL code. Figure 5-1, illustrates such an action.

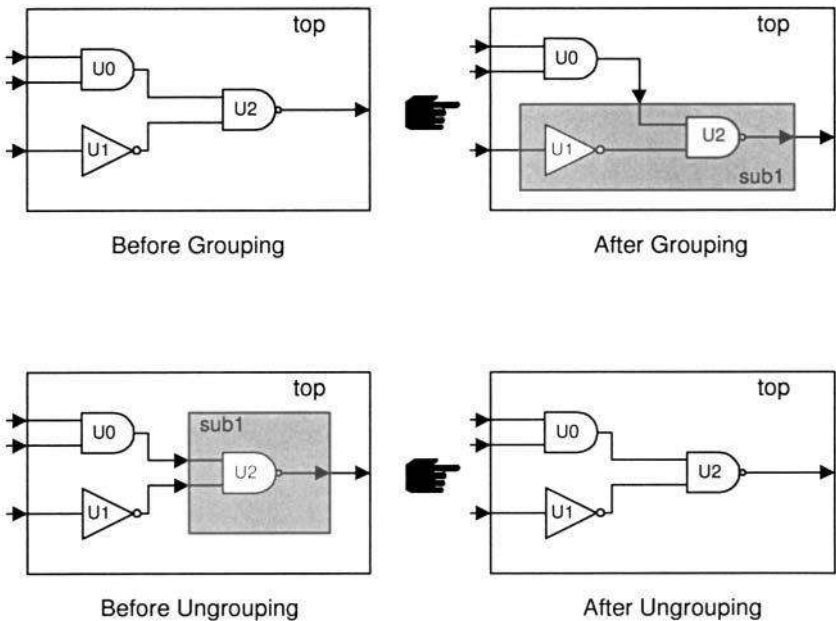


Figure 5-1. Changing Partitions

The **group** command combines the specified instances into a separate block. In Figure 5-1, instances U1 and U2 are grouped together to form a sub-block named sub1, using the following command:

```
dc_shell> current_design top
```

```
dc_shell> group {U1 U2} -design_name sub1
```

The **ungroup** command performs the opposite function. It is used to remove the hierarchy, as shown in Figure 5-1, by using the following command.


```
dc_shell> current_design top
```

```
dc_shell> ungroup -all
```

The designer can also use the `ungroup` command along with the `-flatten` and `-all` options to flatten the entire hierarchy. This is illustrated below:

```
dc_shell> ungroup -flatten -all
```

5.2 What is RTL?

Today, RTL or the Register Transfer Level is the most popular form of high-level design specification. An RTL description of a design describes the design in terms of transformation and transfer of logic from one register to another. Logic values are stored in registers where they are evaluated through some combinational logic, and then re-stored in the next register.

RTL functions like a bridge between the software and hardware. It is text with strong graphical connotations – text that implies graphics or structure. It can be described as technology independent, textual structural description, similar to a netlist.

5.2.1 Software versus Hardware

A frequent obstacle to writing HDL code is the software mind-set. HDLs have evolved from logic netlist representations. HDLs in their initial form (the Register Transfer Level) were a forum to represent logic in a format independent from any particular technology library. A higher level of HDL abstraction is the behavioral level that allows the design to be independent of timing and explicit sequencing.

Frequently, the expectation is that the synthesis tool will synthesize the HDL to the minimal area and maximum performance, regardless of how the HDL is written. The problem remains that at high level there are numerous ways of writing code to perform the same function. For example, a conditional expression could be written using *case* statements or *if* statements. Logically,

these expressions are responsible for performing the same task, but when synthesized they can give drastically different results, as far as type of logic inferred, area, and timing are concerned. A reasonable caveat told to recent adopters of synthesis is – THINK HARDWARE!

5.3 General Guidelines

The following are general guidelines that every designer should be aware of. There is no fixed rule to adhere to these guidelines, however, following them vastly improves the performance of the synthesized logic, and may produce a cleaner design that is well suited for automating the synthesis process.

5.3.1 Technology Independence

HDL should be written in a technology independent fashion. Hard-coded instances of library gates should be minimized. Preference should be given to inference rather than instantiation. The benefit being that the RTL code can be implemented with any ASIC library and new technology through re-synthesis. This is especially important for synthesizable IP cores that are commonly used by many designs.

In cases where placement of library gates is unavoidable, all the instantiated gates may be grouped together to form their own module. This helps in management of library specific aspects of a design.

5.3.2 Clock Related Logic

- a) Clock logic including clock gating logic and reset generation should be kept in one block – to be synthesized once and not touched again. This helps in a clean specification of the clock constraints. Another advantage is that the modules that are being driven by the clock logic can be constrained using ideal clock specifications.
- b) Avoid multiple clocks per block – try keeping one clock per block. Such restrictions later help avoid difficulties that may arise while constraining a block containing multiple clocks. It also helps in managing clock skew at

the physical level. Sometimes this becomes unavoidable, for instance where synchronization logic is present to sync signals from one clock domain to the other. For such cases, it is recommended that designer isolate the sync logic into a separate module for stand-alone synthesis. This includes setting a `dont_touch` attribute on the sync logic before instantiating it in the main block.

- c) Clocks should be given meaningful names. A suggestion is to keep the name of the clock that reflects its functionality in addition to its frequency. Another good practice is to keep the same name for the clock, uniform throughout the hierarchy, i.e., the clock name should not change as it traverses through the hierarchy. This simplifies the script writing and helps in automating the synthesis process.
- d) For DFT scan insertion, it is a requirement that the clocks be controlled from primary inputs. This may involve adding a mux at the clock source for controllability. Incorporate the mux logic within the module that contains all other clock logic. Isolating the clock logic block helps in fine tuning the synthesized logic in terms of gate size and the type of inference. If necessary, this small module can easily be hand tweaked to suit for optimal solution.

5.3.3 No Glue Logic at the Top

The top-level should only be used for connecting modules together. It should not contain any combinational glue logic. One of the benefits of this style is that it makes redundant the very time consuming top-level compile, which can now be simply stitched together without undergoing additional synthesis. Absence of glue logic at the top-level also facilitates layout, if performing hierarchical place and route.

5.3.4 Module Name Same as File Name

A good practice is to keep the module name (or entity name), same as the file name. Never describe more than one module or entity in a single file. A single file should only contain a single module/entity definition for synthesis.

This has enormous benefits in defining a clean methodology using scripting languages like PERL, AWK etc.

5.3.5 Pads Separate from Core Logic

Divide the top-level into two separate blocks “pads” and “core”. Pads are usually instantiated and not inferred, therefore it is preferred that they be kept separate from the core logic. This simplifies the setting of the `dont_touch` attribute on all the pads of the design, simultaneously. By keeping the pads in a separate block, we are isolating the technology dependent part of RTL code.

5.3.6 Minimize Unnecessary Hierarchy

Do not create unnecessary hierarchy. Every hierarchy sets a boundary. Performance is degraded, if unnecessary hierarchies are created. This is because DC is unable to optimize efficiently across hierarchies. One may use the `ungroup` command to flatten the unwanted hierarchies, before compiling the design to achieve better results.

5.3.7 Register All Outputs

This is a well-known Synopsys recommendation. The outputs of a block should originate directly from registers. Although not always practical, this coding/design style simplifies constraint specification and also helps optimization. This style prevents combinational logic from spanning module boundaries. It also increases the effectiveness of the characterize-write-script synthesis methodology by preventing the pin-pong effect that is common to this type of compilation technique.

5.3.8 Guidelines for FSM Synthesis

The following guidelines are presented for writing finite state machines that may help in optimizing the logic:

- a) State names should be described using “enumerated types” in VHDL, or “parameters” in Verilog.

- b) Combinational logic for computing the next state should be in its own *process* or *always* block, separate from the state registers.
- c) Implement the next-state combinational logic with a *case* statement.

5.4 Logic Inference

High-level Description Languages (HDLs) like VHDL and Verilog are front-ends to synthesis. HDLs allow a design to be represented in a technology independent fashion. However, synthesis imposes certain restrictions on the manner in which HDL description of a design is written. Not all HDL constructs can be synthesized. Not only that, synthesis expects HDLs to be coded in a specific way so as to get the desired results. One can say that synthesis is template driven – if the code is written using the templates that are understood and expected by the synthesis tool, then the results will be correct and predictable. The templates and other coding patterns for synthesis are called coding styles. For quality results it is imperative that designers possess a keen understanding of the coding styles, logic inferences, and the corresponding logic structures that DC generates.

5.4.1 Incomplete Sensitivity Lists

This is one of the most common mistakes made by designers. Incomplete sensitivity lists may cause simulation mismatches between the source RTL and the synthesized logic. DC issues a warning for signals that are present in the *process* or *always* block, but are absent from the sensitivity list. This is primarily a simulation problem since the process does not trigger when sensitized (because of the missing signal in the sensitivity list). The synthesized logic in most cases is generally correct for blocks containing incomplete sensitivity lists. However, it is strongly recommended that designers pay special attention to the sensitivity list and complete it in order to eliminate any surprises at the end of synthesis cycle.

Verilog Example

```
always @(weekend or go_to_beach or go_to_work)
begin
  if (weekend)
    action = go_to_beach
  else if (weekday)
    action = go_to_work;
```

VHDL Example

```
process (weekend, go_to_beach, go_to_work)
begin
  if (weekend) then
    action <= go_to_beach;
  elsif (weekday) then
    action <= go_to_work;
  end if;
end process;
```

The examples illustrated above do not contain the signal “weekday” in their sensitivity lists. The synthesized logic may still be accurate, however, during simulation the process will not trigger each time the signal “weekday” changes value. This may cause a mismatch between the simulation result of the source RTL and the synthesized logic.

5.4.2 Memory Element Inference

There are two types of memory elements – latches and flip-flops. Latches are level-sensitive memory elements, while flip-flops in general are edge-sensitive. Latches are transparent as long as the enable to the latch is active. At the time the latch is disabled, it holds the value present at the D input, at its Q output. Flip-flops on the other hand, respond to rising or falling edge of the clock.

Latches are simple devices, therefore they cover less area as compared to their counterparts, flip-flops. However, latches in general are more

troublesome because their presence in a design makes DFT scan insertion difficult, although not impossible. It is also complicated to perform static timing analysis on designs containing latches, due to their ability of being transparent when enabled. For this reason, designers generally prefer flip-flops to latches.

The following sub-sections provide detailed information on how to avoid latches, as well as how to infer them, if desired.

5.4.2.1 Latch Inference

A latch is inferred when a conditional statement is incompletely specified. An *if* statement with a missing *else* part is an example of incompletely specified conditional. Here is an example, both in Verilog and VHDL:

Verilog Example

```
always @(weekend)
begin
    if (weekend)
        action <= go_to_beach;
end
```

VHDL Example

```
process (weekend)
begin
    if (weekend = '1') then
        action <= go_to_beach;
    end process;
```

The above statement will cause the DC to infer a latch enabled by a signal called “weekend”. In the above example, “action” is not given any value when the signal “weekend” is 0. Always cover all the cases in order to avoid unintentional latch inference. This may be achieved by using an *else* statement, or using a *default* statement outside the *if* branch.

A latch may also get inferred from an incompletely specified *case* statement in Verilog.

```
`define sunny 2'b00
`define snowy 2'b01
`define windy 2'b10

wire [1:0] weather;

case (weather)
    sunny : action <= go_motorcycling;
    snowy : action <= go_skiing;
    windy : action <= go_paragliding;
endcase;
```

In the above case statement only 3 of the 4 possible values of “weather” are covered. This causes a latch to be inferred on the signal “action”. Note, for the above example the Synopsys `full_case` directive may also be used to avoid the latch inference as explained in Chapter 3. The following example contains the *default* statement that provides the fourth condition, thereby preventing the latch inference.

```
case (weather)
    sunny : action <= go_motorcycling;
    snowy : action <= go_skiing;
    windy : action <= go_paragliding;
    default : action <= go_paragliding;
endcase;
```

VHDL does not allow incomplete case statements. This often means that the *others* clause must be used, consequently the above problem does not occur in VHDL. However, latches may still be inferred by VHDL, if a particular output signal is not assigned a value in each branch of the *case* statement. The inference being that outputs must be assigned a value in all branches to prevent latch inference in VHDL.


```
case (weather) is
  when sunny   => action <= go_motorcycling;
  when snowy   => action <= go_skiing;
  when windy    => action <= go_paragliding;
  when others=> null;
end case;
```

The above example, although containing the *others* clause will infer latches because the output signal “action” is not assigned a particular value in the *others* clause. To prevent this, all branches should be completely specified, as follows:

```
case (weather) is
  when sunny   => action <= go_motorcycling;
  when snowy   => action <= go_skiing;
  when windy    => action <= go_paragliding;
  when others=> action <= go_paragliding;
end case;
```

5.4.2.2 Register Inference

DC provides a wide variety of templates for register inference. This is to support different edge-types of the clock and reset mechanisms. A register is inferred, when there is an edge specified in the sensitivity list. The edge could be a positive edge or a negative edge.

5.4.2.2.1 Register Inference in Verilog

In Verilog, a register is inferred when an edge is specified in the sensitivity list of an *always* block. One register is inferred for each of the variables assigned in the *always* block. All variable assignments, not directly dependent on the clock-edge should be made in a separate *always* block, which does not have an edge specification in its sensitivity list.

A plain and simple positive edge-triggered D flip-flop is inferred using the following template:

```
always @(posedge clk)
    reg_out <= data;
```

In order to infer registers with resets, the reset signal is added to the sensitivity list, with reset logic coded within the *always* block. Following is an example of a D flip-flop with an asynchronous reset:

```
always @(posedge clk or reset)
if (reset)
    reg_out <= 1'b0;
else
    reg_out <= data;
```

Having a synchronous reset is a simple matter of removing the “reset” signal from the sensitivity list. In this case, since the block responds only to the clock edge, the reset is also recognized only at the clock edge.

```
always @(posedge clk)
if (reset)
    reg_out <= 1'b0;
else
    reg_out <= data;
```

Negative edge-triggered flop may be inferred by using the following template:

```
always @(negedge clk)
    reg_out <= data;
```

Absence of negative edge-triggered flop in the technology library results in DC inferring a positive edge-triggered flop with an additional inverter to invert the clock signal.

5.4.2.2 Register Inference in VHDL

In VHDL a register is inferred when an edge is specified in the *process* body. The following example illustrates the VHDL template to infer a D flip-flop:

```
reg 1: process (clk)
begin
    if (clk'event and clk = '1') then
        reg_out <= data;
    end if;
end process Reg 1;
```

DC does not infer latches for variables declared inside functions, since variables declared inside functions are reassigned each time the function is called.

Coding style template for registers with asynchronous and synchronous resets are similar in nature to that of Verilog templates, shown in previous section.

Negative edge-triggered flop may be inferred by using the following template:

```
reg 1: process (clk)
begin
    if (clk'event and clk = '0') then
        reg_out <= data;
    end if;
end process Reg 1;
```

Absence of negative edge-triggered flop in the technology library results in DC inferring a positive edge-triggered flop with an additional inverter to invert the clock signal.

5.4.3 Multiplexer Inference

Depending upon the design requirements, the HDL may be coded in different ways to infer a variety of architectures using muxes. These may comprise of a single mux with all inputs having the same delay to reach the output, or a priority encoder that uses a cascaded structure of muxes to prioritize the input signals.

The correct use of *if* and *case* statements is a complex topic that is outside the scope of this chapter. There are application notes (from Synopsys) and other published materials currently available that explain the proper usage of these statements. It is therefore the intent of this chapter to refer the users to outside sources for this information. Only brief discussion is provided in the following sub-sections.

5.4.3.1 Use *case* Statements for Muxes

In general, *if* statements are used for latch inferences and priority encoders, while *case* statements are used for implementing muxes. It is recommended to infer muxes exclusively through *case* statements. The *if* statements may be used for latch inferencing and priority encoders. They may also be effectively used to prioritize late arriving signals. This kind of prioritizing may be implementation dependent. It also limits reusability.

To prevent latch inference in *case* statements the *default* part of the *case* statement should always be specified. For example, in case of a state machine, the default action could be that all states covered by the *default* clause cause a jump to the “start” state. Having a *default* clause in the *case* statement is the preferred way to write *case* statements, since it makes the HDL independent of the synthesis tool. Using directives like *full_case* and *parallel_case* makes the RTL code dependent on the synthesis tool. These directives should be avoided.

If the default action is to assign don't-cares, then a difference in behavior between RTL simulation and synthesized result may occur. This is because, DC may optimize the don't-cares randomly causing the resulting logic to differ.

5.4.3.2 *if* versus *case* Statements – A Case of Priorities

Multiple *if* statements with multiple branches result in the creation of priority encoder structure.

```
always @(weather or go_to_work or go_to_beach)
begin
    if (weather[0]) action = go_to_work;
    if (weather[1]) action = go_to_beach;
end
```

In the above example, the signal “weather” is a two-bit input signal and is used to select the two inputs, “go_to_work” and “go_to_beach”, with “action” as the output. When synthesized, the cascaded mux structure of the priority encoder is produced as shown in Figure 5-2.

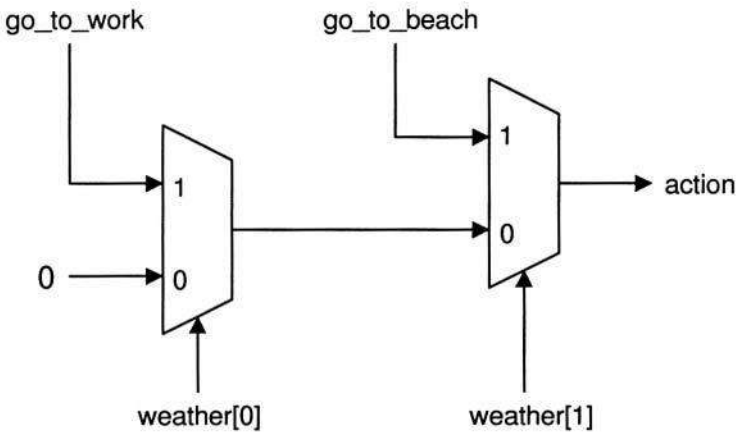


Figure 5-2. Result of using Multiple *if* Statements

If the above example used the *case* statement (instead of multiple *if* statements) in which all possible values of the selection index were covered and were exclusive, then it would have resulted in a single multiplexer as shown in Figure 5-3.

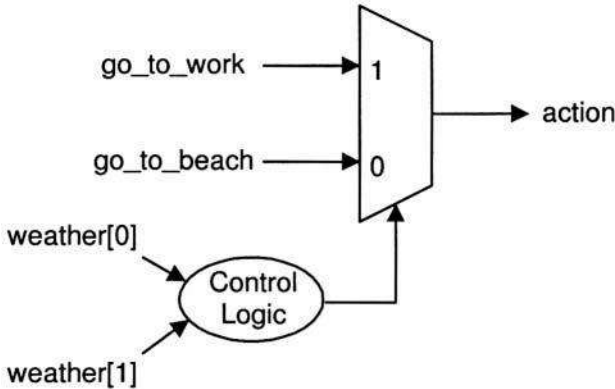


Figure 5-3. Result of using *case* or a Single *if* Statement

The same structure (Figure 5-3) is produced, if a single *if* statement is used, along with *elsif* statements to cover all possible branches.

5.4.4 Three-State Inference

Tri-state logic is inferred when high impedance (Z) is assigned to an output. Arbitrary use of tri-state logic is generally not recommended because of the following reasons:

- a) Tri-state logic reduces testability.
- b) Tri-state logic is difficult to optimize – since it cannot be buffered. This can lead to `max_fanout` violations and heavily loaded nets.

On the upside however, tri-state logic can provide significant savings in area.

Verilog Example

```
assign tri_out = enable ? tri_in : 1'bz;
```

VHDL Example

```
tri_out <= tri_in when (enable = '1') else 'Z';
```

5.5 Order Dependency

Both, Verilog and VHDL provide variable assignments that are order dependent/independent. Correct usage of these produce desired result, while incorrect usage may cause synthesized logic to behave differently than the source RTL.

5.5.1 Blocking versus Non-Blocking Assignments in Verilog

It is important to use non-blocking statements when doing sequential assignments like pipelining and modeling of several mutually exclusive data transfers. Use of blocking assignments within sequential processes may cause race conditions, because the final result depends on the order in which the assignments are evaluated. The non-blocking assignments are order independent; therefore they match closely to the behavior of the hardware.

Non-blocking assignment is done using the “<=” operator, while the “=” operator is used for blocking assignments.

```
always @(posedge clk)
begin
    firstReg    <= data;
    secondReg   <= firstReg;
    thirdReg    <= secondReg;
end
```

In hardware, the register updates will occur in the reverse order as shown above. The use of non-blocking assignments causes the assignments to occur in the same manner as hardware i.e., thirdReg will get updated with the old value of secondReg and the secondReg will get updated with the old value of firstReg. If blocking assignments were used in the above example, then the

signal “data” would have propagated all the way through to the thirdReg concurrently during simulation.

The blocking assignments should generally be used within the combinational *always* block.

5.5.2 Signals versus Variables in VHDL

Similar to Verilog, VHDL also provides order dependency through the use of signals and variables. The signal assignments may be equated to Verilog’s non-blocking assignments, i.e., they are order independent. The variable assignments are order sensitive and correlate to Verilog’s blocking assignments.

Variable assignments are done using the “:=” operator, whereas the “<=” operator is used for signal assignments.

The following example illustrates the usage of the signal assignments within the sequential *process* block. The resulting hardware contains three registers, with signal “data” propagating from firstReg to secondReg and then to the thirdReg. The RTL simulation will also show the same result.

```
process(clk)
begin
    if (clk'event and clk = '1') then
        firstReg    <= data;
        secondReg   <= firstReg;
        thirdReg    <= secondReg;
    end if;
end process;
```

A general recommendation is to only use signal assignments within sequential processes and variable assignments within the combinational processes.

5.6 Chapter Summary

This chapter highlighted the partitioning and coding styles suited for synthesis. Various guidelines and suggestions were provided to help the user code the RTL correctly with proper partitions, to make effective use of the synthesis engine.

The chapter began by suggestions on successful partitioning techniques and why they are necessary, followed by a short discussion on the “what is RTL”. Emphasis was given on “thinking hardware” while coding the design.

Next, general guidelines were covered that encompassed various suggestions and techniques, though not essential for synthesis, have significant impact on successful optimization. Adherence to these suggestions produce optimized designs that are well suited for automating the synthesis process.

An important section was devoted to the coding styles, and numerous examples were provided as templates to infer the correct logic. These included inference of latches, registers, multiplexers and three-state logic elements. At each step, advantages and disadvantages along with the correct usage was discussed.

The last section described the order dependency feature of both Verilog and VHDL languages. Also discussed were appropriate coding techniques to be used by utilizing the order dependency feature of both languages.

6

CONSTRAINING DESIGNS

This chapter discusses the process of specifying the design environment and its constraints. It describes various commonly used DC commands along with other helpful constraints that may be used to synthesize complex ASIC designs.

Please note that the commands described in this chapter only contain the most frequently used options. Designers are advised to consult the DC reference manual for the entire list of options available to a particular command.

This chapter contains information that is useful both for the novice and the advanced users of Synopsys tools. The chapter attempts to focus on “real world” applications, by taking into account deviations from the ideal situation. In other words, “Not all designs or designers, follow Synopsys recommendations”. Incorporated within the chapter are numerous helpful ideas, marked as ☺ to guide the reader in real time application for selected commands.

6.1 Environment and Constraints

In order to obtain optimum results from DC, designers have to methodically constrain their designs by describing the design environment, target objectives and design rules. The constraints may contain timing and/or area information, usually derived from design specifications. DC uses these constraints to perform synthesis and tries to optimize the design with the aim of meeting target objectives.

6.1.1 Design Environment

Up until now, the assumption has been that the design has been partitioned, coded and simulated. The next step is to describe the design environment. This procedure entails defining for the design, the process parameters, I/O port attributes, and statistical wire load models. Figure 6-1 illustrates the essential DC commands used to describe the design environment.

- **set_min_library** This is a new command, introduced in DC98 version. The command allows users to simultaneously specify the worst-case and the best-case libraries. This may be useful during initial compiles, preventing DC from violating the setup-time violations while fixing the hold-time violations.

```
set_min_library <max library filename>
               -min_version <min library filename>
```

```
dc_shell -t> set_min_library "ex25_worst.db" \
               -min_version "ex25_best.db"
```



The above command may be used for fixing hold-time violations during incremental compile or for in place optimization. In this case, the user should set both minimum and maximum values for the operating conditions.

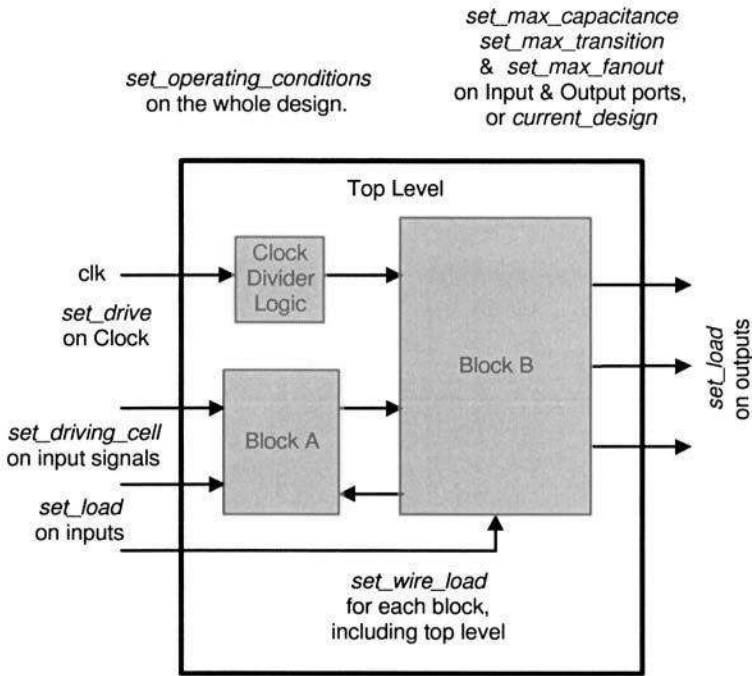


Figure 6-1. Basic Design Environment

- **set_operating_conditions** describes the process, voltage and temperature conditions of the design. The Synopsys library contains the description of these conditions, usually described as WORST, TYPICAL and BEST case. The names of operating conditions are library dependent. Users should check with their library vendor for correct setting. By changing the value of the operating condition command, full ranges of process variations are covered. The WORST case operating condition is generally used during pre-layout synthesis phase, thereby optimizing the design for maximum setup-time. The BEST case condition is commonly used to fix the hold-time violations. The TYPICAL case is mostly ignored, since analysis at WORST and BEST case also covers the TYPICAL case.

`set_operating_conditions <name of operating conditions>`

```
dc_shell -t> set_operating_conditions WORST
```

- ☺ It is possible to optimize the design both with the WORST and the BEST case, simultaneously. The optimization is achieved by using the `-min` and `-max` options in the above command, as illustrated below. This is very useful for fixing the design for possible hold-time violations.

```
dc_shell-t> set_operating_conditions-min BEST -max WORST
```

- ~ **set_wire_load_model** command is used to provide estimated statistical wire-load information to DC, which in turn, uses the wire-load information to model net delays as a function of loading. Generally, a number of wire-load models are present in the Synopsys technology library, each representing a particular size block. In addition, designers may also choose to create their own custom wire-load models to accurately model the net loading of their blocks.

`set_wire_load_model-name <wire-load model>`

```
dc_shell -t> set_wire_load_model-name MEDIUM
```

- ~ **set_wire_load_mode** defines the three modes associated for modeling wire loads. These are **top**, **enclosed**, and **segmented**. Generally, only the first two modes are in common use. The **segmented** wire load mode is not prevalent, since it relies on the wire-load models that are specific to the net segments.

The mode **top** defines that all nets in the hierarchy will inherit the same wire-load model as the top-level block. One may choose to use this wire-load model for sub-blocks, if planning to flatten them later for layout. This mode may also be chosen, if the user is synthesizing the design using the bottom-up compile method.

The second mode, **enclosed** specifies that all nets (of the sub-blocks) will inherit the wire load model of the block that completely encloses the sub-blocks. For example, if the designer is synthesizing sub-blocks B and C that are completely enveloped by block A (which in turn is completely enclosed by the top-level), then sub-blocks B and C will inherit the wire-load models defined for block A.

The last mode, **segmented** is used for wires crossing hierarchical boundaries. In the above example, sub-blocks B and C will inherit the wire-load models specific to them, while the nets between sub-block B and C (but, within block A) will inherit the wire-load model specified for block A.

```
set_wire_load_mode < top | enclosed | segmented >
```

```
dc_shell -t> set_wire_load_mode top
```

☺ It is extremely important that designers accurately model the wire loads of their design. Too optimistic or too pessimistic wire-load models result in increased synthesis iterations, in an effort to achieve timing convergence after post-layout. In general, during the pre-layout phase, slightly pessimistic wire-load models are used. This is done to provide extra timing margin that may get absorbed, by the routed design.

- **set_drive** and **set_driving_cell** are used at the input ports of the block. **set_drive** command is used to specify the drive strength at the input port. It is typically used to model the external drive resistance to the ports of the block or chip. The value of 0 signifies highest drive strength and is commonly utilized for clock ports. Conversely, **set_driving_cell** is used to model the drive resistance of the driving cell to the input ports. This command takes the name of the driving cell as its argument and applies all design rule constraints of the driving cell to the input ports of the block.

```
set_drive <value> <object list>
```

```
set_driving_cell -cell <cell name>  
                 -pin <pin name> <object list>
```

```
dc_shell -t> set_drive 0 {CLK RST}
```

```
dc_shell -t> set_driving_cell -cell BUFF1 -pin Z [all_inputs]
```

- **set_load** sets the capacitive load in the units defined in the technology library (usually pico farads, or pf), to the specified nets or ports of the design. It typically sets capacitive loading on output ports of the blocks during pre-layout synthesis, and on nets, for back-annotating the extracted post-layout capacitive information.

```
set_load <value> <object list>
```

```
dc_shell -t> set_load 1.5 [all_outputs]
```

```
dc_shell -t> set_load 0.3 [get_nets blockA/n1234]
```

- Design Rule Constraints or DRCs consist of **set_max_transition**, **set_max_fanout** and **set_max_capacitance** commands. These rules are generally set in the technology library and are determined by the process parameters. These rules should not be violated in order to achieve working silicon. Previous releases of DC (v97.08 and before) prioritized DRCs even at the expense of poor timing. However, the latest version DC98, prioritizes timing requirements over DRCs.

The DRC commands can be applied to input ports, output ports or on the **current_design**. Furthermore, if the value set in the technology library is not adequate or is too optimistic, then these commands may also be used at the command line, to control the buffering in the design.

```
set_max_transition <value> <object list>
```

```
set_max_capacitance <value> <object list>
```

```
set_max_fanout <value> <object list>
```

```
dc_shell -t> set_max_transition 0.3 current_design
```

```
dc_shell-t> set_max_capacitance 1.5 [get_ports out1]

dc_shell-t> set_max_fanout 3.0 [all_outputs]
```

6.1.2 Design Constraints

Design constraints describe the goals for the design. They may consist of timing or area constraints. Depending on how the design is constrained, DC tries to meet the set objectives. It is imperative that designers specify realistic constraints, since unrealistic specification results in excess area, increased power and/or degradation in timing. The basic commands to constrain a design are shown in Figure 6-2.

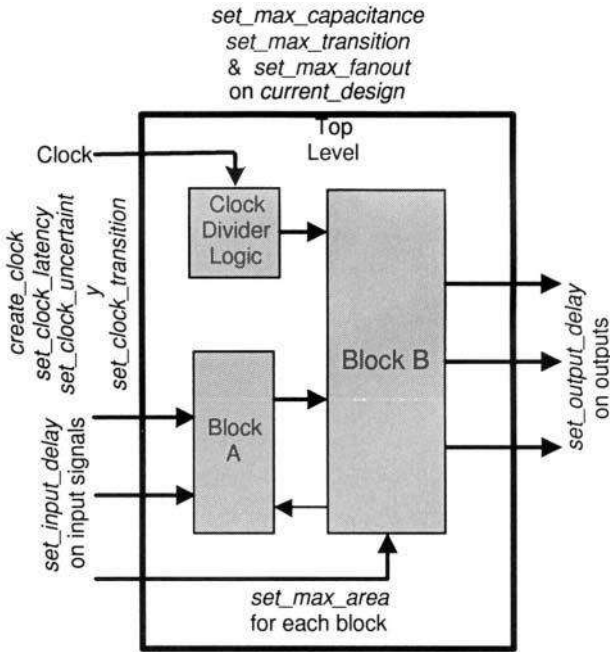


Figure 6-2. Design Constraints for Synthesis

- **create_clock** command is used to define a clock object with a particular period and waveform. The **–period** option defines the clock period, while the **–waveform** option controls the duty cycle and the starting edge of the clock. This command is applied to a pin or port, object types.

Following example specifies that the port named CLK is of type “clock” that has a period of 40 ns, with 50% duty cycle. The positive edge of the clock starts at time, 0 ns, with the falling edge occurring at 20 ns. By changing the falling edge value, the duty cycle of the clock may be altered.

```
dc_shell-t> create_clock–period 40 –waveform [list 0 20] CLK
```

- ☺ In some cases, a block may only contain combinational logic. To define delay constraints for this block, one can create a virtual clock and specify the input and output delays in relation to the virtual clock. To create a virtual clock, designers may replace the port name (CLK, in the above example) with the **–name <virtual clock name>**, in the above command. Alternatively, one can use the **set_max_delay** or **set_min_delay** commands to constrain such blocks. This is explained in detail in the next section.

- **create_generated_clock** command is used for clocks that are generated internal to the design. This is a very powerful command, which until recently only existed in PrimeTime. This command may be used to describe frequency divided/multiplied clocks as a function of the primary clock.

```
create_generated_clock –name <clock name>
                        –source <clock source>
                        –divide_by <factor> | –multiply_by <factor>
                        .....
```

- **set_dont_touch_network** is a very useful command, usually used for clock networks and resets. This command is used to set a **dont_touch** property on a port, or on the net. Note setting this property will also prevent DC from buffering the net, in order to meet DRCs. In addition,

any gate coming in contact with the “dont_touched” net will also inherit the dont_touch attribute.

```
dc_shell -t> set_dont_touch_network {CLK, RST}
```

☺ Suppose, you have a block that takes as input the primary clock, and generates secondary clocks e.g., clock divider logic. In this scenario, you should apply the `set_dont_touch_network` on the generated clock output port of the block. This will help prevent DC from buffering the clock network.

☺ Caution should be exercised while using `set_dont_touch_network` command. For instance, if a design that contains gated clock circuitry and the `set_dont_touch_network` attribute has been applied to the clock input. This will prevent DC to appropriately buffer the gated logic, resulting in the DRC violation for the clock signal. The same will hold true for gated resets.

– **set_dont_touch** is used to set a dont_touch property on the `current_design`, cells, references or nets. This command is frequently used during hierarchical compilation of the blocks. Also, it can be used for, preventing DC from inferring certain types of cells present in the technology library.

```
dc_shell -t> set_dont_touch current_design
```

```
dc_shell -t> set_dont_touch [get_cells sub1]
```

```
dc_shell -t> set_dont_touch [get_nets gated_rst]
```

☺ For example, this command may be used on the block containing spare gates. The command will then instruct DC not to disturb (or optimize) the instantiation of the spare gates block.

– **set_dont_use** command is generally set in the `.synopsys_dc.setup` environment file. The command is instrumental in eliminating certain types of cells from the technology library that the user would not want DC to infer. For instance, by using the above command, you can filter out

the flip-flops in your technology library whose name start with “SDFF” or “RSFF” as illustrated below.

```
dc_shell -t> set_dont_use [list mylib/SDFF* mylib/RSFF*]
```

- **set_input_delay** specifies the input arrival time of a signal in relation to the clock. It is used at the input ports, to specify the time it takes for the data to be stable after the clock edge. The timing specification of the design usually contains this information, as the setup/hold time requirements for input signals. Given the top-level timing specification of the design, this information may also be extracted for the sub-blocks of the design, by utilizing the top-down characterize compile method or the design budgeting method, explained in Chapter 7.

```
dc_shell-t> set_input_delay -max 23.0 -clock CLK {datain}
```

```
dc_shell-t> set_input_delay -min 0.0 -clock CLK {datain}
```

In Figure 6-3, the maximum input delay constraint of 23ns and the minimum input delay constraint of 0ns is specified for the signal *datain* with respect to the clock signal *CLK*, with a 50% duty cycle and a period of 30ns. In other words the setup-time requirement for the input signal *datain* is 7ns, while the hold-time requirement is 0ns.

If both `-min` and `-max` options are omitted, the same value is used for both the maximum and minimum input delay specifications.

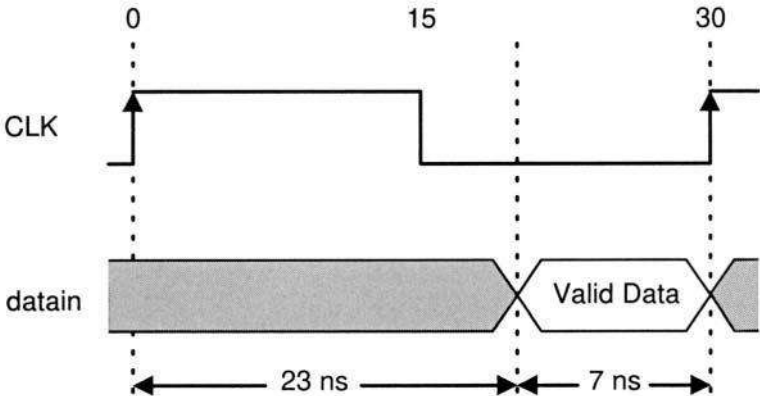


Figure 6-3. Specification of Input Delay

- **set_output_delay** command is used at the output port, to define the time it takes for the data to be available before the clock edge. The timing specification of the design usually contains this information. Given the top-level timing specification of the design, this information may also be extracted for the sub-blocks of the design, by utilizing the top-down characterize compile method or the design budgeting method, explained in Chapter 7.

```
dc_shell-t> set_output_delay-max 19.0 -clock CLK {dataout}
```

In Figure 6-4, the output delay constraint of 19ns is specified for the signal *dataout* with respect to the clock signal *CLK*, with a 50% duty cycle and a period of 30ns. This means that the data is valid for 11ns after the clock edge.

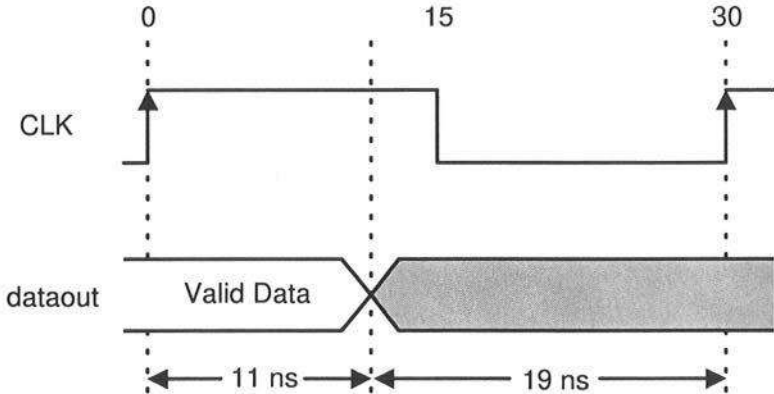


Figure 6-4. Specification of Output Delay

- ☺ During the pre-layout phase, it is sometimes necessary to over-constrain selective signals by a small amount to maximize the setup-time, thereby squeezing extra timing margin in order to reduce the synthesis-layout iterations. To achieve this, one may fool DC by specifying the over-constrained values to the above commands. Remember that over-constraining designs by a large amount will result in unnecessary increase in area and increased power consumption.
- ☺ A negative value (e.g., -0.5) may also be used to provide extra timing margin while fixing the hold-time violations after layout, by making use of the in-place optimization on the design, explained in Chapter 9.
- **set_clock_latency** command is used to define the estimated clock insertion delay during synthesis. This is primarily used during the pre-layout synthesis and timing analysis. The estimated delay number is an approximation of the delay produced by the clock tree network insertion (done during the layout phase).

```
dc_shell-t> set_clock_latency 3.0 [get_clocks CLK]
```

- **set_clock_uncertainty** command lets the user define the clock skew information. Basically this is used to add a certain amount of margin to the clock, both for setup and hold times. During the pre-layout phase one can add more margin as compared to the post-layout phase.

```
dc_shell -t> set_clock_uncertainty -setup 0.5 -hold 0.25 \
               [get_clocks CLK]
```

- ☺ It is strongly recommended that users specify a certain amount of margin both for pre-layout and the post layout phased. The main reason for doing this is to make the chip less susceptible to the process variations that may occur during manufacturing.

- **set_clock_transition** for some reason does not get as much attention as it deserves. However, this is a very useful command, used during the pre-layout synthesis, and for timing analysis. Using this command forces DC to use the specified transition value (that is fixed) for the clock port or pin.

```
dc_shell -t> set_clock_transition 0.3 [get_clocks CLK]
```

- ☺ Setting a fixed value for transition time of the clock signal in pre-layout is essential because of a large fanout associated with the clock network. Using this command enables DC to calculate realistic delays for the logic being fed by the clock net based on the specified clock signal transition value. This is further explained in the “clocking issues” section later in the chapter.

- **set_propagated_clock** is used during the post layout phase when the design has undergone the insertion of the clock tree network. In this case, the latency is derived using the traditional method of delay calculation.

```
dc_shell -t> set_propagated_clock [get_clocks CLK]
```

6.2 Advanced Constraints

This section describes additional design constraints that go beyond the general constraints covered in the previous section. These constraints consist of specifying false paths, multicycle paths, max and min delays etc. In addition, this section also discusses the process of grouping timing critical paths for extra optimization.

It must be noted however, that the use of too many timing exceptions, such as false paths and multicycle paths causes significant impact on the run times.

- **set_false_path** is used to instruct DC to ignore a particular path for timing or optimization. Identification of false paths in a design is critical. Failure to do so, compels DC to optimize all paths in order to reduce total negative slack. Consequently, the critical timing paths may be adversely affected due to optimization of all the paths, which also includes the false paths.

The valid startpoint and endpoint to be used for this command are the input ports or the clock pins of the sequential elements, and the output ports or the data pins of the sequential cells. In addition, one can further target a particular path using the **-through** switch.

```
dc_shell -t> set_false_path -from in1 -through U1/Z -to out1
```

Use this command when the timing critical logic is failing the static timing analysis because of the false paths.

- **set_multicycle_path** is used to inform DC regarding the number of clock cycles a particular path requires in order to reach its endpoint. DC automatically assumes that all paths are single cycle paths and will unnecessarily try to optimize the multicycle segment in order to achieve the timing. This may have a direct impact on adjacent paths as well as the area. Also, the command provides the **-through** option that facilitates isolating the multicycle segment in a design.

```
dc_shell-t> set_multicycle_path 2 -from U1/Z \
               -through U2/A \
               -to out1
```

- **set_max_delay** defines the maximum delay required in terms of time units for a particular path. In general, it is used for the blocks that contain combinational logic only. However, it may also be used to constrain a block that is driven by multiple clocks, each with a different frequency. This command has precedence over DC derived timing requirements.

☺ For blocks, only containing combinational logic, one may either create a virtual clock and constrain the block accordingly, or use this command to constrain the total delay from all inputs to all outputs, as shown below:

```
dc_shell-t> set_max_delay 5 -from [all_inputs] -to [all_outputs]
```

☺ Although, Synopsys recommends defining only a single clock per block, there are situations where a block may contain multiple clocks, each with a different frequency. To constrain such a block, one may define all the clocks in the block using the normal **create_clock** and **set_dont_touch_network** commands. However, it becomes tedious to assign input delays of signals related to individual clocks. To avoid this situation, an alternative approach is to define the first clock (the most stringent one) using the normal approach, while constraining other clocks through the **set_max_delay** command, as shown below.

```
dc_shell -t> set_max_delay 0 -from CK2 \
               -to [all_registers -clock_pins]
```

The value of 0 signifies that a zero delay value is desired, between the input port CK2, and the input clock pins of all the flops within the block. In addition, one may also need to apply the **set_dont_touch_network** for other clocks. This method is suitable for designs containing gated clocks or resets.

- **set_min_delay** is the opposite of the **set_max_delay** command, and is used to define the minimum delay required in terms of time units for a

particular path. Specifying this command in conjunction with the `set_fix_hold` command (described in Chapter 9) will instruct DC to add delays in the block to meet the minimum time unit specified. This command also has precedence over DC derived timing requirements.

```
dc_shell-t> set_min_delay 3 --from [all_inputs] --to [all_outputs]
```

- **group_path** command is used to bundle together timing critical paths in a design, for cost function calculations. Groups enable you to prioritize the grouped paths over others. Different options exist for this command, which include specification of critical range and weights.

```
dc_shell-t> group_path --to [list out1 out2] --name grp1
```

- ☺ Adding too many groups has significant impact on the compile time. Therefore, use it only as a last resort.
- ☺ Exercise caution while using this command. One may find that using this command increases the delay of the worst violating path, in the design. This is due to the fact, that DC prioritizes the grouped paths over other paths in the design. In order to improve the overall cost function, DC will try to optimize the grouped path over others and may degrade the timing of another group's worst violator.

6.3 Clocking Issues

In any design, the most critical part of synthesis is the clock description. There are always issues concerning the pre and post-layout definitions.

Traditionally in the past, big buffers were placed at the source of the clock to drive the full clock network. Thick clock spines were used in the layout for even distribution of clock network delays and to minimize clock skews. Although this method sufficed for sub-micron technologies, it is miserably failing in the VDSM realms. The interconnect RC's currently account for a major part of total delay. This is mainly due to the increase in resistance of the shrinking metal widths. It is difficult, if not impossible to model clocks using the traditional approach.

With the arrival of complex layout tools, capable of synthesizing clock trees, the traditional method has changed dramatically. Since, layout tools have the cell placement information, they are best equipped to synthesize the clock trees. It is therefore necessary to describe clocks in DC, such that it imitates the clock delays and skews of the final layout.

6.3.1 Pre-Layout

For reasons explained above, it is best to estimate the clock tree latency and skew during the pre-layout phase. To do this, use the following commands:

```
dc_shell-t> create_clock -period 40 -waveform {0 20} CLK
dc_shell-t> set_clock_latency 2.5 CLK
dc_shell-t> set_clock_uncertainty -setup 0.5 -hold 0.25 CLK
dc_shell-t> set_clock_transition 0.1 CLK
dc_shell-t> set_dont_touch_network CLK
dc_shell-t> set_drive 0 CLK
```

For the above example, a delay of 2.5 ns is specified as the overall latency for the clock signal CLK. In addition, the `set_clock_uncertainty` command approximates the clock skew. One can specify different numbers for the setup and hold time uncertainties by using `-setup` and `-hold` options as exemplified above.

Furthermore, specification of clock transition is essential. This restricts the max transition value of the clock signal. The delay through a cell is affected by the slope of the signal at its input pin and the capacitive loading present at the output pin. The clock network generally feeds large endpoints. This means, that although the clock latency value is fixed, the input transition time of the clock signal to the endpoint gates will still be slow. This results in DC calculating unrealistic delays (for the endpoint gates), even though in reality, the post-routed clock tree ensures fast transition times.

6.3.2 Post-Layout

Defining clocks after layout is relatively easy, since the user does not need to worry about the clock latency and skews. They are determined by the quality of the routed clock tree.

Some layout tools provide direct interface to DC. This provides a smooth mechanism for taking the routed netlist consisting of clock tree, back to DC. If this information is not present, then the user should extract the clock latency and the skew information from the layout tool. Using the pre-layout approach, this information can be used to define the clock latency and clock skew, as described before. If however, the netlist can be ported to DC, then the following commands may be used to define the clocks. For example:

```
dc_shell> create_clock -period 40 -waveform [list 0 20] CLK
```

```
dc_shell> set_propagated_clock CLK
```

```
dc_shell> set_clock_uncertainty -setup 0.25 -hold 0.05 CLK
```

```
dc_shell> set_dont_touch_network CLK
```

```
dc_shell> set_drive 0 CLK
```

Notice the absence of the `set_clock_latency` command and the inclusion of `set_propagated_clock` command. Since, we now have the clock tree inserted in the netlist, the user should propagate the clock instead of fixing it to a certain value. Similarly, the `set_clock_transition` command is no longer required, since DC will now calculate the input transition value of the clock network, based on the clock tree. In addition, a small clock uncertainty value may also be defined. This ensures a robust design that will function taking into account a wider process variance.

Some companies do not possess their own layout tools, but they rely on outside vendors to perform the layout. This situation of course varies from one company to the other. If the vendor provides the user, the post-routed

netlist containing the clock tree, then the above method can be utilized. In some instances, instead of providing the post-routed netlist, the vendor only supplies the SDF file containing point-to-point timing for the entire clock network (and the design). In such a case, the user only needs to define the clock for the original netlist and back-annotate the SDF to the original netlist without propagating the clock. The clock skews and delays will be determined from the SDF file, when performing static timing analysis.

6.3.3 **Generated Clocks**

Many complex designs contain internally generated clocks. An example of this is the clock divider logic that may be used to generate secondary clock(s) of different frequency, derived from the primary clock source. If the primary clock has been designated as the clock source, then a limiting factor of DC is that does not automatically create a clock object for the generated clocks.

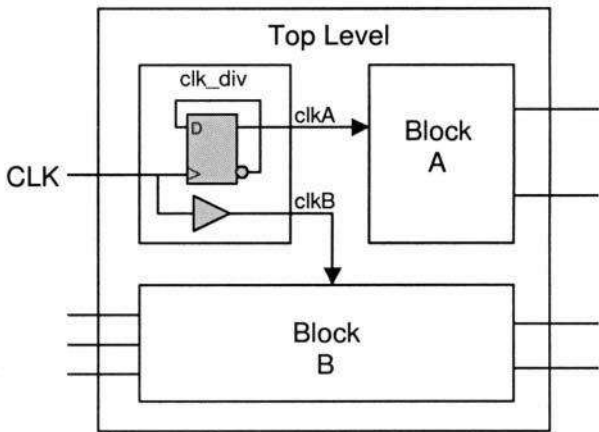


Figure 6-5. Specification of Generated Clock(s)

Consider the logic illustrated in Figure 6-5. A clock divider circuit in *clk_div* block, is used to divide the frequency of the primary clock *CLK* by half, and then generate the divided clock that drives *Block A*. The primary clock is also

used to clock, *Block B* and is buffered internally (in the *clk_div* block), before feeding *Block B*.

Assignment of clock object through `create_clock` command on *CLK* input to the top-level is sufficient for the clock feeding block B. This is because the *clkB* net inherits the clock object (through the buffer) specified at the primary source. However, *clkA* is not so fortunate. DC is unable to propagate the clock object throughout the entire net because the specification of clock object on primary source *CLK* stops at the register (shown as shaded flop). To avoid this situation, the clock object for *clkA* should be specified on the output port of the *clk_div* block. The following commands may be used to specify the clocks for the above example:

```
dc_shell> create_clock -period 40 -waveform {0 20} CLK
```

```
dc_shell> create_clock -period 80 -waveform {0 40}          \
               find(port, "clk_div/clkA")
```

Alternatively, one may use the `create_generated_clock` command to describe the clock, as follows:

```
dc_shell-t> create_generated_clock -name clkA      \
               -source CLK                        \
               -divide_by 2
```

6.4 Putting it Together

Example 6.1 provides a brief overview of some of the commands described in this chapter.

Example 6.1

```
# -----
# Design entry

analyze -format verilog sub1.v
analyze -format verilog sub2.v
```

```
analyze -format verilog top_block.v

elaborate top_block

current_design top_block
uniquify
check_design

# -----
# Setup operating conditions, wire load, clocks, resets

set_wire_load_model large_wl
set_wire_load_mode enclosed
set_operating_conditions WORST

create_clock -period 40 -waveform [list 0 20] CLK
set_clock_latency 2.0 [get_clocks CLK]
set_clock_uncertainty -setup 1.0 -hold 0.05 [get_clocks CLK]

set_dont_touch_network [list CLK RESET]

# -----
# Input drives

set_driving_cell -cell [get_lib_cell buff3] -pin Z [all_inputs]
set_drive 0 [list CLK RST]

# -----
# Output loads

set_load 0.5 [all_outputs]

# -----
# Set input & output delays

set_input_delay 10.0 -clock CLK [all_inputs]
set_input_delay -max 19.0 -clock CLK { IN1 IN2}
set_input_delay -min -2.0 -clock CLK IN3
```

```

set_output_delay 10.0 --clock CLK [all_outputs]

# -----
# Advanced constraints

group_path --from IN4 --to OUT2 --name grp1

set_false_path --from IN5 --to sub1/dat_reg*/*

set_multicycle_path 2 --from sub1/addr_reg/CP \
                    --to sub2/mem_reg/D

# -----
# Compile and write the database

compile

current_design top_block

write --hierarchy --output top_block.db
write --format verilog --hierarchy --output top_block.sv

# -----
# Create reports

report_timing --nworst 50

```

6.5 Chapter Summary

This chapter described all the basic and advanced commands used in DC, along with numerous tips to enhance the synthesis process. Focus was also given to the real time issues facing the designers as they descend deep into the VDSM technology.

A separate section was dedicated to issues related to clocks. This section described various techniques useful for specifying clocks, both for pre and

post-layout. This section also included a topic on specification of generated clocks that are present in almost all designs. Finally, example DC scripts were included to guide the users to perform complex and successful synthesis.

OPTIMIZING DESIGNS

Ideally, a synthesized design that meets all timing requirements and occupies the smallest area is considered fully optimized. To achieve this goal, one must understand the behavior of synthesis process.

This chapter guides the reader to successfully optimize the design to obtain the best possible results.

7.1 Design Space Exploration

To achieve the smallest area while maximizing the speed of the design requires a fair amount of experimentation and iterative synthesis. The process of analyzing the design for speed and area to achieve the fastest logic with minimum area is termed – design space exploration.

Various factors influence the optimization process, primarily the coding style. While coding, designers generally focus on the functionality of the design and may not consider the synthesis guidelines, previously explained in Chapter 5 (This is a fact of life, we just have to live with it). At a later stage modifications to the HDL code are performed to facilitate the synthesis

process. In reality, the HDL is generally fixed and only minor modifications are done, since major changes may impact other blocks or test benches. For this reason, changing the HDL code to help synthesis is less desirable.

For the sake of design space exploration, we can assume that the HDL code is frozen. It is now the designer's responsibility to minimize the area and meet the target timing requirements through synthesis and optimization.

Starting from version 98 of DC (or DC98) the previous compile flow changed. The timing is prioritized over area. This is shown in Figure 7-1. Another major difference between DC98 and previous versions is that, DC98 performs compilation to reduce "total negative slack" instead of "worst negative slack". This ability of DC98 produces better timing results but has some impact on the area. Also, in previous versions area minimization was handled automatically, however, DC98 and later versions requires designers to specify area constraints explicitly. Generally some area cleanup is performed by default even without specifying the area constraints but better results are obtained by including the constraints for area.

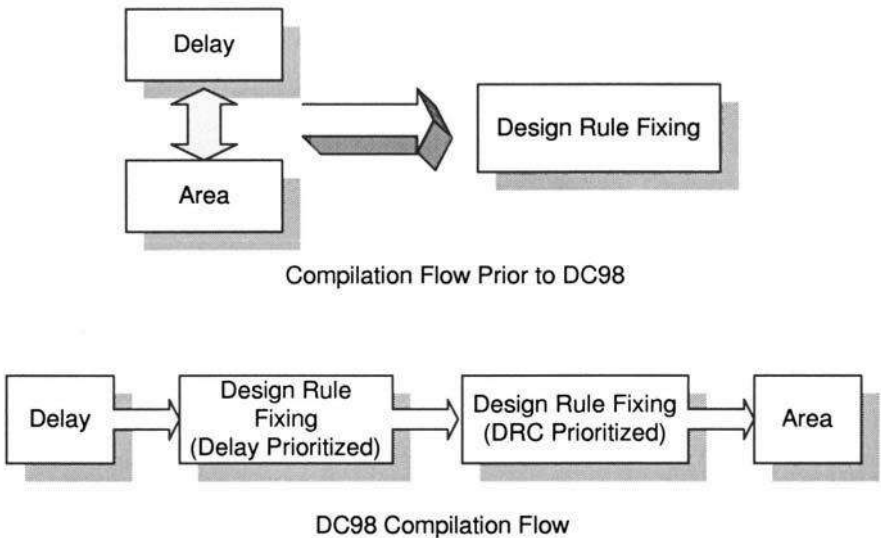


Figure 7-1. DC98 Changes

Although, the delay is prioritized over area, it is extremely important to provide DC with realistic constraints. Some designers while performing bottom-up compile, fail to realize this point and over constrain the design. This causes DC to bloat the logic in order to meet the unrealistic timing goals. This is especially true for DC98 because it works on the reduction of total negative slack. This relationship between constraints and area is shown in Figure 7-2, which emphasizes that the area increases considerably with tightening constraints.

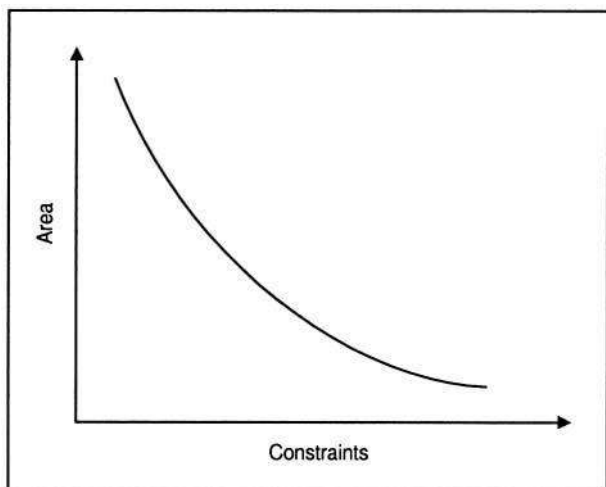


Figure 7-2. Constraints versus Area

Another representation of varying constraint is shown in Figure 7-3. This illustrates the relationship between constraints and delay across the design. It is shown that the actual delay of the logic decreases with tightening constraints, while relaxed constraints produce increased delay across the design. The horizontal part of the line on the left denotes that the constraints are so tight that further tightening of the constraints will not result in reduction of delay. Similarly the horizontal part of the line on the right signifies fully relaxed constraints, resulting in no further increase in delay.

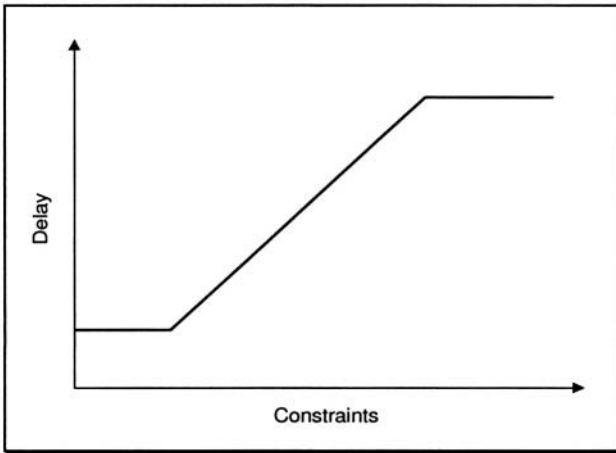


Figure 7-3. Constraints versus Delay

To further explain this concept, consider the diagram shown in Figure 7-4. For overly constrained design, DC tries to synthesize “vertical logic” to meet the tight timing constraints. However, if the timing constraints are non-existent, the synthesized design would result in “horizontal logic”, violating the actual timing specifications.

The idea here is to find a common ground, by specifying realistic timing constraints. It is recommended to over constrain the design by a small amount (maybe 10 percent tighter than required) to avoid too many synthesis-layout iterations. This produces minimum area for the design, while still meeting the timing specifications. For this reason, choose the correct compile methodology, described in subsequent sections.

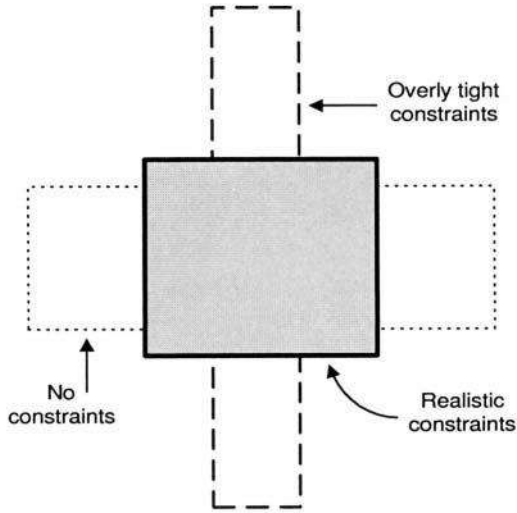


Figure 7-4. Horizontal versus Vertical Logic

7.2 Total Negative Slack

The previous section briefly introduced the phrase “Total Negative Slack” or TNS for short. With the advent of DC98, a lot of importance has been given to this, and designers need to understand this concept to perform successful logic optimization.

Prior to DC98 version, DC would optimize the logic based on “Worst Negative Slack” or WNS. The WNS is defined as the timing violation (or negative slack) of a signal traversing from one startpoint to the endpoint for a particular path. During compile, DC would reduce the WNS one by one, in order to reduce total violations of the block. For this reason, grouping paths and specifying the critical range for timing-critical segments was considered essential.

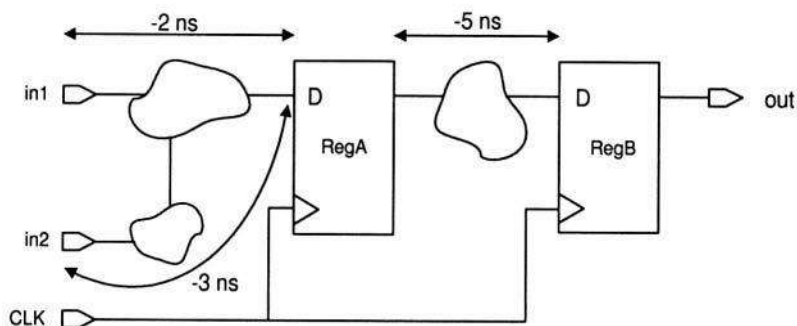


Figure 7-5. Total Negative Slack

DC98 not only, prioritizes delay over area but also targets TNS instead of WNS. To understand the concept of TNS consider the logic diagram shown in Figure 7-5. The WNS in this case is -5ns ; from *RegA* to *RegB*. The TNS is the summation of all WNS per endpoint and in this case equals -8ns , i.e., WNS to *RegA* plus WNS to *RegB*.

There are several advantages to using this technique; primarily it produces fewer timing violations as compared to the previous method. Another benefit is, when using bottom-up compile methodology, the critical paths present at the sub-module may not be seen as critical from the top level. Reducing TNS of the overall design minimizes this effect. By providing smaller number of violating paths to the timing driven layout tool, less iterations between synthesis and layout can also be achieved.

Although, reduction of TNS over WNS produces less timing violations, it does have an impact on the overall area. It is recommended that you set area constraints, regardless of the kind of optimization performed. By default, DC98 prioritizes TNS over area. Area optimization occurs only for those paths with positive slack. In order to prioritize area over TNS, you may use the following command:

```
dc_shell> set_max_area 0 -ignore_tns
```

7.3 Compilation Strategies

Synopsys recommends the following compilation strategies that depend entirely on how your design is structured and defined. It is up to user discretion to choose the most suitable compilation strategy for a design.

- a) Top-down hierarchical compile method.
- b) Time-budget compile method.
- c) Compile-characterize-write-script-recompile (CCWSR) method.
- d) Design Budgeting method.

7.3.1 Top-Down Hierarchical Compile

Prior to the release of DC98, the top-down hierarchical compile method was generally used to synthesize very small designs (less than 10K gates). Using this method, the source is compiled by reading the entire design. Based on the design specifications, the constraints and attributes are applied, only at the top level. Although, this method provides an easy push-button approach to synthesis, it was extremely memory intensive and viable only for very small designs.

The release of DC98 provides Synopsys the capability to synthesize million gate designs by tackling much larger blocks (>100K) at a time. This indeed may be a feasible approach for some designs depending on the design style (single clock etc.) and other factors. You may use this technique to synthesize larger blocks at a time by grouping the sub-blocks together and flattening them to improve timing.

The advantages and disadvantages of this methodology are summarized below:

7.3.1.1 Advantages

- a) Only top level constraints are needed.
- b) Better results due to optimization across entire design.

7.3.1.2 Disadvantages

- a) Long compile times (although, DC98 is much faster than previous releases).
- b) Incremental changes to the sub-blocks require complete re-synthesis.
- c) Does not perform well, if design contains multiple clocks or generated clocks.

7.3.2 Time-Budgeting Compile

The second compilation approach to synthesis is termed as the time-budgeting strategy. This strategy is useful, if the design has been partitioned properly with timing specifications defined for each block of the design, i.e., designers have time budgeted the entire design, including the inter-block timing requirements.

The designer manually specifies the timing requirements for each block of the design, thereby producing multiple synthesis scripts for individual blocks. The synthesis is usually performed bottom-up i.e., starting at the lowest level and ascending to the topmost level of the design. This method targets medium to very large designs and does not require large amounts of memory.

Consider the following design, illustrated in Figure 7-6. The top level module incorporates blocks A and B. The specifications for both of these blocks are well defined and can be directly translated to Synopsys constraints. For designs like these, the time-budgeting compilation strategy is ideal.

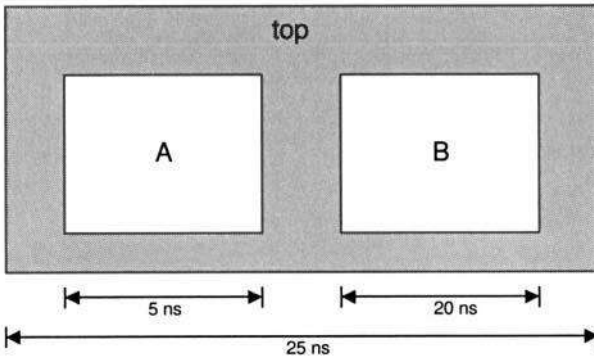


Figure 7-6. Design Suited for Time-Budgeting Compilation Strategy

This advantages and disadvantages of this methodology are listed below:

7.3.2.1 Advantages

- a) Easier to manage the design because of individual scripts.
- b) Incremental changes to the sub-blocks do not require complete re-synthesis of the entire design.
- c) Does not suffer from design style e.g., multiple and generated clocks are easily managed.
- d) Good quality results in general because of flexibility in targeting and optimizing individual blocks.

7.3.2.2 Disadvantages

- a) Tedious to update and maintain multiple scripts.
- b) Critical paths seen at the top-level may not be critical at lower level.
- c) The design may need to be incrementally compiled in order to fix the DRC's.

Figure 7-7 illustrates the directory structure and data organization, suited for this strategy. To automate the synthesis process, a makefile is used (refer to

Appendix). The makefile specifies, the dependencies of each block and employs the user specified scripts (kept in the script directory) to compile the whole design, starting from the lowest level and ending at the top-most level. After the synthesis of each block, the results are automatically moved to their respective directories. The variables used in the makefile are defined in the users .cshrc file, for e.g., \$SYNDB may be defined as: /home/project/design/syn/db

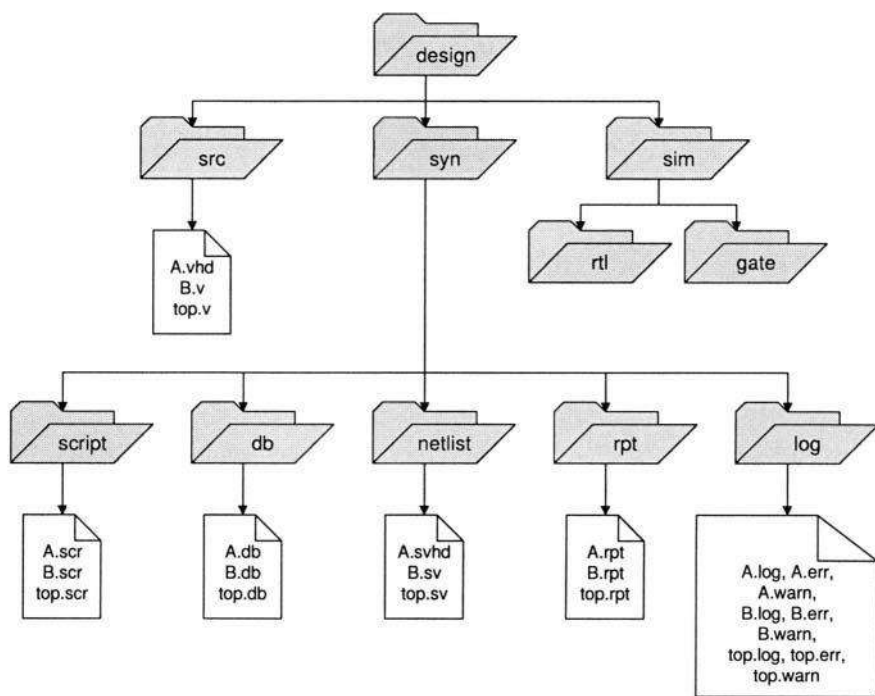


Figure 7-7. Directory Structure

7.3.3 Compile-Characterize-Write-Script-Recompile

This approach is useful for medium to very large designs that do not have good inter-block specifications defined. This method is not limited by hardware memory and allows for time budgeting between the blocks.

This approach requires constraints to be applied at the top level of the design, with each sub-block compiled beforehand. The sub-blocks are then characterized using the top-level constraints. This in effect propagates the required timing information from the top-level to the sub-blocks. Performing a `write_script` on the characterized sub-blocks generates the constraint file for each sub-block. These constraint files are then used to re-compile each block of the design.

Although this approach normally produces good results, it is recommended that designers use the Design Budgeting method the usage of which is explained in the next section.

7.3.3.1 Advantages

- a) Less memory intensive.
- b) Good quality of results because of optimization between sub-blocks of the design.
- c) Produces individual scripts, which may be modified by the user.

7.3.3.2 Disadvantages

- a) The generated scripts are not easily readable.
- b) Synthesis suffers from Ping-Pong effect. In other words it may be difficult to achieve convergence between blocks.
- c) A change at lower level block generally requires complete re-synthesis of the entire design.
- d) Long runtimes if the block becomes over-constrained.

7.3.4 Design Budgeting

This method is by far the most suitable compile strategy for tackling designs that do not have good inter-block specifications. This approach automatically allocates the top-level design specifications to the lower-level blocks. Design budgeting is invoked from within DC or PT, although it can also be invoked by typing `budget_shell`. This shell uses the Tcl interface and cannot be used

with the old non-Tcl `dc_shell` commands. It is recommended to use PT in case you want to take advantage of the PT's GUI interface.

This method cannot be used directly from the RTL stage. The design must be synthesized first to a mapped gate level netlist before it can be budgeted. Once the design is synthesized the budgeter is run on the entire design and scripts are generated for the sub-blocks. The number of hierarchical levels to budget is under full control of the user. In other words, the budgeter will allocate budgets for any number of hierarchical levels that is defined by the user. The generated scripts with accurate constraints are subsequently used to re-synthesize each block in parallel (to reduce runtime) with accurate constraints.

This is a very power method of gate-level optimization. However, it is an iterative process. The user has full control over the scripts, thus the amount of optimization. The generated scripts can be further massaged to suit individual needs. In addition, this method can be used even after the layout stage in order to produce more accurate constraints. This is accomplished by budgeting the back-annotated design.

7.3.4.1 Advantages

- a) Provides accurate constraints across the entire design, thus better QOR.
- b) Does not suffer from the ping-pong effect (as in the Characterize-compile method)
- c) Saves runtime by providing the ability to perform parallel compiles.
- d) Provides ability to customize the scripts to suit individual needs.
- e) Scripts can be used after the elaborate stage (design in GTECH stage)
- f) Less memory intensive

7.3.4.2 Disadvantages

- a) Cannot budget the RTL itself. Design must be a structured gate-level netlist.
- b) Cannot budget using both best and worst operating conditions.
- c) Iterative process (although, this is not a severe limitation)

An example script illustrating this strategy is shown below. The budgeting commands have been highlighted in bold.

```
pt_shell> read_verilog mydesign.sv
pt_shell> source constraints.scr #Top level constraints
pt_shell> allocate_budgets -levels 2 -write_context -format dctl
```

In the above example, the **allocate_budgets** command invokes the Design Budgeter that allocates the budgets to each sub-block of the design descending down to 2 levels of hierarchy. The **-write_context** option instructs the budgeter to generate scripts. The **-format** option specifies the format of generated scripts. The allowed values are **ptsh** (Primetime Tc1 format), **dctl** (DC Tc1 format: **dc_Shell_t**) and **dcsh** (DC format: **dc_shell**). The **ptsh** format is the default.

Several other commands are also available for this method. Users are advised to refer to the Design Budgeting User Guide for full details.

7.4 Resolving Multiple Instances

Before proceeding for optimization, one needs to resolve multiple instances of the sub-blocks of your design. This is a required step, since DC does not permit compilation until the multiple instances present in the design are resolved.

To better explain the concept of multiple instantiations of a module, consider the architecture of a design shown in Figure 7-8. Lets presume that you have chosen the time-budgeting compilation strategy and have synthesized moduleA separately. You are now compiling moduleB that instantiates moduleA twice as U1 and U2. The compilation will be aborted by DC with an error message stating that moduleA is instantiated 2 times in moduleB. There are two recommended methods of resolving this. You may either assign a **dont_touch** attribute to moduleA before reading moduleB, or **uniquify** moduleB. **uniquify** is a **dc_shell** command that in effect creates unique definitions of multiple instances. In this case, it will generate **moduleA_U1** and **moduleA_U2** (in Verilog), corresponding to instance U1 and U2 respectively as illustrated in Figure 7-9.

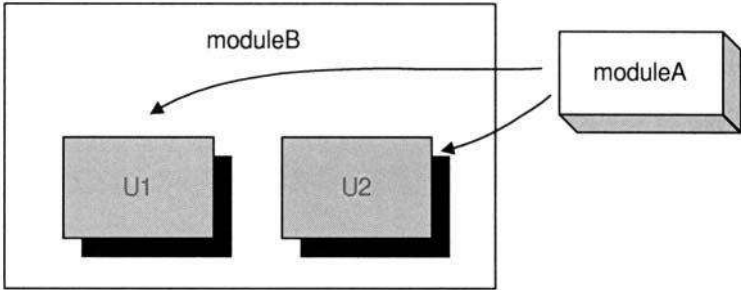


Figure 7-8. Non Uniquified Design

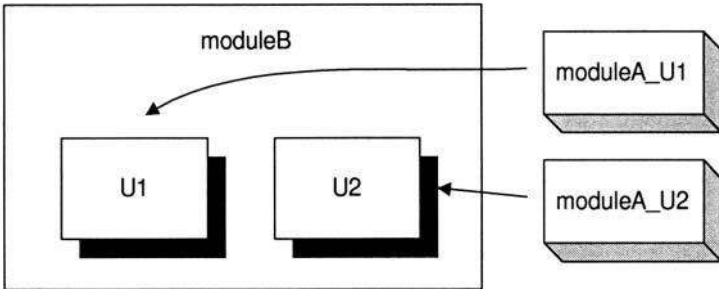


Figure 7-9. Uniquified Design

It is recommended to always uniquify the design, regardless of the compilation methodology chosen. The reason for this suggestion becomes evident while planning to perform clock tree synthesis during layout. This will be explained in detail in Chapter 9.

7.5 Optimization Techniques

This section describes various optimization techniques used to fine tune your design. Before we start on this subject, it is important to know that DC uses cost functions to optimize the design. This topic is thoroughly covered in the DC Reference manual, therefore will not be dealt with here. I would instead like to concentrate more on the practical optimization techniques instead of the mathematical algorithms used for calculating cost functions. Let's just say that DC calculates the cost functions based on the design constraints and DRCs to optimize the design.

7.5.1 Compiling the Design

Compilation of the design or modules is performed by the compile command. This command performs the actual mapping of the HDL code to gates from the specified target library. DC provides a range of options for this command, to fully control the mapping optimization of the design.

The command syntax along with most commonly used options is described below.

```
compile -map_effort <low | medium | high>
        -incremental_mapping
        -in_place
        -no_design_rule | -only_design_rule
        -scan
```

By default, compile uses `-map_effort medium`, which usually produces ideal results for most of the designs. It also uses the default settings for structuring and flattening attributes, described in the next section. The `-map_effort high`, should only be used, if the target objectives are not achieved through the default compile. This option enables DC to maximize its effort around the critical path by restructuring and re-mapping of logic, in order to meet the specified constraints. Beware, this usually produces long compile times.

The `–incremental_mapping` option is used, only after the initial compile (i.e., the design has been mapped to gates of the technology library), as it performs only at the gate level. This is a very useful and commonly used option. It is generally used to improve the timing of the logic and to fix DRCs. During incremental compile, DC performs various mapping optimizations in order to improve timing. Although Synopsys states that the resultant design will not worsen and may only improve in terms of design constraints; on rare occasions, using the above option may actually degrade the timing objectives. Users are therefore advised to experiment and use their own judgement. Nevertheless, the usefulness of this command is apparent while fixing DRCs at the top level of the design. To perform this, you may use `–only_design_rule` option while compiling incrementally. This prevents DC from performing mapping optimizations and concentrate only on fixing the DRCs.

The `–no_design_rule` option is not used frequently and as the name suggests, it instructs DC to refrain from fixing DRCs. You may use this option for initial passes of compile, when you don't want to waste time fixing DRC violations. At a later stage, generate the constraint report and then re-compile incrementally to fix DRCs. This is obviously a painful approach and users are advised to make their own judgement.

To achieve post layout timing convergence, it is sometimes necessary to resize the logic to fix timing violations. The `–in_place` option provides the capability of resizing the gates. Various switches available to designers to control the buffering of the logic govern this option. The usage of this option is described in detail in Chapter 9.

The `–scan` option uses the test ready compile feature of DC. This option instructs DC to map the design directly to the scan-flops – as opposed to synthesizing to normal flops before replacing them with their scan equivalents, in order to form the scan-chain. An advantage of using this feature is that, since the scan-flops normally have different timing associated with them as compared to their non scan equivalent flops (or normal flops), using this techniques makes DC take the scan-flop timing into account while synthesizing. This produces optimized scan inserted logic with correct timing.

7.5.2 Flattening and Structuring

Before we begin this discussion, it must be noted that the term “flattening” used here does not imply “removing the hierarchy”. Flattening is a common academic term for reducing logic to a 2-level AND/OR representation. DC uses this approach to remove all intermediate variables and parenthesis (using boolean distributive laws) in order to optimize the design. This option is set to “false” by default.

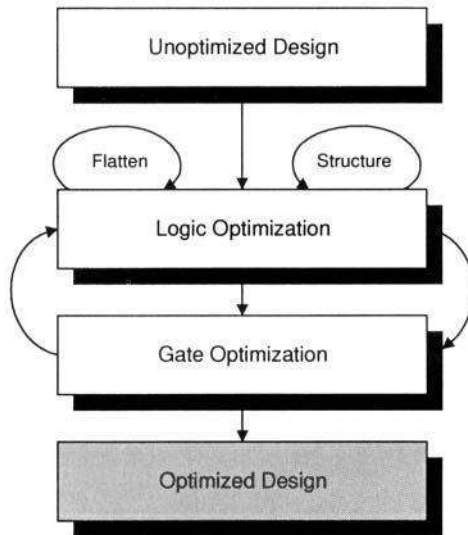


Figure 7-10. Optimization Steps

The optimization of the design is performed in two phases as shown in Figure 7-10. The logic optimization is performed initially by structuring and flattening the design. The resulting structure is then mapped to gates, using mapping optimization techniques. The default settings for *flatten* and *structure* attributes are:

Table 7-2. Default Settings for Flatten and Structure Attributes

Attribute	Value
flatten	false
structure	true
structure (timing)	true
structure (boolean)	false

As shown in the above table, flattening (`set_flatten true`) the design and Boolean optimization (`set_structure -boolean true`) is only performed when enabled.

7.5.2.1 Flattening

Flattening is useful for unstructured designs for e.g., random logic or control logic, since it removes intermediate variables and uses boolean distributive laws to remove all parenthesis. It is not suited for designs consisting of structured logic e.g., a carry-look-ahead adder or a multiplier.

Flattening results in a two-level, sum-of-products form, resulting in a vertical logic, i.e., few logic levels between the input and output. This generally results in achievement of faster logic, since the logic levels between the inputs and outputs are minimized. Depending upon the form of design flattened and the type of effort used, the flattened design can then be structured before the final technology mapping optimization. This is a recommended approach and should be performed to reduce the area, because flattening the design may cause a significant impact on the area of the design. A point to remember, if you flatten the design using `-effort high` option, then DC may not be able to structure the design, therefore use this attribute judiciously.

In general, compile the design using default settings, since most of the time they perform adequately. Designs failing timing objectives may be flattened, with structuring performed as a second phase (on by default). If the design is still failing timing goals, then turn off structuring and flatten only. You may also experiment by inverting the phase assignment that sometimes produces remarkable results. This is done by setting the `-phase` option of the `set_flatten` command to “true”. This enables DC to compare the logic produced by inverting the equation versus the non-inverted form of the equation.

For a hierarchical design, flatten attribute is set only on the `current_design`. All sub-blocks do not inherit this attribute. If you want to flatten the sub-blocks, then you have to explicitly specify using the `-design` option. The syntax for the flatten attribute along with most commonly used options is:

```
set_flatten <true | false>
            -design <list of designs>
            -effort <low | medium | high>
            -phase <true | false>
```

7.5.2.2 Structuring

Structuring is used for designs containing regular structured logic, for e.g., a carry-look-ahead adder. It is enabled by default for timing only. When structuring, DC adds intermediate variables that can be factored out. This enables sharing of logic that in turn results in reduction of area. For example:

Before Structuring

$$P = a x + a y + c$$

$$Q = x + y + z$$

After Structuring

$$P = aI + c$$

$$Q = I + z$$

$$I = x + y$$

It is important to note that, structuring produces shared logic that has an impact on the total delay of the logic. With the absence of specified timing constraints (or structuring is turned off with respect to timing), the logic produced will generally result in large delays across the block boundaries. Therefore, it is recommended that realistic constraints be specified, in addition to using the default settings.

Structuring comes in two flavors: timing (default) and boolean optimization. The latter is a useful method of reducing area, but has a greater impact on timing. Good candidates for boolean type of optimization are non critical timing circuitry e.g., random logic structures and finite state machines. As the name suggests, this algorithm uses boolean logic optimization to reduce area. Prior to version v1997.01, DC used a different algorithm to perform boolean optimization. Synopsys have since introduced another algorithm that

is more efficient and requires less run time. This algorithm is based on automatic test pattern generation (ATPG) techniques to manipulate logic networks. To enable this algorithm, you have to set the following variable to “true”:

```
compile_new_boolean_structure = true
```

As with flattening, the `set_structure` command applies only to the `current_design`. The syntax of this command along with most commonly used options is:

```
set_structure <true | false>  
             -design <list of designs>  
             -boolean <true | false>  
             -timing <true | false>
```

In general, the design compiled with default settings produce satisfactory results. However, if your design is non-timing critical and you want to minimize for area only, then set the area constraints (`set_max_area 0`) and perform boolean optimization. For all other cases, structure with respect to timing only.

7.5.3 Removing Hierarchy

By default, DC maintains the original hierarchy of the design. The hierarchy is in-effect a logical boundary, which prevents DC from optimizing across this boundary. Many designers create unnecessary hierarchy for unknown reasons. This not only makes the synthesis process more cumbersome but also results in an increased number of synthesis scripts. As mentioned before, DC optimizes within logical boundaries. Having needless hierarchy in the design limits DC to optimize within that boundary without optimizing across the hierarchy.

Consider the logic shown in Figure 7-11(a). The top level (Block T) incorporates two blocks, A and B. The logic present at the output of block A and at the input of block B are separated by the block boundaries. Two separate optimizations of block A and B may not result in optimal solution.

By combining block A and B (i.e., removing the boundaries) as shown in Figure 7-11(b), the two logic bubbles may be optimized as one, resulting in a more optimal solution. Designers (not Synopsys) refer to this process as “flattening” the design.

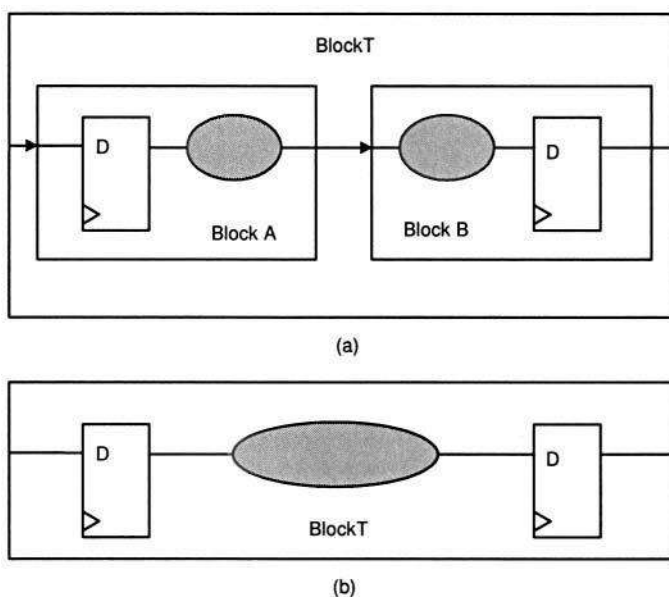


Figure 7-11. Ungrouping the Design to Improve Timing

To perform this, you may use the following command:

```
dc_shell> current_design BlockT
```

```
dc_shell> ungroup-flatten -all
```

7.5.4 Optimizing Clock Networks

Optimizing clock networks is one of the hardest operations to perform. This is due to the fact that as we descend towards VDSM technologies, the

resistance of the metal increases dramatically causing enormous delays from the input of the clock pin to the registers. Also, low power design techniques require gating the clock to minimize switching of the transistors when the data is not needed to be clocked. This technique uses a gate (e.g., an AND gate), with inputs for clock and enable (used to enable or disable the clock source).

Previous methodologies included placement of a big buffer at the top level of the chip, near the clock source capable of driving all the registers in the design. Thick trunks and spines (e.g., fishbone structure) were used to fan the entire chip in order to reduce clock skew and minimize RC delays. Although this approach worked satisfactorily for technologies 0.5um and above, it is definitely not suited for VDSM technologies (0.35um and less). The above approach also meant increased number of synthesis-layout iterations.

With the advent of complex layout tools, it is now possible to synthesize the clock tree within the layout tool itself. The clock tree approach works best for VDSM technologies and although power consumption is a concern, the clock latency and skew are both minimal compared to the big buffer approach. The clock tree synthesis (CTS) is performed during layout after the placement of the cells and before routing. This enables the layout tool to know the exact placement location of the registers in the floorplan. It is then easy for the layout tool to place buffers optimally, so as to minimize clock skews. Since optimizing clocks are the major cause in increased synthesis-layout iterations, performing CTS during layout reduces this cycle.

We still have to optimize the clocks during synthesis before taking it to layout. We cannot assume that the layout tool will give us the magic clock tree that will solve all of our problems. Remember the more optimized your initial netlist, the better results you will get from the layout tool.

So how do we optimize clock networks during synthesis? By setting a `set_dont_touch_network` to the clock pin, you are assured that DC will not buffer up the network in order to fix DRCs. This approach works fine for most designs that do not contain clock-gating logic. But what if the clocks are gated? If you set the `set_dont_touch_network` on the clock that is gated then DC will not even size up the gate (let's assume a 2-input AND gate). This is because, the `set_dont_touch_network` propagates through all the

combinational logic (AND gate, in this case), until it hits an endpoint (input clock pin of the register, in this case). This causes the combinational logic to inherit the `dont_touch` attribute also, which results in un-optimized gating logic that may violate DRCs, hence overall timing.

For instance, suppose the clock output from the AND gate is fanning out to a large number of registers and DC inferred a minimum drive strength for the AND gate. This will cause slow input transition times being fed to the registers resulting in horrendous delays for the clock net. To avoid this, you may remove the `set_dont_touch_network` attribute and perform incremental compilation. This will size up the AND gate and also insert additional buffers from the output of the AND gate, to the endpoints. Although, this approach seems ideal, it does suffer from some shortcomings. Firstly, it takes a long time for incremental compile to complete, and on rare occasions may produce sub-optimal results. Secondly, a lot of foresight is needed, for e.g., you need to apply `set_dont_touch_network` attribute on all other nets (resets and scan related signals that may not require clock tree).

A second approach is to find all high fanout nets in your design using the `report_net` command and buffer it from point to point using the `balance_buffer` command. (Refer to the DC reference manual for actual syntax for this command). Since, the `balance_buffer` command does not take clock skew into account, it should not be used as an alternative to clock tree synthesis.

Another technique is to perform in-place-optimization (IPO), using `compile -in_place`, with `compile_ok_to_buffer_during_inplace_opt` switch, set to “false”. This prevents DC from inserting additional buffers and will only size up the AND gate.

It must be noted that the above mentioned techniques are totally design dependant. Various methods have been provided that may be used for clock network optimization. Sometimes, you may find that you have to perform all the above methods to get optimal results and other times a single approach works perfectly.

Regardless of which method you use, you should also consider what you want to do during layout. For designs without gated clocks, it is preferable

that CTS be performed at the layout level. For other design, with gated clocks, you have to analyze the clock in the design (pre and post-synthesis) carefully and take appropriate action. This may also include inserting the clock tree (during layout) after the AND gate for each bank of registers. Most layout tool vendors have realized this problem and offer various techniques to perform clock tree synthesis for gated clocks.

7.5.5 Optimizing for Area

By default, DC tries to optimize the design for timing. Designs that are non-timing critical but area intensive should be optimized for area. This can be done by initially compiling the design, with specification of area requirements, but no timing constraints. In other words, the design is synthesized with area requirements only. No timing constraints are used.

In addition, one may choose to eliminate the high-drive strength gates by assigning the `dont_use` attribute on them. The reason for eliminating high-drive strength gates is that they are normally used to speed up the logic in order to meet timing, however, they are larger in size. By eliminating their usage, considerable reduction in area may be achieved.

Once the design is mapped to gates, the timing and area constraints should again be specified (normal synthesis) and the design re-compiled incrementally. The incremental compile ensures that DC maintains the previous structure and does not bloat the logic unnecessarily.

7.6 Chapter Summary

Optimizing design, is the most time consuming and difficult task, since it depends enormously on various factors e.g., HDL coding styles, type of logic, constraints etc. This chapter described advanced optimization techniques and how they affect the synthesis process.

A detailed description of the impact on timing and area by varying design constraints is discussed. To reiterate, the best results are achieved by

providing DC with realistic constraints. Over constraining a design results in large area and sub-optimal results.

With the introduction of DC98, the optimization flow has changed with more emphasis given on the timing, rather than area. Although by default area cleanup is always performed at the end of compilation, regardless, it is recommended that area constraints be specified. The timing optimization is performed by DC98 by minimizing the TNS. Prior to DC98, timing optimization was performed by reducing WNS per endpoint. The TNS optimization provides far superior results, although it does have an impact on the overall area of the design.

Various compile strategies are illustrated in this chapter, along with examples to automate this process. To successfully optimize a design, you may choose a single methodology or mix these strategies to get the desired result. All the strategies have their own advantages and disadvantages, which have also been illustrated. Choose the one, which best suits your design.

A separate section was devoted to uniquifying the design. Although, this step may not be needed, as argued by some designers, it is recommended to always uniquify the design because of reasons outlined in Chapter 9.

Finally, other optimization steps are discussed that emphasize on “how to produce optimal synthesized netlists”. Various techniques, including clock network optimization and optimizing designs for area, were described along with recommended approaches.

DESIGN FOR TEST

The *Design-for-Test* or DFT techniques are increasingly gaining momentum among ASIC designers. These techniques provide measures to comprehensively test the manufactured device for quality and coverage.

Traditionally, testability was considered as an after thought, with implementation done only at the end of the design cycle. This approach usually provided minimal coverage and often led to unforeseen problems that resulted in increased cycle time. Merging testability features early in the design cycle was the final solution, creating the name *Design-for-Test*.

8.1 Types of DFT

Various vendors, including Synopsys provide solutions for incorporating testability in the design. Synopsys adds the DFT capabilities to DC through its DFT Compiler (DFTC) that is incorporated within the DC suite of tools. The main DFT techniques that are currently in use today are:

- a) Scan insertion
- b) Memory BIST insertion
- c) Logic BIST insertion
- d) Boundary-Scan insertion

Of all four, scan and logic BIST insertion is the most complex and challenging technique, since it involves various design issues that need to be resolved, in order to get full coverage of the design.

8.1.1 Memory and Logic BIST

Unfortunately, Synopsys does not provide any solution for automatic memory or logic BIST (*Built-In-Self-Test*) generation. Due to this reason these two techniques are not covered in this section. However, there are vendors that do provide a complete solution, therefore a brief overview describing the main function of the memory and logic BIST is included, providing designers an insight into these useful techniques.

The Memory BIST is comprised of controller logic that uses various algorithms to generate input patterns that are used to exercise the memory elements of a design (say a RAM). The BIST logic is automatically generated, based upon the size and configuration of the memory element. It is generally in the form of synthesizable Verilog or VHDL, which is inserted in the RTL source with hookups, leading to the memory elements. Upon triggering, the BIST logic generates input patterns that are based upon pre-defined algorithm, to fully examine the memory elements. The output result is fed back to the BIST logic, where a comparator is used to compare what went in, against what was read out. The output of the comparator generates a pass/fail signal that signifies the authenticity of the memory elements.

Similar to memory BIST, logic BIST uses the same approach but targets the logic part of the design. The logic BIST uses a random pattern generator to exercise the scan chains in the design. The output is a compressed signature that is compared against simulated signature. If the signature of the device under test (DUT) matches the simulated signature, the device passes otherwise it fails. The main advantage of using logic BIST is that it eliminates the need for test engineers to generate huge scan vectors as inputs

to the DUT. This saves tremendous amount of test time. The disadvantage is that additional logic (thus area) is incorporated within the design, just for testing purposes.

8.1.2 Boundary Scan DFT

JTAG or boundary scan is primarily used for testing the board connections, without unplugging the chip from the board. The JTAG controller and surrounding logic also may be generated directly by DC. Boundary-scan insertion is trivial, since the whole process is rather simple and mostly automatic. It is therefore the intent of this chapter to concentrate solely on the scan insertion techniques and issues. Readers are advised to refer to the Design Compiler Reference Manual for boundary scan insertion techniques.

8.2 Scan Insertion

Scan is one of the most widespread DFT techniques used by design engineers to test the chip for defects such as stuck-at faults. It is possible to attain a very high percent fault coverage (usually above 95%) for most of the designs.

The scan insertion technique involves replacing all the flip-flops in the design, with special flops that contain built-in logic, solely for testability. The most prevalently used architecture is the *multiplexed* flip-flop. This type of architecture incorporates a 2-input mux at the input of the D-type flip-flop. The select line of the mux determines the mode of the device, i.e., it enables the mux to be either in the normal operational mode (functional mode with normal data going in) or in the test mode (with scanned data going in). These scan-flops are linked together (using the scan-data input of the mux) to form a scan-chain, that functions like a serial shift register. During scan mode, a combination of patterns are applied to the primary input, and shifted out through the scan-chain. If done correctly, this technique provides a very high coverage for all the combinational and sequential logic within a chip.

Other architectures available along with multiplexed type flip-flops are the *lssd* structure, *clocked scan* structure etc. As mentioned above, the most commonly used architecture is the *multiplexed flip-flop*. For this reason,

throughout this section the focus remains on the *multiplexed flip-flop* type architecture, for DFT scan insertion.

Scan can also be used to test the DUT for any possible timing violations. In order to understand this, we need to dig deeper into the operation of scan technique. Basically, scan uses two cycles: Capture and Shift. Scan data is injected from the primary inputs into the device where it is captured by the flops (going through logic) and is then shifted out to primary outputs where it is compared against the expected results. The signal that selects between the capture and shift cycle is usually called the `scan_enable` signal. In addition another signal is also used which is usually called the `scan_mode` signal. Scan mode signal is used to put the DUT in test conditions. In general, designs are modified such that under test conditions the device behaves different as opposed to the normal functional behavior. This modification is desired in order to achieve greater control and/or observability. Sometimes it is done simply to comply with the strict DFT rules.

The following list summarizes the basic scan operation:

1. Load/Unload scan chain (shift cycle)
2. Force primary inputs (except clocks)
3. Measure primary outputs
4. Pulse clock to capture functional data (capture cycle)

In order to understand how scan can be used to test the device for timing, a basic understanding of shift and capture cycle is necessary.

8.2.1 Shift and Capture Cycles

During the shift cycle the data traverses the entire design through a big daisy chain of registers. These registers are chained together to behave like a shift register (hence the name: shift cycle). The main difference between the shift and capture cycles is that the capture cycle utilizes the functional “D” input of the flop, whereas the shift cycle uses the “SD” input of the flop (Figure 8-1). Toggling the `scan_enable` input performs the selection of which input to use. Thus, data that is captured by the “capture” cycle is simply shifted out for comparison by the tester. In other words, a single clock cycle is needed to

perform the capture operation, while several clock cycles (depending on the length of the scan chain) are needed to perform the shift operation.

Figure 8-1 illustrate the behavior of the shift and capture cycles. Data is injected into the device through primary inputs and is shifted out of the device through the “SD” input port of the flops. Assume `scan_en` port is active high for shift operation. Once the chain has been flushed out and compared, the `scan_en` signal is toggled (driven low). Now a single clock pulse is applied to capture the data into the flops through the “D” inputs, before the `scan_en` is toggled again (driven high) and the data shifted out for comparison.

One interesting thing to note here is that during the capture cycle, data traverses the functional path. In other words, it goes through the logic just as it would if the device was operating under normal conditions. Thus if the clock pulse of same frequency as the functional clock is used to perform the capture cycle, all timing relationships present between flop-to-flop can also be checked during the scan testing. This basically means that functional testing is not needed if the scan coverage is high and the frequency of scan clock is same as that of the functional clock.

The above case is valid for most of the design structures. However in real designs there are generally cases where the functional path may be different than the scan path. One such scenario is depicted in Figure 8-2.

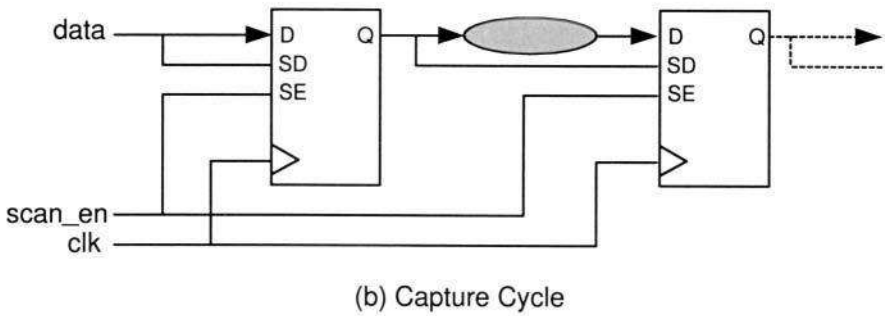
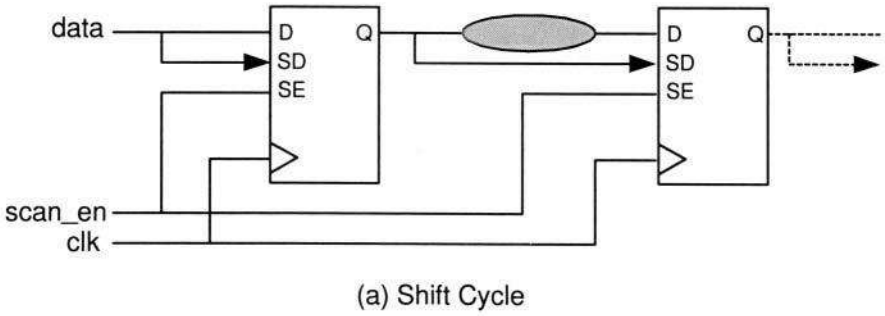


Figure 8-1 Shift and Capture Cycles

In Figure 8-2, during the functional mode of operation, an inverted clock is being fed to the second register. However, to make scan insertion simple, a favorite method used by most designers is to introduce a test signal called **scan_mode**. During test, the **scan_mode** selects the non-inverted clock path, whereas in functional mode the **scan_mode** is set such that the clock is inverted. In this case, the capture cycle differs from the functional cycle. To test the timing for the path originating from the “Q” output of the first flop and ending at the “D” input of the second flop, scan capture cycle cannot be used. Designers will have to manually write test-benches to test this particular path for timing.

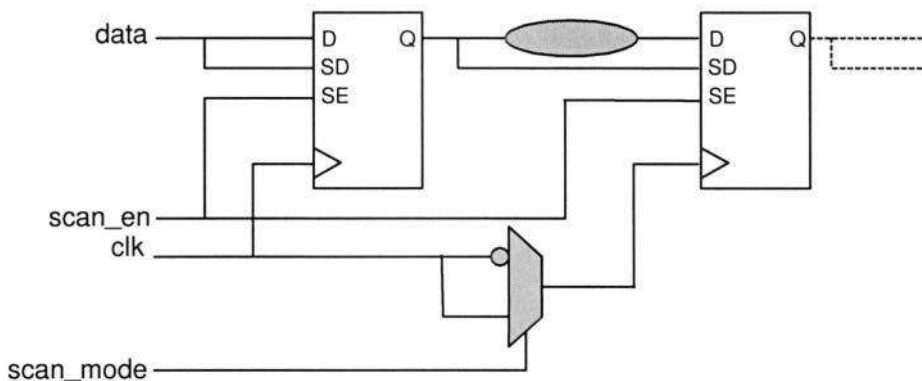


Figure 8-2. Case where functional and capture cycle differ

8.2.2 RTL Checking

This is a new feature recently introduced by Synopsys. It is part of the DFTC and is used to check the RTL for any possible DFT rule violations. This is one of the most powerful and useful features that enables designers to check the RTL design for DFT rules early on in the design cycle. It is invoked through a command called **rtl_drc**. The following script summarizes the use of this feature. Commands related to RTL checking are highlighted in bold.

```
dc_shell-1> analyze -f verilog mydesign.v
dc_shell-1> elaborate mydesign
dc_shell-1> set_scan_configuration -style multiplexed_flip_flop
dc_shell-1> set_hdlin_enable_rtl_drc_info true
dc_shell-1> create_test_clock-period 100-waveform {45 55} clk
dc_shell-1> set_test_hold 1 scan_mode
dc_shell-1> set_signal_type test_asynch_inverted reset
dc_shell-1> rtl_drc > reportfile
```

Setting the variable `hdlin_enable_rtl_drc_info` to "true" informs DC to generate a report that points to the actual line number of the source RTL (possible cause of the DRC violation). Without this variable, the report does not contain any line numbers. The default is "false".

It is important to note that the above script may be included as part of the final script used for synthesis or it may be run stand-alone by the designers to check the validity of the design for DRC rules

8.2.3 Making Design Scannable

Synopsys provides designers the capability to perform scan insertion automatically, through its test-ready (or one-pass) compile feature. This technique allows designers to synthesize the design, and map the logic directly to the scan-flops, thus alleviating potential need for post-insertion adjustments.

Companies that do not use Synopsys tools for scan insertion, instead rely on other means to perform the same task. For such a case, replacing the normal flops in the synthesized netlist, with their scan equivalent flops, before linking the scan chains together, performs the scan insertion. It is strongly recommended that the static timing analysis should be performed again on the scan-inserted netlist, since some difference may exist between the characterized timing of the scan flops and their equivalent non-scan (normal) flops. This difference if not corrected may adversely affect the total slack of the design. To avoid this problem, library developers usually specify the scan-flops timing, to the normal-flops.

To enable the Synopsys test-ready compile feature, the scan style should be chosen prior to compilation. On a particular design, the `set_scan_configuration` command is used to instruct DC on how to implement scan. There are various options available for this command that may be used to control the scan implementation. Among others, these include options for clock mixing, number of scan chains and the scan style. Only some of the most commonly used options with arbitrary arguments are listed below for the sake of explanation. Users are advised to refer to Design Compiler Reference Manual for syntax, and available range of options.

```
dc_shell-t> set_scan_configuration -style multiplexed_flip_flop \
                                     -methodology full_scan      \
                                     -clock_mixing no_mix         \
                                     -chain_count 2
dc_shell-t> create_test_clock-period 100-waveform {45 55} clk
```

```
dc_shell-t> set_test_hold 1 scan_mode
dc_shell-t> set_scan_signal test_scan_enable \
    -port scan_en-hookup pad/scan_en_pad/Z
dc_shell-t> set_scan_signal test_scan_in-port [list PI1 PI2]
dc_shell-t> set_scan_signal test_scan_out-port [list PO1 PO2]
dc_shell-t> compile-scan
dc_shell-t> preview_scan
```

The `create_test_clock` is used to specify the test clock that is used during scan operation. In the above example the test clock is called “*clk*” with a period of 100ns, rising at 45ns and falling at 55ns.

The `set_test_hold` command is used to specify a constant value to the port during test mode. In the above case the *scan_mode* port is held logic high during scan.

The `set_scan_signal` command identifies the scan in/out ports of the scan chain along with the scan enable signal. Here the command specifies a port called *scan_en* to be used as the scan enable port and instructs DC to hook up the SE port of all flops in the design to the Z output of the pad called *scan_en_pad*.

The `compile -scan` command compiles the design directly to scan-flops without linking them in a scan-chain, i.e., the scan insertion is not performed. The design is mapped to the scan-flops directly, instead of the normal flops. The design at this point is functionally correct, but un-scannable.

The `preview_scan` command is used to preview the scan architecture, chosen by the `set_scan_configuration` command.

It is highly recommended that the `check_test` command be used after compilation, to check the design for testability related rule violations. DC flags any violations by issuing warnings/errors. Failure to fix these violations invariably results in reduced test coverage. The violations may occur due to various DFT related issues, encountered during scan insertion. Some of these issues and their solutions are discussed in the next section.

```
dc_shell-t> check_test
```

It is the designer's responsibility to correct the violations. This is mainly achieved by adding extra logic around the "problem" area to provide control to the test logic. In order to fix these problems, modifying the source RTL instead of the netlist is the recommended approach. This approach allows the source RTL to remain as the "golden" database, that may be used for reference at some later stage. On the other hand, if the netlist is modified, then the changes may be forgotten, thus lost, after the design is taped-out.

Although, the scan has not yet been inserted, an additional step at this point is to get an estimate of the fault coverage of the design by generating the statistical ATPG test patterns. This step helps quantify the quality of the design, at an earlier stage. If the coverage number is low, then the only option is to identify and fix the areas that need further improvement. However, if the fault coverage number is high then this is an indication to proceed ahead.

It must be noted that the fault coverage numbers should be considered as best case only, due to the fact that the design may be part of a larger hierarchy, i.e., it may be a sub-block. At the sub-block level, the input port controllability and output port observability may be different when this sub-block is embedded in the full design (top-level). For a full design, this may cause a lower fault coverage number, than expected. The following command is used to generate the statistical test patterns:

```
dc_shell-t> create_test_patterns -sample < n >
```

Once the problem areas have been identified and fixed in the RTL, the design is ready for scan insertion. Using the following command performs the scan insertion:

```
dc_shell-t> insert_scan
```

The `insert_scan` command does more than just link the scan-flops together to form a scan-chain. It also disables the tri-states, builds and orders the scan-chains, and optimizes them to remove any DRCs. This command may insert additional test logic, in order to get better control over certain parts of the design.

After scan insertion the design should once again be checked for any rule violations through the `check_test` command. The `report_test` command may also be utilized to generate all test-related information about the design. Various options exist to control the output of the report. More details can be found in the Design Compiler Reference Manual.

8.2.4 Existing Scan

Designs with existing scans chains need to be treated differently. Such a case may exist when you are importing a design that has been scan inserted by a foreign tool other than Synopsys DFTC. In this case, the “db” file does not exist. The input to DFTC is a scan inserted structured netlist. Thus all scan attributes that were part of the “db” file are also absent. In other words, DFTC does not know anything about the scan ports, resets etc.

The scan attributes can be re-applied to the structured netlist in order to perform further processing (such as scan chain ordering through PhyC). This can be accomplished by using the following script. The items of interest that differentiates this from the original one-pass synthesis approach have been highlighted in bold.

```
dc_shell-t> set_scan_configuration -style multiplexed_flip_flop \
                                -methodology full_scan \
                                -existing_scan true
```

```
dc_shell-t> create_test_clock-period 100-waveform {45 55} elk
dc_shell-t> set_test_hold 1 scan_mode
dc_shell-t> set_signal_type test_scan_enable scan_en
dc_shell-t> set_signal_type test_mode scan_mode
```

For active low reset, use **test_asynch_invert**. Active high use **test_asynch**

```
dc_shell-t> set_signal_type test_asynch reset
```

```
dc_shell-t> set_signal_type test_scan_in [list PI1 PI2]
dc_shell-t> set_signal_type test_scan_out [list PO1 PO2]
```

8.2.5 Scan Chain Ordering

The advantages of scan chain ordering are enormous. However, usually with every good thing there is also something bad associated with it.

The following are some of the benefits of scan chain reordering:

1. Reduces congestion, thus improves timing
2. Less overall area (net length is dramatically reduced)
3. Improves setup time of functional paths due to decreased flop loading
4. Reduces negative hold-times (mainly a simulation vs. static timing analysis issue)
5. Improves timing due to less overall capacitance
6. Improves power consumption by driving less net capacitance
7. Better clock tree (lower latency and fewer buffers), thus improving timing along with low power consumption.

The disadvantages are:

1. Increases the chance of hold-time violations in scan-path
2. Additional runtime in the design cycle.

Scan chain ordering is performed using DFTC, Physical Compiler (PhyC) or your own layout tool. Chapter 10 describes in detail the PhyC approach of ordering the scan chains based on physical proximity of the scan cells. The DFTC approach is very similar to PhyC. Instead of using `physopt`, the following option may be used for the `insert_scan` command to stitch the scan chains more intelligently based on the physical placement location of scan cells.

```
dc_shell-t> insert_scan -physical
```

The above method assumes that the physical information has been back-annotated (in PDEF format) to the design before `insert_scan` is run. The command to back-annotate the physical information in PDEF format is as follows:

```
dc_shell-t> read_pdef <PDEF file>
```

It must be remembered that scan chain ordering increases the chance of hold-time violations. This is due to the fact that the flops are near each other, thus creating a very short route for the shift-cycle scan path. In other words, the Q to SD (Figure 8-1) wire length is very short. Thus, scan data may arrive faster than the clock causing hold-time violation.

Interestingly, DFTC orders the scan chain in an elegant fashion. It analyzes the logic fed by the source register and tries to find a cell that can be used in such a way so as not to alter the functionality of the shift cycle. It taps the output of this cell to connect to the scan-in port of the destination register.

Consider the diagram shown in Figure 8-3. Here the Q output of the source register is feeding a buffer before it encounters the rest of the logic. In this case, `insert_scan` links the output of the buffer to the SD input of the destination flop. By doing this, there is an additional delay of the buffer in the scan path. This delay minimizes the chance of any hold-time violations at the destination flop.

It must be noted that it does not have to be a buffer (as shown in Figure 8-3). In reality, it can be any cell as long as the functionality of the shift cycle is maintained. For example `insert_scan` can tap the QN output of the source cell and subsequently use an inverter's output to link to SD input of the destination flop. In other words, QN of source flop connects to the inverter's input pin with the output of the inverter connected to SD input of the destination flop.

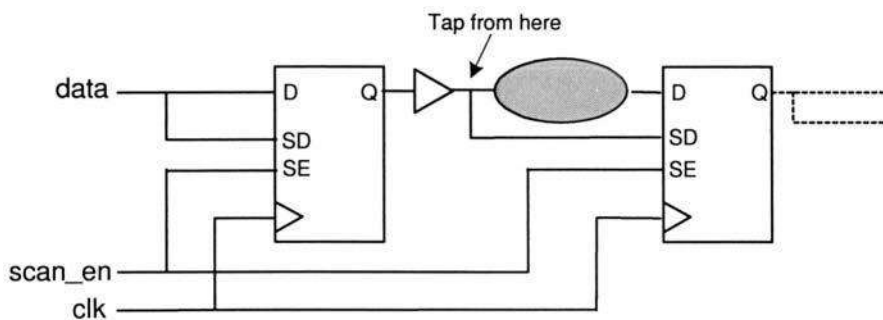


Figure 8-3. Intelligent Scan Chain Ordering

As mentioned before, scan-chain ordering increases the chance of hold-time violations. By utilizing this method, the likelihood of hold-time violations is minimized. This behavior of `insert_scan` is governed by the following variable:

```
dc_shell-t> set test_disable_find_best_scan_out false
```

The default is “false” which means that `insert_scan` will analyze the logic and find the best way possible to stitch the scan chains. If the argument is changed to true, `insert_scan` will tap the output `Q` and link it to pin `SD` of the destination flop. It will not try to analyze the logic fed by `Q`.

8.2.6 Test Pattern Generation

Upon completion of scan insertion in the design, the test patterns may be generated for the entire design using TetraMAX. This is an independent ATPG tool that is used solely for creating test patterns and provides a seamless interface to DC. It also provides enhanced GUI interface for analysis and debugging.

The ATPG discussion and the TetraMAX usage are beyond the scope of this book. Readers are advised to consult the TetraMAX ATPG User Guide for further details.

During dynamic simulation, the test patterns are used as input stimuli to the design to exercise all the scan paths. This step should be performed at the full chip level and preferably after layout.

8.2.7 Putting it Together

Scan insertion is a complicated topic and usually there are many methods of making designs scannable. In order to alleviate the confusion, this section provides an example script that consolidates all the information provided above.

Script for one-pass scan synthesis, insertion & order

```
dc_shell-t> analyze -f verilog mydesign.v
dc_shell-t> elaborate mydesign
dc_shell-t> set_scan_configuration -style multiplexed_flip_flop
                                   -methodology full_scan      \
                                   -clock_mixing no_mix         \
                                   -chain_count 2
dc_shell-t> set hdl_in_enable_rtl_drc_info true
dc_shell-t> create_test_clock-period 100-waveform {45 55} clk
dc_shell-t> set_test_hold 1 scan_mode
dc_shell-t> set_signal_type test_async reset
dc_shell-t> rtl_drc
dc_shell-t> source constraints.scr #clocks, input/output delays etc.
dc_shell-t> set_scan_signal test_scan_enable \
                                   -port scan_en -hookup pad/scan_en_pad/Z
dc_shell-t> set_scan_signal test_scan_in -port [list PI1 PI2]
dc_shell-t> set_scan_signal test_scan_out -port [list PO1 PO2]
dc_shell-t> compile_scan
dc_shell-t> preview_scan
dc_shell-t> check_test
dc_shell-t> read_pdef mydesign_floorplan.pdef
```



```
dc_shell-t> insert_scan -physical
dc_shell-t> check_test
dc_shell-t> write-format verilog -hierarchy-output mydesign.sv
dc_shell-t> write_pdef -v3.0 -output mydesign_scan.pdef
```

Script for ordering scan chains for an existing netlist

```
dc_shell-t> read_verilog mydesign.sv #Gate level netlist
dc_shell-t> current_design mydesign
dc_shell-t> set_scan_configuration -style multiplexed_flip_flop
                                -methodology full_scan      \
                                -clock_mixing no_mix         \
                                -existing_scan true           \
                                -chain_count 2
dc_shell-t> source constraints.scr #clocks, input/output delays etc.
dc_shell-t> set_signal_type test_scan_enable scan_en
dc_shell-t> set_signal_type test_mode scan_mode
dc_shell-t> set_signal_type test_asynch reset
dc_shell-t> set_signal_type test_scan_in [list PI1 PI2]
dc_shell-t> set_signal_type test_scan_out [list PO1 PO2]
dc_shell-t> check_test
dc_shell-t> read_pdef mydesign_floorplan.pdef
dc_shell-t> insert_scan-physical
dc_shell-t> check_test
dc_shell-t> write-format verilog -hierarchy-output mydesign.sv
dc_shell-t> write_pdef -v3.0 -output mydesign_scan.pdef
```

Note: The scripts provided above use DFTC only. PhyC also offers this capability and is superior than using just DFTC. PhyC flow is described in Chapter 10.

8.3 DFT Guidelines

Obtaining high fault coverage for a design depends on the quality of the implemented DFT logic. Not all designs are ideal. Most “real-world” designs suffer from a variety of DFT related issues, and if left unsolved, result in

reduced fault coverage. This section identifies some of these issues and provides solutions to overcome them.

8.3.1 Tri-State Bus Contention

This is one of the common problems faced by the DFT tool. During scan shifts, multiple drivers on a bus may drive the bus simultaneously, thus causing contention. Fixing this problem requires that only one driver be active at a given time. This can be achieved by adding the decoder logic in the design, which controls the *enable* input of each tri-state driver through a mux. The mux is used to select between the normal signal (in the functional mode) and the control line from the decoder. The decoder control is selected only during the scan-mode.

The decoder inputs are generally controlled directly from the primary inputs, thus providing means to selectively turn-on the tri-state drivers, thereby avoiding contention.

8.3.2 Latches

Avoid using latches as much as possible. Although, latches cover less area than flops, they are difficult to test. Testing maybe difficult but is not entirely impossible. Making them transparent during scan-mode can make them testable. This usually means, adding control logic (for the clock) to each latch. If an independent clock, clocks all the latches, then a single test-logic block may be used to control the clock to make the latches transparent during scan-mode.

8.3.3 Gated Reset or Preset

DFT requires that the reset/preset of a flop be controllable. If the reset/preset to a flop is functionally gated in the design, then the flop is un-scannable. To avoid this situation, the reset/preset signal should bypass the gating logic in scan-mode. A mux is generally used to remedy this problem, with the external scan-mode signal functioning as its select line; and bypass reset/preset signal along with the original gated signal, as its input.

8.3.4 Gated or Generated Clocks

The gated clocks also suffer from the same issue that has been described above for gated resets. DFT requires that the clock input of the flop be controllable. The solution again is to bypass the gating logic through a mux, to make the flop controllable.

This problem is prevalent in those designs that contain logic to generate divided clocks. The flop(s) that is used to generate the divided clock should be bypassed during scan-mode. The dividing logic in this case may become un-scannable, but the divided clock can be controlled externally, thus providing coverage for the rest of the design. Small loss of coverage for the dividing logic is offset by the coverage gains achieved for the entire design.

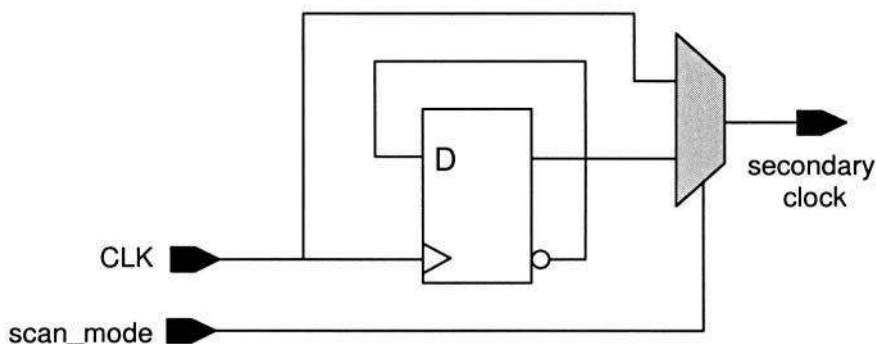


Figure 8-4. Bypassing Generated Clock

In Figure 8-4, the secondary clock is controlled externally, by using a mux that bypasses the CLK signal in the scan-mode. This provides controllability of the secondary clock, for the rest of the design. Depending upon the type of dividing logic being used, some parts of the logic may be un-scannable. The following command may be used to inform DFTC to exclude a list of sequential cells while inserting scan:

```
dc_shell> set_scan_element false <list of cells or designs>
```

8.3.5 Use Single Edge of the Clock

Most designs are coded using a single edge of the clock as reference. However, there are always cases within a design, where both the rising and falling edge of the clock is used. This creates a problem for DFTC, as it is unable to handle such situations. The problem may be avoided by using the same clock edge for the entire design, when the design is in the scan-mode. This is illustrated in the following VHDL example:

```
process(clk, test_mode)
begin
    if (test_mode = '1') then
        muxed_clk_output <= clk;
    else
        muxed_clk_output <= not(clk);
    end if;
end process;
```

The above VHDL code infers a two-input mux. Positive edge of the clock is made use of during scan-mode, while the falling edge of the clock is used during normal mode.

8.3.6 Multiple Clock Domains

It is strongly recommended that designer assigns separate scan-chains for each clock domain. Intermixing of clock domains within a scan-chain typically leads to timing problems. This is attributed to the differences in clock skew between different clock domains. A disadvantage to using this technique is that it may lead to varying lengths of scan-chains.

An alternative solution is to group all flops belonging to a common clock domain, and connect them serially to form a single scan-chain. This requires the clock skew between the clock domain to be minimal. The clock sources should also be accessible from outside (primary inputs), so that the timing can be externally controlled when the device is tested at the tester.

There are other solutions available to this problem. One such solution is to use clock muxing at the clock source, so that only one clock is used during scan-mode.

8.3.7 Order Scan-Chains to Minimize Clock Skew

Presence of clock skew within a scan-chain usually causes hold-time violations. Some designers think that since testing is performed at slower speed as compared to the normal operational speed, the scan-chains cannot have any timing problems. This is a misconception. *Only the setup-time is frequency dependent, while the hold-time is frequency independent.* Therefore, it is extremely important to minimize clock skews to avoid any hold-time violations in the scan-chain.

The scan-chain may be re-ordered with flops having greater clock latency nearer to the source of the scan-chain, while the flops with less clock latency kept farther away. This helps in reducing the clock skew, thereby minimizing the possibility of any hold-time violations.

8.3.8 Logic Un-Scannable due to Memory Element

As explained earlier, the memory itself can be tested by the use of memory BIST circuitry. However, memory elements (e.g., RAMs) that do not have scan-chains (usually built-in) surrounding them, cause a loss of coverage for the combinational logic present at its inputs and outputs.

Let us consider the case for a RAM that is being fed by combinational logic. This logic present at its inputs is being shadowed by the RAM, thus is un-testable. If the inputs to the memory element are not coming directly from sequential elements, then any combinational logic present between the sequential logic and the memory element becomes un-testable. To avoid this situation, one may bypass the RAM in scan-mode. This is achieved by short-circuiting all the inputs feeding the RAM to the outputs of the RAM, through a mux. In scan-mode, the mux enables the short-circuited path and enables data to bypass the RAM.

Another problem that typically arises during scan-mode is that the outputs of the memory element are unknown. This typically results in the ‘unknowns’ being introduced to the surrounding scan-chain, causing it to fail. This situation can be avoided by using the bypass method, described above. The ‘unknowns’ generated by the RAM are blocked by the mux present at its outputs. This is because the mux is selected to bypass the RAM; it will therefore prevent the propagation of ‘unknowns’.

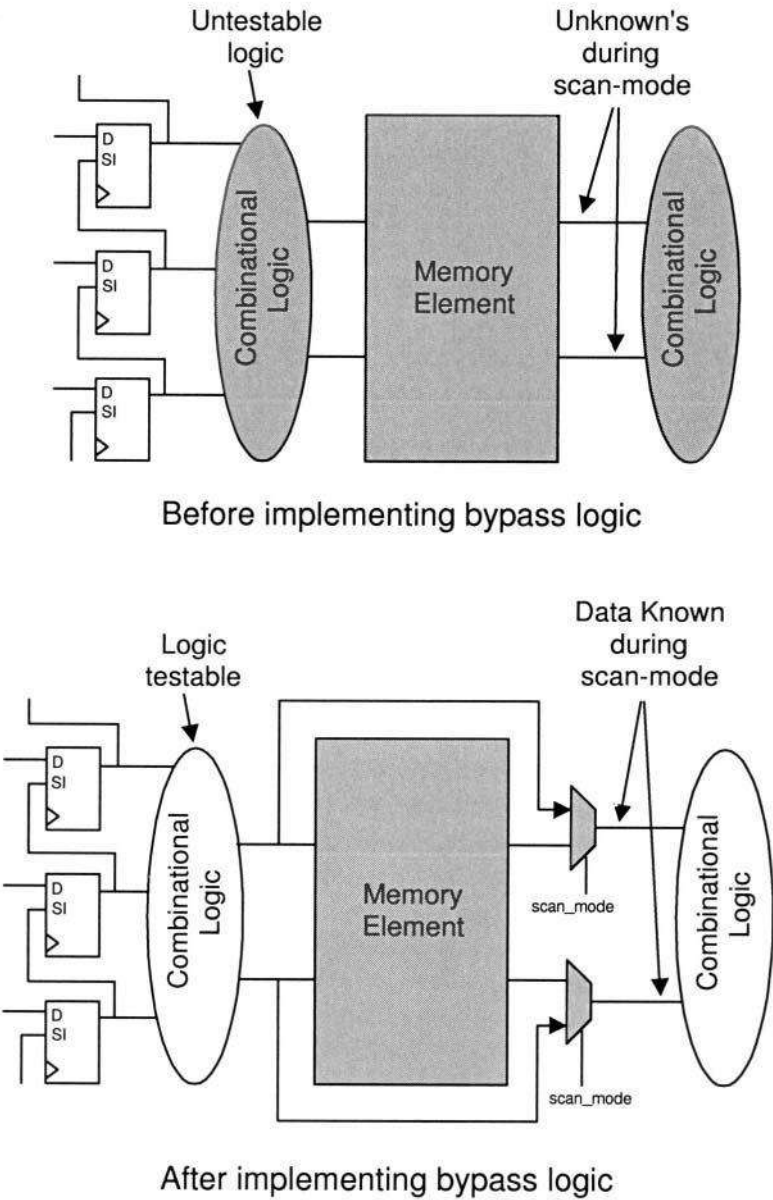


Figure 8-5. Memory Bypass

8.4 Chapter Summary

DFT techniques are essential to an efficient and successful testing of the manufactured device. By implementing DFT features early in the design cycle, full test coverage on the design may be achieved, thereby reducing the debugging time normally spent at the tester after the device is fabricated.

This chapter described the basic testability techniques that are currently in use, including a brief description of logic and memory BIST that is not yet supported by Synopsys.

A detailed description was provided for the scan insertion DFT technique, using the DFT Compiler. Various guidelines and solutions were also provided that may help the user to identify the issues and problems related to this technique.

LINKS TO LAYOUT AND POST LAYOUT OPTIMIZATION

Including Clock Tree Insertion

Until now, a virtual wall existed between the front-end and the back-end processes, with the signoff to the ASIC vendor for fabrication, occurring at the structural netlist level. The ASIC vendor was responsible for floorplanning and routing of the design, and provided the front-end designers the resulting delay data. However, this process was inefficient and often resulted in multiple exchanges of the netlist and the layout data between the designers and the ASIC vendor.

As we move deeper into the VDSM realm, the virtual wall between the front-end and the back-end is destined to collapse. This is because of the tremendous challenges and difficulties posed by the VDSM technologies. In order to overcome these difficulties, it is becoming evident that greater controllability and flexibility of the ASIC design flow is necessary. This requires total integration between the synthesis and layout processes. This means that designers are now compelled to perform their own layout. Instead of providing the ASIC vendors with the structural netlist, they are now given the physical database for final fabrication.

This shift in the signoff has resulted in a well-defined interface, between the synthesis tools and Place & Route tools (referred to as layout tools, from here onwards). Synopsys refers to this interface as Links to Layout or LTL.

This chapter describes the interface between DC and the layout tool. Almost all designs require the LTL interface to conduct the post-layout optimizations (PLO). Also, this chapter provides different strategies used for PLO. Furthermore, for successful layout, a section is devoted to clock tree synthesis, as performed by the layout tool.

Assuming that the user has synthesized and optimized a design. The design meets all timing and area requirements. Now the question arises, “How close are the estimated wire-load models used for pre-layout optimization, to the actual extracted data from the layout?” The only way to find this information is to floorplan and then route the design.

With shrinking geometries, the resistance of the wires is increasing as compared to its capacitance. This results in a large portion of the total delay (cell delay + interconnect delay) being dominated by the delays associated with the interconnect wires. In order to reduce this effect, designers are forced to spend an increased amount of time floorplanning the chip. Therefore, it is imperative for DC to make use of the physical information, in order to perform further optimizations.

Using Links to Layout or LTL for short, one can exchange relevant data (e.g., timing constraints and/or placement information), to and from DC, to the layout tool. This helps DC perform improved post-layout optimizations. It also results in reduced iterations between synthesis and layout.

Note: With the introduction of Physical Compiler (or PhyC), two design flows exist. The traditional flow (described in this chapter) and the PhyC based flow (described in Chapter 10). However, some parts of the traditional flow are still relevant to PhyC based flow. Therefore, in order to obtain full understanding of the entire flow, it is strongly recommended that the reader read this chapter before proceeding to Chapter 10.

9.1 Generating Netlist for Layout

Most layout tools accept only the Verilog or EDIF netlist format as inputs. Many users, who code the design in VHDL, generate the netlist from DC in EDIF format for layout. Although this format is universal, it does have certain drawbacks. Primarily, the EDIF format is not easily readable; therefore modifying the netlist at a later stage to perform ECO is cumbersome. Secondly, the netlist in EDIF format is not simulatable.

So the question is, why should designers route a netlist that cannot be simulated? What happens if DC generates incorrect netlist (bad logic) due to some bug in its EDIF translator? With EDIF, the problem will only be identified at a much later stage, while performing LVS. Therefore, it is recommended that designers generate the netlist from DC in Verilog format, as input to the layout tool. Furthermore, the Verilog format is easy to understand, which considerably simplifies the task of modifying the netlist, in case an ECO needs to be performed on the design. In addition, even if the test-bench for a design is in the VHDL format, one can still simulate the Verilog netlist by using simulators (currently available) that are capable of simulating a mixture of these languages.

Before sending the netlist (of the full design or individual block) to layout, it is recommended that the following procedure be performed on the netlist to facilitate smooth transfer of the design from DC to the layout tool.

- a) Uniquify the netlist.
- b) Simplify netlist by changing names of nets in the design.
- c) Remove unconnected ports from the entire design.
- d) Make sure that all pin names of leaf cells are visible.
- e) Check for *assign* and *tran* statements.
- f) Check for unintentional gating of clocks or resets.
- g) Check for unresolved references.

9.1.1 Uniquify

As mentioned previously, the netlist must be uniquified in DC, in order to perform clock tree synthesis during layout. This operation generates a unique

module/entity definition for a sub-block that is instantiated multiple times in the design. This may seem like an unnecessary operation that reduces the readability of the netlist, and results in increased size of the netlist. However, physically the design is considered flat by most layout tools. In other words, the blocks referenced multiple times, although ideal in all respects, exist physically at separate locations. Furthermore, flops present inside these blocks also need to be connected to the clock source. This makes it obvious that separate clock-net names are required for connecting the clock tree to these blocks.

Non-uniquified netlists pose a problem during clock tree transfer from the layout tool to DC. The problem only occurs, if the clock tree information alone is transferred to DC (through methods described later), that does not involve a complete netlist transfer from the layout tool to DC. In this case, only one module/entity definition for the multiple instanced blocks is present in the netlist, for a non-uniquified design. This causes a problem when the clock tree information is transferred back to DC, i.e., modifying the design database in DC, to include the buffers and additional ports in the sub-blocks. The problem is that two distinct net names (outputs of clock tree) cannot connect to the same port of a single module/entity. Uniquifying the design solves the above problem. However, it also causes the netlist to increase in size, since it creates separate module/entity definition for each instantiation of the block.

Some users prefer to unify the netlist as they traverse the hierarchy to reach the top-level, while others unify the whole chip at once, from the top-level. The recommended approach is to remove the `dont_touch` attribute from all sub-blocks of the design, before unifying the netlist.

The following command may be used to remove the `dont_touch` attribute from the entire design, before unifying the netlist from the top level:

```
dc_shell -t> remove_attribute [get_designs -hier {*}] dont_touch
```

```
dc_shell -t> unify
```

9.1.2 Tailoring the Netlist for Layout

Some layout tools may have difficulty reading the Verilog netlist that contains unusual net names. For instance, DC sometimes produces signal names with “*cell*” or “*-return” appended, or in-between the names. Other times, users may find that some net names (or port names) have leading or trailing underscores. Also, most layout tools have restrictions on the maximum number of characters for a net, or a port name. Depending on the restrictions imposed by the specific layout tool, it is possible for the user to clean the netlist within DC, before writing it out. This ability of DC provides a smooth interface to the layout tool while meeting all tool requirements.

To prevent DC from generating the undesirable signal names, the user must first define rules, and then instruct DC to conform to these rules before writing out the netlist. For instance, one may define rules called “BORG” by including the following in the “.synopsys_dc.setup” file:

```
define_name_rules BORG -allowed {A-Za-z0-9_}      \
    -first_restricted "0-9_[ ]"                  \
    -max_length 30                                \
    -map { {"*cell*", "mycell"}, {"*-return", "myreturnz"} }
```

Instructing DC to conform to the above rule (BORG) is performed at the command line (or through a script), by using the following command:

```
dc_shell-t> change_names -hierarchy -rules BORG
```

In addition to the above, users may also desire to alter the bus naming style in the netlist. DC provides a variable through which the user is allowed to tailor the naming style of the busses, written out in the netlist. The variable may again be set in the setup file as follows:

```
set bus_naming_style {%s[%d]}
```

9.1.3 Remove Unconnected Ports

Many designs suffer from the problem of ports of a block that are left unconnected intentionally, or maybe due to legacy reasons. Although, this practice has no affect on DC in terms of producing functionally correct netlist, however, some designers prefer to remove these ports during synthesis. This is generally a good practice since, if left unconnected, DC will issue a warning message regarding the unconnected ports. Because a design may contain many such unconnected ports, it is possible that a real warning may get lost between the numerous unconnected ports warnings. It is therefore preferable to remove the unconnected ports and check the design, before generating the netlist. The following commands perform this:

```
dc_shell-t> remove_unconnected_ports [get_cells -hier {*}]  
  
dc_shell-t> check_design
```

9.1.4 Visible Port Names

Generally, all synthesized designs result in mapped components that have one (or more) of their output ports not connected to a net. When DC generates a Verilog netlist, it does not write out the unconnected port names. Depending upon the layout tool, a mismatch might occur between the number of ports in the physical cell versus the number of ports of the same cell present in the netlist. For example, a D flip-flop containing 4 ports namely, D, CLK, Q and QN, may be connected as follows:

```
DFF dff_reg (.D(data), .CLK(clock), .Q(data_out) ) ;
```

In the above case, DC does not write out QN the port, since the function of the inverting QN output is not utilized in the design. Physically, this cell contains all 4 ports, therefore, when the netlist is read in the layout tool, a mismatch between the number of ports occurs. Setting the value of the following variable to true in the setup file can prevent this mismatch:

```
set verilout_show_unconnected_pins true
```

Making the port names visible is solely dependent on the layout tool's requirements. However, recently some layout tool vendors upon realizing this limitation have improved their tools so that the above restriction is not imposed.

9.1.5 Verilog Specific Statements

Some layout tools have difficulty reading the netlist that contains *tri* wires, *tran* primitives and *assign* statements. These are Verilog specific primitives and statements that are generated in the netlist for many possible reasons.

DC generates *tri* wires for designs containing “inout” type ports. For designs containing these types of ports, DC needs to assign values to the bi-directional port, thus producing *tri* wire statement and *tran* primitives. To prevent DC from generating these, users may use the following IO variable in the setup file. When set to true, all tri-state nets are declared as *wire* instead of *tri*.

```
set verilout_no_tri true
```

Several factors influence the generation of the *assign* statements. Feedthroughs in the design are considered as one such factor. The feedthroughs may occur if the block contains an input port that is directly connected to the output port of the same block. This results in DC generating an *assign* statement in the Verilog netlist. Also the *assign* statements get generated, if an output port is connected to ground, or is being driven by a constant (e.g., 1'b0 or 1'bl). While writing out the netlist in Verilog format, DC issues a warning, stating that the *assign* statements are being written out.

In case of the feedthroughs, the user can prevent DC from generating these statements by inserting a buffer between the previously connected input and output port. This isolates the input port from the output port, thereby breaking the feedthrough. To perform this, the following variable can be used before compiling the design.

```
dc_shell-t> set_fix_multiple_port_nets -feedthroughs
```

The `-buffer_constants` option may also be used in the above variable in order to buffer the constants driving the output port. However, since there are many other variations that may produce the *assign* statements, it may be safer to use the following for full coverage:

```
dc_shell-t> set_fix_multiple_port_nets-all-buffer_constants
```

Many designers complain that *assign* statements get generated in the netlist, even after all the steps described above have been performed. In almost all cases this is caused by the `dont_touch` attribute present on a net without the users' knowledge. The user can find the presence of this attribute by performing a `report_net` command. The `dont_touch` attribute on the net can be removed from the net by using the following command:

```
dc_shell-t> remove_attribute [get_nets <net name>] dont_touch
```

9.1.6 Unintentional Clock or Reset Gating

It is always a good idea to check and double-check the clocks in the design before handing the netlist over for place and route. Remember that the clock provides the reference for all signals i.e., all signals are directly related to the clock and are optimized with respect to it. If the clock is unintentionally buffered (maybe you forgot to apply a `set_dont_touch_network` attribute on it), it will affect clock latency and skew, which may result in the user not being able to meet the set timing objectives.

Generally, resets are not considered as important as clocks. However, since the `set_dont_touch_network` attribute is also applied for them, it is wise to check their buffering.

To check for unintentional gating for the clocks, the user may use the following command:

```
dc_shell-t> report_transitive_fanout-clock_tree
```

To check for unintentional gating of another signal (say, the reset signal), you may use the `-from` option in the above command. For example:


```
dc_shell-t>report_transitive_fanout-from reset
```

Obviously, the clocks should be defined before the `-clock_tree` option can be used. Alternatively, one may also use the `-from` option for the clocks. This does not require the clocks to be defined first. Note that the `-from` and the `-clock_tree` option cannot be used simultaneously.

9.1.7 Unresolved References

Designers should exercise caution and always check for any unresolved references. DC issues a warning for a design containing instantiations of a block that does not have a corresponding definition. For example, block A is the top level module that instantiates sub-block B. If you fail to read the definition of block B in DC while writing out the netlist for block A, DC will generate a warning stating that block A contains unresolved references. Also, this message is issued for cases where a port mismatch occurs between the instanced cell and its definition.

9.2 Layout

With a clean and optimized netlist, the user is ready to transfer the design to its physical form, using the layout tool. Although, layout is a complex process, it can be condensed to three basic steps, as follows:

- a) Floorplanning.
- b) Clock tree insertion.
- c) Routing the database.

9.2.1 Floorplanning

This is considered to be the most critical step within the entire layout process. Primarily a design is floorplanned in order to achieve minimum possible area, while still meeting timing requirements. Also, floorplanning is performed to divide the design into manageable blocks.

In a broad sense, floorplanning consists of placement of cells and macros (e.g., RAMs and ROMs or sub-blocks) in their proper locations. The objective is to reduce net RC delays and routing capacitances, thereby producing faster designs. Placing cells and macros in proper locations also helps produce minimum area and decrease routing congestion.

Almost all designs undergo the floorplanning phase, and time should be spent trying to find the correct placement location of the cells. Optimal placement improves the overall quality of the design. It also helps in reduced synthesis-layout iterations. For small and/or slow designs the floorplanning may not be as important, as that for large and/or timing critical designs consisting of thousands of gates (>150K). For these designs, it is recommended that a hierarchical placement and routing of the design be performed. For example, a sub-block has been placed and routed, meeting all timing and area requirements. The sub-block is subsequently brought in as a fixed macro inside the full design, to be routed with the rest of the cells or macros.

9.2.1.1 Timing Driven Placement

Finding correct locations of cells and macros is time consuming, since each pass requires full timing analysis and verification. If the design fails timing requirements, it is re-floorplanned. This obviously is a time consuming and often frustrating method. To alleviate this, the layout tool vendors have introduced the concept of timing-driven-placement, more commonly referred to as timing-driven-layout (TDL).

The TDL method consists of forward annotating the timing information of the design generated by DC, to the layout tool. When using this method, the physical placement of cells is dictated by the timing constraints. The layout tools gives priority to timing while placing the cells, and tries not to violate the path constraints.

DC generates the timing constraints in SDF format using the following command:

```
write_constraints -format <sdf | sdf-v2.1>  
                 -cover_design  
                 -from <from list>  
                 -to <to list>  
                 -through <through list>  
                 -output <output file name>
```

The above command generates the constraints file in SDF format. Both versions 1.0 and 2.1 are supported. If the layout tool does not support the 2.1 version, then the user may always use the default version 1.0 by specifying “sdf” instead of “sdf-v2.1”.

The `write_constraints` command provides many more options in addition to the one illustrated in the above example, however, the use of `-cover_design` option is more prevalent. The `-cover_design` option instructs DC to output just enough timing constraints so as to cover the worst path through every driver-load pin pair in the design. For additional information regarding this command and its options, the user is advised to refer to the DC reference manual.

A timing constraint file in SDF version 2.1 format generated by DC with the `-cover_design` option, is illustrated in Example 9.1. The SDF file contains the `TIMINGCHECK` field containing `PATHCONSTRAINT` for all the paths in a design. The last field of the `PATHCONSTRAINT` timingcheck contains three sets of numbers that define the path delay for a particular path segment. The three numbers, although the same in this example, correspond to minimum, typical, and maximum delay values. These numbers and their corresponding paths govern the placement of cells during layout.

Example 9.1

```
(DELAYFILE  
(SDFVERSION "OVI 2.1")  
(DESIGN "hello")  
(DATE "Mon Jul 20 22:59:49 1998")  
(VENDOR "Enterprise")  
(PROGRAM "Synopsys Design Compiler cmos")  
(VERSION "1998.02-2")
```

```

(DIVIDER/)
(VOLTAGE 2.70:2.70:2.70)
(PROCESS "TYPICAL")
(TEMPERATURE 95.00:95.00:95.00)
(TIMESCALE 1ns)
(CELL
(CELLTYPE "hello")
(INSTANCE)
(TMINGCHECK
(PATHCONSTRAINT INPUT1 U751/A3 U751/ZN U754/I1
      U754/ZN REG0/D (1.523:1.523:1.523))
(PATHCONSTRAINT INPUT2 U744/A1 U744/Z U745/A1
      U745/ZN REG1/D (1.594:1.594:1.594))
(PATHCONSTRAINT REG1/CLKREG1/Q U737/I U737/ZN
      OUTPUT1 ( 3.000:3.000:3.000) )
(PATHCONSTRAINT REG2/CLK REG2/Q U1131/A2
      U1131/ZN REG3/D (25.523:25.523:25.523) )
      •
      •

```

It must be noted that depending upon the size of the design, the generation of the timing constraints for the entire design may take a considerable amount of time. Constraints may be generated for selected timing-critical paths (using **–from**, **–to** and **–through** options) in order to avoid this problem. Alternatively, users may perform hierarchical place and route, where small sub-blocks are routed first, utilizing the TDL method. Hierarchical place and route is a preferred approach, since it is based upon the “divide and conquer” technique. Dividing the chip into small manageable blocks makes it relatively simpler for designers to tackle the run-time problems.

An alternative approach of performing TDL is to let the layout tool generate the timing constraints based upon the boundary conditions, top-level constraints and timing exceptions of the design. This is a tool dependent feature and supported by some layout tool vendors, it may not be supported by others. The layout tool uses its own delay calculator to find out the timing constraints for each path in the design in order to place cells. This method is far superior than the others described previously in the sense that this method is considerably faster, however, a major drawback with this approach is that

users are now compelled to use and trust the delay calculator of the layout tool. In any case, timing convergence can be achieved with relative ease, using this approach.

Performing TDL may also have an impact on the overall area. One may find that the area increases when the above approach is used. However, this point is debatable with some users insisting that the total area gets reduced because of the rubber-band effect caused by the TDL method, while others swear by the opposite.

9.2.1.2 Back Annotation of Floorplan Information

Total integration with the back-end tools allows DC to perform with increased efficiency, in order to achieve timing and area convergence. DC makes use of several formats that allow the layout information to be read by DC. For post-layout optimization, it is necessary for DC to know the physical location of each sub-block. Using the physical design exchange format (PDEF) grants DC access to this pertinent information. The PDEF file contains the cluster (physical grouping) information and location of cells in the layout.

Pre-placement, the netlist is optimized using the wire-load models, spread across the logical hierarchy. However, physical hierarchy may be different than the logical hierarchy. Physically, the cells/macros may be grouped depending on the pad locations or some other consideration. Therefore, it is imperative for DC to receive the physical placement information, for it to more effectively optimize the design. This is done by re-adjusting the application of the wire-loads on the design, based upon the physical hierarchy.

DC uses the following command to read the physical placement information generated by the layout tool in PDEF format:

```
read_clusters -design <designname> <pdef filename>
```

Once the netlist has been re-optimized, the physical information may be passed back to the layout tool through the PDEF file. The following command in DC, performs this task:

```
write_clusters -design <design name> -output <pdef filename>
```

9.2.1.3 Recommendations

- a) In general TDL performs well on all types of designs. However, definitely use TDL for timing critical, and/or high-speed designs, in order to minimize synthesis-layout iterations and achieve timing convergence.
- b) When handling large designs, generate timing constraints only for selected nets. This will save you a considerable amount of time. However, if your layout tool is capable of generating its own timing constraints, then it should be given preference over the other approach, in order to save time.
- c) Perform hierarchical place and route for large designs. Although tedious, it will generally provide you with best results as well as better control of the overall flow. Hierarchical place and route also expedites hand editing of netlist that is sometimes required after routing is completed.
- d) Always use physical placement information in PDEF format while performing post-layout optimization within DC, especially for large hierarchical designs.

9.2.2 Clock Tree Insertion

As explained in previous chapters, it is essential to control the clock latency and skew. Although, some designs may actually take advantage of the positive skew to reduce power, most designs however, require minimal clock skew and clock latency. Larger values of clock skew cause race conditions that increase the chance of wrong data being clocked in the flops. Controlling the skew and latency requires a lot of effort and foresight.

As mentioned before, the layout tool performs the clock tree synthesis (CTS for short). The CTS is performed immediately after the placement of the cells, and before routing these cells. With input from the designer, the layout tool determines the best placement and style of the clock tree. Generally, designers are asked for the number of levels along with the types of buffers used for each level of the clock tree. Obviously, the number of levels is dependent on the fanout of the clock signal.

In a broad sense, the number of levels of the clock tree is inversely proportional to the drive strength of the gates used in the clock tree. In other words, you will need more levels, if low drive strength gates are used, while the number of levels is reduced if high drive strength gates are used.

To minimize the clock skew and clock latency, designers may find the following recommendations helpful. It must be noted that these recommendations are not hard and fast rules. Designers often resort to using a mixture of techniques to solve the clocking issues.

- a) Use a balanced clock tree structure with minimum number of levels possible. Try not to go overboard with the number of levels. The more the levels, the greater the clock latency.
- b) Use high drive strength buffers in large clock trees. This also helps in reducing the number of levels.
- c) In order to reduce clock skew between different clock domains, try balancing the number of levels and types of gates used in each clock tree. For instance, if one clock is driving 50 flops while the other clock is driving 500 flops, then use low drive strength gates in the clock tree of the first clock, and high drive strength gates for the other. The idea here is to speed-up the clock driving 500 flops, and slow down the clock that is driving 50 flops, in order to match the delay between the two clock trees.
- d) If your library contains balanced rise and fall buffers, you may prefer to use these instead. Remember, in general it is not always true that the balanced rise and fall buffers, are faster (less cell delay) than the normal buffers. Some libraries provide buffers that have lower cell delays for rise times of signals, as compared to the fall times. For designs utilizing the

positive edge trigger flops, these buffers may be an ideal choice. The idea is to study the library and choose the most appropriate gate available. Past experience also comes in handy.

- e) To reduce clock latency, you may try to use high drive inverters for two levels. This is because, logically a single buffer cell consists of two inverters connected together, and therefore has an cell delay of two inverters. Using two separate inverters (two levels) will achieve the same function, but will result in reduced overall cell delay – since you are not using another buffer (2 more inverters) for the second level. Use this approach, only for designs that do not contain gated clocks. The reason for this explained later (point h).
- f) Do not restrict yourself to using the same type and drive strength gate for CTS. Current layout tools allow you to mix and match.
- g) For a balanced clock tree (e.g., 3 levels), the first level is generally a single buffer driven by the Pad. In order to reduce clock skew, the first level buffer is placed near the center of the chip, so that it can connect to the next level of buffers, through equal interconnect wires. This creates a ring like structure with the first buffer in the center, with the second set of buffers (second level) surrounding it, and the last stage surrounding the second level. Thus, the distance between the first, second and the third level are kept at minimum. However, although a good arrangement, it does result in the first level buffer being placed farthest from the source (Pad). If a minimum size wire is used to route the clock network from the Pad source to the first buffer, it will result in a large RC delay that will affect the clock latency. Therefore, it is necessary to size-up (widen) this wire from the Pad source to the input of the buffer (first level), in order to reduce the resistance of the wire, thereby reducing the overall latency. Depending upon the size of your design and the number of levels, you may also need to perform this operation on other levels.
- h) In order to minimize the skew, the layout tool should have the ability to tap the clock signal, from any level of the clock tree. This is especially important for designs that contain gated clocks. If the same clock is used for other ungated flops, then it results in additional delay, hence the skew. If the clock tree ended at the gate, the additional delay will cause a large

skew between the gated-clock flop and the ungated-clock flop as shown in Figure 9-1 (a). Therefore it is necessary to tap the clock source from a level up for the gated-clock flop, while maintaining the full clock tree for the ungated clock flop, as illustrated in Figure 9-1 (b). However, if inverters are used in the clock tree (point e), then the above approach breaks down. In this case, do not use inverters as part of the clock tree.

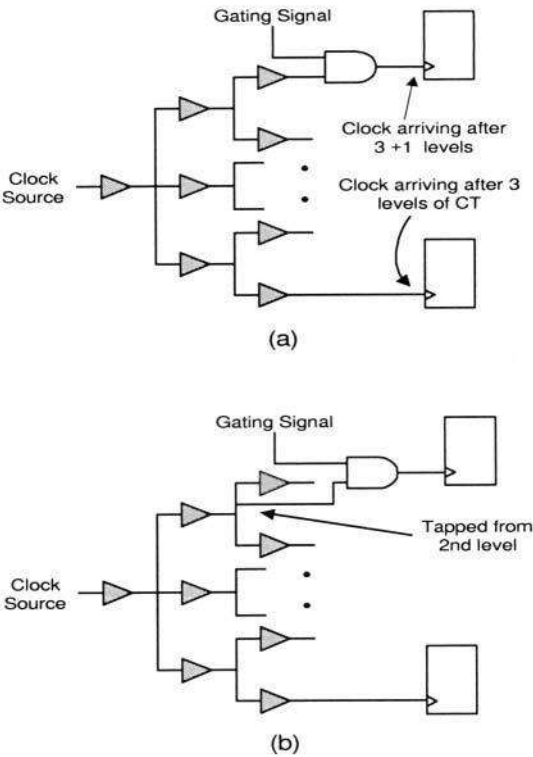


Figure 9-1. CTS for Gated Clocks versus Non-gated Clocks

9.2.3 Transfer of Clock Tree to Design Compiler

Clock tree synthesis done by the layout tool modifies the physical design (cells are added in the clock network). This modification is absent from the original netlist present in DC. Therefore, it is necessary for the user to accurately transfer this information to DC. There are several ways to do this.

- a) Generally all layout tools have the capability to write out the design in EDIF or Verilog format. Since, everything may appear flat to the layout tool, designers may receive a flat netlist from the layout tool. Of course, this netlist will contain the clock tree information, but the enormous size of the netlist itself may be daunting and unmanageable. Furthermore, due to the absence of original design hierarchy, the flat netlist is not easily readable. Another problem with this approach is that the user is now forced to designate this netlist as the “golden” netlist, meaning that all verification (LVS etc.) has to be performed against this netlist. Doing this is comparable to “digging your own grave” because, if the layout tool botches the layout, the same anomalies will be reflected in the netlist. Of course, the LVS will pass without flagging any errors, since the user is checking physical layout data against layout generated netlist i.e., performing LVL instead of LVS. An alternative is to perform formal verification between the original hierarchical netlist and the layout generated flat netlist. This certainly is a viable approach, but has its own limitations regarding the size and complexity of the design. Most formal verification tools suffer from this limitation, i.e., they excel in individual block verification, but fall short in full chip verification. This is especially true for verifying flat netlists against hierarchical netlists.
- b) The second approach is to only transfer, point-to-point clock delay information, starting from the clock source to its endpoints (clock pins of the flops). The delay calculator of the layout tool will perform this task, and upon instruction will provide the designer, the point-to-point timing information of the clock tree in SDF format. Designers may back annotate this SDF file to the original design in order to determine the clock latency and skew. This method does not require the clock tree to be transferred to DC from the layout tool. However, this approach has its own pitfalls. Primarily, this approach does not allow the usage of SPF data from the layout for back annotation to PT. Furthermore, the designer is now

compelled to trust the delay calculator of the layout tool i.e., another variable has been introduced that requires qualification. Since, the layout libraries are separate from Synopsys libraries, in order to get the same delay numbers, the timing numbers present in the Synopsys library need to match exactly to that of the layout library. The dilemma of verifying the original netlist against the layout database still exists, especially since the original netlist does not contain the extra cells and nets due to clock tree insertion. However, one can certainly find work-around and may use this approach successfully.

- c) A solution to all of the above problems is to creatively transfer the entire clock tree to DC without changing the hierarchy of the design. Some layout tools may even generate Synopsys scripts that contain `dc_shell-t` commands like, `disconnect_net`, `create_cell`, `create_port` and `connect_net`. These commands on execution insert the clock tree into the original design database in DC, while still maintaining the hierarchy. Of course, one needs to verify the resulting modified netlist against the original netlist by performing formal verification. Since the design hierarchy is not altered, the formal verification runs smoothly.
- d) Another solution involves brute force modification. Generally the layout tools, upon completion of CTS, produce a summary report of all changes made to the design. One may take advantage of this report and parse it to retrieve the relevant information (e.g., name of clock tree insertion points, type and name of buffers etc.) using scripting languages like Perl or Awk. Once the information is gathered, the original Verilog netlist may be directly modified without going through DC. The modified netlist should be read back into DC to check for any syntax errors. In addition, the modified netlist should also be formally verified against the original netlist.

Recently, upon realizing this problem the layout tool vendors have facilitated this process by generating the hierarchical netlist from the layout database. This netlist contains the clock tree information and should be verified formally against the original netlist. Upon successful verification, the netlist may be declared as “golden”.

9.2.4 Routing

After the clock tree insertion, the final step involves routing the chip. In a broad sense, the routing is divided into two phases:

1. Global Routing, and
2. Detailed Routing.

The first routing phase is called the global route, in which the global router assigns a general pathway through the layout for each net. During global route, the layout surface is divided into several regions. The global router decides the shortest route through each region in the layout surface, without laying the geometric wires.

The second routing phase is called the detailed route. The detailed router makes use of the information gathered by the global route and routes the geometric wires within each region of the layout surface.

It must be noted that if the run-time of global route is long (more than the placement run-time), it indicates a bad placement quality. In this case, the placement should be performed again with emphasis on reduced congestion.

9.2.5 Extraction

Until now, synthesis and optimization was performed utilizing the wire-load models. The wire-load models are based on statistically estimating the final routing capacitances. Because of the statistical nature of wire-load models, they may be completely inaccurate compared to the real delay values of the routed design. This variation between the wire-load models and the real delay values results in a non-optimized design.

The layout database is extracted to produce the delay values necessary to further optimize the design. These values are back annotated to PT for static timing analysis, and to DC for further optimization and refinement of the design.

9.2.5.1 What to Extract?

In general, almost all layout tools are capable of extracting the layout database using various algorithms. These algorithms define the granularity and the accuracy of the extracted values. Depending upon the chosen algorithm and the desired accuracy, the following types of information may be extracted:

1. Detailed parasitics in DSPF or SPEF format.
2. Reduced parasitics in RSPF or SPEF format.
3. Net and cell delays in SDF format.
4. Net delay in SDF format + lumped parasitic capacitances.

The DSPF (Detailed Standard Parasitic Format) contains RC information of each segment (multiple R's and C's) of the routed netlist. This is the most accurate form of extraction. However, due to long extraction times on a full design, this method is not practical. This type of extraction is usually limited to critical nets and clock trees of the design.

The RSPF (Reduced Standard Parasitic Format) represents RC delays in terms of a pi model (2 C's and 1 R). The accuracy of this model is less than that of DSPF, since it does not account for multiple R's and C's associated with each segment of the net. Again, the extraction time may be significant, thus limiting the usage of this type of information. Target applications are critical nets and small blocks of the design.

Both detailed and reduced parasitics can be represented by OVI's (Open Verilog International) Standard Parasitic Exchange Format (SPEF).

The last two (number 3 and 4) are the most common types of extraction used by the designers. Both utilize the SDF format. However, there is major difference between the two. Number 3 uses the SDF to represent both the cell and net delays, whereas number 4 uses the SDF to represent only the net delays. The lumped parasitic capacitances are generated separately. Some layout tools generate the lumped parasitic capacitances in the Synopsys `set_load` format, thus facilitating direct back annotation to DC or PT.

It is worth mentioning that PT can read all five formats (DSPF, RSPF, SPEF, SDF and `set_load`), whereas, DC can only read the SDF and `set_load` file formats. The SDF and `set_load` file formats are not as accurate as the DSPF or RSPF types of extraction, however, the time to extract the layout database is significantly reduced. For most designs this type of extraction provides sufficient accuracy and precision. However, as suggested, only critical nets and clocks in the design should be targeted for DSPF or RSPF types of extraction.

For the layout tool to generate a full SDF (number 3 approach), it uses its own delay calculator to compute the cell delays that are based upon the output loading and the transition time of the input signal. However, there is a flaw in using this approach. The synthesis was done using DC that used its own delay calculator to optimize the design. By choosing to use the full SDF generated by the layout tool, we are now introducing another variable that needs qualification. How do we know that the delay calculator used by the layout tool is more accurate than the one used by DC? Also, upon back annotation of the full SDF to PT, the full capability of PT is also not being utilized. This is because the cell delays are already fixed in the SDF file, and performing case analysis in PT will not yield accurate results, even if the conditional delays are present in the SDF file. This is discussed at length in Chapter 11.

Another problem exists with the above approach. Since only the cell and net delays are back annotated, DC does not know the parasitic capacitances associated with each net of the design. Therefore, when performing post-layout optimization, DC can only make use of the wire-load models to make incremental changes to the design, thus defeating the whole purpose of back annotation. However, if the fourth approach was used (net delays in SDF format + lumped parasitic capacitances), DC makes use of the net loading information during post-layout optimization (e.g., to size up/down gates).

To avoid these problems, it is recommended that only the net RC delays (also called as interconnect wiring delays) and lumped parasitic capacitances are extracted from the layout database. Upon back annotation, DC or PT uses its own delay calculator to compute the cell delays, based upon the back annotated interconnect RC's and capacitive net loading.

To summarize, it is recommended that the following types of information should be generated from the layout tool for back annotation to DC in order to perform post layout optimization:

- a) Net RC delays in SDF format.
- b) Capacitive net loading values in `set_load` format.

For static timing analysis, using PT, the following types of information can be generated:

- a) Net RC delays in SDF format.
- b) Capacitive net loading values in `set_load` format.
- c) Parasitic information for clock and other critical nets in DSPF, RSPF or SPEF file formats.

9.2.5.2 Estimated Parasitic Extraction

The extraction of parasitics at the pre-route level (after global routing) provides a closer approximation to the parasitic values of the final routed design. If the estimates indicate a timing problem, it is fairly easy to quickly re-floorplan the design before starting the detailed route. This method reduces synthesis-layout iterations and avoids wastage of valuable time.

The difference between the estimated extracted delay values after the global routing and the real delay values after the detailed routing is minimal. In contrast, the estimated delay values between the floorplan extraction and detailed route extraction may be significant. Therefore it is prudent that after floorplanning, cell placement and clock tree insertion, the design be globally routed, before extracting the estimated delay numbers.

A complete extraction flow is shown in Figure 9-2. If major timing violations exist after global route, it may be necessary to re-optimize the design within DC with estimated delays back annotated. However, if the timing violations are not severe then re-floorplanning (and/or re-placement of cells) the design may achieve the desired result. The detailed routing should be performed, only after the eliminating all timing violations produced after the global routing phase.

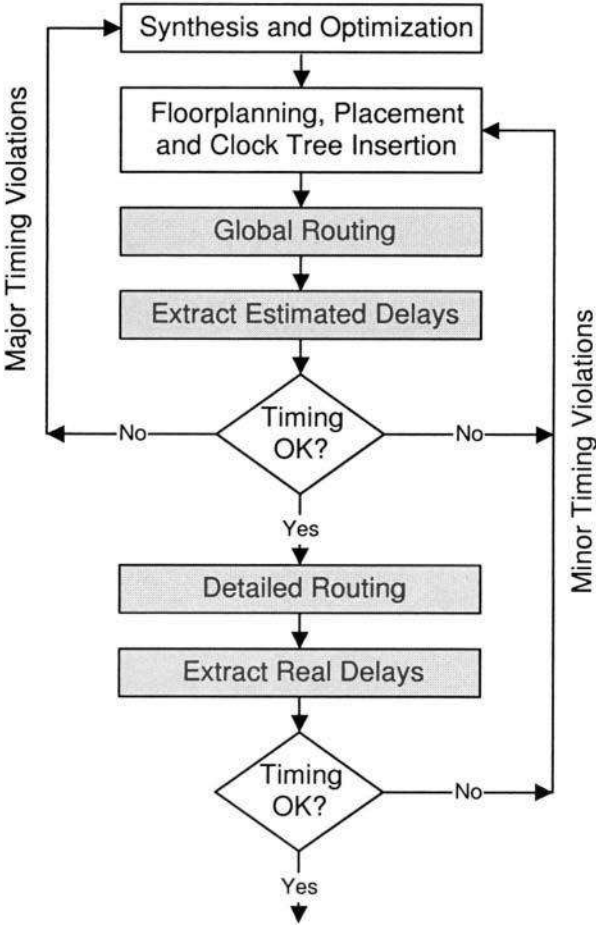


Figure 9-2. Routing and Extraction Flow

9.2.5.3 Real Parasitic Extraction

A full extraction (actual values – no estimation) is performed after the design has been satisfactorily globally routed, i.e., no DRC violations and achievement of required die-size.

This by far is the most critical part of the entire process. The final product may not work, if the extracted values are not accurate. With technologies shrinking to 0.18 micron and below, the extraction algorithm of the layout tool need to take into account the second and third order parasitic effects. Any slight deviations of these values may cause the design to fail.

Consider a case where the extracted values are too pessimistic. Static timing analysis indicates that the signals are meeting hold-time requirements. However, in reality, the signals are arriving faster, causing real hold-time violations, but due to pessimistic back annotated parasitic capacitances, the design is passing static timing. The case for setup-time is similar, if the extracted values are too optimistic.

9.3 Post-Layout Optimization

Post-layout optimization is performed to further optimize and refine the design. The process involves back annotating the data generated by the layout tool, to the design residing in DC. Depending upon the severity of violations, the optimizations may include full synthesis or minor adjustments through the use of in-place-optimization (IPO) technique. As explained in the previous section, the layout related data suitable for back annotation to DC are:

- a) Net RC delays in SDF format.
- b) The `set_load` file, containing capacitive net loading.
- c) Physical placement information in PDEF format.

9.3.1 Back Annotation and Custom Wire Loads

The next step involves analyzing the static timing of the design. Designers may choose to perform this step using PT or DC's internal static timing analysis engine. In any case, post layout optimization can only be performed within DC therefore the layout data needs to be back annotated to both DC and PT.

Depending on the process technology, the layout tool may generate two separate files that correspond to the worst and the best case. If there are two separate Synopsys libraries pertaining to each case, then back annotate the worst case layout data to the design using the worst case Synopsys library. Similarly, best case layout data should be back annotated to the design mapped to best case Synopsys library.

Some vendors provide only one Synopsys library that covers all cases, i.e., the library is characterized for TYPICAL case, with the WORST and the BEST case values derived (derated) from the TYPICAL case. In a situation like this, it is recommended that the designer back annotate the worst case numbers to the design with operating conditions set to WORST, in order to perform the worst case timing analysis. The best case timing analysis should be performed with best case timing numbers back annotated to the design with operating conditions set to BEST.

Use the following `dc_shell-t` commands to back annotate layout-generated information to the design present in DC, before performing post-layout optimization.

```
dc_shell -t> current_design    <design name>

dc_shell -t> source           <set_load file name>

dc_shell -t> read_sdf         <RC file name in SDF format>

dc_shell -t> read_clusters    <cluster file name in PDEF format>
```

Use the following `pt_shell` commands to back annotate layout-generated information to the design in PT, before performing static timing analysis.

```
pt_shell> current_design      <design name>

pt_shell> source              <set_load file name in PT format>

pt_shell> read_sdf            <RC file name in SDF format>

pt_shell> read_parasitics     <DSPF, RSPF or SPEF file name>
```

After back annotation in PT, if the design fails static timing with substantial amount of violations, the user may need to perform re-synthesis (or even re-code certain blocks). Therefore, it is prudent to use the existing layout information for re-synthesis. Discarding the layout data during re-synthesis only wastes the time and effort spent for layout. Furthermore, the layout data is helpful in fine tuning the design. To achieve the maximum benefit, custom wire-load models should be generated through DC, using the existing layout information. The resulting gate level netlist using the custom wire-load models provide a closer match to the post-layout timing results. Use the following `dc_shell -t` command to create custom wire-load models:

```
create_wire_load -design <design name>
                 -cluster <cluster name>
                 -trim <trim value>
                 -percentile <percentile value>
                 -output <output file name>
```

Although, there are other options available for the above command, generally the ones listed above suffice for most designs. The trim value is used to discard data that falls below a certain value, while the percentile value is used to calculate the average value. By altering the percentile value, one may add optimism or pessimism in the custom wire-load models. The cluster name is obviously the grouping name that was used during layout, to group cells or blocks together.

After the creation of the custom wire-load models (CWLM), the library should be updated to account for the new CWLMs. This is because the original technology library contains only the generic wire-load models that are not particular to a specific design. To use the CWLMs that were

generated by the above command, the library must be updated. The following command may be used to update the library present in DC memory:

```
update_lib <library name> <CWLM file name>
```

It must be noted that the above command does not alter or overwrite the source library. It only updates the DC memory to include the new CWLMs.

9.3.2 In-Place Optimization

For designs with minor timing violations after layout, there is no need to perform a full chip synthesis. In-place optimization or IPO is an excellent method to fine-tune a design, in order to eliminate these violations. The concept of IPO is to keep the structure of the design intact while modifying only the failing parts of the design, thereby having a minimal impact on the existing layout. IPO is commonly used to add/swap gates at specific locations to fix setup and/or hold-time problems.

The IPO is library dependent and can be limited to perform only the following:

- a) Resize cells.
- b) Insert or delete existing cells (mainly buffers).

Usually, all Synopsys technology libraries have an attribute defined that enables or disables the IPO. The attribute and its value, that enables IPO in a library is:

```
in_place_swap_mode : match_footprint
```

Along with the above library level attribute, all cells in the Synopsys library also have the `cell_footprint` information. For example, two cells with same functionality, but different drive strengths may have the same `cell_footprint` value. This means that the two cells have identical physical coverage area, therefore replacing one for the other will not impact the existing layout, i.e., adjacent cells will not shift. However, this restriction, along with cell sizing,

area optimization and buffer insertion, is controlled by the use of the following variables:

set compile_ignore_footprint_during_inplace_opt	true false
set compile_ok_to_buffer_during_inplace_opt	true false
set compile_ignore_area_during_inplace_opt	true false
set compile_disable_area_opt_during_inplace_opt	true false

Using these variables allows the designer the ability control the amount of changes made in the design. The appropriate values of the above variables may be set before performing IPO at `dc_shell -t` command line; or in the Synopsys setup file.

The IPO is invoked by using the following commands:

```
dc_shell -t> compile -in_place
```

```
dc_shell -t> reoptimize_design -in_place
```

Both commands are similar in nearly all respects, i.e., both make use of the back annotated layout information, with the exception of physical information. The `reoptimize_design` makes use of the physical location information while re-optimizing the design. Another difference between the two commands is that the “`compile -in_place`” command uses the library wire-load models during IPO, whereas the “`reoptimize_design -in_place`” command makes use of the custom wire-load models. Therefore, it is imperative, that the latter command be used while performing IPO.

It must be noted that the `reoptimize_design` when used on its own makes major modifications to the design. To eliminate this possibility, always use the `-in_place` option.

9.3.3 Location Based Optimization

Location Based Optimization or LBO is an integral part of IPO, and is invoked automatically while performing IPO for designs containing back annotated physical placement location information in PDEF format. In this

chapter the LBO is being identified separately from IPO due to its importance and additional capability.

Performing IPO with LBO improves the overall optimization of the design, since DC now has access to the cell placement information. This allows DC to apply more powerful algorithms during optimization.

Consider a path segment starting from primary input and ending at a flop. Post-layout timing analysis reveals a hold-time problem for this path. As shown in Figure 9-3, LBO optimization will add buffer(s) near the flop (endpoint), instead of adding it at the source (startpoint). Adding buffer(s) at the source may cause a setup-time failure for another path originating from the same source.

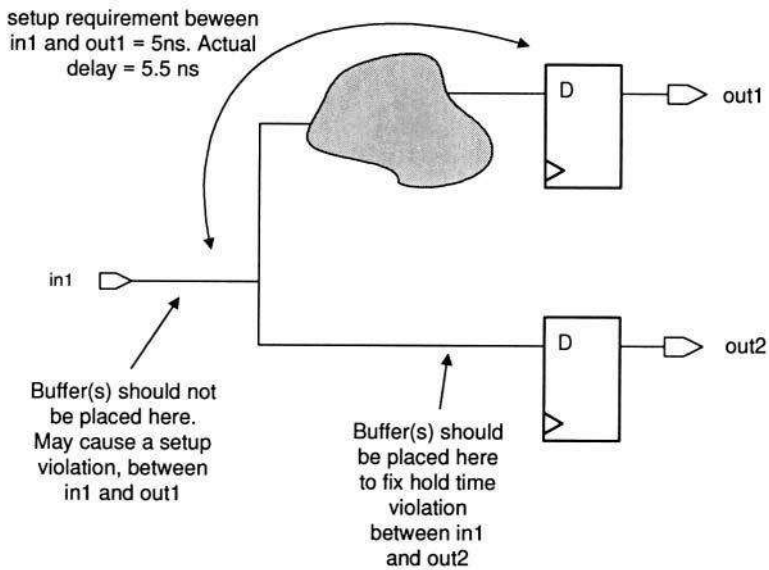


Figure 9-3. IPO versus LBO

In addition to the enhanced capability of inserting buffers at optimal locations, LBO also provides better modeling of cross cluster nets, and the

new nets that were created to connect the inserted or deleted buffers. This is due to the fact that, DC is aware of the location where the cells were inserted or deleted.

In order for the “reoptimize_design –in_place” to perform LBO, the following variables need to be set to true; in addition to the IPO related variable “compile_ok_to_buffer_during_inplace_opt”:

```
set lbo_buffer_removal_enabled true
```

```
set lbo_buffer_insertion_enabled true
```

LBO is not enabled for buffer insertion or removal, if the value of above variables is set to false (default). Location information is disregarded for this case, with only IPO algorithms used to handle the buffer insertion or deletion.

The changes made by performing IPO and/or LBO, using both “compile –in_place” or “reoptimize_design –in_place”, can be written out to a file by using the following `dc_shell -t` variable:

```
set reoptimize_design_changed_list_file_name <file name>
```

If the file already exists, the new set of changes will be appended to the same file.

9.3.4 Fixing Hold-Time Violations

Nearly every design undergoes the process of fixing hold-time violations, especially for faster technologies. Designers tackle this problem using various approaches. For this reason, a separate section is devoted to discuss issues arising from using these methods, and to consolidate them under one topic.

Most designers synthesize the design with tight constraints in order to maximize the setup-time. The resulting effect is a fast logic with data arriving faster at the input of the flop, with respect to the clock. This may

result in hold-time violations due to data changing value before being latched by the flop. Generally, designers prefer to fix the hold-time violations after initial placement and routing of the design, thereby making use of more accurate delay numbers.

Removing hold-time violations involves delaying the data with respect to the clock, so that the data does not change for a specified amount of time (hold-time) after the arrival of the clock edge. There are several methods utilized by designers to insert the appropriate delays, as outlined below:

- a) Using Synopsys methodology.
- b) Inserting delays manually.
- c) Inserting delays automatically, by using brute force `dc_shell -t` commands.

9.3.4.1 Synopsys Methodology

Synopsys provides the following `dc_shell -t` command, which enables the `compile` command to fix the hold-time violations:

```
set_fix_hold <clock name>
```

```
dc_shell -t> set_fix_hold CLK
```

The above command may be used during initial compile (or post-layout), by setting the min/max library concurrently (version DC98 onwards), and specifying the min/max values for `set_input_delay` command. The idea behind setting the min/max library at the same time is to eliminate the two-pass (initial synthesis for maximum setup-time and re-optimization to fix hold-time violations) synthesis needed for almost all designs. Example 9.2 illustrates the methodology of fixing the post-layout hold-time violations using the single pass synthesis approach.

Example 9.2

```
set_min_library "<worst case library name>" \
               -min_version "<best case library name>"

set_operating_conditions -min BEST -max WORST

source net_delay.set_load
read_timing interconnect.sdf
read_clusters floorplan.pdef

set_input_delay -max 20.0 -clock CLK [list IN1 IN2]
set_input_delay -min -1.0 -clock CLK [list IN1 IN2]

set_output_delay -max 10 -clock CLK [alt_outputs]

set_fix_hold CLK

reoptimize_design -in_place
```

Alternatively, the design may be compiled with maximum setup-time using the worst case library, followed by re-optimization after layout, in order to fix the hold-time violations by mapping the design to the best case library. Although, this method uses the two-pass synthesis approach, it is still recommended because of its stable nature. Most designers prefer to use this approach because of the time and effort that has been invested in defining and maturing this methodology.

It must be noted that the above command is independent of IPO commands. The IPO commands are generally used to fix hold-time violations after initial layout with layout information back annotated to the design. Fixing pre-layout hold-time violations is accomplished by compiling the design incrementally, using the “compile -incremental” command.

The `set_fix_hold` command instructs DC to fix the hold-time violations by inserting the buffers at appropriate locations. This again is controlled by the IPO related variables described previously. With buffer insertion disabled,

the cells in the data path can only be swapped (i.e., replacing a higher drive strength gate with a lower drive strength gate) to increase the cell delay, thereby delaying the data arriving at the flop input.

9.3.4.2 Manual Insertion of Delays

If the timing analysis reveals a very small number of hold-time violations (less than 10 to 20 places), it may not be worthwhile to fix these violations using the `set_fix_hold` command. The delays in this case may be manually inserted in the netlist. The designer may chain a string of buffers to delay the data, with respect to the clock just enough, so that it passes the hold-time checks.

A point to note however is that a chain of buffers may not be able to provide adequate delay, since the delay is dependent on the placement of the buffers in the layout. Generally, the buffers will be placed very close to each other therefore the total delay will be the sum of the delay through each cell. The interconnect delay itself will be insignificant, due to the close proximity of the placed cells. To overcome this, it is recommended to link a number of high-fanin gates (e.g., 8 input AND gate) with all inputs tied together, and connected to form a chain. The advantage of using this approach is that now the input pin capacitance of the high fanin gates is being utilized. The input pin capacitance of a high fanin gate is usually much larger than a single input buffer. Therefore, this method of delaying the data provides a solution that is independent of the cell placement location in the layout. This approach is suitable if the technology library does not contain delay cells. If these cells are present, then they should be targeted to fix the hold-time violations.

9.3.4.3 Brute Force Method

This is a unique method, but requires expertise in scripting languages like Perl or Awk.

For instance, if the timing report shows many hold-time violations, and fixing them through Synopsys methodology means large run times. In this case, an alternative approach, to find the amount of slack (setup-time

analysis) and the corresponding violation (hold-time analysis) of the failing paths, is to parse the timing report (both for worst-case and best case) using a scripting language. Using these numbers, the user may generate `dc_shell-t` commands like, `disconnect_net`, `create_cell` and `connect_net`, for the failing paths. Upon execution of these commands in `dc_shell-t`, it will force DC to insert and connect buffers at appropriate places (should be done near the endpoints of the failing paths). This is called a brute force method, done automatically.

This by no means is a clean approach, but works remarkably well. The time taken to fix hold-time violations using this approach is negligible as compared to the Synopsys methodology.

9.4 Chapter Summary

Links to layout is an important part of the integration between the layout tool and DC. This chapter focussed on all aspects of exchanging data to and from layout tools, in order for DC to perform better optimization and fine-tuning the design.

Issues related to transfer of clock tree information from the layout tool to DC were explained in detail. Cross checking the netlist generated by the layout tool against the original netlist remains a major bottleneck. Various alternatives were provided to the user in order to overcome this issue and choose the right solution.

Starting from how to generate a clean netlist from DC in order to minimize layout problems, this chapter covered placement and floorplanning, clock tree insertion, routing, extraction, and post-layout optimization techniques, including various methods to fix the hold-time violations. At each step, recommendations are provided to facilitate the user in choosing the right direction.

PHYSICAL SYNTHESIS

Time-to-market is rapidly shrinking while design complexities are increasing. The problem is further aggravated by shrinking geometries, forcing ASIC designers to think about power and cross-talk along with timing, much earlier in the design cycle. The exchange of data between layout tools and DC (at PT) is certainly not efficient. Time wasted during synthesis-layout iterations is still a major bottleneck.

The main cause of synthesis-layout iterations can be attributed to the traditional synthesis approach of relying on wire-load models to synthesize the design. The wire-load models are just estimates of the final routed design. They may differ considerably from the real extracted delays of the layout surface. Going back and forth from layout to synthesis solves this problem, however at the expense of time.

In order to alleviate this problem, Synopsys recently introduced a novel approach of synthesizing the design without the need for wire-load models. This new tool is called Physical Compiler (or PhyC) and it performs synthesis along with concurrent placement, based on the floorplan information. Combining synthesis and placement provides an accurate modeling of actual interconnect delays during synthesis. In addition this tool

also minimizes the previous headache of passing data back and forth from the layout tool to the synthesis tool.

PhyC is a superset of DC and incorporates all commands of DC along with some of its own. It is invoked by typing: `psyn_shell`

10.1 Initial Setup

PhyC uses the same setup file as DC — `.synopsys_dc.setup`. The only difference being that in addition to logical libraries it also requires inclusion of physical libraries in the same file. A complete description of the syntax and usage is provided in Chapter 3.

10.1.1 Important Variables

Just like DC, the behavior of PhyC is controlled by variables. These variables can be incorporated in the setup file. For a complete listing of PhyC variables type:

```
psyn_shell> printvar
```

Some of the most commonly used variables are described below:

- `physopt_pnet_complete_blockage_layer_names`

```
psyn_shell> set physopt_pnet_complete_blockage_layer_names \  
"metal1 metal2"
```

The above variable defines the power/ground layers that should be treated as blockages by PhyC. In general this is used for power and ground straps that are present in the floorplan. The idea is to tell PhyC not to place any cells underneath the straps. In the above case, both `metal1` and `metal2` layer names are used.

- `physopt_pnet_partial_blockage_layer_names`

```
psyn_shell> setphysopt_pnet_partial_blockage_layer_names\  
"metal1 metal2"
```

This variable allows PhyC limited flexibility in placing the cells under the power/ground straps. The cells will only slide under the straps, if their own layers do not collide with layers used by the power/ground straps. In the above case part of the cells that do not contain metal1 or metal2 layers are allowed to slide under the straps.

10.2 Modes of Operation

Physical synthesis can be performed using the following two modes:

1. RTL to Placed Gates (or RTL2PG)
2. Gates to Placed Gates (or G2PG)

PhyC requires the floorplan information in IEEE PDEF 3.0 format. This format is very similar to the SDF format. It contains the physical coordinates of cells, placement obstructions (such as pre-placed RAM/ROM's), power straps, chip boundary, pad/port locations etc. In other words this file contains all the necessary information required by PhyC to perform optimized placement.

10.2.1 RTL 2 Placed Gates

In this mode the input to PhyC is the RTL design, floorplan information in IEEE PDEF 3.0, I/O timing constraints and the physical libraries. The output is the structured netlist along with the placement data in PDEF 3.0 format. This is shown in Figure 10-1.

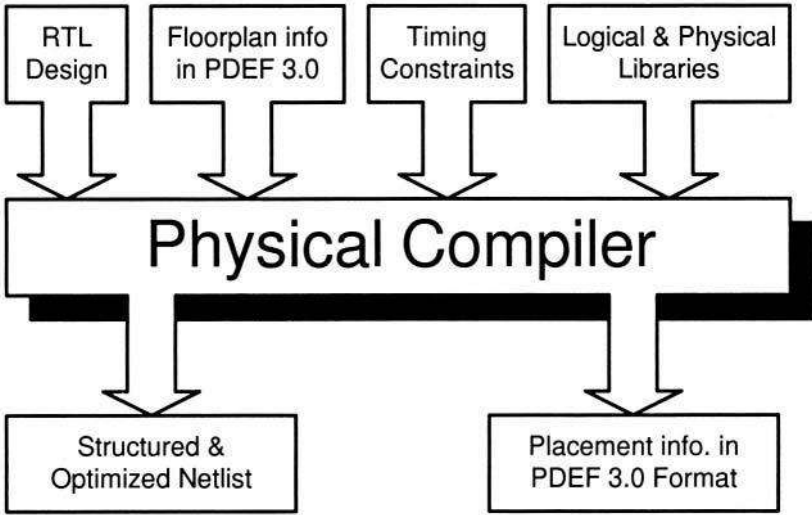


Figure 10-1. RTL to Placed Gates

Similar to DC, PhyC provides a command that compiles the RTL to produce the optimized netlist along with the placement information. This command is called `compile_physical` with similar options as the original `compile` command of DC. The most commonly used options are listed below:

```
compile_physical -congestion -scan
```

The `-congestion` option can be used to invoke further optimization algorithms in order to reduce routing congestion. The `-scan` option is identical to the `-scan` option of the `compile` command in DC. Here also it only maps the design to the scan flops but does not stitch them into a scan chain.

The following script shows the RTL2PG physical synthesis flow. PhyC commands are highlighted in bold.

An example RTL2PG script

```

# Read the source RTL and floorplan information
  read_verilog mydesign.v
  read_pdef floorplan.pdef

# Define operating conditions and timing constraints.
# Note the absence of wire-load models.
# All other steps same as before.
  current_design mydesign
  uniquify
  link
  set_operating_conditions WORST
  set_load 1.0[all_outputs]
  create_clock .....
  set_clock_latency.....
  set_clock_transition.....
  set_dont_touch_network.....
  set_input_delay.....
  set_output_delay.....

# Define attributes for scan
  set_scan_configuration.....
  create_test_clock .....
  set_test_hold 1 .....
  set_scan_signal test_scan_enable .....
  set_scan_signal test_scan_in .....
  set_scan_signal test_scan_out .....

# Synthesize the design using the physical information.
# Also synthesize to scan flops. No stitching done.
  compile_physical -scan
  check_test
  preview_scan

# Stitch scan chains based on physical location of flops.
  insert_scan -physical
  check_test

```



```
# Write out structured netlist along with placement information.  
write -f verilog -h -o mydesign_placed.sv  
write_pdef -v3.0 -o mydesign_placed.pdef  
exit
```

10.2.2 Gates to Placed Gates

In this mode the input to PhyC is the structured netlist instead of the RTL. The rest of the input and output files remain identical to the RTL2PG mode of operation.

In this mode the input to PhyC is a structural netlist that has previously been synthesized using the traditional approach (using the DC compile command utilizing the wire-load models). Only the placement of these gates is desired. Pictorially this is shown in Figure 10-2.

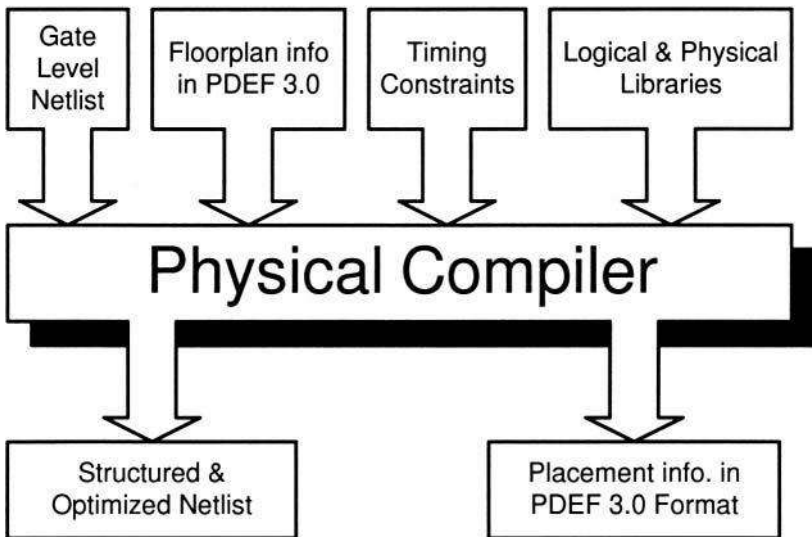


Figure 10-2. Gates to Placed Gates

PhyC provides the following command to perform the G2PG operation:

```
physopt -congestion -timing_driven_congestion -scan_order
```

The `-congestion` option is the same as that described for the RTL2PG method. If the design constraints are such that the only the congestion is paramount (area centric designs with easy to meet timing requirements), this option should be used. The `-timing_driven_congestion` should be used for those designs where both timing and congestion are important. This option performs timing driven placement of cells with congestion in mind.

The `-scan_order` option is used for ordering the scan chain based on the physical location of flops. This also helps tremendously in reducing the congestion.

Within the G2PG mode, there are two sub-modes of operation. These sub-modes are called the "two pass method" and the "integrated method". These approaches relate to the way scan chain stitching is handled by `physopt`.

10.2.2.1 Two pass method

In this mode of operation, the scan chain is hooked up using the `insert_scan` command before `physopt` is run. The `physopt` command is subsequently used not only to perform placement, but also to order the scan chain based on the physical location of each flop.

The following script illustrates the two-pass G2PG flow. PhyC and scan chain linking commands are highlighted in bold.

An example of two-pass G2PG script

```
# Read the synthesized gate level netlist in "db" format.
# Assuming "compile-scan" was used to produce the "db" file.
# In other words, no scan stitching done, only the design has
# been synthesized directly to scan flops.
read_db mydesign.db

# Read the floorplan information
read_pdef floorplan.pdef
```

```

# Define operating conditions and timing constraints.
# Note the absence of wire-load models. The constraints are
# needed by PhyC to perform timing driven placement.
    current_design mydesign
    uniquify
    link
    set_operating_conditions WORST
    set_load 1.0 [all_outputs]
    create_clock .....
    set_clock_latency .....
    set_clock_transition .....
    set_dont_touch_network .....
    set_input_delay .....
    set_output_delay .....

# Define attributes for scan
    set_scan_configuration .....
    create_test_clock .....
    set_test_hold 1 .....
    set_scan_signal test_scan_enable .....
    set_scan_signal test_scan_in .....
    set_scan_signal test_scan_out .....

# Stitch scan chains. No ordering performed yet
    insert_scan
    check_test

# Perform timing driven placement along with reduced
# congestion. Also order the scan chain based on physical
# location of each flop.
    physopt -timing_driven_congestion -scan_order
    check_test

# Write out structured netlist along with placement information.
    write -f verilog -h -o mydesign_placed.sv
    write_pdef -v3.0 -o mydesign_placed.pdef
    exit

```

Note: The assumption in the above script is that "compile -scan" along with other scan attributes was used to produce the starting "mydesign.db" file. Therefore all scan related attributes must already be part of the "db" file. Thus these attributes can be omitted from the above script. They have been provided just for sake of explanation.

10.2.2.2 Integrated method

In this mode of operation, **physopt** command is used not only to perform placement but also to stitch and order the scan chain based on physical proximity of the scan flops. Basically, it unifies the stitching and ordering functions. In other words, the use of **insert_scan** command has been eliminated thus simplifying the flow.

The following script illustrates the integrated G2PG flow. **PhyC** commands are highlighted in bold.

An example of integrated G2PG script

```
# Read the synthesized gate level netlist in "db" format.
# Assuming "compile -scan" was used to produce the "db" file.
# In other words, no scan stitching done, only the design has
# been synthesized directly to scan flops.
  read_db mydesign.db

# Read the floorplan information
  read_pdef floorplan.pdef

# Define operating conditions and timing constraints.
# Note the absence of wire-load models. The constraints are
# needed by PhyC to perform timing driven placement.
  current_design mydesign
  uniquify
  link
  set_operating_conditions WORST
  set_load 1.0 [all_outputs]
```

```

create_clock .....
set_clock_latency.....
set_clock_transition .....
set_dont_touch_network .....
set_input_delay .....
set_output_delay .....

# Define attributes for scan
set_scan_configuration .....
create_test_clock .....
set_test_hold 1 .....
set_scan_signal test_scan_enable .....
set_scan_signal test_scan_in .....
set_scan_signal test_scan_out .....

# Perform timing driven placement along with reduced
# congestion. Also stitch and order the scan chain
# based on physical location of each flop.
physopt -timing_driven_congestion -scan_order
check_test

# Write out structured netlist along with placement information.
write -f verilog -h -o mydesign_placed.sv
write_pdef -v3.0 -o mydesign._placed.pdef
exit

```

Note: The assumption in the above script is that "compile -scan" along with other scan attributes was used to produce the starting "mydesign.db" file. Therefore all scan related attributes must already be part of the "db" file. These attributes can therefore be omitted from the above script. They have been provided just for sake of explanation.

10.3 Other PhyC Commands

Unlike DC, PhyC does not contain many of its own commands. It must be noted that most of these commands are run “under the hood” by **physopt** or

“`compile_physical`”. Users do not need to explicitly run these commands. They are intended for further massaging the layout surface, if needed. Therefore, the commands here are provided to the user with sole intention of “what else is there”. No description is provided. Users are advised to read the Physical Compiler Users Manual for full description and usage of these commands.

Following are some of these commands:

- `create_placement`
- `legalize_placement`
- `check_legality`
- `run_router`
- `set_congestion_options`
- `report_congestion`
- `set_dont_touch_placement`
- `remove_dont_touch_placement`

10.4 Physical Compiler Issues.

Unfortunately as with most EDA tools, PhyC also suffers from several issues. These issues all relate to PhyC version 2001-SP1. It is expected that later versions may have solved some of these problems. Some of the critical ones are listed below:

1. When reading a gate-level netlist using `read_verilog`, PhyC outputs a lot of assign statements in the final verilog netlist. This only happens when performing scan chain ordering, and occurs even after using the following hidden variable:

```
set physopt_fix_multiple_port_nets true
```

When reading a precompiled `db` file and performing the same operations (including using the above variable), no verilog assign statements are generated. If a gate-level netlist is compiled into the `db` format (`read_verilog` followed by `write -f db`), PhyC still produces the verilog

assign statements. The only way to prevent this is to use the following variable in conjunction with the above variable:

```
set_fix_multiple_port_nets -all -buffer_constants
```

This is obviously a nuisance and hopefully will be corrected by Synopsys soon.

2. The heavily promoted “integrated physopt flow” by Synopsys does not work “as advertised”. The idea is to compile the design using one-pass scan synthesis and then run **physopt** with scan order option. **Physopt** is supposed to perform scan stitching, ordering as well as placement. However, **PhyC** aborts and complains that the design is not scan ready even when **check_test** passes. This problem is solved by using the following attribute just before **physopt** is run:

```
set_attribute <design name> is_test_ready true -type boolean
```

For some reason, it is necessary to explicitly inform **PhyC** that the design is scan ready. This problem is well documented in the Solv-Net database and will be corrected in near future.

3. Sometimes for under utilized designs, **PhyC** produces a clumping effect for the placed cells. In other words, if the design is pad limited and logic area is very small compared to the overall chip area, the cell placement is not optimal. They are clumped together in several clusters. The clumping causes a localized routing congestion problem. To following command may be used before running **physopt** in order to spread out the congestion:

```
set_congestion_options -max_util <number>
```

Unfortunately, there is no magic number that works well in all cases. This is a hit and trial method. Users are advised to read the man pages of this command and make their own informed decision based on their design.

4. Placement of multi-row height cells is not supported in the current version. This is an important feature that permits designers to not only place standard cells, but also macros (like RAM, ROM's, PLL's etc.) automatically. The traditional flow is to pre-place these macros before running the **physopt** placement. Synopsys have announced that this capability will be added to PhyC in the near future.

10.5 Back-End Flow

Synopsys recently announced the availability of two new add-on options to the PhyC. These are the Clock Tree Compiler and the Route Compiler. With these options enabled, PhyC becomes the only EDA tool to provide a complete solution starting from RTL synthesis to the final GDSII. Based on a common timing engine across the entire flow and providing additional capabilities such as signal integrity and cross talk analysis, this tool becomes extremely powerful.

At the time of writing this book these new capabilities were not available. Therefore, the rest of the flow based on this technology is not provided. Those users that do not have the clock tree and the route compiler may use their own layout tool and proceed from the clock tree insertion phase (described in Chapter 9).

10.6 Chapter Summary

This chapter described the usage and operation of the Physical Compiler. With the introduction of this capability, Synopsys has solved the long-standing problem of discrepancy between the delays estimated by the wire-load models and the final resulting routed design.

Different flows and techniques were described along with helpful scripts to guide the user in performing successful synthesis, placement and scan chain ordering.

Few problems associated with PhyC were also discussed. Although in time these problem will most certainly be corrected. Still it is the intent of this

chapter to make readers aware of these issues in case they are using this version of PhyC.

Several new add-ons to Physical Compiler have recently been announced (the Clock Tree Compiler and the Route Compiler) that will enhance the capability of this tool enormously. These add-ons have been mentioned in this Chapter, however the usage and operation have not been described due to the unavailability of these options at the time of writing this book.

SDF GENERATION

For Dynamic Timing Simulation

The standard delay format or SDF contains timing information of all the cells in the design. It is used to provide timing information for simulating the gate-level netlist.

As mentioned in Chapter 1, verification of the gate-level netlist of a design through dynamic simulation is not a recommended approach. Dynamic simulation method is used to verify the functionality of the design at the RTL level only. Verification of a gate-level design using dynamic simulation depends solely on the coverage provided by the test-bench, therefore, certain paths in the design that are not sensitized will not get tested. In contrast, formal verification techniques provide superior validation of the design.

The dynamic simulation method for gate-level design verification is still dominant, and is used extensively by designers. Due to this reason, this chapter provides a brief description on generating the SDF file from DC and PT, which can be used to perform dynamic timing simulation of the design. Furthermore, a few innovative ideas and suggestions are provided to facilitate designers in performing successful simulation.

Please note that some of the information provided in this chapter is also described in previous chapters. Since, this is an important topic, it is deemed necessary that a full chapter relating to SDF generation be devoted for the sake of clarity and completeness.

11.1 SDF File

The SDF file contains timing information of each cell in the design. The basic timing data comprises of the following:

- a) IOPATH delay.
- b) INTERCONNECT delay.
- c) SETUP timing check.
- d) HOLD timing check.

Following, is an example SDF file that contains the timing information for two cells (sequential cell feeding an AND gate), along with the interconnect delay between them:

```
(DELAYFILE
(SDFVERSION "OVI 2.1")
(DESIGN "top_level")
(DATE "Dec 30 1997")
(VENDOR "std_cell_lib")
(PROGRAM "Synopsys Design Compiler cmos")
(VERSION "1998.08")
(DIVIDER /)
(VOLTAGE 2.70:2.70:2.70)
(PROCESS "WORST")
(TEMPERATURE 100.00:100.00:100.00)
(TIMESCALE 1ns)
(CELL
  (CELLTYPE "top_level")
  (INSTANCE)
  (DELAY
    (ABSOLUTE
```

```

        (INTERCONNECT sub1/U1/Q sub1/U2/A1 (0.02:0.03:0.04)
                                         (0.03:0.04:0.05))
    )
)
(CELL
  (CELLTYPE "dff1")
  (INSTANCE sub1/U1)
  (DELAY
    (ABSOLUTE
      (IOPATH CLK Q (0.1:0.2:0.3) (0.1:0.2:0.3))
    )
  )
  (TIMINGCHECK
    (SETUP (posedge D) (posedge CLK) (0.5:0.5:0.5))
    (SETUP (negedge D) (posedge CLK) (0.6:0.6:0.6))
    (HOLD (posedge D) (posedge CLK) (0.001:0.001:0.001))
    (HOLD (negedge D) (posedge CLK) (0.001:0.001:0.001))
  )
)
(CELL
  (CELLTYPE "and2")
  (INSTANCE sub1/U2)
  (DELAY
    (ABSOLUTE
      (IOPATH A1 Z (0.16:0.24:0.34) (0.12:0.23:0.32))
      (IOPATH A2 Z (0.11:0.21:0.32) (0.17:0.22:0.34))
    )
  )
)
)
)

```

The IOPATH delay specifies the cell delay. Its computation is based upon the output wire loading and the transition time of the input signal.

The INTERCONNECT delay is a path based, point-to-point delay, which accounts for the RC delay between the driving gate and the driven gate. This

wire delay is specified from the output pin of the driving cell to the input pin of the driven cell.

The SETUP and HOLD timing checks contain values that determine the required setup and hold-time of each sequential cell. These numbers are based upon the characterized values in the technology library.

11.2 SDF File Generation

The SDF file may be generated for pre-layout or post-layout simulations. The post-layout SDF is generated from DC or PT, after back annotating the extracted RC delay values and parasitic capacitances, to DC or PT. The post-layout values thus represent the actual delays associated with the design. The following commands may be used to generate the SDF file:

DC Command

```
write_timing -format sdf-v2.1 -output <filename>
```

PT Command

```
write_sdf-version [1.0 or 2.1] <filename>
```

Note: By default, PT generates the 2.1 version of SDF.

11.2.1 Generating Pre-Layout SDF File

The pre-layout numbers contain delay values that are based upon the wire-load models. Also, the pre-layout netlist does not contain the clock tree. Therefore, it is necessary to approximate the post-route clock tree delays while generating the pre-layout SDF.

In order to generate the pre-layout SDF, the following commands approximate the post-route clock tree values by defining the clock delay, skew, and the transition time.

DC & PT Commands

```
create_clock -period 30 -waveform [list 0 15] [list CLK]
```

```
set_clock_latency 2.0 [get_clocks CLK]
```

```
set_clock_transition 0.2 [get_clocks CLK]
```

By setting (fixing) these values as illustrated above, designers may assume that the resulting SDF file will also contain these values, i.e., the clock delay from the source to the endpoint (clock input port of the flops) is fixed at 2.0. However, this is not the case. DC only uses the above commands to perform static timing analysis, and does not output this information to the SDF file. To avoid this problem, designer should force DC to use the specified delay value instead of calculating its own. To ensure the inclusion of 2.0ns as the clock delay, a `dc_shell` command (explained later in the section) is used to “massage” the resulting SDF.

At the pre-layout level, the clock transition time should also be specified. Failure to fix the transition time of the clock results in false values computed for the driven flops. Again, the culprit is the missing clock tree at the pre-layout level. The absence of clock tree forces the assumption of high fanout for source clock, which in turn causes DC to compute slow transition times for the entire clock network. The slow transition times will affect the driven flops (endpoint flops), resulting in large delay values computed for them.

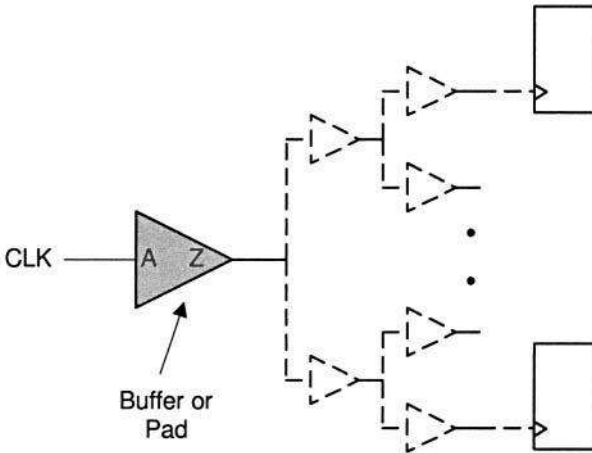


Figure 11-1. Specifying Clock Tree Delay

Consider the diagram shown in Figure 11-1. The dotted lines illustrate the placement of clock tree buffers, synthesized during layout. At pre-layout level, these buffers do not yet exist. However, there usually is a buffer/cell (shown as shaded cell) at the clock source. This cell may be a big driver, instantiated by the designer with the sole purpose of driving the future clock tree, or it may simply be an input pad. Let us assume that this is an input pad (called CLKPAD) with input pin A and output pin Z. At the pre-layout level, the output pin Z connects directly to all the endpoints.

The easiest way to fix the SDF, so that it reflects the 2.0 ns clock delay from the source “CLK” to all the endpoints, is to replace the delay value of the shaded cell (from pin A to pin Z) calculated by DC, with 2.0 ns. This can be achieved by using the following `dc_shell` command:

```
dc_shell -t> set_annotated_delay 2.0 -cell \
               -from CLKPAD/A -to CLKPAD/Z
```

☺ Note: A similar command also exists for PT.

The above command replaces the value calculated by DC, with the one specified i.e., 2.0 ns. This delay gets reflected in the SDF file in the form of IOPATH delay, for the cell CLKPAD, from pin A to pin Z.

Fixing the delay value of the clock solves the problem of clock latency. However, what happens to the delay values of the driven flops? Designers may incorrectly assume that DC uses the specified clock transition for the sole purpose of performing static timing analysis, and may not use the specified values to calculate delays of the driven flops. This is not so. DC uses the fixed transition value of the clock to calculate delays of driven gates. Not only are the transition values used to perform static timing analysis, but they are also used while computing the delays of the driven cells. Thus the SDF file contains the delay values that are approximated by the designer at the pre-layout phase.

11.2.2 Generating Post-Layout SDF File

The post-layout design contains the clock tree information. Therefore, all the steps that were needed to fix the clock latency, skew, and clock transition time, during pre-layout phase, are not required for post-layout SDF file generation. Instead, the clock is propagated through the clock network to provide the real delays and transition times.

As explained in Chapter 9, only the extracted parasitic capacitances and RC delays should be back annotated to DC or PT, for final SDF generation.

The following commands may be used to back annotate the extracted data to the design and specify the clock information while generating the post-layout SDF file for simulation:

DC & PT Commands

`read_sdf <interconnect RC's in SDF format>`

`source <parasitic capacitances in set_load format>`

`read_parasitics <DSPF, RSPF or SPEF file for clocks + other critical nets>`


```
create_clock -period 30 -waveform [list 0 15] [list CLK]
```

```
set_propagated_clock [get_clocks CLK]
```

11.2.3 Issues Related to Timing Checks

Sometimes, during simulation, unknowns (X's) are generated that cause the simulation to fail. These unknowns are generated due to the violation of setup/hold timing checks. Most of the time, these violations are real, however there are instances where a designer may want to ignore some violations related to parts of the design, but still verify others. This is generally unachievable, due to the simulator's inability to turnoff the X-generation on a selective basis.

Nearly all simulators provide capabilities to ignore the timing violations, generally for the whole design. They do not have the ability to ignore the timing violation for an isolated instance of a cell in the design. Due to this reason, designers are often forced to either modify the simulation library or live with the failed result.

Modifying the simulation library is also not a viable approach, since turning off the X-generation can only be performed on a cell. This cell may be instanced multiple times in the design. Turning off the X-generation for this cell will prevent the simulator from generating X's, for all the instances of the cell in the design. This is definitely not desired as it may mask the real timing problems lying elsewhere in the design.

For example, a design may contain multiple clock domains and the data traverses from one clock domain to the other through synchronization logic. Although, this logic will work perfectly on a manufactured device, it may cause hold-time violations when simulated. This will cause the simulation to fail for the design.

Another example is related to the type of methodology used for synthesis. Some designers prefer to fix the hold-time violations only after layout. Failing to falsify or remove the hold-time values from the pre-layout SDF

file may cause the simulator to generate an X (unknown) for the violating flop. This X may propagate to the rest of the logic causing the whole simulation to fail.

To prevent these problems, one may need to falsify selectively, the value of the setup and hold-time constructs in the SDF file, for simulation to succeed. The SDF file is instance based (rather than cell based), therefore selective targeting of the timing checks is easily attained. Instead of manually removing the setup and hold-time constructs from the SDF file, a better way is to zero out the setup and hold-times in the SDF file, only for the violating flops, i.e., replace the existing setup and hold-time numbers with zero's. Back-annotating the zero value for the setup and hold-time to the simulator prevents it from generating unknowns (if both setup and hold-time is zero, there cannot be any violation), thus making the simulation run smoothly. The following `dc_shell` command may be used to perform this:

```
dc_shell-t> set_annotated_check 0 --setup --hold      \
                    --from REG1/CLK \
                    --to REG1/D
```

☺ Note: A similar command also exists for PT.

11.2.4 False Delay Calculation Problem

This topic is covered in Chapter 4, but is included here for the sake of completeness.

The delay calculation of a cell is based upon the input transition time and the output load capacitance of a cell. The input transition time of a cell is evaluated, based upon the transition delay of the driving cell (previous cell). If the driving cell contains more than one timing arc, then the worst transition time is used, as input to the driven cell. This causes a major problem when generating the SDF file for simulation purposes.

Consider the logic shown in Figure 11-2. The signals, *reset* and *signal_a* are inputs to the instance U1. Let us presume that the *reset* signal is not critical, while the *signal_a* is the one that we are really interested in. The *reset* signal

is a slow signal therefore the transition time of this signal is more compared to *signal_a*, which has a faster transition time. This causes, two transition delays to be computed for cell U1 (2 ns from A to Z, and 0.3 ns from B to Z). When generating SDF, the two values will be written out separately as part of the cell delay, for the cell U1. However, the question now arises, which of the two values does DC use to compute the input transition time for cell U2? DC uses the worst (maximum) transition value of the preceding gate (U1) as the input transition time for the driven gate (U2). Since the transition time of *reset* signal is more compared to *signal_a*, the 2ns value will be used as input transition time for U2. This causes a large delay value to be computed for cell U2 (shaded cell).

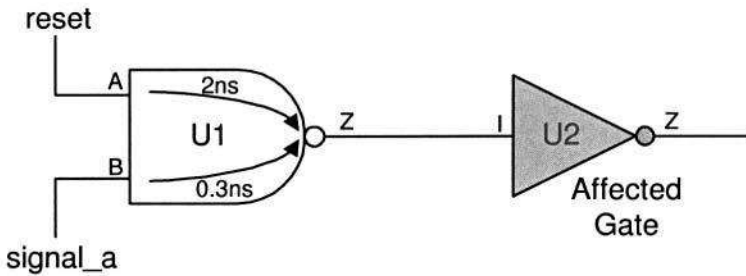


Figure 11-2. False Delay Calculation

To avoid this problem, one needs to instruct DC, not to perform the delay calculation for the timing arc, from pin A to pin Z of cell U1. This step should be performed before writing out the SDF. The following `dc_shell` command may be used to perform this:

```
dc_shell-t> set_disable_timing U1 -from A -to Z
```

Unfortunately, this problem also exists during static timing analysis. Failure to disable the timing computation of the false path leads to large delay values computed for the driven cell.

11.2.5 Putting it Together

The following DC scripts combines all the information provided above and may be used to generate the pre and post-layout SDF, to be used for timing simulation of an example *tap controller* design.

DC script for pre-layout SDF generation

```

set active_design tap_controller

read_db $active_design.db

current_design $active_design
link

set_wire_load_model LARGE
set_wire_load_mode top
set_operating_conditions WORST

create_clock-period 33 -waveform [list 0 16.5] tck
set_clock_latency 2.0 [get_clocks tck]
set_clock_transition 0.2 [get_clocks tck]

set_driving_cell -cell BUFF1 -pin Z [all_inputs]
set_drive 0 [list tck trst]
set_load 50 [all_outputs]

set_input_delay 20.0 -clock tck -max [all_inputs]
set_output_delay 10.0 -clock tck -max [all_outputs]

# Approximate the clock tree delay
set_annotated_delay 2.0 -cell -from CLKPAD/A \
                    -to CLKPAD/Z

# Assuming, only REG1 flop is violating hold-time
set_annotated_check 0 -setup -hold \
                    -from REG1/CLK -to REG1/D

```

```
write_timing -format sdf-v2.1 \
             -output $active_design.sdf
```

DC script for post-layout SDF generation

```
set active_design tap_controller

read_db $active_design.db

current_design $active_design
link

set_operating_conditions BEST

source capacitance.dc # actual parasitic capacitances
read_timing rc_delays.sdf # actual RC delays

create_clock -period 33 -waveform [list 0 16.5] tck
set_propagated_clock [get_clocks tck]

set_driving_cell -cell BUFF1 -pin Z [all_inputs]
set_drive 0 [list tck trst]

set_load 50 [all_outputs]

set_input_delay 20.0-clock tck-max [all_inputs]
set_output_delay 10.0-clock tck-max [all_outputs]

# Assuming, only REG1 flop is violating hold-time
set_annotated_check 0 -setup -hold \
                    -from REG1/CLK -to REG1/D

write_timing -format sdf-v2.1 \
             -output $active_design.sdf
```

11.3 Chapter Summary

The SDF file is used exhaustively throughout the ASIC world to perform dynamic timing simulations. The chapter briefly summarizes the contents of the SDF file that is related to ensuing discussions.

The chapter also discusses procedures for generating the SDF file from DC and PT, both for pre-layout and post-layout simulations. Along with command description, various helpful techniques are described to “massage” the SDF, in order for the simulation to succeed. These include fixing the clock latency and clock transition at the pre-layout level, and avoiding unknown propagation from selective logic of the design for successful simulation.

The final section gathered all the information and put it together in the form of DC scripts for pre and post-layout SDF generation.

PRIMETIME BASICS

PrimeTime (PT) is a sign-off quality static timing analysis tool from Synopsys. Static timing analysis or STA is without a doubt the most important step in the design flow. It determines whether the design works at the required speed. PT analyzes the timing delays in the design and flags violation that must be corrected.

PT, similar to DC, provides a GUI interface along with the command-line interface. The GUI interface contains various windows that help analyze the design graphically. Although the GUI interface is a good starting point, most users quickly migrate to using the command-line interface. Therefore, the intent of this chapter is to focus solely on the command-line interface of PT.

This chapter introduces to the reader, the basics of PT including a brief section devoted to Tcl language that is used by PT. Also described in this chapter are selected PT commands that are used to perform successful STA, and also facilitate the designer in debugging the design for possible timing violations.

12.1 Introduction

PT is a stand-alone tool that is not integrated under the DC suite of tools. It is a separate tool, which works alongside DC. Both PT and DC have consistent commands, generate similar reports, and support common file formats. In addition PT can also generate timing assertions that DC can use for synthesis and optimization. PT's command-line interface is based on the industry-standard language called Tcl. In contrast to DC's internal STA engine, PT is faster, takes up less memory, and has additional features.

12.1.1 Invoking PT

PT may be invoked in the command-line mode using the command `pt_shell` or in the GUI mode through the command `pruntime`.

Command-line mode:

```
> pt_shell
```

GUI-mode:

```
> pruntime
```

12.1.2 PrimeTime Environment

Upon invocation PT looks for a file called “.synopsys_pt.setup” and includes it by default. It first searches for this file in the current directory and failing that, looks for it in the users home directory before using the default setup file present in the PT installation site. This file contains the necessary setup variables defining the design environment that are used by PT, exemplified below:

```
set search_path [list. /usr/golden/library/std_cells]
```

```
set link_path [list {*} ex25_worst.db, ex25_best.db]
```


The variable `search_path` defines a list containing directories to look into while searching for libraries and designs. It saves the tedium of typing in complete file paths when referring to libraries and designs.

The variable `link_path` defines a list of libraries containing cells to be used for linking the design. These libraries are searched in the directories specified in the `search_path`. In the above example there are three elements in the list defined by `link_path` variable. The ‘*’ indicates designs loaded in the memory, while the other two are names pertaining to the best and the worst case, standard cell technology libraries.

Another commonly used method of setting up the environment, if we do not want to use the “.synopsys_pt.setup” file is to use the `source` command. The `source` command works just like DC’s `include` command. It includes and runs the file as if it were a script within the current environment. This command is invoked within `pt_shell`. For example:

```
pt_shell> source ex25.env
```

12.1.3 Automatic Command Conversion

Most of the DC commands are similar to PT commands, the exception being that PT being Tcl based uses the Tcl language format. This promotes the need for DC commands to be converted to the Tcl format before PT can utilize them.

PT offers a conversion script that may be used to convert almost all `dc_shell` commands to `pt_shell`, Tcl based format. The script is called `transcript` and is provided by Synopsys as a separate stand-alone utility. This script is executed from a UNIX shell as follows:

```
> transcript <dc_shell script filename> <pt_shell script filename>
```

12.2 Tcl Basics

Tcl provides most of the basic programming constructs – variables, operators, expressions, control flow, loops and procedures etc. In addition, Tcl also supports most of the UNIX commands.

Tcl programs are a list of commands. Commands may be nested within other commands through a process called command substitution.

Variables are defined and values assigned to them using the *set* command. For example:

```
set clock_name clk
```

```
set clock_period 20
```

Values can be numbers or strings – Tcl does not distinguish between variables holding numeric values versus variables holding strings. It automatically uses the numeric value in an arithmetic context. In the above examples a variable *clock_name* has been set to the string *clk* and a variable *clock_period* has been set to the numeric value 20. Variables are used by preceding the variable name with a \$. If the \$ is not added before the variable name, Tcl treats it as a string.

```
create_clock $clock_name –period 20 –waveform [0 10]
```

Arithmetic operations are performed through the **expr** command. This useful technique provides means for global parameterization across the entire script.

```
expr $clock_period / 5
```

The above command returns a value 4. Tcl provides all the standard arithmetic operators like *, /, +, –, etc.

12.2.1 Command Substitution

Commands return a value that can be used in other commands. This nesting of commands is accomplished with the square brackets [and]. When a command is enclosed in square brackets, it is evaluated first and its value substituted in place of the command. For example:

```
set clock_period 20

set inp_del [expr $clock_period / 5]
```

The above command first evaluates the expression in the square brackets and then replaces the command with the evaluated value – in this case *inp_del* inherits the value of 4. Commands may also be nested to any depth. For example:

```
set clock_period 20

set inp_del [expr [expr $clock_period / 5] + 1]
```

The above example has 2 levels of nesting. The inner command returns a value of 4 obtained by dividing the value of *clock_period* by 5. The outer **expr** command adds 1 to the result of the inner command. Thus *inp_del* gets set to a value of 5.

12.2.2 Lists

Lists represent a collection of objects – these objects can be either strings or lists. In the most basic form enclosing a set of items in braces creates a list.

```
set clk_list {clk1 clk2 clk3}
```

In the above example, the **set** command creates a list called *clk_list* with 3 elements, *clk1*, *clk2* and *clk3*.

Another method of creating lists is through the `list` command that is typically used in command substitution. For example, the following `list` command creates a list just like the previous `set` command:

```
list clk1 clk2 clk3
```

The `list` command is suitable for use in command substitution because it returns a value that is a list. The `set` command example may also be written as:

```
set clk_list [list clk1 clk2 clk3]
```

Tcl provides a group of commands to manipulate lists – `concat`, `join`, `lappend`, `lindex`, `linsert`, `list`, `llength`, `lrange`, `lreplace`, `lsearch`, `lsort` and `split`. For example, `concat` concatenates two lists together and returns a new list.

```
set new_list [ concat [list clk1 clk2] [list clk3 clk4] ]
```

In the above example the variable `new_list` is the result of concatenation of the list `clk1` & `clk2` and the list `clk3` & `clk4`, each of which is formed by the `list` command.

Conceptually a simple list is a string containing elements that are separated by white space. In most cases the following two are equivalent:

```
{clk1 clk2 clk3}
```

```
"clk1 clk2 clk3"
```

In some cases the second representation is preferred because it allows variable substitution while the first representation does not. For example:

```
set stdlibpath [list "/usr/lib/stdlib25" "usr/lib/padlib25"]
```

```
set link_path "/project/bigdeal/lib $stdlibpath"
```

For more details about the syntax of other list commands refer to any standard book on Tcl language.

12.2.3 Flow Control and Loops

Like other scripting and programming languages Tcl provides *if* and *switch* commands for flow control. It also provides *for* and *while* loops for looping. The *if* command may be used, along with *else* or *elsif* statements to completely specify the process flow. The arguments to *if*, *elsif* and *else* statements are usually lists, enclosed in braces to prevent any substitution. For example:

```
if {$port == "clk"} {  
    create_clock -period 10 -waveform [list 0 5] $port  
} elsif {$port == "clkdiv2"} {  
    create_generated_clock -divide_by 2 -source clk $port  
} else {  
    echo "$port is not a clock port"  
}
```

12.3 PrimeTime Commands

PT uses similar commands as DC, to perform timing analysis and related functions. Since all relevant `dc_shell` commands are explained in detail in Chapter 6, comprehensive explanation is not provided in this section for all related commands.

12.3.1 Design Entry

Unlike DC, which can read RTL source files through HDL Compiler, PT being the static analysis engine can only read mapped designs. This performs the basis of design entry to PT. Among others, input to PT can be a file in db, Verilog, VHDL or EDIF format. The following `pt_shell` commands appropriate to each format are used to read the design in PT:

```
read_db -netlist_only <design name>.db    #db format
```

<code>read_verilog <design name>.sv</code>	<code>#verilog format</code>
<code>read_vhdl <design name>.svhd</code>	<code>#vhdl format</code>
<code>read_edif <design name>.edf</code>	<code>#EDIF format</code>

Since the netlist in db format can also contain constraints and/or environmental attributes (maybe saved by the designer), the `-netlist_only` option may be used for the `read_db` command to instruct PT to load only the structural netlist. This prevents PT from reading the constraints and/or other attributes associated with the design. Only the structural netlist is loaded.

12.3.2 Clock Specification

The concepts behind clock specification remains the same as the ones described for DC in Chapter 6. Subtle syntax differences exist due to difference in formats between the two. However, because clock specification may become complex, especially if there are internally generated clocks with clock division, this section will cover the complete PT clock specification techniques and syntax.

12.3.2.1 Creating Clocks

Primary clocks are defined as follows:

```
create_clock -period <value>
             -waveform {<rising edge> <falling edge>}
             <source list>
```

```
pt_shell> create_clock -period 20 -waveform {0 10} \
             [list CLK]
```

The above example creates a single clock named CLK having a period of 20ns, with rising and falling edges at 0ns and 10ns respectively.

12.3.2.2 Clock Latency and Clock Transition

The following commands are used to specify the clock latency and the clock transition. These commands are mainly used for pre-layout STA and are explained in detail in Chapter 13.

```
set_clock_latency <value> <clock list>
```

```
set_clock_transition <value> <clock list>
```

```
pt_shell> set_clock_latency 2.5 [get_clocks CLK]
```

```
pt_shell> set_clock_transition 0.2 [get_clocks CLK]
```

The above commands define the clock latency for the CLK port as 2.5ns with a fixed clock transition value of 0.2ns.

12.3.2.3 Propagating the Clock

Propagating the clock is usually done after the layout tool inserts the clock tree in the design, and the netlist is brought back to PT for STA. The clock is propagated through the entire clock tree network in the netlist in order to determine the clock latency. In other words, the delay across each cell in the clock tree and the interconnect wiring delay between the cells is taken into account.

The following command instructs PT to propagate the clock through the clock network:

```
set_propagated_clock <clock list>
```

```
pt_shell> set_propagated_clock [get_clocks CLK]
```

12.3.2.4 Specifying Clock Skew

Clock skew, or clock uncertainty as Synopsys prefers to call it, is the difference in the arrival times of the clock, at the clock pin of the flops. In synchronous designs data gets launched by the flop at one clock edge and is received by another flop at another clock edge (usually the next clock edge). If the two clock edges (launch and receive) are derived from the same clock

then ideally there should be an exact delay of one clock period between the two edges. Clock skew puts a crimp in this happy situation. Because of variation in routing delays (or gated clock situation) the receiving clock edge may arrive early or late. Early arrival could cause setup-time violations and late arrival may cause hold-time violations. Therefore, it is imperative to specify the clock skew during the pre-layout phase, in order to produce robust designs.

Clock skew is specified through the following command:

```
set_clock_uncertainty <uncertainty value>
    -from <from clock>
    -to <to clock>
    -setup
    -hold
    <object list>
```

In the following example, 0.6ns is applied to both the setup and hold-time of the clock signal, CLK.

```
pt_shell> set_clock_uncertainty 0.6 [get_clocks CLK]
```

The option `-setup` may be used to apply uncertainty value to setup-time checks and while `-hold` option applies the uncertainty value for hold-time checks. It must be noted that different values for setup and hold cannot be implemented within a single command. Two separate commands must be used for this purpose. For example:

```
pt_shell> set_clock_uncertainty 0.5 -hold [get_clocks CLK]
```

```
pt_shell> set_clock_uncertainty 1.5 -setup [get_clocks CLK]
```

Also inter-clock skew can be specified with the `-from` and `-to` options, which is useful for designs containing multiple clock domains. For example:

```
pt_shell> set_clock_uncertainty 0.5 -from [get_clocks CLK1] \
    -to [get_clocks CLK2]
```


12.3.2.5 Specifying Generated Clocks

This is an important feature that is absent from DC. Very often a design may contain internally generated clocks. PT allows the user to define the relationship between the generated clock and the source clock, through the command `create_generated_clock`. This is convenient because pre-layout scripts can be used for post-layout with minimal changes.

During post-layout timing analysis, clock tree is inserted and the clock latency is calculated by propagating the clock signal through the clock tree buffers. Users may opt to define the divided clock independent to the source clock (by defining the clock on an output pin of the dividing logic sub-block). However, this approach forces designers to manually add the clock tree delay (from the dividing block to the rest of the design) to the clock latency of the source clock to the dividing logic block.

By setting up a divided clock through the above command, the two clocks are kept in sync both in pre-layout and post-layout phases.

```
create_generated_clock -name <divided clock name>
                      -source <primary clock name>
                      -divide_by <value>
                      <pin name>
```

```
pt_shell> create_generated_clock -name DIV2CLK \
                                -source CLK -divide_by 2 \
                                blockA/DFF1X/Q
```

The above example creates a generated clock on pin Q of the cell DFF1X belonging to blockA. The name of the generated clock is DIV2CLK, having half the frequency of the source clock, CLK.

12.3.2.6 Clock Gating Checks

For low power applications, designers often resort to gating the clock in the design. This technique allows designers to enable the clock only when needed. The gating logic may produce clipped clock or glitches, if the setup and hold-time requirements are not met (for the gating logic). PT allows

designers to specify the setup/hold requirements for the gating logic, as follows:

```
set_clock_gating_check -setup <value>
                        -hold <value>
                        <object list>
```

```
pt_shell> set_clock_gating_check -setup 0.5 -hold 0.01 CLK
```

The above example informs PT that the setup-time and hold-time requirement for all the gates in the clock network of CLK is 0.5ns and 0.01ns respectively.

Gating checks on an isolated cell can be accomplished by specifying the cell name in the object list. For example:

```
pt_shell> set_clock_gating_check -setup 0.05 -hold 0.01 \
        [get_lib_cell stdcell_lib/BUFF4X]
```

By default, PT performs the gating check with zero value used for setup and hold times – unless the library contains specific values for setup and hold times for the cell used to gate the clock. If the gating cell contains the setup/hold timing checks, then the gating check values may be automatically derived from the SDF file.

The clock gating checks are only performed for combinational cells. Also, the gating checks cannot be performed between two clocks.

12.3.3 Timing Analysis Commands

This section describes a selected set of PT commands that are used to perform STA. Only the most commonly used options are listed for these commands.

- **set_disable_timing:** Applications of this command include disabling timing arc of a cell in order to break the combinational feedback loop, or to

instruct PT to exclude a particular timing arc (thus the path segment) from analysis.

```
set_disable_timing -from <pin name>
                  -to  <pin name>
                  <cell name>
```

```
pt_shell> set_disable_timing -from A1 -to ZN {INVD2}
```

- **report_disable_timing**: command is used to display the timing arcs that were disabled by the user; or by PT. The report identifies individual disabled paths, using the following flags:

Flags:

- u : Timing path disabled by the user.
- l : Timing loop broken by PT.
- c : Timing path disabled during case analysis.

- **set_input_transition**: is an alternative to the **set_driving_cell** command. It sets a fixed transition time is not dependent on the net loading. This command is specified on input/inout ports of the design.

```
set_input_transition <value> <port list>
```

```
pt_shell> set_input_transition 0.2 [all_inputs]
```

```
pt_shell> set_input_transition 0.4 [list in 1 in2]
```

- **set_timing_derate**: is used to derate the delay numbers shown in the timing report. PT provides this powerful capability that is useful in adding extra timing margin to the entire design. The amount of deration is controlled by a fixed value, which is specified by the user. The original delay numbers are multiplied by this value, before the timing report is generated.

```
set_timing_derate -min <value> -max <value>
```

```
pt_shell> set_timing_derate-min 0.2-max 1.2
```

- **Set_case_analysis:** command performs case analysis and is one of the most useful feature provided by PT. This command is used to set a fixed logic value to a port (or pin) while performing STA.

`set_case_analysis [ 0 1 ] <port or pin list>`

`pt_shell> set_case_analysis 0 scan_mode`

Application of this command includes disabling timing paths that are not valid during a particular mode of operation. For instance, in the above example, the *scan_mode* port switches the design between the functional mode (normal operation) and the test mode of operation. The zero value set on the *scan_mode* port is propagated to all the cells driven by this port. This results in disablement of certain timing arcs of all cells that are related to the *scan_mode* port. Since testability logic is usually non timing-critical, disabling the timing arcs of the non timing-critical paths causes the real timing-critical paths to be identified and analyzed. The usage of this command is further explained in Chapter 13.

- **remove_case_analysis:** command is used to remove the case analysis values set by the above command.

`remove_case_analysis <port or pin list>`

`pt_shell> remove_case_analysis scan_mode`

- **report_case_analysis:** command is used to display the case analysis values set by the user. PT displays a report that identifies the pin/port list along with the corresponding case analysis values.

`pt_shell> report_case_analysis`

- **report_timing:** Similar to DC, this command is used to generate the timing report of path segments in a design. This command is used extensively and provides ample flexibility that is helpful in focussing explicitly on an individual path, or on a collection of paths in a design.

`report_timing -from <from list> -to <to list>`

```

    -through <through list>
    -delay_type <delay type>
    -nets -capacitance -transition_time
    -max_paths <value> -nworst <value>

```

The **-from** and **-to** options facilitate the user in defining a path for analysis. Since there may be multiple paths leading from a startpoint to a single endpoint, the **-through** option may be used to further isolate the required path segment for timing analysis.

```

pt_shell> report_timing -from [all_inputs] \
                    -to [all_registers-data_pins]

pt_shell> report_timing -from in1 \
                    -to blockA/subB/carry_reg1/D \
                    -through blockA/mux1/A1

```

The **-delay_type** option is used to specify the type of delay to be reported at an endpoint. Accepted values are **max**, **min**, **min_max**, **max_rise**, **max_fall**, **min_rise**, and **min_fall**. By default PT uses the **max** type, which reports that the maximum delay between two points. The **min** type option is used to display the minimum delay between two points. The **max** type is used for analyzing the design for setup-time while the **min** type is used to perform hold-time analysis. The other types are not frequently used and users are advised to refer to the PT User Guide for full explanation regarding their usage.

```

pt_shell> report_timing -from [all_registers-clock_pins] \
                    -to [all_registers-data_pins] \
                    -delay_type min

```

The **-nets**, **-capacitance** and **-transition_time** options are one of the most useful and frequently used options of the **report_timing** command. These options help the designer to debug a particular path, in order to track the cause of a possible violation. The **-nets** option displays the fanout of each cell in the path report, while the **-capacitance** and the **-transition_time** options reports the lumped capacitance on the net and the transition time (slew rate) for each driver or load pin, respectively. Failure

to include these options results in a timing report that does not include the information mentioned above.

```
pt_shell> report_timing -from in1 \
                    -to blockA/subB/carry_reg1/D \
                    -nets -capacitance -transition_time
```

The `-nworst` option specifies the number of paths to be reported for each endpoint, while the `-max_paths` option defines the number of paths to be reported per path group for different endpoints. The default value of both these options is 1.

```
pt_shell> report_timing -from [all_inputs] \
                    -to [all_registers -data_pins] \
                    -nworst 1000 -max_paths 500
```

- **report_constraint:** Similar to DC, this command in PT checks for the DRC's as defined by the designer or the technology library. Additionally, this command is also useful for determining the “overall health” of the design with regards to the setup and hold-time violations. The syntax of this command along with the most commonly used options is:

```
report_constraint -all_violators -max_delay
                -max_transition -min_transition
                -max_capacitance -min_capacitance
                -max_fanout -min_fanout
                -max_delay -min_delay
                -clock_gating_setup -clock_gating_hold
```

The `-all_violators` option displays all constraint violators. Generally, this option is used to determine at a glance, the overall condition of the design. The report summarizes all the violators starting from the greatest, to the least violator for a particular constraint.

```
pt_shell> report_constraint -all_violators
```

Selective reports may be obtained by using the `-max_transition`, `-min_transition`, `-max_capacitance`, `-min_capacitance`, `-max_fanout`, `-min_fanout`, `-max_delay`, `-min_delay`, `-clock_gating_setup`, `-clock_gating_hold`, `-nworst`, and `-max_paths`.

`max_fanout`, `-min_fanout`, `-max_delay`, and `-min_delay` options. The `-max_delay` and `-min_delay` options report a summary of all setup and hold-time violations, while others report the DRC violations. The `-clock_gating_setup` and the `-clock_gating_hold` commands are used to display the setup/hold-time reports for the cell used for gating the clock. In addition, there are other options available for this command that may be useful to the designer. Full details of these options may be found in the PT User Guide.

```
pt_shell> report_constraint -max_transition

pt_shell> report_constraint -min_capacitance

pt_shell> report_constraint -max_fanout

pt_shell> report_constraint -max_delay -min_delay

pt_shell> report_constraint -clock_gating_setup \
                        -clock_gating_hold
```

- ☺ Initially use the `report_constraint` command to ascertain the amount, and the number of violations. The report produced provides a general estimate of the overall health of the design. Depending upon the severity of violations, a possible re-synthesis of the design may need to be performed. To further isolate the cause of the violation, the `report_timing` command should be used to target the violating path, in order to display a full timing report.

- **report_bottleneck:** This command is used to identify the leaf cells in the design that contribute to multiple violations. For instance, several violating path segments of a design may share a common leaf cell. Altering the size of this leaf cell (sizing up or down) may improve the timing (thus remove violation) of all the violating path segments. The syntax of this command along with the most commonly used options is:

```
report_bottleneck -from <from list> -to <to list>
                  -through <through list> -max_cells <value>
                  -max_paths <value> -nworst_paths <value>
```

The `–from` and `–to` options facilitate the user in defining a path for bottleneck analysis. Since there may be multiple paths leading from a startpoint to a single endpoint, the `–through` option may be used to further isolate the required path segment for bottleneck analysis.

```
pt_shell> report_bottleneck    –from in1                \
                               –to blockA/subB/carry_reg 1/D \
                               –through blockA/mux1/A1
```

As the name suggests, the `–max_cells` option specifies the number of leaf cells to be reported. The default value is 20.

The `–nworst_paths` option specifies the number of paths to be reported for each endpoint, while the `–max_paths` option defines the number of paths to be reported per path group for different endpoints. The default value of both these options is 100.

```
pt_shell> report_bottleneck    –from in1                \
                               –to blockA/subB/carry_reg 1/D \
                               –through blockA/mux1/A1      \
                               –max_cells 50                \
                               –nworst_paths 500 –max_paths 200
```

12.3.4 Other Miscellaneous Commands

- **write_sdf:** command generates the SDF file that contains delays and timing checks for each instance in the design. PT uses the wire-load models to estimate the delays of cells during the pre-layout phase. For post-layout, PT uses the actual annotated delays (from the physical layout) while generating the SDF file. The syntax of this command along with the most commonly used options is:

```
write_sdf –version 1.0 | 2.1
          –no_net_delays
          –no_timing_checks
          <sdf output filename>
```


Unless explicitly specified, by default PT generates the SDF file in SDF version 2.1 format.

The `–no_net_delays` option specifies that the interconnect delays (INTERCONNECT field in the SDF file) are not to be written out separately in the SDF file. In this case, they are included as part of the IOPATH delay of each cell. This option is mainly used during the pre-layout phase because of the fact that the interconnect delays are based upon the wire-load models. However, the interconnect delays after layout are real and are based on the routed design. Therefore, in general this option should be avoided while generating the post-layout SDF file.

```
pt_shell> write_sdf –no_net_delays top_prelayout.sdf
```

```
pt_shell> write_sdf top_postlayout.sdf
```

Specification of the `–no_timing_checks` option forces PT to omit the timing-checks section (TIMINGCHECK field) from the SDF file. As described in Chapter 11, the timing-checks section contains the setup/hold/width timing checks. This option is useful for generating the SDF file that may be used to validate, only the functionality of the design through dynamic simulation, without bothering to check for setup/hold/width timing violations. Once the design passes functional validation, full SDF (no `–no_timing_checks` option) may be generated.

```
pt_shell> write_sdf –no_timing_checks top_prelayout.sdf
```

- **write_sdf_constraints:** This command is similar to the `write_constraints` command in DC and performs the same function. It is used to generate the path timing constraints in SDF format, which is used by the layout tool to perform timing driven layout. The syntax of this command along with the most commonly used options is:

```
write_sdf_constraints –version <1.0 | 2.1 >  
                    –from <from list> –to <to list>  
                    –through <through list>  
                    –cover_design
```

```

-slack_lesser_than <value>
-max_paths <value> -nworst <value>
<constraint filename>

```

Unless explicitly specified, by default PT generates the constraint file in SDF version 2.1 format. The `–from`, `–to` and `–through` options facilitate the user in specifying a particular path to be written to the constraint file.

The `–nworst` option specifies the number of paths to be written to the constraint file for each endpoint, while the `–max_paths` option defines the number of paths to be considered for each constraint group. The default value of both these options is 1. The default settings of these options usually suffice for most designs.

```

pt_shell> write_sdf_constraints –from in1 \
        –to blockA/subB/carry_reg1/D \
        –through blockA/mux1/A1 \
        tdl.sdf

```

The `–cover_design` option is used to generate just enough unique path timing constraints to cover the worst path for each path segment in the design. When specified, all other options such as, `–nworst`, `–to`, `–from` and `–through` are ignored. Although this option is recommended by Synopsys, it should be used judiciously as it may produce long run-times, especially for large designs.

```

pt_shell> write_sdf_constraints –cover_design tdl.sdf

```

An alternative is to use the `–slack_lesser_than` option that specifies that any path that has a slack value greater than the one specified is to be ignored. This means that a negative slack value for a path segment is considered to be most critical and has the highest priority. Thus all critical paths may be universally selected by specifying a low value for this option, hence will be written out to the constraint file. All high slack values (less critical paths) will be ignored.

```

pt_shell> write_sdf_constraints –slack_lesser_than 1.5 tdl.sdf

```

- **swap_cell**: This command may be used to replace an existing cell in the design with another, having the same pinout.

```
swap_cell <cell list to be replaced> <new design>
```

For example, if a path is failing due to hold-time violation and in order to fix the timing violation, you want to see the effect on the reported slack, by sizing down a particular leaf cell in the path, without changing the netlist. In this case the **swap_cell** command may be used at the command line to replace the existing cell with another, containing the same pinout.

```
pt_shell> swap_cell {U1} [get_lib_cell stdcell_lib/AND2X2]
```

In the above example, the instance U1 (say a 2-input AND gate with 8X drive strength) in a design is replaced by the AND2X2 gate (2X drive strength) from the “stdcel_lib” technology library.

12.4 Chapter Summary

Static timing analysis is one of the most critical steps for the entire ASIC chip synthesis flow. This chapter provides an introduction to PrimeTime that included PrimeTime invocation and its environment settings.

PrimeTime is a stand-alone static timing analysis tool, which is based on the universally adopted EDA tool language, Tcl. A brief section is included on the Tcl language in context of PrimeTime, to facilitate the designer in writing PrimeTime scripts and building upon them to produce complex scripts.

The last section covers all relevant PrimeTime commands that may be used to perform static timing analysis, design debugging and writing delay information in SDF format. In addition, this section also covers topics on design entry and clock specification, both for pre-layout and post-layout.

STATIC TIMING ANALYSIS

Using PrimeTime

The key to working silicon usually lies in successful completion of static timing analysis performed on a particular design. PT is a stand-alone tool by Synopsys that is used to perform static timing analysis. It not only checks the design for required constraints that are governed by the design specifications, but also performs comprehensive analysis of the design. This capability makes STA one of the most important steps in the entire design flow and is used by many designers as a sign-off criterion to the ASIC vendor.

This chapter illustrates the part of the design flow where PT is utilized. It covers both the pre-layout and the post-layout phases of the ASIC design flow process.

STA is closely integrated with the overall synthesis flow, therefore parts of this chapter may contain some repetition from elsewhere in this book.

13.1 Why Static Timing Analysis?

Traditional methods of analyzing gate-level designs using dynamic simulation are posing a bottleneck for large complex designs. Today, the

trend is towards incorporating system-on-a-chip (SoC), which may result in millions of gates per ASIC. Verifying such a design through dynamic simulation poses a nightmare to designers and may prove to be impossible due to long run-times (usually days and sometimes weeks). Furthermore, dynamic simulation relies on the quality and coverage of the test-bench used for verification. Only parts of the logic that are sensitized are tested while the remaining parts of the design remain untested. To alleviate this problem, designers now resort to other means of verification such as STA to verify the timing; and formal verification technique to verify the functionality of the gate-level netlist against the source RTL. However, comprehensive sets of test-benches are still needed to verify the functionality of the source RTL. Thus, dynamic simulation is needed to solely verify the functionality of the design at the RTL level. This results in considerable reduction in run-time.

The STA approach is infinitely fast compared to dynamic simulation and verifies all parts of the gate-level design for timing. Due to the similar nature of the synthesis and the STA engine, the static timing analysis is well suited for verifying synthesized designs.

13.1.1 What to Analyze?

In general, four types of analysis is performed on the design, as follows:

- a) From primary inputs to all flops in the design.
- b) From flop to flop.
- c) From flop to primary output of the design.
- d) From primary inputs to primary outputs of the design.

All four types of analysis can be accomplished by using the following commands:

```
pt_shell> report_timing -from [all_inputs] \
               -to [all_registers -data_pins]

pt_shell> report_timing -from [all_registers -clock_pins] \
               -to [all_registers -data_pins]
```

```
pt_shell> report_timing -from [all_registers -clock_pins] \
               -to [all_outputs]

pt_shell> report_timing -from [all_inputs] \
               -to [all_outputs]
```

Although, using the above commands is a cleaner method of generating reports for piping it to individual files for analysis, however, PT takes longer time to perform each operation. PT takes less time to generate the same results, if the following commands are used:

```
pt_shell> report_timing -to [all_registers -data_pins]
pt_shell> report_timing -to [all_outputs]
```

13.2 Timing Exceptions

In most designs there may be paths that exhibit timing exceptions. For instance, some parts of the logic may have been designed to function as multicycle paths, while others may simply be false paths. Therefore, before analyzing the design, PT must be made aware of the special behavior exhibited by these paths. PT may report timing violation for multicycle paths if they are not specified as such. Also, path segments in the design that are not factual, must be identified and specified as false paths, in order to prevent PT from producing the timing reports for these paths.

13.2.1 Multicycle Paths

PT by default, treats all paths in the design as single-cycle and performs the STA accordingly, i.e., data is launched from the driving flop using the first edge of the clock, and is captured by the receiving flop using the second edge of the clock. This means that the data must be received by the receiving flop within one clock cycle (single clock period). In the multicycle mode, the data may take more than one clock cycle to reach its destination. The amount of time taken by the data to reach its destination is governed by the multiplier value used in the following command:

```
set_multicycle_path <multiplier value>
                  -from <from list> -to <to list>
```

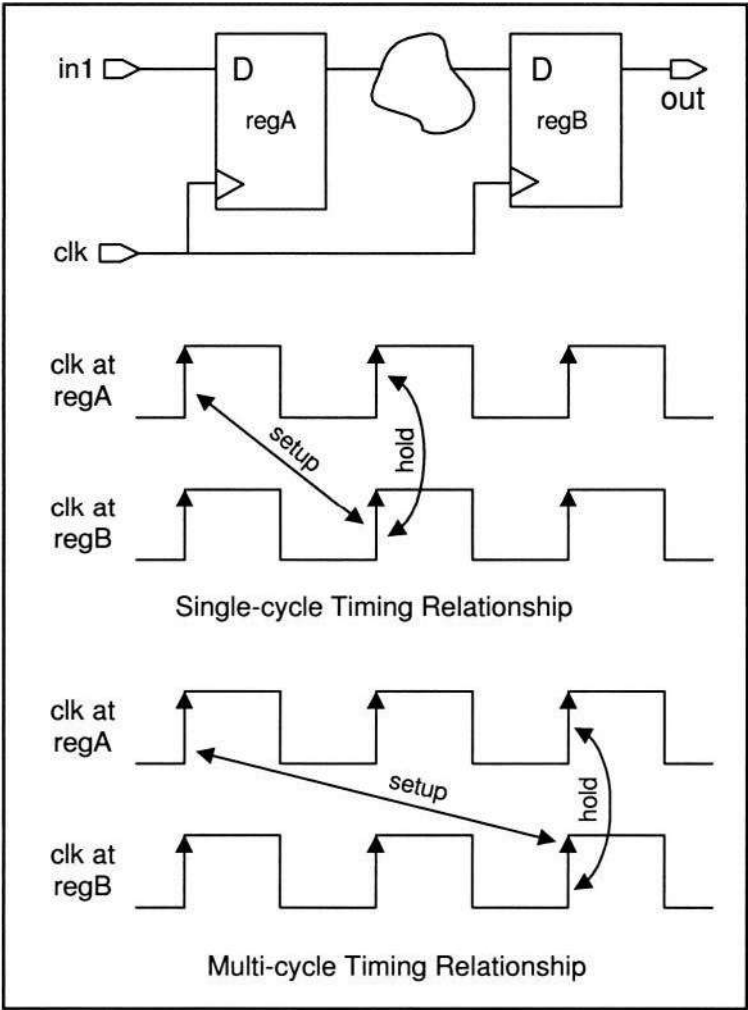


Figure 13-1. Defining Relationship for a Single Clock

Figure 13-1, illustrates the comparison between the single-cycle setup/hold-time relationship and the multicycle setup/hold-time relationship. In the

multicycle definition, a multiplier value of 2 is used to inform PT that the data latching occurs at *regB* after an additional clock pulse. The following command was used:

```
pt_shell> set_multicycle_path 2 -from regA -to regB
```

In case of generated clocks, PT does not automatically determine the relationship between the primary clock and the derived clock, even if the `create_generated_clock` command is used. The single-cycle determination is independent of whether one clock is generated or not. It is based on the smallest interval between the open edge of the first clock to the closing edge of the second clock (in this case generated clock).

For separate clocks with different frequencies, the `set_multicycle_path` command may be used to define the relationship between these clocks. By default, PT uses the most restrictive setup-time and hold-time relationship between these clocks. These may be overridden by using the `set_multicycle_path` command that defines the exact relationship between these clocks.

Figure 13-2 illustrates an example, where a relationship exists between two separate clocks. During the single-cycle timing (default behavior), the setup and hold-time relationship occurs as shown. However, to specify a multicycle path between *regA* and *regB*, the following command is used:

```
pt_shell> set_multicycle_path 2 -setup \  
-from regA/CP -to regB/D
```

The above example uses the multiplier value of 2 to define the setup-time relationship between the two clocks. The `-setup` option is used to define the setup-time relationship. However, this option also effects the hold-time relationship. PT uses a set of rules (explained in detail in PT User Guide) to determine the most restrictive relationship for the hold-time, between the two clocks. Therefore, PT may assume an incorrect hold-time relationship between the two clocks (shown as dotted line in Figure 13-2). To avoid this situation, the hold-time relationship between the two clocks should also be defined. Specification of the hold-time relationship through the

`set_multicycle_path` command is very confusing, therefore not a recommended approach. Designers are advised to use the following command to specify the hold-time relationship between the two flops:

```
pt_shell> set_min_delay 0 --from regA/CP --to regB/D
```

The zero value moves the hold-time relationship from the default value (dotted line in Figure 13-2) to the desired edge (bold line in Figure 13-2).

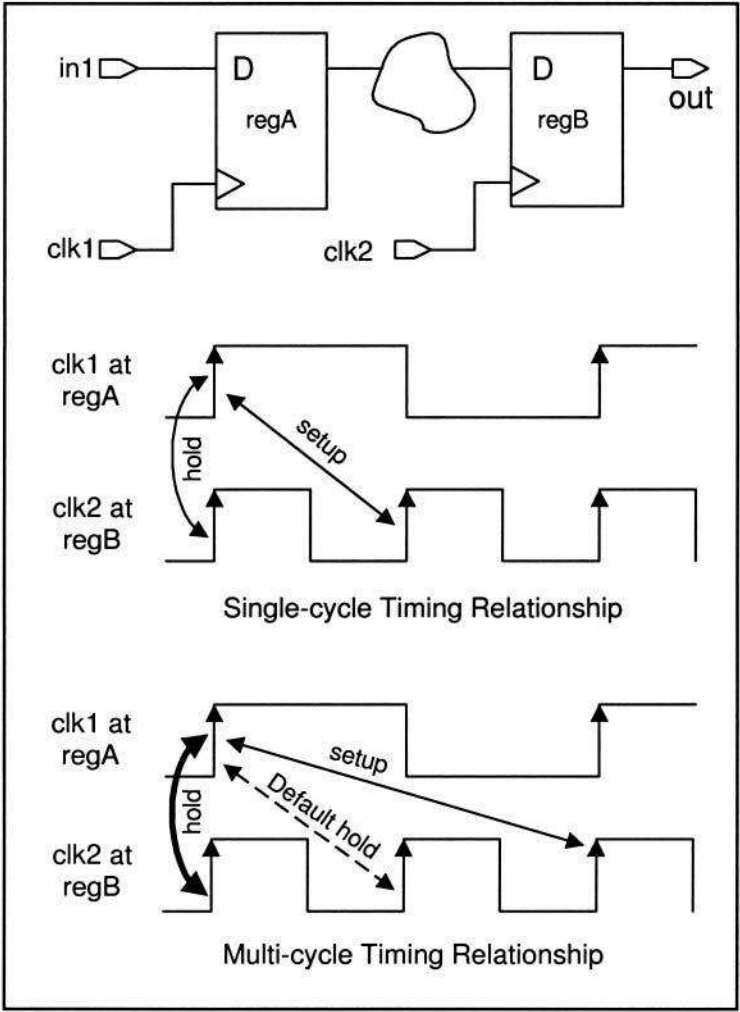


Figure 13-2. Defining Relationship Between Separate Clocks

13.2.2 False Paths

Some designs may contain false timing paths. A false path is identified as a timing path that does not propagate a signal. False paths are created through the following `pt_shell` command:

```
set_false_path -from <from list> -to <to list>
               -through <through list>
```

It must be noted that the above command does not disable the timing arc of any cell, it merely removes the constraints of the identified path. Therefore, if the timing analysis is performed on the false path, an unconstrained timing report is generated.

By default, PT performs STA on all the paths. This results in the generation of timing reports for all the path segments (including the false paths in the design). If the false path segment is failing timing by a large amount then the report may mask the violations of the real timing paths. This of course depends upon the options used for the **report_timing** command.

Lets presume that there are multiple false paths in the design and they are all failing by a large amount during hold-time STA. However, the real timing paths are failing by a small margin. The false paths have not been identified because the user thinks that a large value of **-nworst** and **-max_paths** options will cover all the paths in the design (including the real timing paths), therefore identification of false paths is unnecessary. The user uses the following command to analyze the design:

```
pt_shell> report_timing -from [all_inputs] \
                  -to [all_registers -data_pins] \
                  -nworst 10000 -max_paths 1000 \
                  -delay-type min
```

The above method is certainly a viable approach and may not overly impact the run-time. However, a large value for the **-nworst** and **-max_paths** options (used in the above example) causes PT to generate/display multiple timing reports, covering all the paths in the design, most of which are false paths. Only a selected few timing reports relate to the real timing violations. By using this approach, it becomes tedious to distinguish between the real timing path and the false timing paths. In addition, due to the large amount of timing reports generated, it is easy to mistakenly overlook a real timing path that is violating the timing constraints. To avoid this situation, false path identification is recommended before performing STA.

In addition designers may use the `–through` option to further isolate the false path. It must be noted that the `–through` option significantly impacts the run-time, therefore should be used judiciously and the usage minimized. A better alternative is to disable the timing arc of the cell in the `–through` list, using the `set_disable_timing` command explained later in this chapter.

13.2.2.1 Helpful Hints for Setting False Paths

Timing exceptions impact the run-time. Setting multiple false paths in a design causes PT to slow down even further. Designers inadvertently specify the false paths with no regards to the proper usage, thereby impacting the run-time. The following suggestions are provided to help the designer in properly defining the false paths:

- a) Avoid using wildcard characters when defining false path. Failing to do so may result in PT generating a large number of false paths. For example:

```
pt_shell> set_false_path –from ififo_reg*/CP \
                    –to ofifo_reg*/D
```

In the above case, if the *ififo_reg* and *ofifo_reg* are each part of a 16-bit register bank, PT will generate a large number of unnecessary false paths. Disabling the timing arc of a common cell that is shared by the above paths is a better approach. The timing arc is disabled using the `set_disable_timing` command, explained in the next section.

- b) Avoid using `–through` option for multiple false paths. Try finding a common cell that is shared by a group of identified false paths. Disable the timing arc of this cell through the `set_disable_timing` command.
- c) Do not define false paths for registers belonging to separate asynchronous clock domains. For instance, if there are two asynchronous clocks (say, CLK1 and CLK2) then the following command should be avoided:

```
pt_shell> set_false_path –from [all_registers-clock CLK1] \
                    –to [all_registers –clock CLK2]
```

The above command forces PT to enumerate every register in the design, thereby causing a big impact on the run-time. A superior alternative is to set the false paths on the clocks itself, rather than the registers. Doing this prevents PT from enumerating all the registers in the design, therefore little or no impact on the run-time is observed. This is a preferred and efficient method of defining the asynchronous behavior of two clocks in PT. For example:

```
pt_shell> set_false_path -from [get_clocks CLK1] \
               -to [get_clocks CLK2]

pt_shell> set_false_path -from [get_clocks CLK2] \
               -to [get_clocks CLK1]
```

13.3 Disabling Timing Arcs

PT automatically disables timing paths that cause timing loops, in order to complete the STA on a design. However, designers sometimes find it necessary to disable other timing paths for various reasons, most prevalent being the need for PT to choose the correct timing path at all times. The timing arcs may be disabled by individually disabling the timing arc of a cell, or by performing case analysis on an entire design.

13.3.1 Disabling Timing Arcs Individually

During STA, sometimes it becomes necessary to disable the timing arc of a particular cell, in order to prevent PT from using that arc while calculating the path delay. The need to disable the timing arc arises from the fact that, in order to calculate the delay of a particular cell, PT uses the timing arc that produces the largest delay. This sometimes is undesired and produces false delay values. This is explained in detail in Chapter 4.

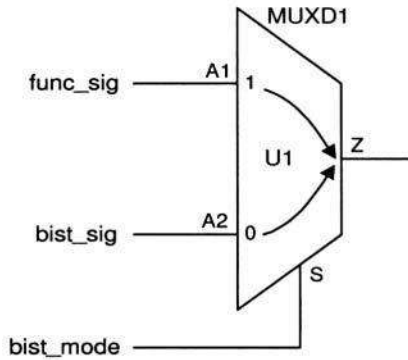


Figure 13-3. Disabling Timing Arcs

Another reason for disabling the timing arc of an individual cell is to prevent PT from choosing the wrong timing path. Figure 13-3 illustrates a case where the control input (*bist_mode*) is used to select between signals, *bist_sig* and *func_sig*, which are inputs to the multiplexer, MUXD1. The *bist_sig* signal is selected to propagate when the *bist_mode* signal is low, while the signal *func_sig* is allowed to pass when the *bist_mode* signal is high. During normal mode (functional mode), the signal *bist_sig* is blocked, while the signal *func_sig* is allowed to propagate. However, during test-mode (for e.g., for testing BIST logic), the *bist_sig* signal is selected to pass through, while blocking the *func_sig* signal. The application of this mux is described in detail in Chapter 8 (Figure 8-5), where it is used to bypass the input signals to the RAM, so that the logic previously shadowed by the RAM, thus unscannable, can be made scannable.

Three timing arcs exist for this cell – from A1 to Z, A2 to Z, and from S to Z. Only the first two arcs are shown in the above figure for the sake of clarity. While performing STA to check the timing in functional mode, unless the user isolates the path using the `-through` option of the `report_timing` command, PT may choose the wrong path (going through A2 to Z), thereby generating a false path delay timing report. Therefore it is prudent that the timing arc be disabled from A2 to Z of the cell MUXD1 (instance name U1)

during functional mode STA. This is performed using the following `pt_shell` command:

```
pt_shell> set_disable_timing -from A2 -to Z {U1}
```

13.3.2 Case Analysis

An alternate solution to the above scenario is to perform case analysis on the design. By setting a logic value to the *bist_mode* signal, all timing arcs related to the *bist_mode* signal are disabled/enabled. In the above case, using the following command disables the timing arc from A2 to Z:

```
pt_shell> set_case_analysis 1 bist_mode
```

The logic 1 value for the *bist_mode* signal forces PT to disable the timing arc from A2 to Z and enables the signal *func_sig* to propagate. By changing this value to 0, the arc from A1 to Z is disabled and the *bist_sig* signal is allowed to propagate.

Although, both the `set_disable_timing` and `set_case_analysis` commands perform the same function of disabling the timing arcs, the case analysis approach is superior, for designs containing many such situations. For instance, a single command is used to analyze the entire design in either the normal mode or the test mode. However, the `set_disable_timing` command is useful for disabling the timing arc of an individual cell, when performing STA.

13.4 Environment and Constraints

Apart from slight syntax differences, the environment and constraints settings for PT are same as that used for DC. The following commands exemplify these settings:

```
pt_shell> set_wire_load_model -name <wire-load model name>
```

```
pt_shell> set_wire_load_mode < top | enclosed | segmented>
```

```
pt_shell> set_operating_conditions <operating conditions name>

pt_shell> set_load 50 [all_outputs]

pt_shell> set_input_delay 10.0 -clock <clock name> [all_inputs]

pt_shell> set_output_delay 10.0 -clock <clock name> [all_outputs]
```

Although, PT provides a multitude of options for the above commands, most designers only use a limited set of options, as shown above. Users are advised to refer to PT User Guide for full details regarding additional options available for each of the above commands.

Since the behavior and function of these commands are same as the commands used for DC, no explanation is given here. The DC commands that are related to each of the above command are explained in detail in Chapter 6.

13.4.1 Operating Conditions – A Dilemma

In general, the design is analyzed for setup-time violations utilizing the worst-case operating conditions, while the best-case operating condition is used to analyze the design for hold-time violations.

The reason for using the worst-case operating conditions to perform setup-time analysis is that the delay values of each cell in the library depict the delays (usually large) of a device operating under the worst-case conditions (maximum temperature, low voltage and other worst-case process parameters). The large delay values cause the data-flow to slow down, which may result in a setup-time failure for a particular flop.

An opposite effect occurs for the data-flow when the design uses the best-case operating conditions for hold-time STA. In this case, the delay values (small) of each cell in the technology library depict the best-case operating conditions (minimum temperature, high voltage and other best-case process parameters). Therefore, the data-flow now encounters less delay for it to

reach its destination, i.e., the data arrives faster than before, which may cause hold-time violations at the input of the register.

By analyzing the design at both corners of the operating conditions, a time-window is created that states – if the device operates within the range defined by both operating conditions, the device will operate successfully.

13.5 Pre-Layout

After successful synthesis, the netlist obtained must be statically analyzed to check for timing violations. The timing violations may consist of either setup and/or hold-time violations.

The design was synthesized with emphasis on maximizing the setup-time, therefore you may encounter very few setup-time violations, if any. However, the hold-time violations will generally occur at this stage. This is due to the data arriving too fast at the input of sequential cells with respect to the clock.

If the design is failing setup-time requirements, then you have no other option but to re-synthesize the design, targeting the violating path for further optimization. This may involve grouping the violating paths or over-constraining the entire sub-block, which had violations. However, if the design is failing hold-time requirements, you may either fix these violations at the pre-layout level, or may postpone this step until after layout. Many designers prefer the latter approach for minor hold-time violations (also used here), since the pre-layout synthesis and timing analysis uses the statistical wire-load models and fixing the hold-time violations at the pre-layout level may result in setup-time violations for the same path, after layout. However, if the wire-load models truly reflect the post-routed delays, then it is prudent to fix the hold-time violations at this stage. In any case, it must be noted that gross hold-time violations should be fixed at the pre-layout level, in order to minimize the number of hold-time fixes, which may result after the layout.

13.5.1 Pre-Layout Clock Specification

In the pre-layout phase, the clock tree information is absent from the netlist. Therefore, it is necessary to estimate the post-route clock-tree delays upfront, during the pre-layout phase in order to perform adequate STA. In addition, the estimated clock transition should also be defined in order to prevent PT from calculating false delays (usually large) for the driven gates. The cause of large delays is usually attributed to the high fanout normally associated with the clock networks. The large fanout leads to slow input transition times computed for the clock driving the endpoint gates, which in turn results in PT computing unusually large delay values for the endpoint gates. To prevent this situation, it is recommended that a fixed clock transition value be specified at the source.

The following commands may be used to define the clock, during the pre-layout phase of the design.

```
pt_shell> create_clock -period 20 -waveform [list 0 10] [list CLK]

pt_shell> set_clock_latency    2.5 [get_clocks CLK]

pt_shell> set_clock_transition 0.2 [get_clocks CLK]

pt_shell> set_clock_uncertainty 1.2 -setup [get_clocks CLK]

pt_shell> set_clock_uncertainty 0.5 -hold [get_clocks CLK]
```

The above commands specify the port CLK as type clock having a period of 20ns, the clock latency as 2.5ns, and a fixed clock transition value of 0.2ns. The clock latency value of 2.5ns signifies that the clock delay from the input port CLK to all the endpoints is fixed at 2.5ns. In addition, the 0.2ns value of the clock transition forces PT to use the 0.2ns value, instead of calculating its own. The clock skew is approximated with 1.2ns specified for the setup-time, and 0.5ns for the hold-time. Using this approach during pre-layout yields a realistic approximation to the post-layout clock network results.

13.5.2 Timing Analysis

The following script gathers all the information provided above and may be used to perform the setup-time STA on a design.

PT script for pre-layout setup-time STA

```
# Define the design and read the netlist only
set active_design <design name>

read_db -netlist_only $active_design.db

# or use the following command to read the Verilog netlist.
# read_verilog $active_design.v

current_design $active_design

set_wire_load_model <wire-load model name>
set_wire_load_mode < top | enclosed | segmented>

set_operating_conditions <worst-case operating conditions>

# Assuming the 50pf load requirement for all outputs
set_load 50.0 [all_outputs]

# Assuming the clock name is CLK with a period of 30ns.
# The latency and transition are frozen to approximate the
# post-routed values.
create_clock -period 30 -waveform [0 15] CLK
set_clock_latency 3.0 [get_clocks CLK]
set_clock_transition 0.2 [get_clocks CLK]
set_clock_uncertainty 1.5 -setup [get_clocks CLK]

# The input and output delay constraint values are assumed
# to be derived from the design specifications.
set_input_delay 15.0-clock CLK [all_inputs]
set_output_delay 10.0-clock CLK [all_outputs]
```

```
# Assuming a Tcl variable TESTMODE has been defined.
# This variable is used to switch between the normal-mode and
# the test-mode for static timing analysis. Case analysis for
# normal-mode is enabled when TESTMODE = 1, while
# case analysis for test-mode is enabled when TESTMODE = 0.
# The bist_mode signal is used from the example illustrated in
# Figure 13-3.
```

```
set TESTMODE [getenv TESTMODE]

if {$TESTMODE== 1} {
    set_case_analysis 1 [get_port bist_mode]
} else {
    set_case_analysis 0 [get_port bist_mode]
}
```

```
# The following command determines the overall health
# of the design.
```

```
report_constraint -all_violators
```

```
# Extensive analysis is performed using the following commands.
```

```
report_timing -to [all_registers -data_pins]
report_timing -to [all_outputs]
```

Also, specification of the startpoint and the endpoint for the `-from` and the `-to` options of the `report_timing` command may be used to target selective paths. In addition, further isolation of the selected path may be achieved by using the `-through` option.

By default, PT performs the maximum delay analysis, therefore specification of the `max` value for the `-delay_type` option of the `report_timing` command is not needed. However, in order to display all timing paths of the design, the `-nworst` and/or `-max_paths` options may be utilized.

As mentioned in the previous chapter, the `report_constraint` command is used to determine the overall health of the design. This command should be initially used to check for DRC violations (`max_transition`,

`max_capacitance`, and `max_fanout` etc.). In addition, this command may also be used to generate a broad spectrum of setup/hold-time timing reports for the entire design. Note that the timing report produced by the `report_constraint` command does not include a full path timing report. It only produces a summary report for all violating paths per endpoint (assuming that the `-all_violators` option is used).

The `report_timing` command is used to analyze the design in more detail. This command produces a timing report that includes the full path from the startpoint to the endpoint. This command is useful for analyzing the failing path segments of the design. For instance, it is possible to narrow down the cause of the failure, by utilizing the `-capacitance` and `-net` options of this command.

13.6 Post-Layout

The post-layout steps involve analyzing the design for timing with actual delays back annotated. These delays are obtained by extracting the layout database. The analysis is performed on the post-routed netlist that contains the clock tree information. Various methods exist for porting the clock tree to DC and PT, and have been explained in detail in Chapter 9. Let us assume that the modified netlist exists in db format.

At this stage, a comprehensive STA should be performed on the design. This involves analyzing the design for both the setup and hold-time requirements. In general, the design will pass timing with ample setup-time, but may fail hold-time requirements. In order to fix the hold-time violations, several methods may be employed. These are explained in Chapter 9. After incorporating the hold-time fixes, the design must be analyzed again to verify the timing of the fixes.

13.6.1 What to Back Annotate?

One of the most frequent questions asked by designers is –What should I back annotate to PT, and in what format?

Chapter 9 discusses various types of layout database extraction and associated formats. Pros and cons of each format are discussed at length. It is recommended that the following types of information be generated from the layout tool for back annotation to PT in order to perform STA:

- d) Net RC delays in SDF format.
- e) Capacitive net loading values in `set_load` format.
- f) Parasitic information for clock and other critical nets in DSPF, RSPF or SPEF file formats.

The following PT commands are used to back annotate the above information:

- **read_sdf:** As the name suggests, this command is used read the SDF file. For example:

```
pt_shell> read_sdf rc_delays.sdf
```

- **source:** PT uses this command to read external files in Tel format. Therefore, this command may be used to back annotate the net capacitances file in `set_load` file format. For example:

```
pt_shell> source capacitance.pt
```

- **read_parasitics:** This command is utilized by PT to back-annotate the parasitics in DSPF, RSPF and SPEF formats. You do not need to specify the format of the file. PT automatically detects it. For example:

```
pt_shell> read_parasitics clock_info.spf
```

13.6.2 Post-Layout Clock Specification

Similar to pre-layout, the post-layout timing analysis uses the same commands, except that this time the clock is propagated through the entire clock network. This is because the clock network now comprises of the clock

tree buffers. Thus the clock latency and skew is dependent on these buffers. Therefore, fixing the clock latency and transition to a specified value is not required for post-route clock specification. The following commands exemplify the post-route clock specification.

```
pt_shell> create_clock -period 20 -waveform [list 0 10] [list CLK]
```

```
pt_shell> set_propagated_clock [get_clocks CLK]
```

As the name suggests, the `set_propagated_clock` command propagates the clock throughout the clock network. Since the clock tree information is now present in the design, the delay, skew, and the transition time of the clock is calculated by PT, from the gates comprising the clock network.

13.6.3 Timing Analysis

Predominantly, the timing of the design is dependent upon clock latency and skew i.e., the clock is the reference for all other signals in the design. It is therefore prudent to perform the clock skew analysis before attempting to analyze the whole design. A useful Tcl script is provided by Synopsys through their on-line support on the web, called SolvNET. You may download this script and run the analysis before proceeding. If the Tcl script is not available, then designers may write their own script, to generate a report for the clock delay starting from the source point of the clock and ending at all the endpoints. The clock skew and total delay may be determined by parsing the generated report.

Although setting the clock uncertainty for post-layout STA is not needed, some designers prefer to specify a small amount of clock uncertainty, in order to produce a robust design.

Let us assume that the clock latency and skew is within limits. The next step is to perform the static timing on the design, in order to check the setup and hold-time violations. The setup-time analysis is similar to that performed for pre-layout, the only difference being the clock specification (propagate the clock) as described before. In addition, during post-route STA, the extracted information from the layout database is back annotated to the design.

The following script illustrates the process of performing post-route setup-time STA on a design. The items in bold reflect the differences between the pre and post-layout timing analysis.

PT script for post-layout setup-time STA

```
# Define the design and read the netlist only
  set active_design <design name>

  read_db -netlist_only $active_design.db

# or use the following command to read the Verilog netlist.
# read_verilog $active_design.v

  current_design $active_design

  set_wire_load_model <wire-load model name>
  set_wire_load_mode < top | enclosed | segmented >

# Use worst-case operating conditions for setup-time analysis
  set_operating_conditions <worst-case operating conditions>

# Assuming the 50pf load requirement for all outputs
  set_load 50.0 [all_outputs]

# Back annotate the worst-case (extracted) layout information.
source capacitance_wrst.pt #actual parasitic capacitances
read_sdf rc_delays_wrst.sdf #actual RC delays
read_parasitics clock_info_wrst.spf #clock network data

# Assuming the clock name is CLK with a period of 30ns.
# The latency and transition are frozen to approximate the
# post-routed values. A small value of clock uncertainty is
# used for the setup-time.
  create_clock -period 30 -waveform [0 15] CLK
  set_propagated_clock [get_clocks CLK]
```



```

set_clock_uncertainty 0.5 -setup [get_clocks CLK]

# The input and output delay constraint values are assumed
# to be derived from the design specifications.
    set_input_delay    15.0 -clock CLK [all_inputs]
    set_output_delay   10.0 -clock CLK [all_outputs]

# Assuming a Tcl variable TESTMODE has been defined.
# This variable is used to switch between the normal-mode and
# the test-mode for static timing analysis. Case analysis for
# normal-mode is enabled when TESTMODE = 1, while
# case analysis for test-mode is enabled when TESTMODE = 0.
# The bist_mode signal is used from the example illustrated in
# Figure 13-3.

set TESTMODE [getenv TESTMODE]

if {$TESTMODE==1} {
    set_case_analysis 1 [get_port bist_mode]
} else {
    set_case_analysis 0 [get_port bist_mode]
}

# The following command determines the overall health
# of the design.
    report_constraint -all_violators

# Extensive analysis is performed using the following commands
    report_timing -to [all_registers -data_pins]
    report_timing -to [all_outputs]

```

As mentioned earlier, the design is analyzed for hold-time violations using the best-case operating conditions. The following script summarizes all the information provided above and may be used to perform the post-route hold-time STA on a design. The items in **bold** reflect the differences between the setup-time and the hold-time analysis.

PT script for post-layout hold-time STA

```
# Define the design and read the netlist only
    set active_design <design name>

    read_db -netlist_only $active_design.db

# or use the following command to read the Verilog netlist.
# read_verilog $active_design.v

    current_design $active_design

    set_wire_load_model <wire-load model name>
    set_wire_load_mode < top | enclosed | segmented >

# Use best-case operating conditions for hold-time analysis
    set_operating_conditions <best-case operating conditions>

# Assuming the 50pf load requirement for all outputs
    set_load 50.0 [all_outputs]

# Back annotate the best-case (extracted) layout information.
    source capacitance_best.pt #actual parasitic capacitances
    read_sdf rc_delays_best.sdf #actual RC delays
    read_parasitics clock_info_best.spf #clock network data

# Assuming the clock name is CLK with a period of 30ns.
# The latency and transition are frozen to approximate the
# post-routed values.
    create_clock -period 30 -waveform [0 15] CLK
    set_propagated_clock [get_clocks CLK]
    set_clock_uncertainty 0.2 -hold [get_clocks CLK]

# The input and output delay constraint values are assumed
# to be derived from the design specifications.
    set_input_delay 15.0-clock CLK [all_3inputs]
    set_output_delay 10.0-clock CLK [all_outputs]
```

```
# Assuming a Tcl variable TESTMODE has been defined.
# This variable is used to switch between the normal-mode and
# the test-mode for static timing analysis. Case analysis for
# normal-mode is enabled when TESTMODE = 1, while
# case analysis for test-mode is enabled when TESTMODE = 0.
# The bist_mode signal is used from the example illustrated in
# Figure 13-3.
```

```
set TESTMODE [getenv TESTMODE]

if {$TESTMODE==1} {
    set_case_analysis 1 [get_port bist_mode]
} else {
    set_case_analysis 0 [get_port bist_mode]
}

# The following command determines the overall health
# of the design.
report_constraint -all_violators

# Extensive analysis is performed using the following commands.
report_timing -to [all_registers -data_pins] \
               -delay_type min

report_timing -to [all_outputs] -delay_type min
```

13.7 Analyzing Reports

The following sub-sections illustrate the timing report generated by the `report_timing` command, both for pre-layout and post-layout analysis. A clock period of 30ns is assumed for the clock named *tck* of an example *tap_controller* design.

13.7.1 Pre-Layout Setup-Time Analysis Report

Example 13.1 illustrates the STA report generated during the pre-layout phase. The ideal setting for the clock is assumed using the pre-layout clock specification commands.

The following command was used to instruct PT to display a timing report for the worst path (maximum delay), starting at the input port *tdi* and ending at the input pin of a flip-flop.

```
pt_shell> report_timing -from tdi -to [all_registers -data_pins]
```

The default settings were used i.e., the `-delay_type` option was not specified, therefore PT performs the setup-time analysis on the design by assuming the `max` setting for the `-delay_type` option. Furthermore, PT uses the default values of `-nworst` and `-max_paths` options. This ensures that the timing report for a single worst path (minimum slack value) is generated. All other paths starting from the *tdi* input port and ending at other flip-flops will have a higher slack value, thus will not be displayed.

Example 13.1

```
*****
```

```
Report : timing
```

```
    -path full
```

```
    -delay max
```

```
    -max_paths 1
```

```
Design : tap_controller
```

```
Version : 1998.08-PT2
```

```
Date    : Tue Nov 17 11:16:18 1998
```

```
*****
```

```
Startpoint: tdi (input port clocked by tck)
```

```
Endpoint: ir_block/ir_reg0
```

```
    (rising edge-triggered flip-flop clocked by tck)
```

```
Path Group: tck
```

```
Path Type: max
```

Point	Incr	Path

clock tck (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
input external delay	15.00	15.00 r
tdi (in)	0.00	15.00 r
pads/tdi (pads)	0.00	15.00 r
pads/tdi_pad/Z (PAD1X)	1.32	16.32 r
pads/tdi_signal (pads)	0.00	16.32 r
ir_block/tdi (ir_block)	0.00	16.32 r
ir_block/U1/Z (AND2D4)	0.28	16.60 r
ir_block/U2/ZN (INV0D2)	0.33	16.93 f
ir_block/U1234/Z (OR2D0)	1.82	18.75 f
ir_block/U156/ZN(NOR3D2)	1.05	19.80 r
ir_block/ir_reg0/D (DFF1X)	0.00	19.80 r
data arrival time		19.80
clock tck (rise edge)	30.00	30.00
clock network delay (ideal)	2.50	32.50
ir_block/ir_reg0/CP (DFF1X)		32.50 r
library setup time	-0.76	31.74
data required time		31.74

data required time		31.74
data arrival time		-19.80

slack (MET)		11.94

It is clear from the above report that the design meets the required setup-time with a slack value of 11.94ns. This means that there is a margin of at least 11.94ns before the setup-time of the endpoint flop is violated.

13.7.2 Pre-Layout Hold-Time Analysis Report

Example 13.2 illustrates the STA report generated during the pre-layout phase. The ideal setting for the clock is assumed using the pre-layout clock specification commands.

In order to perform hold-time STA, the following command was used to instruct PT to display a timing report for a minimum delay path, existing between two flip-flops.

```
pt_shell> report_timing--from [all_registers--clock_pins] \
          --to [all_registers --data_pins] \
          --delay_type min
```

In the above case, the `--delay_type` option was specified with `min` value, thus informing PT to display the best-case timing report. The default values of all other options were maintained.

Example 13.2

```
*****
Report : timing
       _path full
       -delay min
       -max_paths 1
Design : tap_controller
Version : 1998.08-PT2
Date    : Tue Nov 17 11:16:18 1998
*****
```

Startpoint: state_block/st_reg9
 (rising edge-triggered flip-flop clocked by tck)
Endpoint: state_block/bp_reg2
 (rising edge-triggered flip-flop clocked by tck)
Path Group: tck
Path Type: min

Point	Incr	Path

clock tck (rise edge)	0.00	0.00
clock network delay (ideal)	2.50	2.50
state_block/st_reg9/CP (DFF1X)	0.00	2.50 r
state_block/st_reg9/Q (DFF1X)	0.05	2.55 r

state_block/U15/Z(BUFF4X)	0.15	2.70	r
state_block/bp_reg2/D (DFF1X)	0.10	2.80	r
data arrival time		2.80	
clock tck (rise edge)	0.00	0.00	
clock network delay (ideal)	2.50	2.50	
state_block/bp_reg2/CP (DFF1X)		2.50	r
library hold time	0.50	3.00	
data required time		3.00	
data required time		3.00	
data arrival time		-2.80	
slack (VIOLATED)		-0.20	

A negative slack value in the above report implies that the hold-time of the endpoint flop is violated by 0.20ns. This is due to the data arriving too fast with respect to the clock.

To fix the hold-time for the above path, the setup-time analysis should also be performed on the same path in order to find the overall slack margin. Doing this provides a time frame in which the data can be manipulated.

For the above example, if the setup-time slack value is large (say, 10ns) then the data can be delayed by 0.20ns or more (say 1ns), thus providing ample hold-time at the endpoint flop. However, if the setup-time slack value is less (say 0.50ns) then a very narrow margin of 0.30ns (0.50ns – 0.20ns) exists. Delaying the data by an exact amount of 0.20ns will produce the desired results, leaving 0.30ns as the setup-time. However, the minute time window of 0.30ns makes it extremely difficult for designers to fix the timing violation – delay the data just enough, so that it does not violate the setup-time requirements. In this case, the logic may need to be re-synthesized and the violating path targeted for further optimization.

13.7.3 Post-Layout Setup-Time Analysis Report

The same command that is used for pre-layout setup-time STA also performs the post-layout setup-time analysis. However, the report generated is slightly different, in the sense that PT uses asterisks to denote the delays that are back annotated.

Example 13.3 illustrates the post-layout timing report generated by PT to perform the setup-time STA. The same path segment shown in Example 13.1 (the pre-layout setup-time STA) is targeted to demonstrate the differences between the pre-layout and the post-layout timing reports.

Example 13.3

```
*****
Report  : timing
         -path full
         -delay max
         -max_paths 1
Design  : tap_controller
Version : 1998.08-PT2
Date    : Wed Nov 18 12:14:18 1998
*****

Startpoint: tdi (input port clocked by tck)
Endpoint:  ir_block/ir_reg0
          (rising edge-triggered flip-flop clocked by tck)
Path Group: tck
Path Type: max

Point  Incr  Path
-----
clock tck (rise edge)          0.00      0.00
clock network delay (propagated) 0.00      0.00
input external delay           15.00     15.00 r
tdi (in)                       0.00     15.00 r
pads/tdi (pads)                0.00     15.00 r
pads/tdLpad/Z (PAD1X)          1.30     16.30 r
```


pads/tdi_signal (pads)	0.00	16.30 r
ir_block/tdi (ir_block)	0.00	16.30 r
ir_block/UI/Z (AND2D4)	0.22*	16.52 r
ir_block/U2/ZN (INV0D2)	0.24*	16.76 f
ir_block/U1234/Z (OR2D0)	0.56*	17.32 f
ir_block/U156/ZN(NOR3D2)	0.83*	18.15 r
ir_block/ir_reg0/D (DFF1X)	1.03*	19.18 r
data arrival time		19.18
clock tck (rise edge)	30.00	30.00
clock network delay (propagated)	2.00	32.00
ir_block/ir_reg0/CP (DFF1X)		32.00 r
library setup time	-0.76	31.24
data required time		31.24

data required time		31.24
data arrival time		-19.18

slack (MET)		12.06

By comparison, the post-layout timing results improve from a slack value of 11.94 (in Example 13.1) to 12.06. This variation is attributed to the difference between the wire-load models used during pre-layout STA and the actual extracted back-annotated data from the layout. In this case, the wire-load models are slightly pessimistic as compared to the post-routed results.

Another difference between the pre-layout and the post-layout results is the propagation of the clock. In the pre-layout timing report, an ideal clock was assumed. However, during the post-layout STA the clock is propagated, thereby accounting for real delays. This is shown in the above report as “clock network delay (propagated)”.

In the pre-layout phase, an ideal clock network delay of 2.5ns was assumed. The post-route STA results indicate that the clock is actually faster than previously estimated, i.e., the clock network delay value is 2.0ns instead of 2.5ns. This provides an indication to the post-routed clock network delay values. Therefore, the next time (next iteration, maybe) the design is

analyzed in the pre-route phase, the clock network delay value of 2.0ns should be used to provide a closer approximation to the post-routed results.

13.7.4 Post-Layout Hold-Time Analysis Report

The same command that is used for pre-layout hold-time STA also performs the post-layout hold-time analysis. However, the report generated is slightly different, in the sense that PT uses asterisks to denote the delays that are back annotated. In addition, the clock network delay is propagated instead of assuming ideal delays.

Example 13.4 illustrates the post-layout timing report generated by PT to perform the setup-time STA. The same path segment shown in Example 13.2 (the pre-layout hold-time STA) is targeted to demonstrate the differences between the pre-layout and the post-layout timing reports.

Example 13.4

```
*****
Report  : timing
        -path full
        -delay min
        -max_paths 1
Design  : tap_controller
Version : 1998.08-PT2
Date    : Tue Nov 17 11:16:18 1998
*****

Startpoint: state_block/st_reg9
            (rising edge-triggered flip-flop clocked by tck)
Endpoint:  state_block/bp_reg2
            (rising edge-triggered flip-flop clocked by tck)
Path Group: tck
Path Type: min
```

Point	Incr	Path
clock tck (rise edge)	0.00	0.00
clock network delay (propagated)	1.92	1.92
state_block/st_reg9/CP (DFF1X)	0.00	1.92 r
state_block/st_reg9/Q (DFF1X)	0.18	2.10 r
state_block/U15/Z (BUFF4X)	0.04*	2.14 r
state_block/bp_reg2/D (DFF1X)	0.06*	2.20 r
data arrival time		2.20
clock tck (rise edge)	0.00	0.00
clock network delay (propagated)	1.54	1.54
state_block/bp_reg2/CP (DFF1X)		1.54 r
library hold time	0.50	2.04
data required time		2.04
data required time		2.04
data arrival time		-2.20
slack (MET)		0.16

In the above case, the hold-time for the endpoint flop is met with a margin of 0.16ns to spare. Notice the difference in clock latency between the startpoint flop (1.92ns) and the endpoint flop (1.54ns). The difference in latency gives rise to the clock skew. Generally, a small clock skew value is acceptable, however a large clock skew may result in race conditions within the design. The race conditions cause the wrong data to be clocked by the endpoint flop. Therefore, it is advisable to minimize the clock skew in order to avoid such problems.

13.8 Advanced Analysis

This section provides an insight to the designer to perform advanced STA on the design. Depending upon the situation, designers may analyze the design in detail, utilizing the concepts and techniques described in the following sections.

13.8.1 Detailed Timing Report

Often, in a design a path segment may fail setup and/or hold-time and it becomes necessary to analyze the design closely, in order to find the cause of the problem.

Consider the timing report shown in Example 13.2. The hold-time is failing by 0.20ns. In order to find the cause of the problem, the following command was used:

```
pt_shell>report_timing -from state_block/st_reg9/CP \
                    -to state_block/bp_reg2/D      \
                    -delay_type min                 \
                    -nets -capacitance -transition_time
```

In the above command, additional options namely, `-nets`, `-capacitance` and `-transition-time` are used. Although the above command uses all three options concurrently, these options may also be used independently.

The timing report shown in Example 13.5 is identical to the one shown in Example 13.2, except that it uses the above command to produce the timing report that includes additional information on the fanout, load capacitance and the transition time.

Example 13.5

```
*****
Report   : timing
          -path full
          -delay min
          -max_paths 1
Design   : tap_controller
Version  : 1998.08-PT2
Date     : Tue Nov 17 11:16:18 1998
*****
```

Startpoint: state_block/st_reg9
(rising edge-triggered flip-flop clocked by tck)
Endpoint: state_block/bp_reg2
(rising edge-triggered flip-flop clocked by tck)
Path Group: tck
Path Type: min

Point	Fanout	Cap	Trans	Incr	Path
clock tck (rise edge)			0.30	0.00	0.00
clock network delay (ideal)				2.50	2.50
state_block/st_reg9/CP (DFF1X)			0.30	0.00	2.50 r
state_block/st_reg9/Q (DFF1X)			0.12	0.05	2.55 r
state_block/n1234 (net)	2	0.04			
state_block/U15/Z (BUFF4X)			0.32	0.15	2.70 r
state_block/n2345 (net)	8	2.08			
state_block/bp_reg2/D (DFF1X)			0.41	0.10	2.80 r
data arrival time					2.80
clock tck (rise edge)			0.30	0.00	0.00
clock network delay (ideal)				2.50	2.50
state_block/bp_reg2/CP (DFF1X)					2.50 r
library hold time				0.50	3.00
data required time					3.00
data required time					3.00
data arrival time					-2.80
slack (VIOLATED)					-0.20

By analyzing the timing report shown in Example 13.5, it can be seen that the cell U15 (BUFF4X) has a fanout of 8, with a load capacitance of 2.08pf. The computed cell delay is 0.15ns. As stated before, the hold-time violation is fixed by delaying the data with respect to the clock. Therefore, if the drive strength of the cell U15 is reduced from 4X to 1X, it will result in an increased delay value for the cell U15, due to the increase in transition time. This increase in delay value will contribute towards slowing the entire data

path, thus removing the hold-time violation. The resulting timing report is shown in Example 13.6.

Example 13.6

Report : timing
 -path full
 -delay min
 -max_paths 1
 Design : tap_controller
 Version : 1998.08-PT2
 Date : Tue Nov 17 11:16:18 1998

Startpoint: state_block/st_reg9
 (rising edge-triggered flip-flop clocked by tck)
 Endpoint: state_block/bp_reg2
 (rising edge-triggered flip-flop clocked by tck)
 Path Group: tck
 Path Type: min

Point	Fanout	Cap	Trans	Incr	Path
clock tck (rise edge)			0.30	0.00	0.00
clock network delay (ideal)				2.50	2.50
state_block/st_reg9/CP (DFF1X)			0.30	0.00	2.50 r
state_block/st_reg9/Q (DFF1X)			0.12	0.05	2.55 r
state_block/n1234 (net)	2	0.04			
state_block/U15/Z (BUFF1X)			1.24	0.40	2.95 r
state_block/n2345 (net)	8	2.08			
state_block/bp_reg2/D (DFF1X)			1.25	0.10	3.05 r
data arrival time					3.05
clock tck (rise edge)			0.30	0.00	0.00
clock network delay (ideal)				2.50	2.50
state_block/bp_reg2/CP (DFF1X)					2.50 r
library hold time				0.50	3.00

data required time	3.00
data required time	3.00
data arrival time	-3.05
slack (MET)	0.05

In the timing report shown above, by reducing the drive strength of the cell U15 from 4X to 1X, an increase in transition time, and therefore an increase in the incremental delay of the gate is achieved. This impacts the overall data path, which results in a positive slack margin of 0.05ns, thus removing the hold-time violation for the endpoint flop.

13.8.2 Cell Swapping

PT allows the ability to swap cells in the design, as long as the pinout of the existing cell is identical to the pinout of the replacement cell. This capability allows designers to perform what-if scenarios without leaving the `pt_shell` session.

In Example 13.6, the cell BUFF1X (having a lower drive strength and identical pinout to BUFF4X) replaced the cell BUFF4X. However, the process of replacement was not discussed. There are two methods to achieve this. The netlist could be modified manually before performing STA; or designers may use the cell swapping capability of PT to perform the what-if STA scenarios on the violating path segments, before manually modifying the netlist.

Manually modifying the netlist before performing the what-if scenarios is certainly a viable approach. However, it is laborious. For the case shown in Example 13.6, first the `pt_shell` session is terminated, then the netlist modified manually (BUFF4X replaced with BUFF1X), and finally the `pt_shell` session is invoked again to re-analyze the previously violating path segment (from `st_reg9` to `bp_reg2`). If the modifications to the netlist do not produce the desired results (the path segment is still violating timing) then the whole process needs to be repeated. This approach is certainly tedious and wasteful.

A preferred alternative is to use the following command to replace the existing cell with another:

```
pt_shell> swap_cell {U15} [get_lib_cell stdcell_lib/BUFF1X]
```

The above command can be used within the `pt_shell` session to replace the existing cell `BUFF4X` (instantiated as `U15`), with `BUFF1X` (from the “`stdcell_lib`” technology library), and the path segment re-analyzed to view the effect of the swapping. This provides a faster approach to debugging the design and visualizing the effect of cell swapping without terminating the `pt_shell` session.

Note that the cell swapping only occurs inside the PT memory. The physical netlist remains unmodified. If the path segment and the rest of the design passes STA, this modification should be incorporated in the netlist by modifying the netlist manually.

13.8.3 Bottleneck Analysis

Sometimes a design may contain multiple path segments that share a common leaf cell. If these path segments are failing timing then changing the drive strength (sizing it up or down) of the common leaf cell may remove the timing violation for all the path segments. PT provides the capability of identifying a common leaf cell that is shared by multiple violating path segments in a design. This is termed as bottleneck analysis and is performed by using the `report_bottleneck` command.

In Example 13.2, a hold-time violation exists for the path segment starting from `state_block/st_reg9` and ending at `state_block/bp_reg2`. However, the hold-time violation also exists (shown in Example 13.7) for the path segment starting from the same startpoint (`state_block/st_reg9`) but ending at a different endpoint, `state_block/enc_reg0`.

Example 13.7

Report : timing
-path full
-delay min
-max_paths 1
Design : tap_controller
Version : 1998.08-PT2
Date : Tue Nov 17 11:24:10 1998

Startpoint: state_block/st_reg9
(rising edge-triggered flip-flop clocked by tck)
Endpoint: state_block/enc_reg0
(rising edge-triggered flip-flop clocked by tck)
Path Group: tck
Path Type: min

Point	Incr	Path
clock tck (rise edge)	0.00	0.00
clock network delay (ideal)	2.50	2.50
state_block/st_reg9/CP (DFF1X)	0.00	2.50 r
state_block/st_reg9/Q (DFF1X)	0.05	2.55 r
state_block/U15/Z (BUFF4X)	0.15	2.70 r
state_block/enc_reg0/D (DFF1X)	0.07	2.77 r
data arrival time		2.77
clock tck (rise edge)	0.00	0.00
clock network delay (ideal)	2.50	2.50
state_block/enc_reg0/CP (DFF1X)		2.50 r
library hold time	0.50	3.00
data required time		3.00
data required time		3.00
data arrival time		-2.77
slack (VIOLATED)		-0.23

In the timing report shown above, the hold-time violation is 0.23ns. Visual inspection of the two timing reports (Example 13.2 and 13.7) reveal that a single cell BUFF4X (instanted as U15) is common to both path segments (**st_reg9 → bp_reg2**, and **st_reg9 → enc_reg0**). Thus, reducing the drive strength of this cell may eliminate the hold-time violation for both the path segments.

However, this process involves careful visual inspection of all the path segments in the design in an effort to identify the common leaf cell between the startpoint and the endpoint of all the violating path segments. This method can be extremely tedious for a large number of path segments.

The recommended method of identifying a common leaf cell between the startpoint and the endpoint of all the violating path segments is to perform the bottleneck analysis. For the above case (in Example 13.7), the following command was used to identify the common leaf cell shared by the violating path segments.

```
pt_shell> report_bottleneck
```

Example 13.8 illustrates a report that was generated by PT, identifying the cell U15 (BUFF4X) as the common leaf cell shared by the two path segments mentioned above.

Example 13.8

```
*****
Report  : bottleneck
         -cost_type path_count
         -inax_cells 20
         -nworst_paths 100
Design  : tap_controller
Version : 1998.08-PT2
Date    : Tue Nov 17 12:09:09 1998
*****
```

Bottleneck Cost = Number of violating paths through cell

Cell	Reference	Bottleneck Cost
U15	BUFF4X	2.00

Once the cell has been identified, it can be swapped with another in order to fix the timing violation of multiple path segments. Once again, a complete STA should be performed on the entire design. Any required changes (due to cell swapping etc.) should be manually incorporated in the final netlist.

13.8.4 Clock Gating Checks

Usually, low power designs contain clocks that are enabled by the gating logic, only when needed. For such designs, the cell used for gating the clock should be analyzed for setup and hold-time violations, in order to avoid clipping of the clock.

The setup and hold-time requirements may be specified through the `set_clock_gating_check` command explained in Chapter 12. For example:

```
pt_shell> set_clock_gating_check--setup 0.5 --hold 0.02 tck
```

Example 13.9 illustrates the clock gating report that utilized the setup and hold-time requirements specified above for the gated clock, “tck”. The following command was used to generate the report:

```
pt_shell> report_constraint --clock_gating_setup \  
--clock_gating_hold \  
--all_violators
```

Example 13.9

```
*****  
Report : constraint  
--all_violators
```

```

        -path slack_only
        -clock_gating_setup
        -clock_gating_hold
Design   : tap_controller
Version  : 1998.08-PT2
Date     : Tue Nov 17 12:30:07 1998
*****
    
```

clock_gating_setup

Endpoint	Slack

state_block/U1789/A1	-1.02 (VIOLATED)
state_block/U1346/A1	-0.98 (VIOLATED)

clock_gating_hold

Endpoint	Slack

state_block/U1789/A1	-0.10 (VIOLATED)
state_block/U1450/A1	-0.02 (VIOLATED)

It is important to note that the `-all_violators` option should be used in addition to the `-clock_gating_setup` and the `-clock_gating_hold` options. Failure to include the `-all_violators` option will result in a report displaying only the cost function of the failures, instead of the identifying the failed gates.

The `-verbose` option may also be included to display a full path report for the purpose of debugging the cause of the violation, and how it may be corrected. Example 13.10 illustrates one such report that was generated by using the following command:

```

pt_shell> report_constraint -clock_gating_hold \
                        -all_violators          \
                        -verbose
    
```

Example 13.10

```
*****
Report : constraint
        -all_violators
        -path slack_only
        -clock_gating_hold
Design  tap_controller
Version 1998.08-PT2
Date    Tue Nov 17 12:32:10 1998
*****
Startpoint: state_block/tst_reg11
            (rising edge-triggered flip-flop clocked by tck)
Endpoint:  state_block/U1789
            (rising clock gating-check end-point clocked by tck)
Path Group: **clock_gating_default**
Path Type: min
```

Point	Incr	Path
clock tck (rise edge)	0.00	0.00
clock network delay (propagated)	2.25	2.25
state_block/tst_reg11/CP (DFF1X)	0.00	2.25 r
state_block/tst_reg11/Q (DFF1X)	0.05*	2.30 r
state_block/U1789/A2 (AND4X)	0.12*	2.42 r
data arrival time		2.42
clock tck (rise edge)	0.00	0.00
clock network delay (propagated)	2.50	2.50
state_block/U1789/A1 (AND4X)		2.50 r
clock gating hold time	0.02	2.52
data required time		2.52
data required time		2.52
data arrival time		-2.42
slack (VIOLATED)		-0.10

In the above example, the AND0X gate is used to gate the clock, “tck”. Pin A2 of this cell is connected to the enabling signal, whereas the clock drives pin A1 of this cell. As can be seen from the report, the hold-time is being violated by the gating logic. In order to fix the hold-time violation, the cell AND0X may be sized down to slow the data path.

13.9 Chapter Summary

Static timing is key to success, with working silicon as final product. Static timing not only verifies the design for timing, but also checks all the path segments in the design. This chapter covers all the steps necessary to analyze a design comprehensively through static timing analysis.

The chapter started by comparing static timing analysis to the dynamic simulation method, as the tool for timing verification. It was recommended that the former method be used as an alternative to dynamic simulation approach. This was followed by a detailed discussion on timing exceptions, which included multicycle and false paths. Helpful hints were provided to guide the user in choosing the best approach.

A separate section was devoted to disabling the timing arcs of cells and to perform case analysis. The case analysis was recommended over individual disabling of timing arcs, for designs containing many cells with timing arcs that are related to a common signal. An example case of DFT logic was provided as an application of case analysis.

In addition, the process of analyzing designs both for pre-layout and post-layout was covered in detail, which included clock specification and timing analysis.

Finally, a comprehensive section was devoted to timing reports followed by advanced analysis of the timing reports. At each step, example reports were provided and explained in detail.

Appendix A

Appendix A

A New Timing Closure Methodology using Physical Compiler

Presented at Boston SNUG 2001 by:

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ABSTRACT

Business success is increasingly dependent upon the ability of development teams to deliver a shortest “time-to-market” product that meets customer requirements. To mitigate the “time-to-market” pressure, several tools and

techniques have evolved over the past few years. As design complexities increase along with the need for higher speeds, timing convergence is one of the biggest challenges faced by the design teams. This promotes “timing closure” to the forefront of the design cycle.

Due to added design complexity, testing the device has also become one of the major issues. Designers now must not only code the design for accurate functionality, but also code with “test” in mind.

This paper explores new techniques using Physical Compiler that may be used in order to achieve early timing closure. The paper also describes the design-for-test (or DFT) scan insertion and reordering using Physical Compiler.

INTRODUCTION

Timing closure is one of the biggest forces driving the EDA vendors today. Designs ready to be synthesized get stuck in an infinite loop of synthesis and layout in order to converge on timing. The situation is further aggravated by separation of front-end and back-end tools. The separation enforced by this artificial wall between the front-end and the back-end causes the design engineer and the layout engineer “in-effect” to talk two different languages. The bottleneck occurs when the netlist is thrown over the wall to the layout engineer. The layout engineer who is responsible for floorplanning, placement and routing may not have intricate idea of the complexity associated with the design. To make matters even worse, different delay calculators of each tool (synthesis and layout) gives varying results, adding to timing convergence problem.

To alleviate this issue, Synopsys introduced Physical Compiler (or PhyC). This tool sits between the front-end synthesis and the back-end layout tools. The idea is to use the same timing constraints, libraries etc. and perform cell placement of the design. Instead of relying on wire-load models, the tool uses Steiner route as basis for calculating cell delays. This method provides a more accurate delay computation.

This paper will explain a new method of using PhyC (version 2000.11-SP1) in order to achieve an early timing closure. In addition, this paper also details, scan insertion techniques as well as some shortcomings of PhyC.

DESIGN FLOW IN A NUTSHELL

In order to reduce the design cycle starting from RTL coding to final tape-out, we analyzed each step in the design cycle and then devised ways to reduce each step.

The following is a broad outline of a design cycle.

- RTL coding
- RTL verification
- One-pass synthesis with scan insertion
- Floorplanning
- Placement
- Clock tree insertion
- Routing
- Static timing analysis

The design cycle step of RTL coding and simulation requires a lot of patience, designer intervention and tremendous amount of verification. Using sophisticated techniques like formal verification, state-of-the-art simulators running on faster machines with ample memory may certainly reduce cycle time. However, the process of coding the design along with verification is not automatic. The rest of the steps can be automated to reduce the time-to-market.

In the past, Synopsys and other users have provided various guidelines in order to automate the synthesis process. These methods provide a comprehensive solution of synthesizing and timing the design. Although not perfect, due to their dependence on wire-load models, they still provide an adequately optimized netlist.

DFT scan insertion is another major bottleneck that requires designers' intervention. Not only do designers have to code for accurate functionality

they must also comply with DFT rules while coding the design. In the past scan insertion was performed on the synthesized netlist. However, due to increasing complexity of the design and problems in timing convergence, the designs are being made DFT rule friendly in the source RTL itself.

After synthesis the design is thrown over to wall to the layout engineer where it enters the floorplanning step. In general, this stage suffers the most mainly because of the lack of understanding of the design by the layout engineer. If the floorplan is does not provide a good starting point, it can cause routing problems due to congestion. This severely impacts the timing of the design.

KEY TO ACHIEVING EARLY TIMING CLOSURE

In order to reduce time, we identified two areas where design cycle time can be reduced tremendously.

- Make RTL, DFT rule compliant early in the design cycle
- Perform timing driven floorplanning

Making designs DFT rule compliant (reducing step3 in section 2.0)

Recently, Synopsys introduced a much-needed command called “`rtl_drc`”. This command runs on the source RTL and identifies problematic test areas, which may violate DFT rules. Usage of this command greatly simplifies the burden of knowing all the DFT rules by design engineers. With this command, designers can simply code the design for functionality and then run it through `rtl_drc`. If violations occur, they then modify the design immediately before proceeding further. By doing this, any surprises detected at the end (which may have required RTL changes and re-synthesis) are completely eliminated. This benefit of using this method is that it saves a tremendous amount of time by reducing the iterations involved. In other words, the need to change the synthesized netlist in order to make it DFT friendly is removed.

The following is an example of running `rtl_drc`:

```
dc_shell -t> analyze -f verilog my_design.v
dc_shell -t> elaborate my_design
dc_shell -t> create_test_clock -p 100 -w {45 55} my_clk
dc_shell -t> set_test_hold 1 my_test_mode
dc_shell -t> set_signal_type test_async_inverted my_reset
dc_shell -t> rtl_drc
```

After the design is `rtl_drc` clean, you may proceed with full one-pass synthesis and produce a fully optimized netlist, which is scan ready.

Timing driven floorplan (reducing step4 in section 2.0)

The traditional approach to floorplanning consists of defining the chip area with macro (such as RAM/ROMs etc) placement along with routing power and ground straps by hand. Layout engineers create the floorplan based on the following:

1. Connectivity using fly lines in the layout tool
2. Designers suggestions on block/macro placement

Both of these approaches may not yield optimal timing results. The designer's view of the logic blocks connectivity and what the layout tool "sees" can differ dramatically. If the floorplan is not optimal, the poor quality of result in timing can only be realized during post-route static timing analysis. In other words, the design has to be placed, clock tree inserted and then routed before the static timing analysis can be performed. If timing analysis fails, the whole process starts again. This methodology wastes valuable time.

Upon realizing this, we created a better solution that utilizes PhyC's capabilities in placing not only the standard cells, but macros also. We used the following flow to perform this:

1. Defined chip area/aspect ratio and placed the IO pads.
2. Modified the LEF files for macros, so they “look” identical to standard cells.
3. Converted the LEF’s to pdb’s (physical equivalent of logic db’s), using `lef2pdb` conversion utility
4. Ran `physopt` and wrote out the PDEF.
5. Read the PDEF into the layout tool.
6. Made a proper floorplan (power straps, shuffle the macros, added blockages etc.)
7. Wrote out the new PDEF
8. Ran `physopt` using the new PDEF and placed standard cells only (used original pdb’s for macros)

Only a single line in the LEF file for macros needs to be modified in order to perform step 2 as shown below:

Original LEF

Modified LEF

CLASS RING

CLASS CORE

The advantage of using the above flow is that we are using the timing driven placement capabilities of PhyC to place both the standard cells and the macros. Once all the cells have been placed, we moved the macros slightly in order to remove minor overlaps (see note below) while still keeping their relative positions. We then proceeded to add power/ground rings around macros along with blockages and power straps before writing out the final floorplanned PDEF. In other words, we beautified the layout surface after the all cells have been timing driven placed.

Using the above flow, we were not only able to converge on timing in a single iteration, but we also significantly reduced the time needed in floorplanning the chip.

Note: While placing macros, PhyC complains that it cannot place multi-row height cells optimally. It places them, but the placement may not be optimal. We saw some overlaps, however the relative positioning of macros was excellent. Synopsys informed us that this ability of PhyC would be available in the upcoming 2001.08 release.

SCAN CHAIN ORDERING

The advantages of scan chain ordering are enormous. However, usually with every good thing there is also something bad associated with it.

The following are some of the benefits of scan chain reordering:

1. Reduces congestion, thus improves timing
2. Less overall area (net length is dramatically reduced)
3. Improves setup time of functional paths due to decreased flop loading
4. Reduces negative hold-times (mainly a simulation vs. static timing analysis issue)
5. Improves timing due to less overall capacitance
6. Improves power consumption by driving less net capacitance
7. Better clock tree (lower latency and fewer buffers), thus improving timing along with low power consumption.

The disadvantages are:

1. Increases the chance of hold-time violations in scan-path
2. Additional runtime in the design cycle.

Using Clock Skew to reduce Hold Time Violations

With the advantages far outweighing the disadvantages, it seems that scan chain ordering is a must. However, what do we do with the increased hold-time violations? One alternative is to insert a perfectly balanced zero-skew clock tree. Generally, the ASIC community prefers this approach and tries to shoot for the minimum skew clock tree. The hope is that the Clock-to-Q delay of flops + additional RC delay of the wire, will exceed the clock skew, thus preventing hold-time violations. In general this approach works best and eliminates most of the hold-time problems. The remaining hold-time problem areas can be individually targeted by adding delay cells along the scan path after the zero-skew clock tree insertion.

The other alternative to minimizing the hold-time violations is to ignore the clock skew completely. The motto here is -- “SKEW IS GOOD”. There are

two reasons for this. First, skew may actually prevent hold time problems. Secondly, skew reduces the power spikes caused by all clock buffers switching at the same time. Of course, it depends on the design speed on how much skew can be tolerated by the design. Too much skew may lead to setup time issues.

The way the placement tool works is that the flops are generally clustered together in concentric circles (figure 1). When the clock tree is inserted, the clock buffers are placed inside each circle. This arrangement provides a zero clock skew for the flops within a cluster. Zero clock skew for a set of flops within a cluster means that there cannot be any hold-time violations for these flops. In other words, the Clock-to-Q delay of flop itself will prevent any chance of hold-time violations.

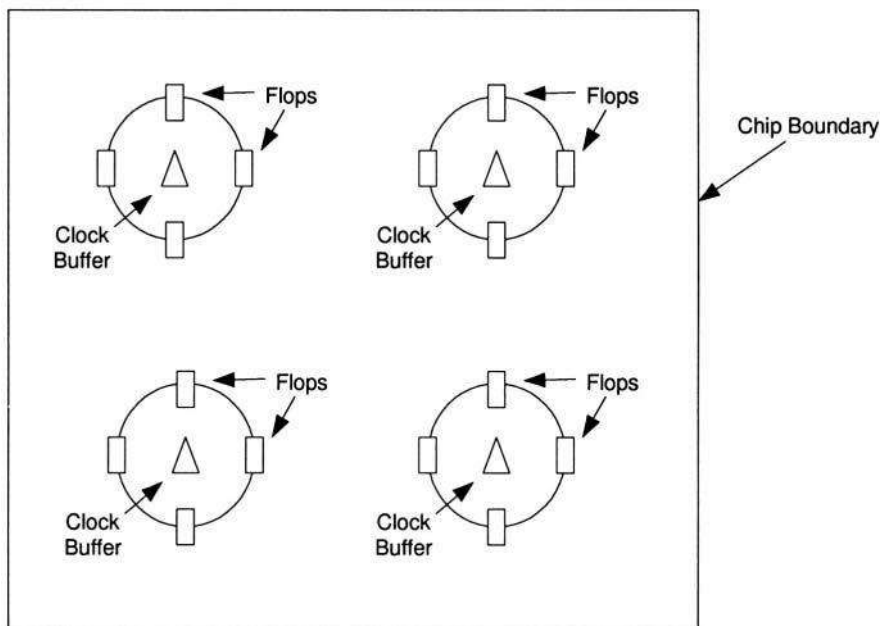


Figure 1: Flop Clusters

The potential of hold-time violations increases when the data signals cross the cluster boundaries. A separate clock buffer is driving another cluster and may lead to clock skew between the first cluster and the second. In this case, two flops will be affected. The last flop in the scan chain belonging to the first cluster, and the first flop in the scan chain of the second cluster. Here, the skew can be positive or it may be negative.

Figure 2 illustrates a case of two registers with their clocks arriving at different times. If the arrival time of clock signal at the RegB, is less than the arrival time of clock signal at RegA (negative skew), there cannot be any hold time issue for RegB. However, if the clock signal at RegB arrives later than the arrival time of clock at RegA (positive skew), there is a potential of hold-time violation.

Skew is unbiased, in other words, it can be both positive and negative. Thus 50% of the flops sitting at cluster boundaries will not have hold-time violations while the other 50% may experience them. Now consider the delay of the data path from one cluster to the next. If the clusters are spread apart, the RC delay itself may be larger than the clock skew between clusters. This placement arrangement itself will prevent the hold-time issues. This leaves a very small percentage of flops that are susceptible to hold-time problems. These can be targeted individually by inserting delay cells in their scan paths to fix the hold-time violations.

Note: If you're going for a minimum skew clock tree, then balanced rise/fall times clock buffers and inverters greatly reduce the skew. If you've gated clock trees, then make sure that the gating logic also has balanced rise/fall times.

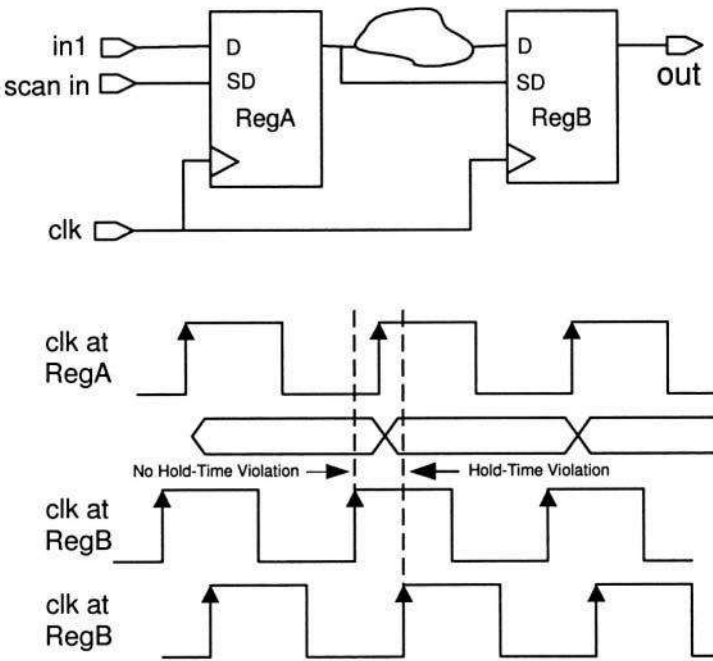


Figure 2: Positive and Negative Clock Skew

insert_scan and Hold-Time

The way insert_scan works is that it tries to find the best scan-chain route in order to minimize hold-time violations. It evaluates the logic being driven by the flop and if it finds any gate like buffer or inverter then it will connect the scan chain to the output of this buffer instead of connecting the Q output directly to the scan-in port of the next flop. This is an immense help in minimizing the hold-time violations. The following switch controls this behavior:

```
dc_shell -t> set test_disable_find_best_scan_out false
```

By setting the above variable to “false” `insert_scan` will try to find the best scan out. Setting it to true will disable this behavior and it will tap the output Q of the flop and link it to the scan-in of the next flop. Default value of this value is “false”.

PHYSICAL COMPILER ISSUES

- When reading a netlist using `read_verilog`, PhyC outputs a lot of *assign* statements in the final verilog netlist. This only happens when performing scan chain ordering, and occurs even after using the hidden variable: `set physopt_fix_multiple_port_nets true`. When reading the db file and performing the same operations (including using the above variable), no *assign* statements are generated. If the verilog netlist is compiled into the db format, PhyC still produces *assign* statements. The only method that does not produce the *assign* statements is when the db is written out with scan insertion operation done by the DFT compiler.
- The heavily promoted “integrated physopt flow” by Synopsys does not work “as advertised”. The idea is to compile the design using one-pass scan synthesis and then run `physopt` with scan order option. `Phyopt` is supposed to perform scan stitching, ordering as well as placement. However, no matter what we did, PhyC complained that the design was not scan ready. Running `check_test` came out clean, but `physopt` kept on complaining that the design was not test-ready. This problem was solved by the Synopsys AE (along with a solv-net article) who told us that we have to set an attribute telling PhyC that the design is test-ready (DUH!!). After using this attribute we were able to run `physopt` successfully.

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REFERENCES

1. Physical Compiler Users Guide
2. DFT Compiler Users Guide

Appendix B

Example Makefile

```
# =====
# General Macros
# =====

MV=\mv-f
RM=\rm-f
DC_SHELL=dc_shell

CLEANUP = $(RM) command.log pt_shell_command.log

# =====
# Modules for Synthesis
# =====

top:
    @make $(SYNDB)/top.db
$(SYNDB)/top.db:      $(SCRIPT)/top.scr    \
                    $(SRC)/top.v          \
                    $(SYNDB)/A.db         \
```

```

$(SYNDB)/B.db
$(DC_SHELL) -f $(SCRIPT)/top.scr | tee top.log
$(MV)  top.log    $(LOG)
$(MV)  top.sv     $(NETLIST)
$(MV)  top.db     $(SYNDB)
$(CLEANUP)

```

A:

```

@make $(SYNDB)/A.db
$(SYNDB)/A.db:      $(SCRIPT)/A.scr    \
                   $(SRC)/A.v         \
                   $(SYNDB)/A.db
$(DC_SHELL) -f $(SCRIPT)/A.scr | tee A.log
$(MV)  A.log      $(LOG)
$(MV)  A.sv       $(NETLIST)
$(MV)  A.db       $(SYNDB)
$(CLEANUP)

```

B:

```

@make $(SYNDB)/B.db
$(SYNDB)/B.db:      $(SCRIPT)/B.scr    \
                   $(SRC)/B.v         \
                   $(SYNDB)/B.db
$(DC_SHELL) -f $(SCRIPT)/B.scr | tee B.log
$(MV)  B.log      $(LOG)
$(MV)  B.sv       $(NETLIST)
$(MV)  B.db       $(SYNDB)
$(CLEANUP)

```

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